4th Applied Active Inference Symposium (Part 1, Nov 13, 2024) ~ Full Upload

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hello and welcome everyone it is November 13th 2024 we're live and kicking off the fourth applied active inference Symposium it will be a great program and we will begin with Carl friston and Lance dcosta discussing scale-free active inference we discussed in live stream 58.1 with Lance last week and in 58.0 with Arun two weeks ago a bit about this work and today Carl thank you so much for joining and presenting on it so looking forward to it and throughout in the live chat please feel free to write any comments thanks again everyone for joining and looking forward to this well thank you Daniel it's a great pleasure to uh kick off the Symposium this year um I'm going to rehearse um a presentation I made at the um active inference Workshop um a few months ago but this time I hope in a slightly more relaxed uh and unpack way so I'm fondly hoping for lots of questions and uh embellishments from the uh from the audience and uh and from Lance um this is a bit of a colloquial presentation it's just going to focus on something specifically that we've been working on for the past few months which is effectively addressing the question how can one scale active inference for high-dimensional real world problems and the solution that we've been pursuing is to adopt a scale-free approach um this work has been written up um and put on archives um however it's also being submitted for uh publication peer reviewed publication to a special collection um in uh memory of Herman hen um and we've equipped the peer reviewed version with a little forwards I just wanted to frame or preempt the scientific part of this presentation just by motivating why why the work of Herman haken is foundational and underwrites many of the um ideas and procedures that underly this scale-free approach to active inference so Herman hen was famous for synergetics um a physics Le approach to self-organization um and in particular the importance of of understanding the coupling between different scales of Dynamics in self organization and if you go to the Wikipedia entry you will read about macroscopic and microscopic variables and how they are coupled to each other in a in the fashion of circular causality um so there's both bottom up and top down causation where the top down causation is sometimes read in terms of um the slaving principle I'm going to present this approach um to harnessing this um notion of a scale-free universe uh in terms of installing that scale-free aspect in the generative models that we bring to the table to solve decision-making to uh simulate or um emulate active inference under discreete State space models so let me start at the beginning um the nature of things and the free energy principle um and this is just to rehearse the fundaments of the free energy principle it all rests upon the notion of a markof blanket where we can separate internal and external states by boundary States or states that constitute the mark of blanket namely the sensory States and the active States and for those people not familiar with um this particular partition of states of any random dynamical system it's just specifying systemically um a system say a brain in terms of its inputs and outputs and um just formalizing the conditional dependencies that Define what is an input to a brain or an agent and what is its output in terms of the sensory and active states that thereby render the system of interest in this instance a brain open to the environment the econ Nish um the heatbath if you're a physicist or a chemist um in virtue of the fact that the inside can influence the outside via the um active States the outputs and the outside can influence inside via the sensory States or the um or the inputs and with this setup we can elaborate um a kind of physics namely uh um basing mechanics that is almost well forly um related um and inherit exactly the same mathematical and functional form as all the other mechanics that we find in physics including quantum mechanics sta statistical mechanics and classical mechanics the the only thing that separates the um the basy mechanics is the fact that we have in mind explicitly this partition between internal States and external States and the uh blanket states that intervene and couple the internal to the external and that basic mechanics is is effectively summarized here in terms of um variational and expected free energy um where we can read the expected free energy as effectively PRI beliefs about the kind of outputs or actions that um in this instance a brain could exert on the external States um and the variational fre keeping implicitly the um the internal States a good model so it's tracking in a probalistic sense the um the Dynamics on the outside and we can combine this to uh create a generalized free energy um that now theologically or or perhaps even anth anthropomorphically we can now read the gradient flows that subtend this phasing mechanics in terms of action perception just by identifying the internal and active states of any agent as autonomous States and noting that they effectively um um flow on the same free energy functionals so that's the basic setup um but what I want to do before um applying the free energy principle in the service of things like active inference is just back up and reflect upon the fundamental role of the Markoff blanket and how that leads to a scal invariant picture or view of Dynamics and self-organization so what I'm going to assume is that ultimately we will assume that basic mechanics um means that the Dynamics of anything that is defined in terms of its boundary or Mark of blank it can be described in terms of Act of inference but before that we have to think about the nature of things um and in particular um if one goes back to the particular physics paper um the nature of the states that constitute things or particles um and there's a bit of a subtlety here because before you can start talking about States you really have to Define what is a state is it the state of something is it the state of a particle um and that presents an issue because you're defining a particle in terms of this particular partition into internal external uh and Boundary or blanket States so there's a um um there's a slight toor not topology but there's a slight um problem in resolving how you define states of things or particles and particles that possess um States and the resolution of that is to adopt um a recursive definition so that the particles constitute a set of microscopic blanket and internal States while the states are a set of macroscopic igen functions igen modes or just simply mixtures of the blanket States so I'm here deliberately um using the notion of microscopic and microscopic States just in uh as a um a nod to synergetics and in particular kind of synergetics that um Herman hen developed and subsequently uh championed so this uh is cartooned or Illustrated um on the right hand side here so we sort of take some states um and um uh and then um apply a a particular partition on the basis of which states are coupled to which other states and the conditional independences that ensue um and then having done that we can then take mixtures zon mixtures of the boundary states that then generate new States um and then we can um effectively by taking those mixtures we effectively reduce the dimensionality so that the higher scale um coar grains the lower scale and then we reduce it and then we just rinse wash and repeat and we do that and every time we sort of pass through this cycle of RG operators um reduction and course graining operators we move to one scale above successively and this is the scale invariant aspect of um that I want to leverage in in the remainder um of this presentation um I use the word RG uh deliberately because um one way of looking at this or articulating this is to use the apparatus of the renormalization group so this is where renormalization gets into the game that you're recursively um applying these operators course graining things um in a way that conserves their Dynamics and that's an easy um um uh that's an easy criteria to meet for us because we know that anything that has a particular petition can be described in terms of a gradient flow on a variational generalized free energy so we have everything that we require in order to apply um the apparatus of the renormalization group so that what that basically does for us once we recognize that particles are composed of states where the states are ion mixtures of particles in this scale invariant fashion gives us a view of any universe that comprises things of things of things um as um inherently uh resolving a chicken and egg problem through scal invariance by application of the renormalization group so here's just another illustration of how that actually works in practice when you write down the equations I'm not going to bother going through the equations this is just used iconically um just to make the observation that as you move upscales um so for example in an evolutionary context this would be the sort of the Dynamics of phenotypes and then we move up to the uh an evolutionary scale um by um effectively Co graining in a way that Herman hen um would exactly prescribe in terms of taking those slow macroscopic variables that decay very very slowly um which are technically the igen mixtures equipped with very small um igen values whose real part approaches zero From Below effectively saying that these patterns Decay very very very slowly and they have a kind of persistance that lends higher and higher scales a temporal dilation so things change more and more slowly as you increasingly uh C grain things and that this falls out of the maths um when you simply apply a particular partition and put that partition in um as the um the course grading operator in this uh in these in this scale free uh formalism so in short basy mechanics entail a generative model of external Dynamics but we've just seen that just by defining things in terms of particles that have States and defining States in terms of the I functions of particles that external Dynamics are inevitably scale invariant it can be no other way and therefore the generative model must be scale invariant or when expressed as a graphical model scale free and I just slip that in as a um a nod to the fact that people often talk about scale freeness um generally speaking when people apply the notion of scale invariance to anything if that thing is a graph then you can call it scale free so a scale free thing is a um is a graph that has has this scal inv variance crucially in both time and in space because we are dealing with dynamical systems and we're doing this um Dimension reduction or course graining um using these um igen functions that necessarily pick out these slow macroscopic sometimes called stable modes they're stable because they they don't Decay or dissipate um almost infinitely quickly so they're stable in the sense that they Decay very very slowly so how can we harness that observation that truism at least truism under the world as described by hen um in um the construction of plausible generative models and in particular um generalized Markoff decision or partially observed Markoff decision processes um well what we can do is we can just look at the different kinds of depth um and ask at what point would we um be able to leverage um this renormalizing aspect of the cause effect architecture of Any

Given world that we're trying to navigate and predict so here's uh the normal um depiction of a um a generalized um markof decision process and by generalized what I mean is that there's um an explicit representation of not just the states of things that are generating outputs or things that can be measured or observed on the um on the sensory sector of a marov blanket but also their Dynamics in terms of the paths that um specify the transitions between states as we know normally encoded in terms of a transition tensor B here um so just by putting in the paths as explicit random variables that accompany for each given Factor um the hidden States we have a generalized kind of uh Markoff decision process uh encoded in the usual way in this in um in terms of tensors um that themselves can be parameterized as dis distributions so um that parameterization equips these kinds of models with a parametric depth in the sense that if something is deep parametrically that means that the parameters of various probability distributions are themselves random variables that have their own probability distributions so for example the likelihood mapping that Maps from sensory states to outcomes here uh usually encoded by an a tensor here mapping from latent states to observations is itself encoded in terms of um derish parameters um blending this uh kind of model A parametric depth and um one can ask what the implications in that are well the implications are another kind of depth in terms of um you now have to op optimize or the B apply basic mechanics to the latent States the hidden States and that would be inference and then you also have to um consider at a slightly slow time scale the optimization of the derish parameters and that would correspond to learning and so on um if I wanted to go to the structural learning of the model in of it in and of itself we also have um implicitly a temporal depth in the sense that we're going to be raring out into the future and that there are dynamics here um and this Temple depth is generalized in in virtue of having um these explicit uh path variables that we're going to uh exploit later on we can also have hierarchical depth in the sense that we can put um compose many markof decision processes um um in the in the sense that the outputs of one process now constitute the inputs or the empirical prior the top- down constraints the inductive biases on the Dynamics and states of the level below and the Dynamics are um um the kind of inputs or sorry outputs of the uh upper level that provide empirical PES on the paths and the initial conditions are supplied or mediated by a mapping between higher levels slower States um and the uh initial conditions or initial states of the lower uh of the lower level by this DET tensor here that you can lump together um in terms of sort of high you can lump together the D and the E tensors they play the role effectively of the a tensors in sort of um mapping from one level to the uh the next level there's also factorial depth which many people will be familiar with in terms of um for any one of these um lump States here there may be many many different factors that have an a certain Independence structure such that you have interactions between factors in determining outcomes um you can look at this in terms of uh a generative model when equipped with factoral depth effectively entangles latent states to generate um outputs or outcomes which themselves can be prior constraints on the states and dynamics of of the level below such that the inversion of these models effectively disentangles the data or the content that is being observed to explain it or um in terms of a disentangled representation which is the um the latent State and dynamics of the generative model so what we have here then is um a generalized discrete State space model with Paths of as random variables um just to um return to this um fundamental aspect that as one Ascend scales in real life in um you know in terms of uh the external States things slow down in virtue of the um the the the the synergetic um mapping from fast microscopic States uh to um slow macroscopic States um the same thing can now be implemented in these um Markoff decision processes that have a hierarchical depth um and the way that is done is effectively to discretize or quantize time such that there there are more updates at any given level than any um than the level Above So for every um update at this level there are uh say three updates or two updates at the level below and that um affords a temporal scaling um in accord with the scal inv variance um implied by the synergetic view of self-organization I've tried to cartoon that here in terms of if you imagine lots of these um little markof decision processes populating this uh circumference here and that as we go deeper and deeper and deeper into a model with hierarchical depth as we consider updates in Universal or clock time there are more updates at the peripheral level exposed say to the know the um external states of entire brain um relative to the ative to the updates that as you get deeper and deeper and deeper in into the model so that's where the temporal depth um becomes if you like intimately tied to the hierarchical depth the higher the hierarchy the slower the updates which is why um we generally parameterize the initial conditions um or generate the initial conditions and the initial path from the States above and that path can pursue um um a trajectory over multiple time steps until a certain um Horizon uh and which case it then sends messages back to the level above likelihood messages um that then cause belief updating um and the subsequent Step at the level above and then generating priors about the subsequent path the next PATH at the level below and then so on recursively uh to the depth of the model um so that's the time bit what about the spatial bit what about the S spatial core scening the lumping together of various States um and of course this depends upon the particular partition so in the General Physics formulation You' be looking at uh

Markoff blankets and Markoff blankets in Practical applications one can take a shortcut um and that shortcut effective rests upon um the simplifying assumption for most systems that are um composed of um coupled um coupled uh dynamical systems um the interactions are usually local so if we um just think about um some um um system that um can be described in terms of local interactions so for example um in a um a ukan world um that comprises some ma massive objects that bounce around and if we ignore gravity for the moment then they're only going to influence each other or depend upon each other when they're touching when they're um when they're proximate um effectively what that means is the the the mark the particular partition that defines all the Markoff blankets in um in this particular State space um is composed of lots of local little tiles um so what we can do is we can use a um a device um very prevalent or indeed ubiquitous uh applications of the renormalization group which is called a uh a block or a spin operator that just groups together local tiles or quadrants of a say two-dimensional arrangement of tile of of States so I've Illustrated that here just by um in terms of um How We Do the spatial core graining in these um what will refer to as a renormalizing generative uh model with spin block uh Transformations um in terms of um this little um um this hidden state or hidden factor with its states and paths here um is responsible for generating predictions about the Dynamics of these two um factors um at the um at the lower level um and in so doing what we are effectively doing is successively grouping together sets of sets of sets of sets so that any set or group at one level is accountable for or trying to generate the Dynamics of a small local group at the at the lower level um this is quite fundamental that's kind of Architecture is quite fundamental because what that means is anything at the lower level only has one parent and if it only has one parent and if remember the mark of blanket is constituted by the the the um the parents and the children the parents of the children because there are no co-parents there are no parents of the children and what this means practically is that we can now ignore dependences between or effectively resolve or dissolve any dependences between the factors at any given level because each factor is only responsible for its children and it doesn't share any children in virtue of this um particular spin block transformation so this leads to um a very efficient kind of generative model um or certainly an efficient kind of inversion of the gened model um where we've got all these converging streams but that the streams don't have to um uh talk to each other until at the point that they come together through these blocking or spin block um um uh Transformations so at some point all the tiles or the groups of the lowest level will actually come together at in um uh at the top or very deep in the model at which point we're now encoding quite long trajectories technically 2 to the^ of the depth minus one um in terms of updates um but at the lower level um there are no interactions between the blocks as we Ascend them and this is just to um remind you we can treat these d and e the um the initial States and initial paths effectively as likelihoods that couple um one scale or level to the scale below so how uh with if that's the kind of model that we aspire to because we think that's appropriate for any world that shows this scale uh invariant aspect in space and in time or in state space and in time um how are we going to build one of these things um and this is the sort of second um if you like um thing that we have been working on um um because it's going to be very difficult to specify by hand these um scale-free or re normalizing gered models we really want the model to learn itself so basically how does one build a renormalizing generative model and what we're going to do we're going to use very fast structure learning and this particular kind of structure learning is necessarily recursive and uh in the sense that we're going to build these models through the recursive application of these um RG operators or renormalizing operators so what's fast structur leing so fast structureal basically equips the model with unique states and transitions as they are encountered so to put that simply what we're going to do is grow a model from scratch from nothing and how are we going to do that well we're effectively going to take one observation and say okay that was generated by some latent State and then we're going to take the next observation um and say okay I haven't seen that particular observation before um so I'm going to induce a second latent State and furthermore I've seen these two observations in sequence therefore I'm going to um encode that with a transition from the first to the second state and then I just repeat I take the the third one and if I haven't seen that before I induce a third latent State and a third a second transition um accumulating all the Unseen or unique instances of um um of observations in a likelihood M mapping that just grows and grows and grows until we have seen something before so we don't induce a new state or implicitly column of the likelihood Matrix if we've seen uh the previous state that's just a um um an instance of something that has been seen before um it may well now come along with a path so the next state may not be the same as a the subsequent State following the previously seen instance of this state and therefore I'm going to induce A New Path and slowly grow my B tenses and my probability transition matrices until I can um basically encode all the dynamical structure of the effectively the training sequence of observations summarized in a a maximally efficient way and I mean maximally efficient in a technical sense um in the sense that the implicit mapping that we are growing from from scratch um has the highest Mutual information between the latent States and the observations that are at hand um the way that you can think of that in terms of um free energies that the expected free energy um is just the mutual information in the absence of any expected cost cost so the expected free energy is just the mutual information of any mapping um um um plus um or minus an expected Cost U where cost is defined the usual way in terms of constraints or prior preferences so one can actually look at this uh fast structure learning um as a particular kind of active inference but not about um it's not active inference Over States or Active Learning over parameters it's active selection of the right model in terms of the size of the likelihood tenses um and the transition uh tensors um and the criteria for whether we accept a new Laten state or a new transition is simply does it optimize or improve the expected free energy namely does it preserve or conserve or maximize the the the mutual information so we can actually look at this structure learning as just another instance of active inference but here deployed in terms of um um in terms of basic model selection or so some people call this uh structure learning that has exactly the same rules as um planning um and um uh and um accepting um parameter updates provided they they maximize expected free energy as well um the application of this kind of uh uh fast structure learning to um these renormalizing models um looks complicated here's the algorithmic description of it um again please don't worry about the details here it it's relatively simple you all we're doing basically is ingesting um observations but we're just doing it for local groups and um and then we are combining the initial conditions and the paths into new outcomes for the next level um and so because we are doing this for local groups the number of groups just get smaller and smaller and smaller as you as you get higher and higher and higher but obviously the number of combinations of paths and state initial States increases as you get deeper and deeper into the model and you may be asking well you know how do you ban that you know how can you um possibly assimilate all the data without having an extremely large number of latent States very deep in the model that are now encoding effectively evolving patterns or paths over quite a large number of time points into the future well it's trivial to do that because the size of the highest level I'll call them generalized states that basically generate paths and initial conditions for the lower levels um is upper bounded by the number of elements in your training set um so that you can control and upper bound the size of these renormalizing operators or likelihood mappings and transition um um tensors simply by giving it canonical training data the kind of data that you want this um agent to generate or this um um agent to recognize and thereby predict um so here's a a particular example of that um what we've done here is just ask can we very very quickly learn a Jed model that will generate a um a simple movie and this movie is simple because um it it begins and ends in the same state so it just Loops over and over and over again um so we only have to supply one Loop um to this fast structure learning so that the model builds itself and can then be used in generative mode to generate the movie time and time and time again um so here's one depiction of um the results of the learning the the F structure learning um and subsequent infants when exposed to um this kind of image so the course graining here at the very bottom in in fact um uses um um a slightly finessed version of the spin block Operator by um harvesting um local igen modes or IG uh IG images um of little patches of of the video to get it into a reduced or coar grained rep presentation that then is just passed up using these spin block operators and in this instance we just need uh two levels to the model to uh have the highest level see the entire image and then if we present the image um um after discretization using the singular Val decomposition or the sequences of images to a model it learns the model and then we can generate um we can generate um little movies simply by moving through these paths or episodes I use the word episode um because remember these generalized states of the highest level encode paths or trajectories into over um um 2 to the N minus one time steps in the future when n is the um is is the uh the depth of of the model and these episodes themselves have Dynamics so we just go through a little Loop here go back to the beginning and thereby generate um this image at the bottom what else can we do with these kinds of models well one thing we can do is um present not the entire data but just u a little prompt um so in this example what we've done is actually show it just the upper right quadrant of the movie and the and see what this um this generative model um believes is going on in terms of the um the predictive posterior in output space in image space in this instance now because during training it's only L to recognize a dove flapping her wings um that's all it can believe come is the cause of its Sensations even if those Sensations are very partial so basically what we have here is a kind of pattern completion that inherits from the fact that this generative model only knows about latent causes that constitute an entire Dove flapping her wings so um even though this is what the dove is actually um exposed to it's sensing this is the predicted after a couple of frames it very quickly um is able to predict um what it would what it should be seeing or what it would have seen had it been given the entire uh data and another way of thinking about this um remarkable ability to sort of fill in the gaps um is from the point of view of compression so just a little digression here um a lot of the basing mechan mechanics and um variational inference that uh ensues from variation mechanics can also be likened to um the to um the same fundaments underwrite um um efficient information transfer so we've talked about maximizing Mutual information in the context of uh um algorithmic complexity and um the associated Universal computation um then what the objective function looks like is basically how much can I compress this content um to represent it in the most efficient way so you can look at this past structure learning under these renormalizing or scale-free generative models as the most efficient way of compressing whatever your data you've been exposed to um and because we've compressed it in such a way that it can only now recognize a dove flapping her wings um then it can if you like fill in the gaps in a very efficient way when it's given partial or noisy data and I use the word noisy here because you can regard the missing three quadrants here as extremely imprecise data data that has extremely low signal to noise for example so um here's another example in the example we we um took a um effectively um a movie or a damic or um a a system that had a periodic orbit um um which was each flap of the Wings will this work for um a periodic orbits and and generally for um stochastic um uh chaos um yes it works it works relatively efficiently so this example um we um used training data that was generated um um through creating images of a white ball that moved with a chaotic trajectory governed by a uh a len system so this is the kind of system that governs or is used to describe uh chaotic events um in meteorological systems or in fluid convection um so by applying exactly the same procedures as in Pre in the previous example um we now uh generating this as a three-level um um hierarchal or renormalizing model um that now accommodates the um stochastic or Randomness induced by not only random fluctuations on the Dynamics of this stochastic uh random dynamical system uh but also if there were any determin IC aspects the deterministic uh um um chaos that ensues from exponential Divergence of trajectories that now has been summarized stochastically in terms of probalistic um uh transition matrices here which means that um we can having compressed the training data we can now withhold not a quadrant but just simply withhold the data to stop the data uh but the model will keep on generating um what it predicts would have happened given the statistics of the Dynamics that have been installed in this renormalizing generative model um and what we've done here is um um for the first um few hundred time bins at the lowest level we basically rendered the images or the input very very imprecise so that everything is governed now or everything is generated from this top- down Prize or posterior um um posterior um predicted posteriors that inherit from from precise prior because there's no precise likelihood um and then we've actually uh removed the U um removed the um imprecise input completely and just allowed the um the agent to generate its own beliefs about what's uh about what's going on um so here this is fairly precise in the first instance and then we remove completely the um stimulus but it still uh imagines or generates its or predicts um what's going on in terms of this chaotic um Dynamics um and then um with a degree of uncertainty then um continues to then generate um the um the Dynamics associated with this kind of stochastic chaos here's a preter example um exactly the same uh technology and approach um but here explicitly um talking about um video compression um so we literally just presented a video of naturalistic Dynamics in this instance a little bird a robin um feeding um itself um and um compressed that using these this renormalizing um generative model um such that we can again simply remove stimuli at certain points um in the video um and yet the uh the in in the agent's mind nothing has really happened uh because all it knows is that if it was like this um previously then this must have happened um and so on through this in this instance um orbit that doesn't have a closure it's just a trajectory through from the beginning to the end of the sequence and here's the posterior prediction after just being exposed to a couple of frames right at the beginning um in the first um six uh time frames of um of about 12 sorry of um 18 um this example moves applies exactly the same um approach um as previously but now we're not talking about um ingesting or learning or compressing images we're talking now about compressing or ingesting sand files um and that is if you like a simple or limiting case of this two-dimensional uh compression that we talked about before so in this instance now what we're doing is compressing um one-dimensional frequency summaries of a particular um wave file or uh um file um that um describes flu situations and the intensity of uh of a sound that then is um summarized in terms of a continuous wavelet transform um and then the succession of different trajectories in this um time frequency space is then ingested or assimilated or learned by the generative model um now unfortunately you can't hear this um so I'm not going to bother playing it but um it um this is what the agent was trained on it's basically Jaz piano um and these are the um uing predictions in the absence of any inputs it's been used to generate music um using um the stochastic transitions um through various bars of Music um that it has learned um returning to one or other bars of the munic of the music when it reaches uh the end of the orbit to generate something that um sounds at least Jazzy if not oh started it but you can't hear that so I'm not going to pursue that so to summarize what we've done so far um um we've gone beyond generative AI in the sense of um um having a a true gened model um uh that entails active inference in agency um and in the setting of these um well sorry what we are going to do now is go beyond just simply generating uh content um having been uh trained on the basis of some training data um and we're going to now consider um agency by putting action into the mix um and in the setting of um the simplified setting of optimal State action policies where we can effectively um ignore the imperative to maximize expected Information Gain or um um explore in the sense of uh responding to epistemic affordances um we can um do one of two things we can either use some Universal function approximator say deep learning to optimize the mapping from sensory to active states to maximize the um expected cost or the expected utility part the expected free um or we can use active inference to realize predictions under a normalizing generative model of rewarded events so what I'm doing here is deliberately setting up a dichotomy that tries to distinguish the active inference approach to purposeful Behavior where the purpose is provided purely by the prior preferences or the um um or the the uh the constraints usually encoded in the c um uh tensor um against a sort of reinforcement learning approach and this is a rather busy slide but um I think it's probably worth just um uh briefly uh rehearsing the distinctions between active inference and reinforcement learning so the you know the story with respect to action as a gradient flow on free energy which basically reduces to um pick those actions that maximize the predictive accuracy um in the sense that if you look at the expression for the variation free energy the only thing that you can change um is the accuracy in the sense that free energy can be written as accuracy minus complexity minus accuracy and the complexity um is does not depend upon action whereas the accuracy does so that picture basically says um I can describe behavior in terms of a generative model that um is optimized through um perception uh in the usual way um cast in terms of say variational inference um that basically generates predictions about what should happen next and then I can pick my actions to fulfill those predictions in continuous state for formulations this would be start minimizing the prediction error usually the proceptive domain but you can generalize this to any any consequence of action I'm just going to pick that action that brings about or is most likely to bring about what I predict will happen next and what I predict will happen next um is optimized through perception through variational inference based upon my learned and selected um generative model so this is very similar to sort of um model predictive control um um and neurobiology would could be cast in terms of um um perceptual control theory um which puts action in the game basically of realizing evidencing um predicted States so I've tried to summarize that here just in terms of um you know our markof blanket where we're optimizing our beliefs about the world um noting that planning in this instance so con um planning as inference uh is all about the paths it does not have to be about action it's just about how does this world unfold um and from the point of view of these scale-free generative models we're really thinking about how do episodes follow each other which episode is going to follow this going to U um this episode and we can say well you know you are in control of that um you know imagine a um a benevolent World in which everything happens according to um your beliefs about how to be an expert in this particular world and the agent can plan a series of episodes generating predictions all the way down to the bottom which then slavishly action come fulfill just by fulfilling those predictions um so one could in some sense think of this in terms of control as inference at the bottom level in the spirit of model predictive control and planning as inference on the top level so belt and braces in terms of realizing active inference in this particular example where I repeat um we are really just worried about um State action policies um why are we worried about State action policies well these are the kind of policies that reward learning schemes um are aimed at um and have a similar architecture um but um to my mind a slightly more um over engineered architecture first of all you actually need explicit inputs which are rewards um there are some proxy for um the constraints uh you know um um under which um the basic mechanics would normally operate um but more importantly you're you're you're basically sending um your inference scheme is basically about selecting the right policy I've used Q learning here um um to maximize the discounted reward um so similarities and differences but we're going to focus on uh active inference um uh in this in the particular setting of these renormalizing models where repeat planning and thinking or imagining a future can proceed at the very highest level that has no notion of action they're just patterns that unfold they may or not be caused by me I don't know I don't care um but the action at the lowest level the reflexes as it were um do um are in a position to make my um post predictive posters come true simply by picking those actions that realize the predicted outcomes um so how are we going to leverage that kind of merely reflexive active inference um in the context of building or learning from scratch um renormalizing or scale free generative models that would be apt for doing something implementing a state action policy um and I'm going to um appeal here again to this notion of um active selection as one way of um looking at um action which can always be at least phys mathematically expressed in terms of um of um doing something if it minimizes expected free energy um and in this instance um what we can do is actually apply it not to the selection of the model but the selection of the training data so if we just um uh apply the the dictat that you only select something if it um um minimizes expected free energy that includes um the expected cost what that basically means is I'm only going to select those data features or those training sequences that don't come along with a high cost namely those that are rewarded um and that's what we're going to do here we're basically going to um learn to play um a very simplified um game of pong where where this ball bounces around inside a circle um and we can move the bat um in a way to uh return the ball so that we can avoid penalties when the ball hits the upper boundary um and secure a reward um whenever um there's contact of the ball and the bat and how can we learn this kind of dynamic these episode in exactly the same way we learned the video of the robin or the the um the dove flapping her wings well what we can do is we can just um select um instances of random play that conform with um our um dictat that we cannot um include those costly events and so what we've done here is just generate training data training sequences using random play but omitted anything that does not that contains a costly outcome and furthermore just to make things more efficient we've restarted the random play after a reward is secured and then just use the subsequent play if it actually attained a subsequent reward so basically chaining together bootstrapping together or splicing together sequences of rewarded uh play um and then use that to um uh to um or submitting that for fast structure learning and now we have a generative model of expert play effectively or a generative model that has that entails a state action policy that can be realized through um um reflexive active inflence so active data selection um and um um sort of cartoon that here in terms of applying a certain kind of Maxwell's demon that only lets through non-costly episodes um basically yes admit this sequence for fast structure learning if there is a reward if not ignore this ignore this sequence um and this effectively learns um or furnishes a generative model of events that lead to a reward and nothing else so now this agent can only recognize expert play therefore all of its predictions have to be uh the predictions of an expert player and because action is fulfilling the predictions it will look as if it's an it is an expert player and here are the um the Learned trajectories um at the highest level of different episodes that we'll see uh um shown in in an alternative format uh in the next slide so here's the initial starting episodes little events here uh the red dots correspond to reward so you know first hit second hit and then it falls into a nice little orbit and this com completes um this orbit indefinitely by in it's the way that it has learned so remember it's only been trained on random data so this particular style of play is unique to this agent but it is sufficient now to play with the 100% um um uh performance um simply because it's now going round and round its orbit very much like the wing flapping her dug flapping her wings um these are just different ways of portraying the same um the same Dynamics and the same the things that have been learned um the first eight frames and then subsequent frames um of expert play um this is the usual format showing the posteriors at different levels the posteriors over the states and the posteriors or predicted posteriors over the paths and then rewarded play as a function of the uh free energy at different levels here illustrated by the by the red dots every time there's a hit up until 512 time frames here and here's this rather long orbit through 50 episodes um each four uh time steps in in length that subtends this kind of play this is an interesting one and speaks to the fact that we've actually switched on inductive inference at the highest level um so what does that mean well what we've done is just um um basically take each of these highest level States worked out the sequence that it encodes as by um um Computing or projecting down through the model and if that sequence includes a hit um we said that's an intended state that's the kind of state that I want to work towards through the successive um four um time step epochs um and if I want to be at some point in the future in one of these intended States in this instance rewarded States or episodes um what states must I avoid in order to preserve or conserve the possibility of getting in or getting to one of these intended States um basically that's the technology behind inductive inference and it's sort of summarized here in terms of the latent States of uh you know 50 or so of these latent States in terms of how many time steps would it would it what's the shortest number of time steps until the intended State um and basically um we um as we move from state to state to state we want to keep in the white areas here avoiding these dead ends if you like from which there is no possibility of reaching an intended State and it's really trivial to evaluate these um areas that you have to avoid because you know the probability transition matrices so you just multiply it by itself a given number of times and you can work out can I get from this state to this state and if you can't um then you get this um black area here and you know you have to move from here to here to here to here to to get to um your um your intended slate so that's inductive inference in brief and we've applied at the top level so we just the you just tell the agent this world um is under your control imagine the you know imagine what you want um noting that these particular episodes or events contain you are things that you should pass through or should experience um and then deploying inductive inference to basically predict what will happen all the way down at the bottom and then using reflexes to make that happen um we can apply exactly the same um technology or the same procedure to different kinds of games there a slightly more um complicated game um SM to emulate breakout um where we've introduced uh certain stochasticity in terms of the initial uh conditions and uh sticky action but the the overall behavior is the same is basically choosing paths through episodes into the future given what it knows that can be achieved in order to get to uh rewarded uh States so effectively what we've done is create um an attracting set of episodes at the highest level of these scale-free um um generative models where all the aable paths and paths now in this generalized State space or event space lead to reward um we can go a little bit further and just um add in uh various episodes by merging uh or assimilating new data new training data if and only if um the path actually leads to an existing um part of this attracting set and I've just Illustrated that here um in this final example so this is the path or the orbit or the attracting set that was learned at the highest level by the um the ping pong playing uh agent but now we can um fill in um fill in this um effectively um allowable Dynamics to include all states that lead to a path on the orbit um and or thereby all paths will eventually lead to reward or to roam so we can augment the model of expert play with paths that lead to an orbit namely this attracting set um and paths that lead to event that lead to the orbit um recursively and get quite a dense set of allowable transitions that we can be guaranteed always keep within the attracting set that will contain these preferred States or um states that we have um constraints uh where um we have constrain those states that are not visited simply because they cannot be recognized and that's it thank you very much indeed amazing Carl thank you awesome okay Arun if you'd like to lead in with a first question and Lance too thank you but go for run first uh yeah thanks very much Carl um so my first question is um based on a sentence in the paper which talks about any model being Spar must be renormalizable could you say a little bit more about why that must be oh I can't remember give me a clue uh I I'm presuming it has to do with Markoff blankets and yes yes yeah that's as far as I got I see yes no I I remember now um I it may have been in reference to some work done by Dalton um showing that with probability one provided the system is large enough there will be Markoff blankets and as soon as there are Markoff blankets there is now an opportunity to take Markoff blankets or Markoff blankets in the spirit of um that applying the renormalization group that that I started off with um so as soon as you've got um as as soon as any um sparsely coupled random dynamical system um can now support a um a particular partition into Markoff blankets you've now got the opportunity to take Markoff blankets of Markoff blankets and then apply that renormalizing procedure that I said before so all you need to posit is any system um that has a Markoff blanket at any given level and as soon as you've got that you've got Markoff blankets all the way up um not necessarily all the way down but um youve certainly got them all the way up um the the physics approach would would actually um also actually um require for for there to be markup blankets all the way down as well but that's that's that's another issue um so um what that basically means is that um all you require is a sufficient degree of sparity that would um lead to the conditional Independence that are definitive of Markoff blankets and in one sense much of the sort of um the um the maths behind the free energy Principle as it inherits from random dynamical systems or uh in physics say Lan um equations is simply showing that a sparse coupling a sparse causal coupling in a physics or control theoretic sense not a philosophical or Granger um or um cause effect structure sense but in the physics sense of a coupling so that the change um this state um influences the rate of change of that state so that kind of um which could be written as an influence graph so if there's a sparse coupling on the influence graph the equivalent basian graph or probabilistic graphical model induced by that sparse coupling um now uh can be interpreted in the in the sense of um um pearls Markoff blankets um so all you need to get to the to markof blankets is sparse causal coupling hence sparse Rand random dynamical system that is sparely coupled just out of interest um the spin block um the motivation for the spin block applying the spin block um or blocking Transformations that basically comes from the assumption that um many systems that people deal with both in computer vision and uh and in um idealized um models in physics um has an exquisitly sparse distribution and that's because the causal inferences are all local so as soon as you in um you look at a a coupled map l ltis or a globally coupled map you'll notice let's take the latice for example let's take a sort of idealized um um um Crystal so what you're saying when you write down a ltis is I'm only connected to my neighbors to my Mo neighborhood or to my Markoff blanket um that means that in a large latice where I could be connected to N squared neighbors I'm only connected to 1 2 3 4 5 6 7 8 so a fantastically small almost empty set of possible connections and I use that phrase because there I can't remember who said it now but I really liked it the notion that if one looks at the connecto in the brain in terms of the connections it's almost empty you know the number of actual connections you know white matter connections external connections and indeed intrinsic um um external connections are almost zero in comparison to the total number of possible connections between the 10 to the 11 or whatever uh neurons in the brain so that's not surprising and you it's one of those nice examples that the the anatomy the functional architecture of the brain recapitulates the anatomy of the lived World in it has this incredible sparcity um um and that just means um the universes in which there is um the action at a distance is quite rare um um means that that you you're just interacting you're just coupled to your neighbors and that's and that gives you if you then look at the mark of blanket structure or the particular partition for these ltis structures you get to the spin block um blocking Transformations as a way of carving up nature you know in terms of little patches or Markoff blankets I do have a follow-up question on that but then I'll hand it back to Daniel or Lance um so the ltis example is really interesting because I think there's a symmetry there that maybe we lose when we talk about blankets that only have one parent so in a normal lce as you say you're just connected to your eight Neighbors in 3D space when we looked at the spin block

Transformations before we were going let's look at this square and everything in this square has one parent but then the square along is connected to a separate parent so now if you're at the boundary we are breaking some symmetry there yes technically or practically a very good point um it doesn't really you it doesn't really um uh have any material impact upon the the the algorithms or the partitioning um so just practically what happens is you you you you you you just take your tiles and then you have a half tile or a partial tile um until you get to the boundary so these these groups in the grouping operation or the or the blocking operation are not necessarily the same um you know they're not the same size um and indeed in uh um current um work we're actually dealing with sort of multiple streams so you can have sort of audio files and video files separately tiled and then converging right right at the top um ideally what you'd want to do is is um not use this um um blocking transformation but to try and identify the Markoff blanket structure at the particular scale in question and actually use the Markoff blankets based upon um well there are a number of ways of doing that you can either use a markof blanket Discovery algorithm of the kind that Jeff Beck um Works upon or you can use um empirical um analysis the the dynamical coupling um uh by estimating effective the jacobians you know for those people are interested there's a worked example of that in the particles and Parcels in the brain paper in network Neuroscience um it's really framed in terms of data analysis but you can use the same principles to um in to identify by a particular partition from a large number of um um States or boxes or or or or pixels um um you know in three and 2D um respectively um there's another thing though that I think your question speaks to um which is um you know you've got you got you know one block of four then another block of four and then um because of the sparcity they're separately uh modeled at at the level above so I thought you were going to ask how in Earth now do you start to repair the obviously rather file assumption that you've got just local action because of course in our world there's lots of action to distance you we are um talking to each other over over over thousands of miles or Kil kilometers and you know we have Vision uh so all the and we have uh linguistic communication via sound so you know it is interesting that we have actually violated much of the local action if I was a virus this would be perfectly okay because the only thing that's really going to affect me as a virus is the things that Arch molecular contact with me but for you and me um and the worlds in which we live in um we can't there is no such thing as just uh the action distance now starts to really matter um and effectively because the two um latent States generalized latent States generating stuff um initial conditions and paths for the two blocks below because they themselves now constitute a single uh contribute to a single um latent State at the at the level above as you move deeper and deeper you start to now repair the simplifying Assumption of conditional Independence between the groups so it is at the higher levels that you now become you're able to see these action this action at a distance is coupling your what what happens over here really matters in terms of what's going on over here but you only ever see that you only have that as part of the generative model right at the top which is why all the inductive inference is implemented right at the top level where where you can see everything um so that it becomes you know if you like you are now you are now in a position to um um model and thereby generate appropriate predictions that rest upon um these long range dependencies um and of course you know using the word long range I mean in the sense of the the coupling distance uh you you in a dynamical sense is that what you had in mind uh that was exactly what I had in mind thank you thank you Lance do you have any remarks or want to add a question or I can yes please go for it yeah I have lots of questions uh thanks a lot Cal for the talk so I'm just wondering about this all Road lead to Rome slide that you had at the end and I'm wondering whether you have some sort of a curse of dimensionality as you increase the dimensionality of the system so it sounds to me like the manifold of expert play because um because you're taking temporal trajectories that's going to be a onedimensional manifold now as the state space of the game um increases in dimensionality you're going to have an ambient space of trajectory that's much much higher dimensional and so it sounds like the amount of trajectories they need to you know construe to bring me back to your um onedimensional expert manifold is is just going to grow and grow and grow and it sounds like it might be hard to fill that space so could you could you comment a little bit on that and whether it's a problem right yeah in general in my world you know these things aren't problems but they're challenges um and I think that challenge is yet to be met to be met um so what I imagine people will do um is that you will um ingest extra episodes or sequences that bring you onto your attracting set up until some bound um and then you will switch off active selection by which I mean the structure learning um so you stop growing the model and then you switch on parameter learning so that now you're forcing the Transitions and the um expression of various generalized states to accommodate variations around the training data that you have been previously exposed to so that fixes the size of the tenses and hence the dimensionality of the state Bas State spaces in question um um but will in principle now accommodate by preserving by using sort of um Act learning in the technical sense that you only update a parameter if it um does not uh increase expected free energy or decrease Mutual information um then um you you you're now in a position to become more robust um and perhaps I should qualify this and the whole point of this adding in extra if you like insets onto an attracting set is just to equip the agent with a certain robustness should itself find find itself outside or off that attracting set so um I know Lance knows this but I'll just make this explicit for for everybody else um this kind of learning um and state action policy um implementation is very much like learning to ride a bike um so you you basically um just accumulate and remember those experiences that were successful so for example I you know managed to um rotate the pedals by one cycle and I don't fall off and I remember that and then I have another go and I wait until I've actually done two cycles and I don't fall off and I remember that I forget everything every time I I I fall off and then I keep on going until at some point I connect back my last cycle was identical to the first cycle and then I've learned to ride the bike but because I've forgotten all the falling off I can't recognize or know where I am when I when I've fallen off so I've got a very brittle expertise that will not allow me to recover if I fall off the edge so another example which makes it slightly clear I think is like mountain climbing you know if I want to be an expert mountain climber I cannot learn by my mistakes provided I'm climbing a sufficiently high mountain should I fall off it then I will die another example example here is learning how to cross the road as a child you know you cannot learn by um um experiencing being run over but the cost you um you pay or the price that you pay is you can't recognize when you are being run over um and that I repeat lends a certain brittleness and a lack of robustness to agents who may be exposed to Danger that doesn't actually terminate them so putting this robustness back in by growing the um by growing the attracting set or basically putting these insets on um in this sort of generalized um phase space um you know it's just to make it more robust um in numerical studies it it does seem to um converge it doesn't seem to grow indefinitely yeah so those are likely violations are moving off the attracting set are themselves quite small relatively small in number um you know practically speaking uh if if you've controlled the sources of stochasticity or unpredictability so it may not be always the case that you need to have an explicit bound on the upper the size of the um the very high level tensors if if you do then one has to now think of a principled way of making that upper bound an emergent property of um um expected fre energy minimization it may be that you know you violate now the mutual information that certain rare events are just so rare you're actually compromising the mutual information by including them as a particular option um or a particular column of of a likely likelihood mapping or yeah but I don't know but these are all really interesting questions to pursue thank you Carl all right Arun go for it um yeah so following on from that I think it's an interesting question on the uniqueness part of ingesting data and the question is are there many more ways to fail than there are to succeed so if you're passing your data into sort of Max demon Maxwell's demon uh with a gate saying accept this data reject this data if this uh improves my Mutual information if there are many more ways to fail but then get back onto the uh successful path but there only a few paths that actually keep you there would it not make sense to just continually accumulate these sort of recovery States because the mutual information will keep increasing and then you stop there then you parameter you learn parameter wise because we're not including the sort of Prior distribution of How likely you are to be succeeding at any given moment because with the uniqueness thing you don't know oh 10 times out of 10 I'm cycling or like nine times out of 10 I'm cycling and one times out of 10 I've falling off the bike yeah I think it depends um again I don't have any explicit numerical analyses or analytic analys mathematical analyses of this but intuitively um I think during learning there may well be um um good mileage in ingesting failures um or roots to success if you like or to the attracting set that that actually Traverse regimes that would normally be excluded by my constraints or my PRI preferences um you know just taking the view of the role of of the C tenses or prior preferences from the point of view of physics what that basically is saying is it's actually carving out vast regimes of State space latent State space that you should not be in because the it would be uncharacteristic of you to be in these things so it it's really not so much about the rewards and the preferences uh it's really what you where you shouldn't be and that's what defines the attracting set um however before you've learned that attracting set it may be useful to include constrained regions of State space that offer a root into or onto that attracting set so I can certainly see that that would be a useful device but notice that once you're on the attracting set you will never visit the constrained regions because you now keep yourself on the attracting set um um and therefore if you've encoded those but you never visit them your Mutual information will actually fall so what that means operationally is what you'll probably do is have this sort of slightly over inclusive structure learning learn how to cope with all the necessary sources of Randomness so that your attracting set is sufficiently robust and you do actually visit from time to time but not um not never um s of excursions from the attracting set drawn back onto the attracting set um and then use basing model reduction to eliminate all the states that you don't want to recognize simply because you never go there um you know once in a you know when I was younger I would certainly be able to recognize what it's like to fall off my bike but now as an expert bike rider I really don't want or need that so you know and that's just basically getting old and wise and you know implementing basing model reduction to remove the Redundant parameters which would in this instance just be the columns um and um Associated slices or or rows of the of the transition majy so I think you could perhaps another way of then of answering L's question is yeah you just keep accumulating um but um start to engage Bas in model reduction at some point you should reach an equilibrium which will be an attracting set that allows you to survive in the character states that Define you in this particular environment with this particular volatility thank you awesome okay I'm going to bring in a question from the live chat okay Javier wrote so stochastic plus dynamical probabilities learned by the agent itself are scale free will this hierarchical learning also be applied to the goals of the agent different abstractions granularities of goals self-learned [Music] my answer would be yes but I'm going to I'm G to ask Lance to to to to answer one oh that that's cheeky uh yeah I would say for sure um because typically when we have when we oper operation operationalize sorry the goals in our gen model they need to have the same structure at our gen of model itself so imagine then we would have a hierarchical raliz gos which kind of make sense cool okay I'll bring in oh yeah go for it Carl no I just GNA say um you know just to um unpack what Lance just said in terms of practical things you know if you're trying to um you're trying to say um what my goal is is to go uh is to have dinner um that naturally unpacks hierarchically into I've got to move from the living room to the kitchen then I have to find the you open the fridge and sorry I have to find the food which then unpacks into I have now open the and all the way down to to the finest finest muscle movement so almost by definition um your intended narratives can always be um decomposed in this scale-free way you um there there's a this is very similar to sort of um motor chunking and hierarchal decomposition in in um um theoretical uh motor control for example and you probably find it a lot in in robotics that there's a global long-term plan that that provides empirical priz of constraints on shorter little chunks that themselves provide constraints on shorter and shorter chunks and that's exactly the um if you like the the architecture implicit in these scale free models awesome and totally ties back to the synergetics okay I'll ask two questions from ml Dawn he wrote how can we inject the same separation of time scales by imposing smoothness constraints on say the parameters as opposed to playing with proportional update rates at different levels and this relates to something Arun and I talked a lot about in thezero and leading up to it is identifying which Windows to use you could imagine picking the right separation of time scales if the classes on campus really do shift on the hour then the hour course scening is going to be great and effective but the 59 or the 61 minute my Alias you with the real variability of the system and so there's not necessarily even a smooth landscape for determining the windowing or the binning of coar graining and coar graining may have dependences on each other so how do you go about playing and finding viable approaches and robust ones amidst all these different levels of of meta optimization that are opening up right I'll take this one and I could take take the next one um then that's a really good question because there is an implicit separation of time scales between um so vanilla implementations of inference and learning however in these schemes there isn't um so the the know the um the Matlab version or implementation um of um active inflence under these and all uh Markoff decision process models um unlike um the um procedures that you might find in data analysis where you acquire you basically do some filtering on a on a an Epoch of data and then at the end you update your parameters and then you rinse wash and repeat um in um in active inference um simulations and also in some um applications of what's called General filtering you have continual learning so the update of the parameters actually occurs at every time step um at each level of the model so the likelihood parameters at the lowest level are updated every time that routine is called um so there is continual learning the learning um is yes you're absolutely right is smooth in the sense that you're slowly accumulating drish parameters but the temple scale of that learning is the same as the inference um and you could also argue well could one also put Bas in model reduction um into play at the same time scale so basically can you at every time point at each level in the model um identify redundant parameters and remove them uh using Bas and model reduction in effect that's what's done uh using Active Learning well I said before Active Learning will only proceed um if the expected free energy of mutual information improves that's actually also implemented so this the kind the the the rate the rate at which it looks as though things change um between inference learning and model selection is certainly distinct I don't think that's quite the same kind of separation of time scales that is implied in the renormalization group that's as you were intimating uh Daniel exactly specified by the number of ticks or updates at level n per single update at the uh level n plus one um and that is I think you know the you the key question you're asking how do you get right um and of course the answer to that is the the model that has the greatest model evidence or smallest free energy or greatest marginal likelihood for the kind of data generated by this Environ in this world so um are there any um sort of useful PRI you can put over this um well one one useful prior is um that um things are changing as uh to to well one prior which is actually used in operation is you take the lower limit on on What's called the uh the temporal RG flow which is basically the the length of the window at each and every level um at the moment that's two what does that mean how can you motivate that well it's the most expressive in the following sense that if I um just imagine um what imagine what this dis discrete discretization of time means from the point of view of classical Notions in continuous State space like position and momentum or position and velocity if um if I use a a um a scaling of two which is basically means there are two time steps or every time step of the um the level above what I'm effectively doing is um um modeling two time steps for every state at the level above and that basically means I can model the initial condition and the subsequent condition basically in terms of the change which of course is the velocity um and what does that mean for the level above that well it now means that I'm modeling the chain in the velocity so I'm now modeling the acceleration and then so on up until um you know the highest order of motion so what that means is by taking this limiting case of T the temp temporal horizon or the the you the ratio of time time steps from one level to the other level um as two means that I got a very expressive model that can handle um can recognize um things that change very very quickly in the ex ation their jerk and all and the implicit um context sensitivity that that that gives you if I didn't do that I'd have to assume that I C grain over long periods of time and therefore the path is consistent during that time now that sometimes is a useful approximation and indeed there's a whole industry called you know called linear switching linear dynamical systems that makes that approximation and it means you can um approximate all the nonlinear complicated high order motions in some evolving pattern or system with a series of um um linear trajectories that basically go you know that intercede between a series of points at which you which you which you switch the problem with that is that the length of the linear trajectories for as for example I'm committed to one Path U so one slice of my B Matrix for the next eight time steps the problem with this is that you don't know when you when you deviate from that transition Dynamics so then you get into the game or perhaps I could make my scheme reactive so sometimes I do three steps sometimes I do eight steps sometimes I do one step and I have some criteria for saying your predictions about my trajectory have now been met please um I'm going to tell you give you the evidence that I've been pursuing this path you update your priv level above and tell me what you expect me to do next so this now speaks to um a particular kind of reactive message passing um that starts to have implications for the computer science and the sort of CR the crud um um or um as um as I think we've rehearsed before um The Institute you the the original conception uh behind um the actor model in computer science from the 1990s so it's all about reactive message passing you I I will send you a message when you send me I will return a message to you if you send me a message um and what this um what these um renormalizing time models have and I should say that this also applies in the absence of the the spatial renormalization so any deep temporal model will have this aspect what that means is that from the point of view of Any Given level I will um be responding to messages more quickly from my subordinates than from my superiors so once in a while I'll get a request from a superior my response to that is to ask my juniors to do a series of things they report back to me and when they report I give them the next bit bit of request or guidance uh until they complete their job and when all my juniors have completed their job of course in the re normalizing spal re normalizing context I'll have a whole group of of Juniors to worry about um then um I have gathered enough evidence for me to then return the the message to my to my Superior so you get the in terms of the reactive message passing in this sort of actor model of um of computer message passing um you get the the separation of temporal scales inherits from the number of messages or re um requests and uh responses um um exchange with the people lower down in the hierarchy relative to the number immediately subordinate um and in principle they don't have to be fixed so there could be a criteria um before I pass the message back back up to the to the higher level I think that's Daniel probably what you were getting at um I'm not there may be somebody who's done that um um but I'm I haven't done that and I I I don't know I can't give you an example of somebody who has done that you may find you may find somebody um in Bert de's group um well so of you know some of the world experts um in reactive message passing who have considered these kinds of things um you know you in in principle um you know there should be a way of um each level testing whether I have fulfilled the predictions of my superiors um and now it's time to get more guidance you know uh based upon the free energy awesome the organizational technology I didn't know that my PhD needed okay I'll ask a question as we get close to the end and and this will be for for anyone ml Dawn wrote how does the training time of active inference compared to that of reinforcement learning and how does their scalability compare and it would be interesting just to hear anyone's very takes where are we at with that entire Arc from problem conceptualization on through the training and the inference how do we even grapple and describe with similarities and differences in the training time and resources and all of this that's definitely Lance for for at least five minutes I think I'm I'm a bit incapacitated in terms of talking um I think this actually a good question for Aron um but yeah I mean I can say a few things um so uh scaleability in active inference I would say it has been a challenge and there has been um much less man work hours that have been put into that than in reinforcement learning now for the past few years we we've had a lot of work um a lot of it led by C actually um trying to scale the structure learning in active inference and and I think we've done a lot of mileage on that so there's been the supervised structure learning paper there's this paper uh there was also some interesting work um presented at the international workshop on AC inference um I personally think there are still some uh challenges ahead and and I think uh one of them is is the one that we just talked um a few questions ago about you know if you if you learn an expert orbit how do you get back on the orbit as the dimensionality of the system increases so this is definitely um one challenge um now I know much less about reinforcement learning so um yeah maybe Arun has more things to say there um but but what I will say is a lot of reinforcement learning is powered by Deep neural networks when we use deep neural networks we sacrifice uh the interpretability of our generative model this is one of the strength of active inference is having more interpret um Dynamic clal models where we can actually here into them and see okay well this representation enclos this this other one enclos that and so forth um the fact though that reinforcement learning typically uses deep n networks means that it has been much more scalable uh just because there has been so much work done in like training deep n networks this is a bit my uh my perspective I don't think it fully answers the question though I can come in with a little bit on that but I must admit that I've not trained any reinforcement learning agents myself I've worked with people who have uh I'd agree Lance that the infrastructure generally is much more advanced for training RL agents for doing stuff uh and completely agree with as you say the explainability and the interpretability of why did an RL agent pick a particular action is very take uh and I think the deeper those networks become the harder it is to understand what's going on in the hidden layers and uh generally why they do what they do I think with active inference yes we do have more explainability but I do wonder if we go the far structure learning approach I think KL had a particular uh comment earlier about looking at the highest level um identifying the desirable States learned by the agent from what I understand the identification has to be done by a human they have to look at the model that's been learned by this uh renormalization approach look at those top levels and then say ah what do these levels mean so an example in the paper is the Mist uh classification it's the case of a human being looking at those top levels and going ah these top levels each correspond to a digit in the handwriting task so we are uh sacrificing some explainability I think by doing this fast structure learning compared to what might be the oldfashioned way of creating uh G models By Hand by thinking sitting in my armchair and thinking this is how I might model the world these structures these likelihoods these states and when you do that I think just by hand you have 100% explainability that model just might not perform very well so I think there's always going to be that tension there um but in terms of how quickly these things learn um having not trained either i' would be having a reckon and at the moment I would say probably an RG um active inference agent would train faster simply because it's able to use the structure in the data much more effectively that would that would be my take and I think as well really interesting thinking about the active data selection part of this too I know that there is some stuff in the paper about the efficiency being both a curse and a burn but I'll hand over to Lance and Carl to give more details on that yeah Lance yeah just to um echo echo what Arun said um also so so the structure learning in active inference and in reinforcement learning is is also very different um in reinforcement learning we typically have a a model with deep known networks that's massively overparameterized but has some fixed representational capacity from the outside because the structure of that of that model will be specified and then typically the model will learn and then probably you would prune it using some techniques like Dropout or or other things if we looked at that from the perspective of active reference it would be like specifying a massively overparameterized model and including it with Bas model reduction the um but but now in active inference we have this Bas and model expansion for example in this paper and what this Basin model expansion does is it learns from the Geto the minimal model for explaining the data and so in that sense it's extremely efficient and also provably because it extremes the um the marginal likelihood or the mutual yeah I just wanted to Echo that last Point that's a really important Point um so Lance is drawing the distinction and there are many distinctions you can draw with um deep RL um and um deep active inference read in a more in in its um generic sense um and that that distinction at the level of structure learning or model selection I I think is absolutely crucial so you know deep RL starts with an overly expressive model with a known functional form um that is deliberately overparameterized and then shrinks it or reduces it to um maximize the mutual information the expected free energy sometimes they um refer to as a cross entropy in machine learning so the reward function now becomes uh there's no explicit reward in unsupervised learning it's just replaced by the mutual information or the cross entropy um which should be distinguished between supervisory enforcement learning such as you know digit classification for example um whereas the you know active inference um under these re under fast structure learning is exactly as Lance says it starts from nothing and stops when it's minimally complex which means it's maximally efficient and that efficiency translates in many guises through to sample efficiency through to um um statistical efficiency through to thermodynamic efficiency and this matters because from the point of view of climate change this is the right way to do it as opposed to building power stations to to power large language models and Transformers that need big data so um the efficiency um on all levels whether it's the the sample efficiency or using the right data uh or whether it's right through to how much electricity do you need to actually train your model and how many millions of dollars do you need to train a foundation model um all of these are reflections of the failure of um machine learning to write in the efficiency into the ultimate objective function which as Lance said is just the marginal likelihood of the evidence for this model of of this kind of content or or World awesome okay final question and a very apt one as we head into the rest of the Symposium so just give any short thoughts Tintin wrote hi all thanks for the incredible event does the sequence does the sequence in which the training data is fed to the AI structure learning model impact what the resulting learn structure would be so the learning curriculum AKA Symposium program challenge well the answer is sorry I'm mindful of lances he was recovering from his um jaw surgery um yeah absolutely Ely um and this is um again just in relation to to to to the last question something I think that eludes uh conventional RL certainly that rests upon um um classification procedures um because um the active inference as an application of the free energy principle is all about the Dynamics of self-organization and Dynamics means time and time means sequence matters so inevitably at some level the order in which you see things is absolutely crucial for getting the right kind of generative model um that is even in the case um actually of static image recognition because you're assuming that there's no ordinal structure or correlation structure to the presentation say of M digits you're assuming that they are selected at random as opposed to seeing all the tens and then all the all the ones and then all the all the sevens um so even in these edge cases um um which are you know the focus of um 20th century RL um you know static image classification for example um you know order does matter but they they it matters absolutely acutely because of course this the order in which you um see uh um content in um this infrast structure learning determines how you populate the um the transition tensors and it's the transition tensors that enable separation temporal scales that enables the ability to um encode narratives and plans that far transcend the actual rate at which you samp sample the environment you could also to argue that you know this issue has emerged um in a in a strange kind of guise in Transformer models so the attention heads operate on the past so the the where you select the from the past in terms of which token is going to uh predict the next token most efficiently um again is exquisitly dependent upon and sensitive to the order in which you train train train models So Daniel have you thought carefully about your ordinal approach today I I let the structure I let the structure of the niche and others' constraints provide the first pass and make requests from there so thank you Carl thank you Lance for joining again thank you run for all the contributions with azero Carl great to see you so next up we'll take a 30se second break and we'll be back with Michael Lennon so thanks fellows see you soon e okay welcome back from that brief brief break we're here with Michael lenon and the session is co- inferencing organizational symbiosis so Michael thank you for joining take it away hello yes um I am uh so grateful to be here I would like to um approach today's conversation in um in a conversational style where um I share what led me to active inference and how the discovery of the principles of free energy principle and and active interest changed how I understood as a practitioner and and how that has led both to both personal professional and organizational um shifts in conversations um one of which will conclude in an invitation to this community to um experiment with an uh an organizational structure uh designed for um building both the global n knowledge comments um as well as um um uh building individual Ventures so similar to the way in which uh for example Linux is open source uh and then you have Ventures building off of Linux to do things in the world that that experimenting with such a an organizational model uh for uh the active inference comments uh is some an experiment worth taking on so I I I don't want to let the the final invite get away but but I'm going to backtrack and do this very much in a storytelling fashion so um so um I have some slides but I'm not going to uh spend too much time uh well in fact let me just um skip that just just to recap what this um uh what this conversation is about is different ways in which um not just active and but um science has collided with internal intuition and led to transformational journeys of different kinds and so um and so um I would describe the um you know I I started out as an individual uh as a uh Economist um and um and as a technologist working at um institutions such as the World Bank and IBM and um doing large scale um it implementation projects and um and that work often although there was a lot of um engineering um it was often the human factors that got in the way of success or failure and so um that has been a kind of orienting um how to both be informed by what the the the science and the math provides and uh then build uh how does the biologically based if you will way of sense making um affect or um alter the the the the evolution of of successful project implementation so um just um a little bit about um about where this uh project um prior to coming to active inference um I say project my journey um uh led me through was was um I had a uh I had as I said been trained as a as an economist and um one of the things that economists have is a point of view about what it means to grow the culture to grow the society and um primarily denominated in dollars and using um GDP as a way of um of uh tracking whether or not one was impacting the experience of flourishing and um I came across a uh a chart by um by an economist that showed that GDP had tripled in the last 60 years both in the UK and the us and that well-being had not changed at all and this economist who had been in office in the UK at the time when when he left office was kicked out um he went back to Academia and started looking into what does Science Show moves the needle of well-being and how do we make this socially more contagious and um and I both U loved the idea but I also was um uh shocked by that information because as someone who was trained in a particular lens a particular perspective what I had been trained to do and how to um cause effect and impact was in fact not moving the needle of wellbeing and so there was this kind of personal and professional crisis um that arose from this uh from this knowledge and um as I say that that led to this this journey of unlearning in some ways and um and and in both personal interpersonal and organizational ways uh recalibrating the experience of what does it mean to pay attention to thriving and so um that Journey has been very satisfying um not only do I understand how I need to be to experience uh more goosebumps in my life uh I understand uh how I need to be to experience connection and belonging and to co-create with others uh a sense of um purpose and of meaning that um that I had not had previously and it's uh one of the ways I sometimes uh descri Des cribe this relationship between the subjective uh internal experience and scientifically objective let's call it was that I might know how to go to the gym and flex uh my arms to get a workout but by understanding Anatomy I could learn um the bicep had a particular placement in the body a particular Arrangement and I still had to do the workout but I could more effectively and um skillfully uh do the work that needed to be done of working out and so similarly the study of the science of well-being um and social uh social uh Sciences of of connections of groups and of um and of um how how for example um one of the surprising findings for me uh in in the realm of Happiness was um a finding by an epidemiologist and um basically he was studying how do emotions spread and um and he in studying this realized that your happiness um uh was not only more heavily INF uh influenced by the people that you're surrounded by but also by the next degree of separation the people to whom they are connected and even the third degree of separation and so that these three um Degrees of Separation predicted your happiness more than uh the way that I explained I had been explained what happiness was which was that uh it is an inalienable right like life and Liberty the pursuit of happiness and that it's subject to individual traits of my own that if I just do me the best I can then I'm going to be happier and so all of a sudden there's this disorienting um Insight in that it's not about being my best but it's just as important to inter be my best and so it's like well what the heck does that mean what does it what does it look like to to inhabit one's own experience and one's own skin in in a completely different way and and I know a conversation that uh you and I Daniel once had about um about um the impact of active inference you were describing how you might have been driving on the right hand side of the road your old life and then all of a sudden you um you uh go to a place where everybody drives on the left and so some intuitions still work exactly as as they used to but others are completely counterintuitive and so this developing the capacity to to go against intuition against the The Familiar is um is part of what sometimes these these these new forms of knowledge there's there's knowledge that that adds to what we know and then there's knowledge that changes how we understand and and happiness first did that for me intuitively at the personal experiential level right it first was like okay I I am playing the game of my own life uh inappropriately um do I how do I begin to inter be my best instead of be my best and this is where um uh some subsequent work in working personally and in with groups and and organizationally led to um paying attention to what was happening in in a different way and uh so I uh as I say this this um crisis LED to a very fruitful um Learning Journey and um and uh excuse me I um I got distracted um some of this led to writing different ways of describing how to do work and so for example one of the spaces that I've been working in was in um government Consulting and in government Consulting top down um uh uh is often the way to direct the effectiveness of the organization um you come up with the right strategy you figure out all the components and you then execute on the plan and uh and so in a sense uh if there are two forms of of uh probabilistic prediction a basian being one at the heart of of active inference and another one being a fisher fisher is one that that says it knows where you need to go and the way that you relate to to surprises and to mistakes and to errors is you correct back to what you felt your original plan was whereas basian probabilism probability obviously um can have similar results but also can update instead of reacting to surprise as something to be fixed it can sometimes be something to be adapted to and uh it changes the way that uh manage you manage um the the uh emergence of differing information from what you originally predicted and so in the same way that I described that happiness uh had a disorienting effect on how to practice it active inference has a similar um disorienting effect about how co-regulate um um how to supervise the um the evolution of of differences between what you expect and what you what you get and um one way in which I I describe this uh to to folks is it changes your understanding or changed my understanding of how cause and effect works that um you know as as as I arrived to um to um active inference there were there were a couple of intuitions about uh how the world operated that active inference helped um evolve the first was that instead of instead of the world being composed of objects um that were fixed and stable that um that the mathematics of how to explain things was much more of a dynamic verb and so to look at the world as being not nouns but verbs and and and a way that I describe this for myself when talking to people about it is that instead of seeing myself as a self a noun that I'm a a be a being and so um that paying attention to the continuous verbing and the relating uh to to to others is is a is a different um uh a different game and so um the interesting thing about Act of inference was that there was a mathematics for describing this that um that related the internal States and the external States um through uh mathematics of information that the way in which um the the way in which um uh science Works um is is is familiar and intuitive to to to many of us um Let me let me let me rephrase this in a different way that you know when I say the inside and the outside are informationally connected um that that um that the outside is all the information all the stimulus of the world and that the inside creates a map of that world and that the mathematics of creating that map not only um reduce the world and is a mathematically um efficient way of describing how to make the how to get the most signal for for all the information that's being received but that it also um it it also um enables like I was describing earlier with the understanding of the bicep a more accurate way of of sensing the world of um orienting to the world and um and of responding to the world um I I should have uh shared my my slides instead of trying to talk through them but I'm going to take a a pause and ask any any questions or shall I proceed yeah uh I'll ask one question and then looking forward to slides dsk wrote there has been a lot of talk around GDP not being a good enough metric of national development has your research informed you on a better metric if so does it have a working name so yes and it's a different um answer than a single metric so the metric a single metric was one of the problems and the fact that it was denominating currency was one of the problems so one of the things that uh has shifted um some of the work of uh the UN of of really a group led by Bill Boi and Ralph thurm on um on what are called contextual indicators are that um are that instead of an indicator being um GDP is both currency and it's um it's set to a single the denominator and the nominator of the indicator are both set in in the same thing so um let me describe this in a in a different way um there has been a diversification of what are the things to pay attention to so not only um dollars but also subjective experience of well-being for example better predicts the outcome of Elections than does Financial income and so um the idea that dollars are are predictive accurately predictive of um of outcomes has been often relied upon and turns out to be maladapted right and so that's um whether whether you're talking about biological phenomena whether you're talking about social phenomena there's a whole range of extra indicators that are now seen as um not a single one but but better visibility into these different nested levels of of life the second what I would call breakthrough type of indicator Beyond GDP that I was describing denominator and uh not numerator is that um most organizational reporting for example is is um for example um Starbucks might say we are using 10,000 gallons less of water than we did last year so we are improving um our consumption and the way that we are related to um the water shed that we might be related to and and all that um reporting is if you will individual Centric and when you Center your just like as as with happiness you know where if you're just looking at the at the individual you're missing an important part of the picture when when measuring your relationship to resources to Commons to um natural Commons and social Commons um if you if you put your um uh individual Behavior within the context of the collective so let's say Starbucks says uh we used 10,000 gallons last year out of the total water shed then you can begin to draw comparisons about how everybody is using the Watershed as a whole and how the percentage the percentage behavior of of usage of the water has shifted that can be compared from one to the next to the next so if you will you're putting these indicators in a in a similar language in a similar uh they're speaking in a in a similar base than um than when they are um reported individually and so you know when when looking at at well-being it's it's important not only to to add additional things Beyond money but it's also important that these indicators be um set to the same um denominator or or you wind up not having the visibility that you need and there's a huge cat fight about this because uh there have been a whole series of um of Privileges and a whole series of um social practices that are grounded in the way in which um uh individual corporations are um are structured and so this this kind of points to a different kind of natural Mee that I wanted to bring up that might be of of Interest so so um we talked about how with happiness um you know we we're doing it individually and that we're missing the relational um when you look at uh cells that um are part of an organism if they are traumatized uh they become more individualistic they become uh they consume more they reproduce more and they grow into what we call a cancer because their if you will relationality is traumatized and they become more and more selfish and so there's there's both a huge importance of paying attention to relationality at the level of the cell at the level of the individual human and some would say at the level of clusters of humans such as organizations and that with cancer part of the um New new form of treatment has been to call the cells informationally back into relation so instead of punitive burning those cells and killing them it's to remind them of their relatedness to restore them from a traumatized State back into relationship with other cells and they begin to behave again in better relation to the body around them and so that metaphor speaks to the importance of how do we call each other back into relation if the language of individualism is permeating both our individual Behavior and the language of Corporations is equally um traumatized because originally when when uh corporations were first formed in the 1600s in uh in the Netherlands there were really people coming together to venture and do together what they could not do alone so trading with um the Far East was was something that no individual uh Capital uh no individual business owner could could afford to venture but by pulling their funds they were able to do and they were able to become a great trading power globally this model of groups behaving as groups so a different cluster of um of uh of existence groups um and then behaving um pro-socially with each other is what grew the next level of um economic performance and economic um Effectiveness um however over time some would say that organizations have become traumatized and have become disconnected both from the places where they originated as well as from all the stakeholders that they were serving and so for example here in the US when when the Supreme Court last century two centuries ago now almost decided that financially Financial stakeholders were the foremost important um stakeholders in corporations they distorted the relationality of of Corporations and so it has become a kind of cancer that that has become extractivist both upon the its its employees upon the the the ecosystems in which it inhabits by architecture so there are plenty of people that will tell you in in an organization we want to be more attentive to these um other considerations Beyond Finance but but we don't even have permission because the agreements of the corporation have evolved from mutualism to um to selfishness if you will to more consumerism more productivity more Surplus and those all have a place they're they're they're important but they've become like like the cells in the body that become cancerous they've become overly um overly uh they become individually uh oriented alone and and not sufficiently skillful at at mutualism and so this all comes back to a new basis for equipping organizations to pay attention to how they're interrelating within between and to things beyond their current purview and it is much of a disorienting and challenging um uh um uh bit of information as I was just as I was sharing the idea of learning how to be more skillful at happiness was to me personally you know it's like oh I have to pay more attention to you to for for us to be happier or to be the third degree of separation as I was describing earlier means that not only the people to whom I'm directly connected but that really that it's my dumbar number the dumbar number being presumably the 150 to the 250 people that you have meaningful relationships in your life the dumbar cubed so it's beyond your direct sensory observation and it's like how do you co- Haron eyesee across groups how do you sing to use a different kind of language songs in common to which you're Co cooriented and um and these are some of the things that um active inference provides a mathematics for how this works you know that is fascinating of how the synchronization of the individual to the group and the nesting and and how the signaling happens across the nested groups and so it's it's um it's fascinating the mathematics can be daunting but the promise of it and the ability to do these things more skillfully and reliably is just it gives me goosebumps it gives me goosebumps and this is part of um why I'm trying to tell this somewhat ineffectively I would say today in a story Manner and through experience so that people can pull on the intuitions that they have the memories that they have and say I dare to tackle this uncertainty building upon these intuitions and feelings that I have and instead of the classic hierarchical um corporation that um that has as part of its founding documents when you when you plant a seed and you use the template that says we are we are organizing according to these rules even if you want to change it's it's like it's a lot harder to change the Constitution than it is some minor policy in a back office in a you know in your in your um uh you know in some app that you might be uh writing the code for you know and and similarly it's important to to incubate um uh organisms that have relationality the mutualism and the holistic perspective codified into and and the intering codified into how they behave individually inter intercluster and then in relation to the commons that they might serve and so before I take a turn into the case for um the experiment that I was hoping to invite people to uh do together any any any more questions popping up sure I can ask some short questions then slides and invitation okay lightning round okay Arun asked isn't there a cyclic dependency of economic factors and happiness for example High inflation made everyone sad over 2024 so there there is um uh there are um time lag factors uh that impact um uh when we how the economy operates and how individuals operate and these are not necessarily in sync one of the things that um that uh I found from the um from the science of Happiness was um more effective ways to carry the suffering and so um some have that this change that just happened politically for example um is a Bad Thing often when things break open there are whole new Realms of possibility to to be had and so um happiness is not just about having all positive emotions it's about taking well-being is the word actually should be using is about absorbing all of the signal that the universe is is giving all that's coming in and then optimizing within that and so um the breaking down can also be a breaking open for what there is to to uh become and so that is not to say that there is a lack of struggle or pain or friction one of the great intuitions for me um that that arose from from studying people explaining Act of inference was um how things emerge and the person explaining it was using water and uh they they were describing still water as being the way in which current conventions of how things got done and how people calculate it so if you're if you're in Still Water if you're in a lake and you're trying to get across um the way you Rod your boat or the way you swim is all set by the conditions of the water in um um in flowing water um even though those that same water has the same physical properties uh as Still Water it weighs the same no matter whether it's moving or not um when when the friction of the water over rocks um creates new dynamics that emerge it creates whirlpools right and that those structures um that persist um you know the creation of order out of disorder um allow for the emergence of also meta patterns that are again described by informational mathematics rather than the physics of the water itself and and this is this is again trying to get it what is so hard to Intuit it it's like oh these the whether whether it's a single Whirlpool coming into existing or whirlpools into whirlpooling into into greater movements that um that these frictions are also creative and so to really um uh be willing to settle into the into the dis into the what you were calling the cyclical disconnect in a symmetry um as as something that is a signal also that can be driving learning driving better calibration um can be driving um new learning uh because at the heart of all of active inference is a mathematics of continuous learning um yes I I love that and just flows like water with our previous sessions okay next question and then back into the flow we go social Design Studio wrote so curious about the transition away from individualistic and mechanical into Collective and organic where is the most levered locality to start well that's a a great question and um um here's um a way in which I would answer this in the in the world of project evaluation so there's a point of view um in the world of systems thinking there's a certain order of what to uh what of the you know Donella Meadows is well known for for uh the way in which she created this categorization of ways to intervene in a system and um uh in the world of evaluating um projects and programs complex projects and program over time uh there's been some some meta studies done about what are the things that best predict uh that if if the supervision is done uh attending to certain factors what are the things that seem to produce effective and positive results because we're talking about complexity you could pick a thousand things but in in this research they identified three things and I'll add two the three things things that they said you know when people look at the following things they they tend to not only manage what they thought they were looking for but also what emerges in other words managing the surprise and the unexpected more effectively and these three things were looking at the inter relationality surprise surprise right we're talking paying attention to the interbeing and the interbeing across not just the individual or whatever whatever our our individual unit of focus was originally but but who were seeing the system more holistically second um multiperspectivity um and the way that I in my the way I uh describe this for myself is um I say looking within looking between and looking beyond having one or more um disciplines and perspectives and ways of knowing about out uh those those lenses of of attention on what is happening um uh if we look to if we look within we often become a better compass for our own lives um uh if we um if we look between we often develop and harness the intelligence uh that collectively and the Insight that we have and when we look Beyond we often detect uh patterns that are beyond our Collective purview that might uh um break open and um enable us to to transform in in powerful ways so again to return to that second Point multiperspectivity um and I suggest with those three uh lenses which is what I call the me to we to all you know um and it's it's kind of the label that I'm putting on on this invitation I'm going to bring up later um is um is a is a good starting point is it enough no but it's a great starting point and then thirdly and here's where it gets messy and is the non-answer to your original question is um scope definition um because the moment you pick what are the boundaries what is my mark of blanket you know what what what is the what are the boundaries of the map that I'm going to use as my lens you begin to abstract everything else into what fits in that lens so so if you say we're going to have this conversation in English the way that we everything we describe will be denominated mostly in nouns instead of in verbs and with Concepts that exist in English but don't exist in other languages and so um the attending to what are the boundaries that we will use to focus is is a gigantic mechanism for seeing um how you're holding your attention and so again it's not a fixed answer but it's a way of attending to perception again repeating relationships diversity of perspectives and scope definition uh the other two that I like to add in that um were not a part of that research so that research by the way was um a 100 Years of developmental evaluation um you know hundreds of studies and and so it's very robust and you can you can go to the bank that that's a good place to start building your dashboard or not the dashboard your conversations about the dashboard um the other two that I would add and and these in part come from active inference and from other systemic view places and Views is looking at patterns how are we you know what are we clustering and looking at flows you know um what what are the core flows are we looking at energy if we're looking at biological systems are we looking at the water flows you know life organizes around water um so do do we understand what are the core flows of information of energy whatever uh in the system that we are observing that that that gives us a a key insight into the way in which causality unfolds um and causality uh organizes itself around and that it's not the mechanistic you know geometry that we we may have intuited in uh for for the longest time but that by looking at Source by looking at flow we we develop a better compass for for the world thank you it's it's awesome and the live chat is super lit you'll see it after um in the last 20 minutes do you want to share your slides and open the invitation letter yes so uh let me um switch here to share I share are you seeing my screen okay uh me so um the invitation uh was um was to test a new way of organizing and a fair share Commons is a kind of um a mutualistic Co-op that um first originated in the UK and has been endorsed by the financial I I I should have read it down here because I can't remember like the finance trust accounting Board of of the UK it's a spread through many countries of the Commonwealth over the last uh 15 20 years and it's primarily an adapted form of organizational structure that is intended to both cultivate the practice of mutualism within the group so it's kind of like a co-op um in that everybody has power uh to to participate it's self-governing um and that it attends to the relationality of the founders the workers and uh and external whether they investors or they they self-organize it's really uh there is a a pattern language about forming co-ops and one can use that pattern language to to um to to form these Commons and that these whether they are to create Ventures um or to create um full businesses that the pattern language is useful for that effect um and so as I described earlier a kinda Linux um you know it's it's intended to grow the commons but also to support the application of those Commons and build the organizational traits uh for successful Ventures and what I am uh hoping to find allies uh around um oops let me um so it um the these are again suggested and there are many ways to to go about it but um you know there there's certain core agreements that differ from if you will the extractivism and the foundation is uh relationality um which drives mutualism symbiosis um and uh the the invit ation um is to um is to constitute one in service of the active inference uh Community for growing both the commons and the capacity to um to prototype and develop both the the experience of mutualism but also projects and that uh there' be two starting experiments um the first around prototyping uh agents for accelerating access to money so often um folks incubating a new project are looking to find funding for that project um find uh funding for what that project might serve and often they're people willing to fund those projects and the idea of writing up what are the sources of different funding what are the sources of what what are the needs that they have that they're looking to fund how do I um discover and engage that um funding is an important agent that can be used in common after common after common and so possibly developing uh um an agent or uh there are plenty of methods but but agen toying if you will uh that component as one of the key flows you know that in business the flow of money is is an important element to to inactivate and and that the second is about um finding um resources of a different kind competence and complimentarity and um what are the what are the components often for example a u uh the active inference Community has a wide diversity of competencies um the but but it is highly highly Technical and so to engage with people who who um have complimentary sko skills um uh and and um and to support the accelerated discovery of allies not only based on competencies but also on um uh other holistic considerations is um is the attending to the flow of mutualism um and so those those three um actions is kind of a starting point is um is something that I would like to um I'm I'm already in action on the uh with with other groups um and a variety of groups on the constituting the legal structure of a of a fair sharing Commons um and I would like to enroll um the active infrance Community to become an ally with with kind of other competencies and other types of mutualistic eco-minded socially minded groups to grow um the capacity for mutualism and it's early stage I literally um this this is germinating you could say and really would like to um would like to find allies who who knowing that we don't have all the answers but that we're willing to grow not just the methods but the relational soil um is um is what this is really about so um yeah I'll stop there awesome exciting I can ask some questions from live chat and we can explore just a little bit of a shall I leave this on the screen or keep going sure you can leave it up I'll ask there's a lot of great I'll just start with a direct question so people can add more in this last 15 minutes okay social Design Studio wrote does the commons use emergent hierarchy so as to accommodate contextual expertise and thus relational power so um great question and um there are several definitions of Commons um but I'm going to go with elrom she won the Nobel Prize for and um and she she basically defined it as it's whatever Markov blanket people choose and then self-organize around so so um there are many variations and um and in order she she identified eight principles for common to not just be a nice idea but to be able to persist sustainably right um just to do a little sidebar she she was trying to make a business the reason reason she won the Nobel Prize is because in the profession um everybody was explaining most of the economy's activities as either being driven through hierarchies or through marketplaces the Dynamics of those two Collective groups and she said there's a third kind of group a common and it has the following properties and um the socially mutualistic uh side of the world was just in love with her and this is why she won the Nobel Prize because she developed not only a body of Science and a series of eight principles that predictably describe those that succeed these eight include you know a sense of a boundaries that they have in common and whether they decide to self-organize around a fishery whether they decide to organize around um a data comons around uh a knowledge comment that is up for the group to to decide and then um the the way in which they co-regulate um she there there were many variations but but basically there was a um there had to be agreements on where to play shared attention and as people diverged from the Norms that there were ways to call them back into relationship and to have escalating um consequences if people deviated um now all of that sounds lovely but um but that was not enough either there had to be the ability to enforce one's boundaries not only internally but externally and so I'll give you an example um I I uh have some First Nation friends with whom I've been meeting every Friday for years to work on decolonizing our minds and they have a point of view about a piece of land that um that is their Garden of Eden and sacred and how it should be used or not be used and they use all these principles but they cannot enforce their agreements on other people and so they really what they have is a wishlist I mean I know that's kind of rude and they'll be very pissed to hear me say this but without the capacity to enforce your your boundaries you um you don't have a Commons and so um this idea of uh how enforcement happens and whether that is done in a hierarchical manner or whether that's done by self organizing you know like when when uh in Valencia recently there was a a a a gigantic flash flood you know the government um did a very poor job for the first week and the response was done by self-organizing ant behavior of neighbors helping neighbors coming in doing whatever they and that happens in crisis after crisis after crisis and um and so there is a a capacity to self-organize if you will and get things done done um often what a government structure can do is help prioritize bring order to the what are the most pressing needs how to discern what will make the biggest difference collectively and so there is a value to having a a coordinating role but um but uh you know how that emerges or evolves or unfolds is um can either be by by cultural precedent of what the rules are or it can emerge through and from the group so um so that's my long comment about Commons and and commoning and how to be in relationship to the common awesome I I'll read a comment from Scott even before you mentioned Ostrum Scott had written Commons Elanor Ostrom is a co-management regime framing nicely consistent with active inference in human organizational contexts C Scott Shackleford at all at U of Indiana for awesoms Commons programs Commons analyses have historically been focused on quote asset based Commons this nicely anticipates common co-management Beyond Asset Management I.E risk management skills coordination Etc y i I should have made the point that um one of the things that Commons do that uh that the marketplaces and the institutions have not been able to do is for example it is financially rational to fish to Extinction to deplete the forest because the price just keeps on going up so so so um the logic of markets is insufficient for sustainability and this is why organizations that don't have a logic of of of that friction of of um of considering other factors other than profit maximiz ation will inevitably um deplete the resource um and so what she created with this um case for Commons was that there were sustainable organizations and that these existed both in indigenous you know in in original people cultures they exist uh throughout developed economies they exist in a variety of places and are unrecognized and what what you're hearing this in this invitation is like let's let's let's use commoning even though it's unfamiliar even though it's different it's actually much more widespread than we ever have experienced you know it's it's like believing that there might be something other than McDonald's you see McDonald's everywhere so you think I have to eat burgers but there are other ways to eat than McDonald's and not trying to pick a fight with h fast food but how are we going to carry this forward into 2025 and Beyond yes so um so the invitation to join um a conversation is is the beginning one one of the Allies is um that is both co-incubating other Commons and that I wanted to use for if there's energy here is um a a group called uh evolute 6 uh based in Germany and and uh South Africa and UK uh it's a small firm dedicated to incubating uh and helping Commons type organizations grow um it's kind of the we're uh um and and so uh trying to both invite the interested to begin join the conversation sharpen uh a vision and an understanding of what it is that we think we can be in action on uh what is it that how that active inference can be applied to particular problems the search for funding there might be a better way I had to I had to scramble to get something plausible but if as we have conversations we might very well identify much more um important points of intervention with the question that somebody else was asking earlier we might decide that there are certain flows and certain dynamics that are in fact a great a better starting point than what I imagine right now but uh the invitation is come let's convene and do this in a systematic way and both be transformed individually interpersonally and ripple out into the world awesome okay few more questions while we're still here Frasier wrote I'm not sure what this language of trauma means I'd like to hear this elaborated yes excellent sorry um I forgot to explain that so um uh lately in Psychology and in organizational psychology uh there has become a modified use of the word trauma and um and what it refers to is the equivalent of memory in in computation so you can have a traumatic event and be traumatized by it and then the trauma as as I'm using it is the memory the scar that's left either in you or the um if it's it's if if it's a collective trauma that has a different um property I'll describe it in a moment but but when I say it's the memory the scar is what remains in you and how you carry that whether you heal fully or whether it becomes a a debilitating Dynamic is is when the word trauma can be used to describe it Collective trauma is um and this is an important distinction because sometimes the same language can create confusion um uh when I'm acting as an individual cognitive system I am all the trauma happens within my body so I feel something I fix something or I have a story about something happening to me all of that is personal trauma if I'm part of a hive the the um hiving is happening in the signals that we are sending back and forth to each other and so um the uh the this the the language of how we signal and the meaning of what that means is what can become traumatized and let me give you a simple little example uh and and those are called trauma agreements ways in which you agree to carry the the significance the pain the feeling about the situation that you're looking at so so my father took me fishing when I was eight years old for the first time and we caught a fish with fish and Hook and the fish was flapping and resisting you know and I was kind of shocked I'm like oh my God that fish is suffering my father's like no no no no no you just grab you know whack them on the head knock them out and they don't feel pain and in that moment I entered into a trauma agreement a different understanding of how to be with what I was observing and as a group we would talk about the our reality assuming that shared trauma agreement that shared way of of understanding that same kind of dynamic whether it is how what it means to be a certain gender what it means to be an organization that is um beneficial to the world is it because it makes a profit is it because it makes a contribution there's a whole series of assumed agreements that that it's like oh we need to surface them that they're not the truth they are an agreement into which we stepped and we're now willing to re-evaluate and this is both the the the gateway to disorientation but also transformative adaptation um because it changes what has been a habitual way when when when I started considering that maybe animals are still an omnivore but when I started considering um that maybe animals had feelings I had to start thinking differently about what had been my ordinary eating patterns and so this is a language for surfacing um anti- symbiotic ways of being that we are now developing a pattern and a way of recognizing them and renegotiate aing them awesome great answer last question music minded wrote how does Michael see the future of businesses or more simply do you have Exemplar businesses that are commons type in your view yes there's a whole range of Commons businesses and and and so even though we're talking here about incubating a new one in a sense um I I believe that the the important work is about what I was describing about treating cancer calling back into relation and so um some of us and this you know if there's interest in this I've been working with with global standard setting groups about how to measure and monitor well-being and then how to so there's the sensing it and then there's the what do I how do we digest it you and this was another benef another you know benefit from from active inference was the way of framing the cognitive process not just think but sensing integrating predicting responding adapting and and really helping organizations think about in that cognitive flow in that Collective cognitive flow how to recalibrate them their their attention so that they see more of the world with with a denominate a planetary denominator rather than an organizational denominator and how do they then more skillfully interrelate with the different factors in which they're they're they're engaging perfect thank you Michael great presentation a lot of people appreciated it in the chat so see you next time thank you thank e e okay welcome back everyone we are here in part one of the workshop that John clippinger and our partners at first principles first are coordinating so with that for the next session please John to you and take it away well thank you very much um I would like to just to share a a deck with the the group here if I could um and uh and provide a sort of an overview of of um what we'll be talking about in this particular session can you see that now is this is this good for you okay great um and we'll be anticipating with Andrew is uh sure will be coming on and providing the tutorial but what I would like to do is in the first sense of this is is being able to provide an overview of why active inference uh is important why active inference agents this whole perspective that we're taking here is worth people's attention and pay uh and being concerned about um and what it has to offer to address some of the the the the limitations of the current uh AI systems for example large language models and some of the uh agenda architectures have come out of them um and and in that in that context what I looking at is said what is the current state of large language models and and agent uh and agents and I think you know there's a lot of activity that's been talked about but there there is still no Mission critical applications yet there's a lot of an exploration you have people like um Salesforce developing Little Einstein that can do uh various kinds of uh workflow automation you have co-pilot and Microsoft but it's not it's not it's not really core critical not to say that it won't be but it it is still uh is still on the experimental phase it's still a back box we do not know rela how it works hence you have issues of data Pro Providence you have a question of how to you govern it um and there's more recently there's a concern about you've invested a trillion dollars but there's an unknown business model coming out so there's a lot of money that's going into it but it's not clear that the current framing of uh of uh AI um is is now known as really is going to be viable and part of the issue here is is that we and I love this example from The New Yorker is like who's modeling who we're actually putting data into the models we're draing representations and we have that interaction but there's not necessarily an abstraction that's coming out so it's just it's kind of this uh relationship of of seeing ourselves mirroring ourselves and going into this sort of cycle um and what we have now is a certain hype cycle so it's around the peak of the this is 2023 from the uh Gartner group and it's the generative AI is at the top of the hype hype cycle and then if it's hard to read this particular uh graph but but you'll see that principle AI is is U basically um on the early cycle it's on the early phase of it but it's not seen as taking very long where something like uh you're going to have AGI is seen as being like 20 years out it's the top of the hyp cycle I think if we update this there now I think it's starting to be appreciated that the generative models and the expectations and the kind of problems and the costs associated with them and the lack of Improvement the the the lack of degree of improvement through successive models is starting to show through so the the Next Generation open AI shift strategies a rate of GPI Improvement slows in other words the new or Orion model is not did not have the level of improvements they were expecting I think that is something to take in account um and hence the cost of of of training and inferencing is extremely high and and with that it it the the M the effect you have this enormous consumption of energy uh that Microsoft is going to have to recommission Three Mile Island and drive one of their own data s this shows that impracticality of having to um have so computation um in order to have work make these models work and I I think that this is starting to be recognized within the the the limitations of large language models and sort of the the the parametric approach to to uh modeling uh AI modeling and I think to just start to step into what what we have now what we're looking at I love this is another cartoon from the the New Yorker it say says to think it all began with letting autocomplete finishing our sentences and well autocomplete is a sort of form of markof chain so we are we are having the sort of chains on top of chains that we've been able to to develop some pretty rich representations of of of human uh activity text and actually forms a cognition but there's not an underlying model there and so what we're looking at there may be some foundational issues that are coming up in in for large language models and that the fact that it is a black box is not explicable uh it locks internal causal models um there's not a way of reducing the complexity it's sort of a batch processing it's there's no real time updating and that batch processing we see is extremely expensive and and so we we're we're caught on on on that that issue um it's not atic it has no no no means of a spontaneous adaptation and correction on its own um and it certainly doesn't know what it it doesn't know that has no way of representing its completeness of its knowledge and and adapting uh coherently to that and and and also I would say it's not it's not derived from really sort of fundamental scientific principles is really sort of an engineering phenomena on a case-by casee base uh basis now this is changing um and I think the same issue if you start to see foundational issues for multi-agent uh llms in other words the transition to to making agentic architectures where you can um create a particular agent that's optimized to do a particular task or role a lot of the success of that agent is depending on the quality of the prompts of that and sometimes those prompts are good sometimes the prompts are bad but there's not a really princip way of doing that currently and so they have a very fixed way of of of and specialized kind of goals it's sort of a a a mechanical it's it's so the instantiation of an industrial model um that we see um now what we we rather than having a a a sort of biological self-correcting model it is sort of these fix static models now what I would say the active inference difference here um is is that it's really developed from first principles and I think and and I think you're going to see that Andrew is going to go into this much more detail but it really C it cuts across different disciplines that themselves have been maturing fairly quickly in the last two or three decades and so it is going and basic basic physics Quantum information Theory the whole idea of computational Science Biology and cognitive science all these are intersecting into creating a new way of looking at what intelligence and life are and how they're interconnected um and it also builds in well established statistical methods including Markoff blankets blazi and belief models the stuff the work has been done with JD a Perl um and basically what you look at active infine does and and just sort to highlight is it codifies a predictive brain thesis that's been nurturing in in neuroscience and so the pr and then that in turn sort of mimics the processes of the scientific method and when I get so we think about what the method is that you're you're trying cify and and I turn to Richard Fan's comment well quoted comment that I can't what I can't create I can't understand so there's a new kind of criteria for what it means to have a um to understand something is be able to model to create a digital twin if I can recreate something then I understand what it is and if I can replicate that behavior and that that's sort of the the the principles that you find an active inference from the very beginning is ah how do I grant a generative model how does a living agent have a generative model of it s in its world and this this really owes his work his Origins back to to uh the work that's done with J Pearl on his understanding causal reasoning and also Markoff blankets you how does some what is an autonomous thing and this is where

Carl fritzen an work how how does some maintain its separation from the world in which it is in and so it is not governed by the things but maintains its autonomy and there's a certain kind of topology there that and something has certain kinds of inter internal States it has external States observ states and action States and there so there's a lot of deep Concepts that's in it I'm sure Andrew will go into that much more detail um and then what we have is is the active invers agents themselves as sentient live things so you're creating something and is very different than say a large language model you're taking something that has agency and it has a representation of itself in the context in which is trying to stay alive um and so it's self-organizing is self-healing is self-replicating and symbiotic in the sense that it evolves complexity Mutual synchrony with other living things in its World in its Niche it's a joint Niche construction uh hence sentience is is sort of a is a kind of situational intelligence and awareness that's granted in the survival of that particular agent and the person that's sort of the the context of the person who's really behind this is Carl friston and and um my own experience of being someone who is very interested in sort of in artificial intelligence from from many many years ago and self-organizing system of cybernetics it really hasn't been until the last five or six 10 years that this has come together in a coherent uh not only Coan Theory but a whole Cohan formalism um and through what I what Carl would call a physics of living things and he's one of the most scied scientists in the world it's a neuroscientist that's done um basically developed a reputation neuroimaging and inferring the structure from the brain from Neuro Imaging uh and he applied the free energy principle which we'll go into more discussion to later um and so one of the I love this idea of a Tre foil not it's sort of like okay is it is sort of this autocatalytic biotic system it's really important it's a living system and it has certain kinds of characteristics that are all intertwined with itself one cycle synchronizes is exception or it's a representation of the external world to minimize uncertainty the other does it internally it has to align that with the external world and then you have another cycle that has expected free energy minimization it makes predictions about how to change itself in its World um and so what we have here is basically a a a what I think active iners offers is a really a soundness a grounding in bi in biology and physics and so you have a sort of a ground truth of evidence base that do I make predictions and do those predictions allow me to act and survive in the world um so it can be explicit it can be explainable and and we'll see we'll talk about this later but you can have since it has CA of models and language and knowledge graphs it can explain why it doing what is doing and also it can talk about explain why how much it doesn't there's degrees of uncertainty associated with it and there's also the potential of self- policing self-correcting mechanism and I think this this is really important in other words you can design a active inference agent to to live within certain constraints and Bounds and you can build in certain me me mechanisms that have for self-governance that cannot be captured uh so it's actions with the limits of his designed Mission and not to have generate negative externalities so this I will sort of leave with this one and then just to quote William Blake I mean who talks about this in a poetic form as a fractal nature reality see the world in a grain of sand and heaven and a wildflower and hold Infinity in the palm of your hand and turn an hour this just sort of pulls it together in a more poetic sense so on this I would hope to uh hand off to uh to uh Andrew if uh M thank you I need still great thanks John um and so hi everyone I'm Andrew um Andrew pase I'm uh currently involved with the active inference Institute it's been great partnering with with uh Dr clippinger with the director of the Institute you know we've we've been doing some really exciting work lately um and I want to address a lot of the points that that John has made although most of that will come tomorrow when we give some major updates on our biofirm so today in the spirit of like an initial workshop and kind of introduction to active inference I'm actually going to refer back to uh this tutorial that I gave a couple months ago uh at a International Conference for computational social science meaning folks who are involved in sociology economics mix uh you know a broad range of fields and uh the context was applying active inference a lot of these principles that John is referring to to social science research uh and and I I basically make the the claim and the argument this is already being done so I'm not necessarily providing a bunch of new information but it's just to kind of extend the dialogue to computational social scientists like here's how we can apply active inference to you know your particular setting in different ways so uh we just kind of cover some of of what I went over and you know I apologize if there's any Whiplash from how quickly I go through these slides um but we're just going to move a little quickly um so the context again of this this uh this tutorial that I gave was it was split into three parts was a 3.5 hour Workshop where we actually we started with talking about agent based modeling you know how it's traditionally been done and then followed by that something that I would call a cognitive turn in agent based modeling that's happened over the past couple decades as you know there are these kind of calls for you know we have the computational resources now we have an immense amount more of you know Empirical research and validation of different kinds of principles related to human cognition coming from Neuroscience neurobiology uh you know cognitive modeling uh studies of neuronal Dynamics uh organisms and their environment all those kind of things so so there's a shift to that and then the most recent you know means of trying to establish that in the broader state-ofthe-art area is uh using reinforcement learning agents so we'll touch on what those are and kind of how that works and you know anyone who's kind of been looking at the the agentic uh framework for for what's being done today they'll probably already be familiar with reinforcement learning then after that we'll go into AC conference and how that kind of kind of bridges together a lot of these different things um and apologies I want to just mention real quick uh you can actually go to this this link uh the quickest way of following that is just by going to Google type you know my name Andrew pashe and then you could type GitHub or ic2 S2 in any case you'll come up to uh the landing page for this you'll see links to all previous slides I've made uh slides that I'm making for tomorrow uh I'm also making a lot of this Open Access uh so so there's code that people can you know play around with and things like that um so agent-based modeling uh this has been done you know this goes back to roughly 70s 80s with a lot of you know uh initial developments of computer technology and being able to realize that oh we can we can simulate uh people right in some loose sense in in some kind of simplistic sense uh how can we simulate people and why would we want to do that we would want to do that so that we could maybe imagine uh hypothetical scenarios imagine you're designing some kind of policy or want to test a policy uh and see you know oh if I recreate that in a simulation and have people in the environment where I'm implementing that policy uh what happens what's the collective outcome you know what occurs so all kinds of fascinating work from uh shelling and the the traditional uh way of looking at uh housing segregation for example th handling those kinds of problems how where do people land where how do they settle you know in in uh like a residential environment uh even more interesting work later after that Santa Fe Institute recreating an artificial stock market uh where you where you have agents who are basically interacting with this kind of financial environment where they're looking at stock prices changing they make choices to you know buy or sell Etc see how that plays out Recreation of housing booms and bust excuse me not housing financi so this can be applied in many different ways point and so what what is Agent based modeling you know in it's fundamental principles and how does that relate back to active inference which we'll get to um key components of agent based modelings that we have agents kind of the catchword of the day um traditionally agents are you know that they just represent some entity uh a living organism you know this this can be used for for biological and Zo iCal studies trying to understand you know how ants or or insects you know interact in environment or at a broader scale how humans and entire you know societies and cultures might interact with their environment and and so what's really important about this with agents environments simulations that know we just kind of run a simulation over a series of time steps what and so we let Dynamics play out and then what is it that that composes those Dynamics it's the assumptions we make about agents what is their behavior what do we think that they do you know how do we program them in this way or in that way uh and then also what are the aspects or characteristics of the environment you know and so again given that a lot of this work was done starting in the 80s going forward with with with very you know we're nearly archaic at this point uh forms of like computational resources it's like things were kept as simple as as possible and there are a variety of guidelines that need to be used to develop agent-based models for running these kinds of simulations uh you know and so so some of them even running up to to 2007 here this is the work of of drors laser and fredman or respectively uh have been involved with Northeastern University in Stanford um you know that they give some guidelines for how to do this um guideline one and two uh in some is you need realistic uh assumptions that you're making about your agents but not too realistic in the sense that they're too complex one because we don't have computers that are Advanced enough to run highly sophisticated complex agents and then also you know if we make if we make too many assumptions then the situation won't generalize well you might be making an agent based model for one very very particular scenario where it it's not adaptable across different real world circumstances right so we need some kind of balance in terms of the complexity of the assumptions we're making and the Simplicity of the assumptions we're making the rest are a bit more plain which are one uh reproducibility you should publish your code you should document it very well other researchers should be able to kind of take up the torch and be able to experiment with your code um you know we at the Active inference Institute we're constantly putting out new GitHub repositories and documenting or processes so that other people can come get involved so I just you know want to mention that there's a big problem in reproducibility in science today and it's very well known you know so uh just wanted to touch on that the non-triviality it's like why do agent-based modeling uh if it doesn't help us uncover anything you know why why not just sit down and have a conversation about you know what what do we expect to happen what are our thoughts about what's going to happen why you know do all of this um with with setting up these programs and and agent based models and signing them if we can just come up to the same conclusions through speaking so you you want your results to be interesting you want to make realistic assumptions that you know we can kind of have all these things synthesized um together so I made some points about balancing Simplicity and complexity this is just a representation of uh you know this these These are common problems in machine learning and deep learning you know you you go and get uh you know University education these are the core comp components of all of your courses you know this isn't specific to active inference like wide ranging you know concern how do you uh create a model that is realistic enough to capture uh and and explain the en the phenomena you're looking at um but uh simple enough to where it's adaptable to different circumstances you know there you know these are kind of in part uh you know just as uh uh Dr clippinger referred to Richard feeman you know just some reminders from uh you know famous statistician George EP box you know all mod all models are wrong but some models are useful right we're always trying to approximate a reality and so the question is how do we come closer to approximating uh our reality and so how do we actually you know using modern computational resources using all these new advances and all these fields how do we actually make you know new forms of what we'll call cognitive agents who exhibit the kind of mental Dynamics and social dynamics between them that actually better approximate real human behavior so I give some definitions the differences between traditional agent based models and what agent uh assumptions are are being made and uh more often than not for the sake of Simplicity an agent is pretty one-dimensional if they observe X then they do y otherwise they do Z these very simple pre-program rules they're not autonomous they're they're kind of like you can think of them as like thermostats or something like that right the temperature is too high you you turn it down if it's too low you turn it up um that kind of reactive framework uh there's no beliefs about what's going on there's no internal goals per se aside from a single homeostat that is you know maintain the temperature at what you set um cognitive ages however the expectation for those is throughout the the recent uh agent-based modeling literature is like how do we achieve cognitive agents to exhibit internal cognitive mechanisms they exhibit perception action they actually learn and update their models I'll make a point that you know as John reference with in the case of llms which I'll cover much more deeply tomorrow llms do not I mean it's so expensive to train them and you you typically just train them once right and so um they don't learn in real time they're kind of stuck with knowledge that they have they have a lot of it to be fair they've been trained on you know thousands thousands thousands of documents uh and so um but but that's it like that's what you have and then and then we have to figure out how to work with lenss from there with cognitive active inference agents as well as other kinds of cognitive agents the ideas that they can learn in real time meaning in the moment and adjust their expectations and their beliefs to what's actually going on now um and then also planning so the capacity for being able to infer or guess using evidence that they take in like what's going to happen in the future right and that's immensely important in today's climate right where we're that I mean that's in part the reason we're doing H mod is to attempt to plan to test things to to come up with their own inference about what's going to happen in the future and then act in actionable ways to to realize our own goals in the future right so that's that's you know ideally that's what an agent would do and be capable of um and then with you know this cognitive turn we we're also looking at a broad synthesis of all these fields you know as John mentioned we're looking at things like cybernetics and machine learning we're looking at physics information Theory um people are starting to it's great it's it's fascinating that so many uh interdisciplinary conversations can come up and collaborations between researchers of various backgrounds because it's like oh you know once we once we realize what it is we're trying to do here we can start to pull from maybe principles or relate principles from other fields to this field what works what makes sense what's empirically you know what can be validated so on so it's it's it's um you know it's a very rich field right now and then furthermore a lot of traditional agent based modeling has been done using uh a software called net logo which is very specific uh it's it's good if you want to do agent based modeling like you just learn this one programming language there's a GUI like an interface for using it it's highly simplistic it is not integratable with other tools not integratable with you know llms with natural language all those kinds of things um and so it's it's very narrow you know it's good at what it does uh with cognitive agents we would want to be able to pull from all kinds of other resources that are you know available including open source ones uh being able to use apis being able to pull in other kinds of real world data not just purely simulations you know so um there's just a lot of richness behind this idea of moving to cognitive agents then there's some basic you know uh ideas of what makes a cognitive agent again perception they take in observations and they infer what's going on in the world and kind of like we what we do my stomach growls and I infer that I'm hungry for it's a very simplistic example um and then agents are are generative the sense that they actually come up with uh their own plans or a you know choices of what actions to take based upon what they believe such as I observe my stomach is growling I iner I'm hungry I will now choose to go eat food as like an affordance that's available to me in my environment so this is a new way in a certain sense of doing Age based modeling where before we were looking at modeling an environment with a bunch of onedimensional Agents you know predictable agents on the inside if they see X and they do y otherwise they do Z but now we have models in a model in the sense that we're actually able to internally inspect our agents right by having this beliefs based framework where agents observe real data or information and then infer beliefs about it we're able to then retroactively look inside the agent and see what its beliefs are as opposed to these kind of you know very difficult to to interpret uh blackbox models that we find very common and deep learning and and and you know very very highly performative yet very uh uh difficult to interpret like neural networks and architectures like that with with uh agents we would want to be able to see in a in almost like a human sense like what are their beliefs that they're holding and why are they doing what they're doing so we need to be able to accomplish that in order to actually realize all the additional benefit of doing this which is being able to look at you know internal cognition what are agents learning whenever I whenever I Implement a new policy or something right I want to be able to know what's going on with them internally I want to know the full impact of of what it is you know this plan I I'm considering like actualizing so let's bring it back um you know how how do we address all these traditional criteria how do we make realistic assumptions how do we um you know how do we make our code reproducible that's an easy one because like well we're still coding the whole time uh and then you know how do we how do we derive useful insights from this um and what are other problems that are going on like now that we're getting into to cognitive architectures um and and where we are starting to brush up against the potential of recreating more blackbox models right we we need to keep things simple enough like how how do we navigate all this common problem in in social science research in general but also in data science machine learning in those kinds of fields uh reinforcement learning common issue is what is uh how do we handle What's called the explore exploit tradeoff this this is famously an issue uh if you have an agent who comes up with a suboptimal uh solution to a problem you know let's say an agent who's you know they're trying to figure something out they find something that they deem to be good they get a positive you know reward signal or positive feedback from that but it's suboptimal there there's actually a better solution exists they just don't know about it yet how do you get an agent to actually be willing to explore and same thing for humans right we we don't always stick with the you know the first decent looking you know option available to us sometimes we we explore in the sense that we seek better solutions to problems and this is especially important and something to be addressed in agentic Frameworks where if you're trying to use agents for say software application or a use case like you want them to be able to better Solutions ideally you know at any given moment especially in a changing context that might call going forward for better Solutions you know say agents who are maintaining an environment that's subject to uh natural disaster or or or or climate change right like that that might call for a whole new set of of of policies or actions that need to be taken uh you know that that couldn't be relied upon in the past so we need agents to seek new information um unfortunately in reinforcement learning you know that this is still an issue um people are trying to figure out how to kind of manually Define what what defines exploratory behavior and it tends to boil down to kind of simple mathematic of of defining like okay well we'll s we'll we'll modify these agents so that 90% of the time they go with the best solution they have they know about and then 10% of the time they'll just something random that way they're at least exploring 10% of the time you know you could put it that way active inference again I'm sorry I'm kind of zooming through a lot of this call this is a over threeh hour tutorial um but uh rein with active inference agents this kind of gets naturally naturally resolved in terms of What's called the free energy principle which we'll be getting into um but basically the idea is if you're able to to have agents who naturally seek information perhaps in line with how we do right that when they come up against something they see oh there's value in seeking new information to find even better Solutions the ones I already have and being able to to do that in a way that isn't some kind of ad hoc rule of you know the 90% versus 10% as I mentioned that's that's um kind of a key component you know a key consideration designing agents um so this brings us to the active AC inference which I I describe here as the framework for doing agent-based modeling it uh begins with the free energy principle uh I I'll make a lot of mentions to this nice textbook that uh Thomas par Carl friston and uh you know their other co-authors published a couple of years ago um but you know there perhaps even some more updates to be made because it's such a lively field but uh that textbook really lays out a lot of the core principles and and ways of getting started with active inference um the the free energy principle everything that changes in the brain must minimize free free energy now despite all of this interdisciplinary communication dialogue between you know again cybernetics with psychology with behavioral economics um that actually sounds quite simple right to be able to boil everything down to oh we have a brain it's trying to minimize free energy that that that almost sounds simple enough to where we might be able to deoy that as a kind of you know simple assumption that we make in agent based modeling it's quantifiable uh it's it's a presume presumed is the wrong word uh but but something like an a principle that can be applied in any given circumstance whenever we're looking at you know the these organisms who are self-organizing such as humans such as you know living organisms such as you know plants in an environment that's being taken care of uh you know that they exhibit this this kind of uh characteristic of minimizing free energy as a as a form of quantifiable uncertainty so we'll touch back on that again in a moment but the the the what active inference provides is a kind of normative holistic framework for what I argue should be applied to agent based modeling it it itself addresses agent based modeling that phrase is used in the textbook this is not a foreign you know out of left field kind of invasion of active inference and agent based model and proposing there's so much overlap and I think it's just an underutilized connection that's already been made right um active inference is based in you know it provides neural process theories meaning that we're not just you know assuming in the same way with agent based modeling we might assume oh all people are utility maximizers um in this case we're not just assuming it we're we're looking at neuronal Dynamics we're looking at you know Decades of search into using neuroimaging studies and looking at how you know how does the brain really function how does it handle uncertainty different kinds of Behavioral Studies and tasks where you can you know record uh fmri or EEG along the way and look at the Dynamics of those um uh of that data that's being collected along the way and um and then from there we can start to kind of model how that works so um as I already mentioned uh we're still looking at a kind of belief-based framework which is to say uh you know we're looking at uh for for agents who have beliefs and they update their beliefs and they do that by way of the free energy principle which is minimizing uncertainty uh we we start we start to ex we start to see these organisms that we're modeling humans or otherwise they start to have naturally emergent adaptive Behavior right by by trying to figure out what's going on in your environment and then adapting your actions in order to better understand it to better realize your own goals you're effectively minimizing your uncertainty and uh and the free energy principle basically uh is is a kind of free energy is like a proxy for that uncertainty that that renders a lot of these things computationally tractable meaning that we're able to quantify it we're able to record it whenever we run agent based modeling we're able to like actually look on a chart and a graph how much a free is being reduced uh over time you know and and look at the agent's internal beliefs about you know what what is it that they're figuring out uh within their environment over the course of simulation um the view from from this perspective is that uh another aspect I I already addressed the explor exploration exploitation balance and tradeoff uh self-evidence this means that agents in order to minimize their uncertainty um one way that they can do that is is um to try and act to realize their goals there's something they want something that they expect right and so it's as if you have this prior set of beliefs of what you want to see in reality this is the you know it's this is the kind of framework that active inference deals with but also referring back to you know Juda Pearl as clippinger mentioned uh you know this is a kind of basian idea you know this is rooted in something like like basian statistics or even mechanics where the idea is that you you have your prior knowledge you bring it to the situation and then that will impact how you update your beliefs based on new information right so we're not always starting from some blank State some some starting from scratch we're always coming to a situation with what we thought before and then that gets updated how does that get updated from active inference that standpoint is through free energy principal um so you know it's I made some points in this tutorial because I realized we're we're we're really uh uh you know I want to make sure we have time to to kind of bring this back um but uh with active inference that we already have so many aspects of uh reinforcement learning and active inference which overlap we're trying to create cognitive models you know active inference I don't want to say it's necessarily competing I don't know if that's the right word but it's you know it's it's it's making advances that reinforcement learning is trying to make itself um and uh and with active inference it's like well we with active inference we're actually basing this on you know a variety of of of principles you know that I'm again sorry zooming through you know we're able to like cognitively model you know what's what's going on in the mind in various ways that reinforcement learning is really lacking um today it's it's really lacking that it's it's uh you know these models are being built to be highly performative but they're missing the mark whenever it comes to something like realism and basing uh basing these reinforcement learnings and something like a you know science that extends Beyond just the mathematics of machine learning right we we want to be able to find biological physiological substrates for how all these things happen right um we want to be able from there to be able to apply everyday words and interpretability so what is it that defines what a habit is what what process is involved whenever a goal is formed and made and then from there uh you know attempted to be achieved through action uh you know what defines planning what is prediction um and so I just want to see if there's anything else I might be able to touch on this is a nice um you know for for for anyone and maybe I should have started with this but this is a little bit more of an intuitive glance and this is actually these are not words from uh Carl friston surprisingly uh they're from uh Sarah Feldman Barrett who's at Northeastern she's uh you know nearly as rep reputable as Carl friston in the Neuroscience field and I just personally uh as someone who's also interested in in in neuroscience and cognition you know I find her work fascinating she's done a lot of work on um emotions and uh you know kind of affective neuroscience has to do a lot with looking at what's going on inside the mind of you know especially humans and so this is kind of trying to provide an intuitive glance that you know how can we think about basian Frameworks about um um inference about active inference as well so it's like uh this is a long quote from a podcast that she did your brain receives sense data from the world sights and sounds smells and so on receives sense data from the body so when there's a loud bang or a tug in your chest your brain receives that information as the outcomes of some set of causes but what caused that loud bang what caused that tug in your chest your brain actually doesn't know it only receives the outcomes it has to guess at what the causes are this is called an inverse problem you know what the outcome is you don't know what the cause is so your brain has what uh you know another source of information available to it that his past experience memory um apologies for just like doing a slideshow karaoke here but you can read it for yourself I suppose but the point is and with some of this highlighting it's like we're actually taking account of a variety of things you know we're taking account of interoception meaning observations that arise from within the body or for an active inference agent uh you know it it it it has to kind of model itself you know I mentioned earlier you know if I if I'm hungry this goes beyond the five senses in that kind of perspective right touch taste feel all of that um with this it's like well I need to understand what's going on within my body I need to understand what's going on outside of my body as clippinger you know referred to earlier the idea of a Markov blanket it's kind of like I need to figure out what defines the boundaries between myself and what's around me you know on the one hand I need to have a model of myself in order to maintain it uh otherwise I succumb to the environment around me to use some of the language used in uh you know the textbook it sounds a little Bleak but you know we're referring to real world circumstan is if I you know if I don't take care of myself I will naturally Decay like that is kind of the sort of the principle the world is entropy so to speak right it's like if I don't eat food you know I I I fall apart so I I need to have a model of myself what's going on within I need to have a sense of what's going on without uh and then and then I need to be able to kind of synchronize those things in such a way that I can continue to exist as an entity within this environment around me that's a that's a that's a very intuitive you know perhap perhaps not very well technically described but for now an intuitive sense of of kind of what defines a markup like it so you know for this is a sort of foreshadowing for tomorrow we'll talk about this this really interesting project we're working on for the biofirm uh with with with uh you know Daniel and and and under guidance of of John um you know in that in that sense we're we're trying to develop homeostatic agents uh homeostatic meaning agents who you know their own sense of their survival is predicated on uh the the the homeostasis of their environments around them so we're trying to create agents who actually take care of and attend to the uh the environment around them as if their own Survival depended upon it you know this is a very interesting kind of uh kind of flip you know where where where you actually want to match and model the environments around you but also take care of it in a way that you can maintain M it within those bounds just as you know I have to eat in order to maintain my energy levels you know that the agents need to take care of their environment you know within some kind of range to where it has you know enough caloric energy so to speak um so um I think for now John is there anything that you think I have not addressed that might be useful to to our audience or if anyone else would like me to touch on anything maybe something I skipped or or otherwise would be happy to to speak more no I I think you you you you've uh covered some of really key points that um that are that what I I think is you've done is what differentiates active inference from the more traditional agent-based modeling I think is important because when people think about agent-based modeling they they sort of default to what the traditional models are and I think that you've done an excellent job as sort of uh highlighting what the difference is and what the the the having having that broader perspective that more grounded interdisciplinary perspective you you're able to more not only represent the complexity of it but also provide a a way of modeling agents in within that complexity um and I think that the the the notion of a the the homeostatic nature of of of an agent trying to live in in conjunction or really harmony with the world around itself is depending upon its survival and its ability to stay alive and differentiate and have complexity but the synony of the environment is really another key point because I uh I think a lot of people they think oh well you may say like navigating a fitness landscape you're just trying to optimize different levels but in fact the landscape itself is shaped doing joint Niche construction so it's a much more complex notion than traditionally understood General modeling and I think in in consideration what we be looking at tomorrow this whole idea of the btic character of model having it based upon principles and biology and nature and having being able to create an economic agent that acts that way I think is is is really quite important um and so maybe some people have some questions about that that we want to to focus on or not but that I I that's something that that I think we it's important to highlight yeah awesome presentation Andrew Ernesto if you have a question also anyone in the live chat is welcome to ask a question no questions thank you very much this is so inspiring well and as just to to put it in context I know this is something that you're very interested in in this whole idea of a btic economic system and you've been looking at this yourself so this is really trying to lay down the principles really the the scientific principles and computational principles of doing this and and and uh and we'll get in more to that tomorrow but that that's the the intent here yeah a lot more to say again in our remaining 40 minutes I'm happy to read questions from the live chat I'm also happy to screen share and look at some of the registered participants backgrounds and interests sure okay okay you can unshare Andrew and I'll pull up some word clouds all right okay okay so let's let it load okay these are word clouds that are generated from the participants registration information and we'll we'll zoom in on them and and uh I I'll just kind of mention them we'll we'll Zoom it in and uh if anyone has a remark or something that they want to uh like highlight or call out about it just go for it okay so this is a word cloud of the uh 150 or 200 registered presenters overall asking about what is your personal background so the biggest and and and the this is scaled just by number of uses of each term so one person mentioning a term a ton of times might make it large so just to be taken as sort of a representative we see those tiny two letters AI which could mean almost anything at this point and a lot of other terms like computational research systems Neuroscience cognitive philosophy social academic robotics technology practice this kind of speaks to the broad range of topics that that that leads people into joining and at least entering that information on the uh form any thoughts on this or we'll keep on exploring these word clouds okay okay all right again just just raise your hand or or or go for it interest okay well there's I'm interested in the notion of community and systems I mean uh applications the community seems very strong yeah yeah and that that's the the sensor of a of a later one so here we asked what what would be useful and by asking what would be useful and what would be interesting we alluded a little bit to the Dual composition of the expected free energy what would be useful what would be utility what would be pragmatic and then what would be interesting what would be epistemic so heing what would be useful you know here we are in in applied active inference Symposium wondering about how does active inference go from being a Ying framework for ecosystems of shared intelligences and all these different kinds of things that we've heard already and we're going to hear in the Symposium to come so interesting systems a way to operationalize and really move forward from framings of complex adaptive systems to having the implementations and also the community and the people understanding research models collaborators and then a lot of different topics that are being explored in more mainstream computational science Fields like multi-agent algorithms Andrew alluded to that and kind of brought that up in the context of the the Goldilocks parameterization question the agent based model can be so simple it gives you a great intuition it shows at a a demonstration layer how simple Behavior can lead to complex outputs on the other hand the agent based modeling can go the other extreme where it's so sophisticated then it loses its transferability so that's what people would find useful and then here's what they put for what would they would find interesting mhm applications makes sense as that's what this Symposium is all about and again a lot of other ideas thinking about different specific systems research ideas architectures synergies toolkits insights implementations support Collective directions here's asking what would help you in your learning and applying active inference and Community looms large other people as well as some of the more tool resource textbook education examples Frameworks interactive tutorials different languages python what does anyone think about this I'm uh a little surprised that we not seen the phrase llm anywhere um I just uh kind of noticed that so one of the one of the points that that I you know I've been trying to work out and I hope to speak more on tomorrow is um you know integrating active inference with llms and uh you know there have been attempts to to do this and even even with priston and Par and other folks you know that there there's been a attempts at trying to pull out an agent's beliefs and then use an llm to process those beliefs like kind of decode them and re-encode them in natural language so to speak it's as if the agent is speaking and saying what it thinks and believes and why it did what it did is you know another layer of interpretability and uh I don't know I'm slightly surprised to not see LMS anywhere um you know it maybe it's a little too maybe that's a little too on the frontier right now um I know it is in many other areas you know it's uh sorry in the couple minutes we have it just uh you know something that I've been working on personally and this is feeding into the the biofirm is um how how do we how do we use llms in a way that we can actually integrate them with you know all these major priorities that we have uh where where we want to have agents who actually have defined goals right and and and ways of trying to stay alive you know in a way that they interact with their environment in a way that it's as if their own Survival depends upon it we don't want llms which you know again I'm just saying this quickly you know I don't to sound aring but llms in one sense can act as like functionally as Talking Heads where you just send in some text and they try to predict what would be the best text and response you're still subject to hallucination to confabulate making things up so to speak even though the answer looks structurally sound you know and and and so the my point is that rather than viewing LMS as uh as agents I don't I don't I don't think they are I think that uh you could have an active inference agent who is augmented with the capacity for natural language by having it interact with an llm right so yeah sorrys my own yeah scan scanning across all the registration fields for llm mentions so H how does active inference apply to the design of llm systems AI outside of llms better ways of Designing internet skill skill protocols and governance systems and should converge with tiny ml to challenge lm's deep learning so it's it is interesting that that it's uh referenced with only a glancing blow and and uh We've even seen earlier like in the responses to similarities and differences with reinforcement learning with frisen earlier about the increased amount of attention and Tool development and the the later and more professional maturity of training environments and this is like one of the key fosi for the open- source active inference ecosystem and for the Institute is to develop those kinds of methods and trainings and all onboarding all these different aspects of of increasing the professionalization so let's look at the challenges so this was what what what are the challenges facing your learning and applying active Influence People interesting the problem problem and the solution models the math software maths again mathematical understanding difficult physics scaling simple examples you know it's it's it's interesting interesting um because in the larger conversation about what AI is I mean it's it's that that conversation is so dominated by uh large language models and quote generative models within large language models um and and well at least I've been observing I tried to refer to my my initial slides is that they're starting to appreciate the limitations of of the large language models moved towards more agentic models and more towards causal models and hence creating aent architectures that do work with large language models but they're still there's not they're still not based upon biological principles that we're discussing in active inference there more of the mechanical models being applied over and I I think to me to from from a point of view of sort of the um the sociology of science soci there seem to be these different worlds that that I mean the number of people that i' I've known sort of in the open AI world they're they're very unaware of active inference and some of the critiques that are if you if you listen to the Gary Marcus he's got a very pointed critique of of um of of large language models and not being able to achieve abstractions but that doesn't extend to appreciation appreciating what the the opportunities are in active inference and so there are there there sort of these islands of understanding that they still exist um and I think that's one of the challenges and when you talk about developing an infrastructure me you need there's huge financial resources are going into the the the large language models infrastructure and where I think that the totally bias on this I think all the the opportunity is totally within the active ERS of free energy principle and I think that's where the science is going and yet is it's still it's sort of in in this isolate World um and I think the only way to get out of that is start to to really start to showing the applications and showing the with results and and uh because uh but it it's it's it's it's interesting I me to say they're they're different they're different worlds um uh but I I think that what's the there's a peing and so the hype curve around uh of the large language models and and it's going to they're going to descend my bet is they're going to descend into a valley of doubt and you suddenly you're going to see that showing up in the in the Nvidia stock going down and then people are saying you can't you know destroy all the natural resources or run these mods you don't have to you know the other and Carl alluded to this in his earlier comments today you know I mean he was talking about climate change and models compression I and I think that's going to be a driver of this um so but that that's some my my observations on on these these clouds is yeah I'll bring in a question from the live chat also welcome Ma just thought I'd bring you on if you wanted to participate in this panel or just chill so Steven select wrote is the lack of llms mentioned because large language models are different to Dynamic agent belief modeling and you want to take that could you could you repeat that uh yeah question lack of llms in the word clouds here but just more generally thinking about similarities and differences between active inference generative models and llms is is that because large language models are different than Dynamic agents or belief modeling I I don't want to approach this this question in the in the wrong way but I I will say that that that but as far as you know whenever it comes to llms and and dealing with you know their internal parameters are obviously highly quantitative and then they're dealing with with natural languages tokens and then switch back to looking at active inference agents which you know are highly quantitative in nature and uh typically you know if you're making continuous time models uh you're looking at again quantitative information and then even with discrete models you're looking at you know these kind of discrete values that you could potentially move in the direction of trying to you know make it I think this has already been done uh but the uh you know there there are agents who um you know have been used to try and understand like oh we can read individual tokens then move up to another layer and then we have entire sentences then move up to another layer we have entire you know paragraphs or ideas that kind of thing is done but I think this kind of like llm with quantitative uh uh grounded agents like that connection uh is still in my view that's kind of the frontier of where things are at and so it's difficult for people to conceive like how can we how can we translate encode decode re-encode natural language in a way that's readable by these quantitatively oriented agents and then vice versa how do we transition quantitative beliefs into natural language that's the very thing that I've been uh you know looking at thinking about uh recently I made like a active inference like PDP agent who uh you know they what they do is that they use their beliefs about which action to take and then that action that they carry out is actually corresponding to a prompt to an lln and then you can actually take those beliefs the quantitative beliefs find some way of meaningfully interpreting them and then putting that into these blank spaces in the prompt is as if you can have the agent say in real time I currently believe this with probability this give me you know a solution to this problem you know based on so so being able to find this kind of numeric versus the quantitative and the qualitative you know it's it's historical problem when working with with data that goes back centuries at this point so um it's it's really exciting though it's really exciting to think how to bridge that kind of Gap and I I still haven't seen anyone you know nail that down this this is the frontier this is where we're at and I think that's kind of the next thing for everyone to really be considering thinking about working on um and I'll give my view on how that could be done tomorrow well well um yeah I mean effectively llms don't really encode explicit beliefs or really maintain a persistent model of uh the world um even when we sort of give them a sort of loop by which we communicate with them that Loop in itself doesn't actually factor into the predictions that it's going to do um so they they just lack a framework for recursively updating beliefs which is probably why it's not being thought of here um but also mainly that you know llm agents are only agents by virtue of sometimes being connected to stuff and having um explicitly stated goal but I don't think that actually qualifies them to be agents you know yeah I I totally concur with that I from my perspective I I think that the reason the LMS are are are not looking that are are not looking at or that whole mindset is not looking at at belief structures in in subjective experience is I think is a product of a whole sort of objectivist reductionist methodology that um is has been tuned not to look at agentic systems I mean in my generation if if you if you were in biology you in and you talked about agentic systems you you were you were outed I mean that was that that's that was verboten and I think the the there's a whole school of thought there's a whole set of of in aonian sense a normal science built around um a sort of objectivist reductionist non- agentic notion and and and uh and so it's it's a new generation that's coming in in both in biology and in and modeling and cognition there's recognizing there this notion of an agentic perspective of beliefs and updates and and and that's a huge shift I think in in sort of the epistemological perspective of of a lot of what has been traditional science that's really interesting there's also just operationally the difference between taking a very large fixed number of parameters in a given model 70 or 7 or 405 training on existing data and then finding patterns or igen modes or activation um summaries that have secondary interpretability but there's not really an effort of trying to find interpretability at the level of activation of a given neuron it's about looking at patterns and trying to find Collective level interpretability like in in the AI interpretability field contrast that with designing from the ground up an example agents where the beliefs are explicitly encoded numerically and so that's a different mode of building and also one that's hinted at in in this 2023 paper with Ma and others that highlights that element of explicit semantic interpretability and also explicit semantic interpretability of uncertainty estimates whereas a string of tokens emitted by an llm that says I'm not sure it's sure about saying that but that does mean that it has a given certainty about the referent so understanding what the communication semantics of llms are is incredible and what they do and I think when we look at this Cloud for people's responses how can active inference make a difference in 2025 where there there is llms here many many kinds of systems and that's sort of like a microcosm of what the Symposium is different people with different systems of Interest asking H how do these different attributes of the model that we're talking about the first principles grounding semantic interpretability surprise and uncertainty and its bounding as a central component as opposed to reward how do all those things really play out what does that demonstration look like in in different systems yeah here's go for it Andrew no I was just gonna say on that that previous one you were just looking at I I I really appreciate the kind of a broad-mindedness that the people seem to be exercising I like seeing terms like a symbology scientist governance um industry I mean those are some very I'm very excited to see people who who want to you know so much work has been put into kind of the theoretical and then sort of like in in in vitro and and in lab and and in simulation kind of work that's been put into active inference so I'm just really excited to see you know with the the um thinner as the the conference the the back of inference conference that was given recently and being able to see like a drone that's flying based upon the free energy principle and things like that I'm just really excited to see more that kind of real world applications right and and kind of creating uh you know systems and and applications for like real world use cases it's it's really cool to see some of those terms there yeah I think that's going to be really critical see see yeah here's the cloud for how are you applying active inference again systems I think just reflecting people talking about variety of different systems types but and also reminding me of my own journey to active inference being excited about complex adaptive systems however not really having a software toolkit or a formal way to proceed past okay we think about nested multi-agent systems and then where's the software toolkit and then we're right back to where Andrew showed us on the slide it's like well we can do a really simple one in net logo or we can engage on this Enterprise scale engineering research program and yeah here's a more incremental interoperable way to have portfolios of motifs and generative model components and be assembling and standing across these different kinds of compositions like three ant nestmates in the desert three ant mate nestmates in the forest and these different kinds of environments being able to compose them and over the last several years seeing a lot of the advances in the category theory in the first principles physics surrounding all these sort of research things and it's like that's kind of the water drawing out and then that's the question about how how it looks when it applied and that's what I guess we convene around in this Symposium and and are seeing differently in 2024 than in previous years that's curious so what what's the change that you've seen in pre previous years versus this year and then so looking forward more of an application Andrew or ma what what what do you think I was yeah but absolutely that that that is I oh yeah please yeah I mean I feel like um the work of Maxwell ramstead C Casper HP I mean obviously Carl but um a lot of people who perhaps came from more Dynamics uh decided to take a real interest in what we could do with something that was so scale-free and I find that it's been really really rich in terms of the different kinds of theories we've tried to sort of reforme I know that's what most of my work is is a Reformation of existing social sciences um and to your point Daniel the the problem we still have is that we're missing like these key little plugs um that yes you could couch as you you know just different schemes of message passing but you still need some interesting tools to to have a a very good description of the vast vast variety of types of structures that we have Which social sciences have done a really great job at if not intuiting at least describing properly but we still don't have the best tools to make accurate predictions and that's part of the reason why most of our polls when we make them are are n useless and that should be the main thing we apply them to is just literally just understanding where we're at not even making predictions just understanding where we're at and we still fail at that so I'm very um I'm very encouraged by by seeing this and I hope that we will start valuing more and more interdisciplinary perspectives that are sometimes more technical but also sometimes uh you know more philosophical or um or social awesome also while we're in this in in the coming workshops tomorrow will be more of a Biore Regional biofirm focus and then day three will be entirely with clippinger at all on this topic and and I want to show one really operational open science Crossover with llms and active inference which is and I'll put it in the live chat as well here I used perplexity which does an internet search summarizes and translates information using llms so here we'll pick on Alexi there's a a profile a learning plan and a set of small medium and large project proposals about active inference so this is just summarizing expertise and experience here we have a learning plan it's just a starting position but this brings a new low bar to the kinds of resources that everyone can take on in their learning and use as like a syllabus or a curriculum and even as an interactive tutor you know it will not get bored with you asking to explain Concepts again and again or using different metaphors or making it applicable making it like this poetry format putting it in this narrative context and so even asking how do I start with this or can you help me understand what I don't know that puts such a centrality and and highlights metacognition in learning and then the project proposals each of them structured according to essentially what a grant officer would be looking for and what a what a project manager would be thinking about in terms of small medium and large projects so this unprecedented High throughput Way to fuse even scanty input from a participant in a inclusive open science setting into real projects that if enough people think hey that looks kind of interesting then we're well down the road of actually making it happen using different kinds of of funding which is something Michael Lenin mentioned a little bit earlier so it's sort of like we can get really detailed and Technical and Architectural thinking and and and finessing the differences between reinforcement learning deep learning active inference and we saw slides to that like when Carl showed the action perception Loop similarities and differences okay active inference doesn't have a reward function it's not optimizing based upon this trained auxilary reward concept it's just doing something direct with likelihoods like that is critical that is what makes the active inference generative models what they are but then looking at something like this which wasn't so accessible one to several years ago that just feels like opening up and unleashing what people can do and about using these as Tools in our Niche and I think it's a total open and unknown what happens when you can get to level of project and learning plan for for an arbitrarily complex field with the patience of an AI That's just going to accept questions all day long like that's almost the bigger applications question again just zooming out from how the architectures and the parameters are similar and different to how are we going to do what is authentic and right and best given the tools we have there's a bigger picture Beyond just holding the magnifying glass to different computational artifacts it's well said Daniel I really agree it's transformative if anyone has questions in the chat or here or any other things they want to mention yeah what's going to happen in the next two sessions JN or what would be your goals or preferences for for this Arc of the Symposium for you yeah so I I I think as as

Andrew alluded um this the session tomorrow is going to be really on the application of of U of active agents uh to a a design of a economic agent of new kind of firm biofirm this homeostatic um and the objective of that is to develop a new kind of incentive structures a point the collective action um that are based upon memetic principles that I think we'll see address a lot of the issues that people are having in the whole idea of B Regional Finance and how to deal with climate change and how to create those incentives so there one of the things I'll kick off with I've been talking to a number of people in the whole fields of of um um regenerative Finance by where they go bofi um this it ties into the work that was done in Elan ostrom's work on uh how common pole resource how do you manage a common P resource how do you incentivize people different scales I really think that what you identify some the requirements that people are looking for in order to address these problems to scale and I think what we're be talking about is to the implementation of active inference uh principles into an economic agent that it provides I think a a a reasonable way of of addressing these issues um and that uh um and and so that's part of the conversation that we want to have and and and bring together these two these two worlds um then the third session is to really look at this um from a a broader point of VI is saying where is this going and they so the people that we have in the panels are very of very much involved with Enterprises um applying it to nonprofits to applying it to SAR problems to looking at what they can do with LMS currently and what the certain current agentic architectures are and what happens when when you s what happens when you go to full autonomy what does it mean to have autonomy um in in a in a in a biological sense something keeping alive and what does it mean to build systems that have autonomy I this issue of autonomy is sort of fundamental uh when when when people think of of autonomous organiz autonomous cars or decentralized autonomous organizations or the whole idea of U Central Bank the the the the the the principles behind something like a Bitcoin is is like oh that that that is autonomous but is it really autonomous what is this new notion of autonomy if we base it on biological principles um and I I think that's that opens the door to a very different set of discussions than than we're currently having and it also opens the door to new Notions of governance and how we can build governance in to maintain the homeotic states um and Andrew is alluding to that so I mean that that that that that's where I become a an optimist and I'm not an optimist around the extension llms and O open Ai and the and the tech the tech Giants but I am an optimist in being able to apply these principles to to some of the fundamental issues we having around climate and social equity and economic equity awesome okay in the closing minutes if anyone has last comments and David's question

question too if you want to address it Andrew or anyone else any thoughts about how we start to develop a multiscale approach to ethics um I want allow time for other people so I'll briefly say uh so so tomorrow um you know we'll we'll we'll present uh various aspects of the biofirm there will be a little bit of a Redux to some of the things I went through today just to kind of thread The Narrative um and I'll kind of talk about you know homoeconomicus and all these kinds of assumptions about human behavior and things like that but uh um I mean multiscale governance I what we're trying to do with our biofirm is to uh you know and clippinger referred to um uh Elanor ostrom's work on the commons and so I I'm gon to address that a little bit too but the idea is to be able to have a kind of like adaptable system of multiple agents uh you know and so we have a kind of small Society so to speak and uh you know they're kind of governing the commons around them and by being able to have something like the capacity for one shared goals uh and two adaptive learning such that they can actually adapt what they do and what they believe and actions they carry out have those be localized to the current context which is an essential part of asam's like work is like it's not so simple to just give a singular top down instruction that applies in all places I mean to be able to have that kind of adaptivity and awareness of the local um you know while at the same time having agents who are kind of synced up in these various ways I think that's a one way of trying to approach governance and then from there you might be able to have kind of layered you know you could have individual hubs you know and then you could have some kind of like interconnecting system between them that that extends that to an even broader scale of multiple hubs and interaction between them so just some just some breadcrumbs there um thanks yeah Ma and then John with the last word on this session yeah so I mean the question of Ethics is a bit it's obviously complicated but the the the key part about whether we coordinate around multiscale goal sharing Etc share potention that's intrinsic to power active inference right it just is the question that isn't going to be solved is what is good uh because you could you can always say well you know what is good is good by virtue of being contextually sort of stable but there may be other ways to be stable or that might prioritize other kinds of individuals or entities Etc so I don't think we're going to solve that question however what we can solve is the is the global versus local coherence of certain kinds of beliefs relative to what might constitute goodness at given scales and find where the friction points might arise because this version of goodness is incompatible with this version of goodness and therefore Uh something's got to give somewhere an error is going to have to propagate so I think that's going to be answered but the the larger broader question you you'd have to look into different schools of thought around what constitutes uh good because we don't necessarily want to go towards um moral realism um because there's a lot of things that are baked into that uh but you might want to go into the um ethical implications of neom materialism and how it it is about interactions in locality and at that point it becomes about how do certain pockets of ethical framework sort of form and allow to just coexist away from other Pockets what happens when they start interacting uh which again we may be seeing right now in the US yes good point John any last comments on the session oh this is great uh um and I think U we we now surfacing some one of the issues that are really important to to talk about and and um I think Le least what we're going to be presenting is is a is a framework um that at least we can have these discussions in a more rigorous way and testable way um and uh I I think that's part of the way moving it forward so yeah awesome thank you okay we'll take a one minute break see you soon okay thank you e welcome back welcome back hello son and Ma again we are in the session physics and metaphysics of social Powers so thank you both for joining and take it away thank you so much for having us I was just asking if we should introduce ourselves which uh which we will um so I will introduce myself and then ma can introduce herself so that I don't make any mistakes uh because my brain is on a hyper hyper abstractions at the moment anyway I am Sonia I'm a PhD student uh based in rdam at the araso school of philosophy and uh we've been working on this paper with ma recently and Ma please uh yes so I'm a PhD candidate at the University dubec amoral in cognitive Computing and I'm also a director of research strategy ad versus thank you so yeah as Daniel said we're going to be talking about the the Spectrum the big spectrum of between the physics and the metaphysics of uh social powers uh very inspired by the work of uh of ramstead here hesp Hines and uh by mil's work on on power recently with Sarah Grace mansky um and then we're trying to combine that with some work that I'm doing with two other colleagues uh Wilkins and clav Vasquez on the concept of active ignorance which we will be um continue continuing to work on in the coming time um so yeah and then generally inspired by by by ideas of semantic attraction from Hinton and by Major names in in philosophy and sociology who have been treating these Concepts through different Avenues uh less kind of formal Avenues but still relevant nonetheless now that we're actually capable of joining um yeah the physical with the metaphysical that's what we're going to we're going to try here with presentation so the concept of power we we argue can be explored at at several scales from physical action and process effectuation all the way to complex social dynamics so this kind of spectrum wide analysis of power requires attention to the fundamental principles that constrain these processes and in the social realm the acquisition and maintenance of power is intertwined with both social interactions and cognitive processing capacities so socially facilitated empowerment grants empowered agents more information processing capacities and opportunities either because they're relying on others to bring about desired policies and or because they're able to enjoy more information processing possibilities as a result of relying on others for the reproduction of material tasks material is in Brackets because these tasks need not be material so the effects of social empow empowerment imply an increased ability to compute information thereby augmenting the evolution of a specific State space specific is key word here uh really commenting back on what Ma just uh just mentioned in at the end of the previous talk so empowered individuals attract the attention of others attention is one the central Focus uh of of our paper they attract the attention of others who contribute to increasing the scale of their access to various policies effectuating these these State spaces that we mentioned so the presented argument posits that social power in the context of active inference is a function of several valuable variables among them an individuals or groups ability to attract the attention of others with access to or the ability to facilitate desired policies and an individuals or group's capacity to effectively process information as a result of its power amplifying effects this extended computational ability also buffers against potential loss or or possible vulnerabilities semantic social artifacts so narratives um ideologies histories representation at large are attentional scripts which ensure the coordination of social patterns so you can think of anything from psychological schemata habits group think insulation policies Etc and these are the structures under and by which agents mediate and leverage power and they're because of that the main focus of attention here so uh thinking about this the linking between the physical and metaphysical Concepts and the world gravitational metaphors abound in language this is like really prominent in the work of of Shephard Lake of and Johnson of course with special metaphors um I've been I've done some writing on it at the beginning of my PhD and also um laand Kent uh did some really nice work that's also up on the on the activ inference archive and then in his papers course um so the idea of power being pulled towards complex social attractors uh was also famously presented by Pier bord who linked various concepts of capital cultural capital political Capital Social Capital to power as this kind of social attraction uh and also traction and he borrowed the metaphor from from physical field theory in fact so there is already the connection is really present and obvious there uh only in in a metaphoric way um which we're kind of trying to expand on here uh so expanding also on his legacy to to you know take the epistemological and the empirical as totally Inseparable things that we're analyzing and his desire to actually ground sociological analysis in in in a scientifically tractable manner we're trying to frame uh these kind of attentional attractors that move systems toward certain configurations through activ inference and this framing allows for a new take on the dialectics between the individual in the collective again linking back to what was just mentioned at the end of the previous talk so in terms of um questions of information processing capacity not only our resource distribution and you know the division of labor a systemic fact but also the modulation and projection of future possibilities cannot be computed at the level of the individual clearly so this is something rather obvious in our context and our framing power this is how and why the minimization of uncertainty for for social systems is in fact social right so agents seek alliances and policies which ensure that cognition so information processing and future projection can be distributed and this allows for efficient practical traction because and here I quote from Steinberg at all 2023 epistemic confidence in knowing interpreting and acting together is that traction that happens when we when we're socially distributed uh and this in contrast with this kind of individualized agency where one's estimation of and here I quote again Divergence in mental States between self and others is what allows for the sort of prediction and interpretation of others Behavior which we might kind of want to frame as something that has a shorter range traction on on the future so now we're going to look at the theoretical framework and I pass it over to Ma so um obviously you have heard this 15 times but I'm still going to go through it because there are some Concepts in here that will become relatively silent after but to go over it briefly active inference explains how biological agents really minimize varal free energy to align their internal beliefs with external reality so effectively theyve reduce discrepancies between expected and actual sensory inputs and we sort of Leverage The Descent on free energy to to to couch model uh perception cognition and action through basian belief updating and so free energy is the upper bound on surprise or the unexpectedness of sensory data relative to an agent's predictions and so direct computation of surprise is complex so agents use variational free energy uh to to approximate based on their internal in generative model and so um you can see that for example the K Divergence term measures how closely the internal beliefs match the true posterior of and so agents align their beliefs with actual data as they minimize it so the log likelihood term ensures predictions align with observations so to minimize free energy agents update their beliefs they allow for adaptive action selection that supports coherent gold directed behavior and so there are two strategies perceptual inference to adjust your beliefs to match the sensory inputs and active inference which is um acting to change the environments and align new sensory data with internal models um active inference interestingly and as we saw earlier with the the whole word cloud spans or can span multiple scales from cellular processes maintaining homeostasis to cognitive systems using predictive coding to update beliefs based on hierarchical prediction errors and so this really gives us a broad biological principle where minimizing free energy promotes systemic efficiency and optimal homeostasis across scales and so um reducing free energy really means that you minimize entropy as well and you align your beliefs with reality to reduce this uncertainty important to note is that we also Factor um action and so expected free energy can be decomposed to extrinsic goal oriented and epistemic uncertainty reducing components that balance exploration and exploitations and so this dual strategy insert ensures that agents achieve immediate goals while gathering information to adapt effectively in un certain environments and so Central to active inference we really have this concept of a generative model which allows agents to predict sensory inputs and their hidden causes the generative process represents the real world states that produce observations and generative models reflect the agent's hypotheses about these processes uh the agent actions correct mismatches between its model on real sensory inputs uh to align model with reality going quickly over uh the the different terms in a hierarchical structure each level predicts the states of the ones below and it corrects predictions based on sensory feedback and there are conditional dependencies that Cascade between layers that support uh belief updating and adaptation we have observable states that are the sensory inputs the agent receives and tries to predict ch its model the hidden States uh latent variables that represent underlying causes inferred by the agent to explain sensory inputs there's a series of parameters that Define the statistical dependencies between hidden and observable States likelihood function which indicates how well the hidden States and parameters predict observe data we have a prior distribution which is the initial beliefs about the hidden States and the posterior distribution which is the updated beliefs after observing data and combining likelihood and prior then obviously we use the marginal likelihood which is the probability of observed data under the model to integrate over hidden States and so we use this for model evaluation and comparison to determine how well a model explains sensory inputs so the generative model really enables the agents to align its beliefs with the sensory data sometimes through hierarchical predictions realistically most often through hierarchical predictions and feedback loops and they refine their understanding of the environments to improve action selection for adaptive behavior and so this picture is really uh very individual so um we can see though that within this picture it's really really Rife to uh to couch power dynamics so we start from Individual agents as predictive systems that minimize uncertainty through prediction uh through perception and action and so brinberg really made a salian point about this with his paper on the Primacy of I can where agents develop you know empowerment through embodied interactions which enable them to affect the environment so if we scale this up to Social Power agents use generative models for self- prediction but also to predict others intentions they form the basis of social understanding in Collective action Carl has lots of work about this um Connor Hines as well Axel constant maxel ramstead Casper hes Natalie Castell which I believe will give a talk as well and so attention uh is driven by Deep preferences and it shapes what becomes Salient which influences both individual and group Behavior at the group level shared predictive models facilitate coordinated actions so stigmergic processes as explored by our very own Daniel uh environmental modifications that guide Behavior allow distrib did control while deont cues as explored by Axel constants enforce social norms and so cultural practices further shape Collective attention it creates shared perceptual Landscapes which is something that we explored with Toby Sinclair Smith I think I said his name wrong I call him Toby so I don't really know how to say his name and finally we have to know um how to couch joint attention and this is work that and at the University of Quebec in Montreal have also positive that posited that this mechanism basically enables efficient coordination through mutual recognition of action possibilities and so power here power structures emerge through self organizing subgroups based on belief similarity so again this work that has been explored by Natalie Castell which we can call decentralized influenced spreads within Network structures and they amplify certain narratives While marginalizing others and so these these processes create cultural capital where shared norms and knowledge facilitate efficient social interaction um attention here is thus shadeed by those preferences but it shows that the importance uh of critically constructing shared narratives without homogenizing scripts that stifle diversities uh is an important thing to consider so it's a point made by Anna AR simplistic one-dimensional fuse like there is no alternative constrain attention and reduce complex systems to narrow goals which is why fascism tends to be very attractive especially in times of high uncertainty and so distributed generative models over time including historical narratives and generational Dynamics give us the tools to Envision power as possibilistic which is a term that Sonia introduced here this this shifts the focus from isolated agency to Collective Dynamics and hierarchical organization we get a richer analysis of how agents and groups interact within constraints to shape future possibilities so active inference really gives us a frame for social power beyond very simplistic atomized ideas of agency and focus instead on this emergent distributed process that structure Collective behavior and potential thank so yeah to return to the concept of narrative because it's sort of the thing that that um frames something that might go from from from the small scale to the large scale in different ways and something that we can all colloquially uh uh sort of latch onto narratives are are are social mechanisms which have the power to make or break power you can say uh the cognitive exploration of of of the not yet engaged right of of scripts that are not yet there but one may kind of uh uh latch onto sort of effectuate anything you can think of here things like speculations fictions future possibilities of All Sorts um can be these can be understood as semantic attractors that enable kind of relatively risk-free learning right so you're you're projecting to imagine what might be the case if and the simpler The Narrative usually the easier it is to sort of minimize like vast uncertainty so these possibilistic Explorations in the form of narratives allow agents to explore counterfactual scenarios and minimize uncertainty without the sort of exposure to the to the real world consequences of this um this this safety mechanism frames both The evolutionary advantage and persistent appeal of narrative engagement in human culture and in the context of power you know leveraging strong narratives as as Ma just pointed out such as US versus them that's like the the the one that you know the biggest semantic attractor that we seem to be keep returning to all the time these types of reductionisms that greatly minimizes uncertainty about an agent or group's current predicaments and they become because of this very very appealing and sometimes self-fulfilling unfortunately policies that bring about their effects resulting in Collective Collective behaviors which can benefit sorry I'm so like which can be Ben beneficial or detrimental to the greater whole the uh the effects of of variational free energy minimization can be observed of course at the level of the basic structures of language in this sense so semantic attractors like redundancy rhyme alliteration Etc can be understood as patterns which Aid memory communication informational retrieval policies because they permanently reappear right so pulling attention to the same phenomena and here I use scare quotes very very strongly the the same phenomen uh thus greatly reducing the need to interpret and organize linguistic interaction and cognition a new every time which is very costly so we can understand for example the possible emergence of rhythmic patterning in communication as enabling the storage and access of information collectively by connecting a sound pattern to a phenomenon we reduce uncertainty about reality once that sound pattern is witnessed again in in sort of true pavlovian fashion by way of the same uncertainty minimizing mechanisms simple simple or complex narrative constructs create cognitive situations where agents reduce comple complex aspects of the world down to Preferred scenarios and now we can look at how active inference is a useful applied framework for this pass it over toor again thank you so I'm I'm here drawing on work that is currently ongoing from other people as well with who I'm collaborating and I want to make sure that these ideas are appropriately attributed to them we're going to talk about individual agency and empowerment and so autonomy uh can be defined as a system's capacity to self-govern using internal Norms uh so again this is a topic explored by Jaclyn Hines Alex kefir inesi poo in part adversive um and so empowerment here is measured by the maximum Mutual information between an agent actions and resulting States so this reflects how much influence an agent has over future States uh higher empowerment means greater potential for desired outcomes and so empowerment maximizes the entropy of actions and States while ensuring strong action State correlations autonomous agents are thus more driven by intrinsic motivation and this is a premise that again Alex Kefir is one of the Pioneers uh here it's thus connected to drives and control to systems that optimize future action State paths and so under active inference agents minimize expected free energy to balance exploration information seeking and exploitation goal Pursuit so this involves perceptual inference to update beliefs and generate model refinements through action outcom and so here power operates as an attractor in Social systems it it draws agents towards influential entities through access to resources and strategic advantages Happ it leads through it leads therefore to amplification through feedback loops um powerful agents help others minimize their expected free energy by providing some degree of stable predict Frameworks in reshaping the informational landscape so high power entities have privileged access to information and can model their environment more effectively and they also influence what can be modeled and how impacting the collective predictive landscape powerful agents can offload cognitive demands as well processing complex information more efficiently and directing narratives to manage error absorption ion and uncertainty so power here enables agents to shape others Focus directing attention to favorable States while diminishing access to Alternatives and this leads to feedback loop where control over the environment enhances influence and perpetuates power the whole story around 1% so power grants a broader range of operational capabilities and extends influence over others State space policies within its domain and so here active inference reveals power as this emerging property um of the connection between perception action and social structures or powerful agents maintain and expand their control by shaping both informational and physical environments so let's go over three main compon components Precision modulation belief alignment and ultimately minimizing Divergence first Precision modulation powerful agents shape expected outcomes by altering the social or informational environment as we saw they create narratives or scripts that effectively reduce the dimensionality of this state space forming stable attractors such as social nors or shared stories and these attractors direct Collective attention and ensure that other agents align their actions with these preferred States um but here basically the posterior belief about States uh has uh an alpha that modulates the impact of precision on these beliefs and so in belief alignments the power to direct attention and utilize Precision modulation really enables the leaders to make specific aspects of reality more Salient for others and so by establishing these attractors within the belief space the leaders guide others interpretations and behaviors um these attractors act as a focal points reducing ambiguity and encouraging agents to align their beliefs and actions around them thus fostering a shared understanding so the the alignment process naturally emerges from minimizing the expected free energy uh and so for example ex Le um where P of O given s and Pi is the likelihood of outcomes uh given States under policy pi and Q of s given o and Pi is the posterior belief of about States and so this decomposition shows us the intrinsic and extrinsic components which balance uncertainty reduction with coal oriented behavior that is um shaped by this Precision that we mentioned earlier so finally uh minimizing Divergence uh can be represented just using a coolback liary Divergence uh which is a measure of the difference between two agents belief distributions and so a lower kale Divergence indicates closer belief alignments and so leaders influence this alignment by modulating this Precision which amplifies confidence in specific interpretations and so the process minimizes belief Divergence across agents which promotes synchronized understanding and cohesive grou Behavior without the necessity for direct coercion although that is yet another way to minimize the state space and therefore is an available practice so Precision modulation sets the framework for control belief alignments ensures Collective cohesion and minimizing Divergence uh formalize as this alignment so basically we capture the Dynamics uh and influence uh the Dynamics dynamics of power and influence within social systems and so finally we come to the conclusion we have a lot more in the paper that we're still working out so look out for that um yeah so what we're we're thinking about here in terms of this kind of shift um attentional shift uh from from ideas of territorial and material power to this new realm of of informational and computational power um well this this for us requires attention to attention uh for sure something that we can we can look at in uh in ways that we can formalize and of course we propose that active inference provides crucial theoretical tools for for making power structures visible and therefore analyzable and so allowing possibly for the potential to rethink current social asymmetries so again given this sort of apparent transition between the organization of life around the centrality of physical work and matter and things moving in space towards the centrality of attention and information uh especially considering the whole attention economy idea we kind of try to think about a possible sustainable social future which requires the addressing of both resource and information distributions and the latter is best analyzed through the power of attention um as we've tried to to show as a major factor or one major Vector influencing social dynamics and of course this is very important to mention hierarchies are inevitable in complex thermodynamic systems but we propose if we can begin thinking about this uh fostering transparent plastic hierarchies that remain contestable and thus open to novelty open to learning right this approach um has to acknowledge uh historic and thermodynamic constraints while also acknowledging space for systemic Evolution and while hierarchies are inevitable social scripts should be read writable by those with an interest in survival within them right and this helps reorient our understanding of an unfolding globalizing interconnected and interdependent life and participation in a system is only participatory if those involved are able to track the consequences of the variables and thus future State spaces at stake a true distribution of information processing cap capabilities is essential for systemic sustainability and proper Equitable development and when thinking through evolutionary biology lenses for example let's try to polish them clearly and not fall into social Darwinism narratives and to quote from the xenop feminist Manifesto um lovely colleagues there Patricia Reed for example if nature is unjust then we change nature which is what we are all in the business of doing here a little bit all right that was um that was our presentation thank you awesome okay well will'll be back in a few minutes e Carl hello welcome back everyone I'm here with Alex aoria we may have Carl join we may have someone else join if you want to join maybe write it in the YouTube live chat and maybe I'll email you a link otherwise write some questions however to kick things off Alex feel free to say hello and start in and and I know that there's things that we can explore together sure you just picked it back to me and I was saying to you all right well I I'll just introduce myself because I'm not entirely sure how most of the audience if they know of who I am so I'm Alex roia I am a computational neuroscientist and cognitive scientist who has worked for active inactive inference and specifically predictive coding versions of it for quite some time I'm Carl's friend and collaborator uh some of you might be aware of more of our recent arguments of mortal computation which is a coraly or a new coraly or another coraly of the free energy principle and uh yeah and you know I'm all about what's wrong with deep learning and machine intelligence and I'm usually the guy you go to if you want to attack back propagation of Errors large language models Transformers kind of relates to a lot of the discussion I was able to catch Snippets of earlier today so if you want me to help dismantle what's what's the problem with deep learning you know that that can be fun too so that at least if nothing else everyone now knows who I am Carl needs no introduction maybe start off with that while I try to figure out why he cannot join oh okay uh what part of that do you want me to I mean I can go into Manifesto but I'm not sure that's what this panel should be about out maybe beginning with a five minute Manifesto and then there will be some questions and uh we'll see where we get well maybe to try to make it useful to the audience uh for the moment although I wasn't intending the panel to be focused about that um maybe one thing that we can talk about is there was this discussion during the active inference Workshop about metabolism and sort of in steing a homeostatic metabolic drive a survival almost orientation to machine intelligence and so uh Carl and I have now been arguing for probably now about more than a year since uh our paper came out like on yeah around November actually start of November talking about uh another way to redesign reformulate to use Ma's earlier uh uh wording uh intelligence itself uh particularly machine intelligence and so mortal computation without again if for those that might not be familiar actually my friend Jeff also two years ago introduced this phrase the idea that computation or intelligent calculations and substrate uh in biological natural systems are entangled and it's very hard to divorce those two concepts uh whereas computer science and a lot of like what cat GPT is or large language models or your favorite GPT structure is Immortal which is the idea that we can copy that program including its weight tensors and it's encoded knowledge and place it on GPU server you want and it'll behave roughly the same there's nothing inherent about its physical instantiation uh in this universe uh that dictates or shapes or uh necessarily embodies it right and so there's this like move to uh embodiment by the way active inference in the free energy principle has all of these characteristics mortal computation which is again as the concept I explain earlier Carl and I sort of uh recast it we strongly redid it and said that um uh because while Jeff didn't want to necessarily directly explore it we put it as it's just basically thinking of the free energy principle and taking it to its natural extreme conclusion which is that machine intelligence systems uh because they are inherently B to their substrate must focus on this survival objective right and the free energy principle and one of its ways to talk about it uh basically is saying you know we have a markup blanket or a nested system of marup blankets and our goal in this universe is to protect those internal States uh yet interact vicariously with our external States and so that eventually gives us our ultimate goal which is to prevent ourselves from disintegration into a heat bath it's a very physics uh form of thinking about identity and identity construction um but this is kind of what's centrally missing in basically all of machine intelligence um is this the completely right path to go there's plenty of criticisms and I'm not one to say that there's only one direction we need to go if we ever want something like artificial general intelligence um but this might be a path worth exploring it also addresses the Energy Efficiency arguments I was hearing earlier discussed uh Transformers particular particularly in large language models are uh grossly negligent of their energy costs they have often times have the carbon footprint of a large city uh and that's actually now out of date as you know Carl and I wrote that about a year ago um there's even discussions I find kind of humorous by individuals I'm not going to particularly name saying we should be powering now uh these Transformers and large language models with nuclear power plants and so we're kind of going down the rabbit hole of let's just burn as much of our resources as we can to just get this particular type of intelligence I argue very strongly narrow AI uh and we're cheating ourselves of looking and embracing actually Energy Efficiency uh which would lead us towards the real grand goal of green AI uh because clal GL global climate change is clearly and empirically a thing and we should be a little worried about that so uh mortal computation sort of helps steer everything into sort of what I would argue hopefully maybe in multiple years time the emergence of a whole new field because I consider at least my personal take I'd love to have you know heard Carl wherever he might be um but I Envision actually mortal computation is not just like the framework and definitions that Carl and I talked about which kind of take free energy uh and active inference to the strongest conclusion to a naturalistic machine intelligence but maybe perhaps a new domain of thought of biomedic intelligence and where everyone else can come together and play in this really interesting space and there's a lot to unpack there but I don't think we want to make this about that uh there's plenty of talks i' given including active inference Institute we've had a podcast actually I think it was right when the paper came out so for those that are interested I just recommend you to go to the YouTube channel and watch wonderful videos that Dan LED oh is there is there a new link okay sure was was I just blabbing there to no one no that was perfect we we we just hot swapped okay I was trying to buy Carl welcome Carl we talking about okay thank you the overview on Mortal computation and and and it was a highlight of the year um Carl with the evening Sunset now how um goes it wait I don't hear you oh oh continue sorry it's all good Carl continue can you hear me now yep excellent yeah I'm sorry about that I got locked out I was uh paying the price for enjoying for the first two hours right wait just I really enjoyed the um the intervening sessions um and look forward to comenting some the cross cutting SE yes okay um Alex mute it's a little noisier on jitsy and then uh Carl um also a little quiet but we hear you so um what were the what were the themes that that uh connected the presentations that you were in or otherwise one thing that um struck me can you hear me right now is this y excellent I mean there are lots of lots of little uh golden nuggets um throughout all the presentations um but one um one crosscutting theme was this notion of community that emerged in various guises such as the notion of Commons that was the subject of uh a lot of the exchanges in the first um in the first um uh presentation um and then the notion of community that was a Dumont of the the workshop um and then the of coherence and synchronization and narrative social science accounts of that and how that might be formalized in terms of um joint free energy minimization ensembles of Agents you know finding a coherent communal um and empowered free energy Minima so you know I thought I I thought that that was a common theme sort of focusing much less upon um individual agents but what happens when you have a an ecosystem of Agents um and what principles um can be brought to the table to understand interactions or ensembles of Agents so I I have a number of U you know lots of things came to mind I was trying as a physicist to sort of um rewrite all of these wonderful Concepts um in a mathematical way or or a physics way um so for example I was think about the importance of Commons and it struck me that one way that we could motivate the choice of capital c for prior preferences uh would be that these are the common goals and the and the C characteristic States um that are shared between um multiple agents uh and of course um the last presentation took us through some of the mechanisms and the capital c circular capital c causality um that leads to this Capital C coherence so you know which originally was meant to be constraints in the sense of a constraint Maxim entry principle um or cost in the you in a reinforcement learning uh reinforcement learning setting but coming back of course all this started when I was hearing about Commons and how it underwrote communities it it is very interesting how development and the presentations have proceeded a lot more about the communication and somewhat of the multi-agent coordination as opposed to a longer memory train for individuals or deeper prospection for one individual um and in a way the renormalization method brought a a Clarity or a rigor to what had already been qualitatively explored from 2013 and Life as We Know It And all these other works that we're talking about multiscale systems and about blankets of blankets and within this year to make it timely we start to see some of the ways in which that compositionality can actually be brought into the generative models that we've seen I mean it looks like the textbook figure 4.3 and the nested models and now that's actually the motif that gives a way of looking at a multiplicity at the more inactive level as having a Unity at a higher level Alex in your designing or work how how does the single agent or multi-agent design come into play now yeah yeah can hear okay good yeah I was having mic troubles earlier um well so it's kind of interesting I think I've spent most of my career at the single agent level in the sense that I want to build I don't think we have right the single agent and I and I think it's kind of important to sort of scope that out from an architectural learning perspective I'm very much a fan of like Carl's arguments and you know I Echo them myself a lot which is the idea of the inference learning and structural time scales uh I still think we have a lot to do but if I were to say how in my research the multi-agent approach really comes into play well I do think a lot in terms of assemblies or ensembles right putting together multiple circuits I'm a I'm a huge proponent of another type of this comes from cognitive science called the common model of cognition uh I'm not sure how many in the audience are familiar with it but it's sort of like a generic reprint that has emerged over the last several decades of cognitive science to say that we can think about and take our current Knowledge from neuro physiology and cognitive studies that brain has these rough levels of organization different types of memory we have the motor CeX we have phas of gangu we have all these very important structures and we can start appending to them some functionality and so in that sense that's where I think of multiple agents right because you can start breaking apart using again Carl and I've have argued this the mean field approximation uh there's natural mean field approximations that emerg through the result of Evolution where you have these little substructures that communicate with each other um obviously the sparse connectivity that Carl mentioned in his talk this morning is uh kind of indicative of how these little substructures emerge and then communicate and perform sort of like what uh Marvin Minsky would have said you know sort of a society of mind right uh so it's kind of a call back to that and I think in that regard I often uh one branch of my work often tries to highlight emphasize bring forth that notion of cognitive Blueprinting or cognitive structure um because then we can start filling in the pieces with what we know about neurobiology like for example I use predictive coding quite a bit uh you know to fill in those gaps and so I think in that sense I think a lot of interacting subsystems you could think of those as agents and then my final comment uh without letting myself BL too much uh I have worked in neuroevolution so there is another scale Above This which is you know you do need to account for how multiple mortal computers or multiple agents interact with each other um and so I think in that regard I have done some work with very simplified agents where you can kind of abstract away the neuroscience and just say I want to look at these little tiny systems that interact and then we can you know modify them through genetic algorithms and neuroevolution so there is a little bit of that like higher level uh societal kind of learning I have looked at that and I think a big challenge just to bring it to the audience for the panel is uh merging those time scales in a in a coherent and efficient way I still think well there has been good work I know Daniel you work on M Colony optimization as well and other types of kind of aggregate algorithms uh I think bringing those time scales together and really interacting with what we currently know about single agents even like the common model of cognition and going back to the high level which is this multi-agent evolutionary time scale is a Grand Challenge and how do we of course effectiv simulate those in a way that anyone can play with this I think is a big Challenge and yeah and I think I'll kind of stop there before I go down more radicals neuroevolution and neural Darwinism something Carl and and others have also explored it's like they're all and then the multiple nested time scales multiple nested spatial scales it's like there's this attractor to want to come back and find the unifying patterns and and it's and how those can be applied is kind of an exciting question I'll read one question in the chat so feel free to give a thought or take it any direction David Highland wrote do you view mortal and IM do you view mortal and Immortal computation as binary categories or are they more like two points along a spectrum actually want Carl to answer that I mean I could give something but I really want to hear your thoughts on that it's your favorite subject though design I know but I'm actually really curious if you think of it as a spe because remember the conversation we've had with Mike Levan and Chris fields and we've been sort of debating you know how important it is to lean into mortality versus immortality do you think that there could possibly be anything in between because I honestly you know I think of it kind of more rigidly I'll just give my aners that and I firmly think we need to like focus on Mortal computation but do you think that there can be a hybrid in between so sorry to turn it on you but I'm very curious to know your thoughts um my my instinctual answer is no um having said that I am sure that you can present some um description of mortal computation in terms of immortal algorithms I mean I'm just thinking back to sort of um um the trilogy due to David Mah uh but does that mean that these kinds of descriptions are apt um for simulating reproducing understanding um or describing the actual process I I I suspect not they may at one level of description they may be but when it comes to um applying the principles usually in computer simulations I suspect that

that just would not work I mean because Daniel and I were wrestling with zoom and other um um communication things I I I may we may have missed whether you've taken people through the the the key distinction between Mortal and Immortal uh computation so Al do you just want to briefly rehearse the sort of the two Cardinal views of the of the distinction um between Mortal and Immortal uh computation I did I just wasn't sure if anyone heard me so I wasn't sure what was going on Zoom yeah I mean certainly from the point of view the free energy principle the free energy principle is a description of a random dynamical system and as such um it is um a system that constitutes the world the you know the lived World um perhaps not the lived world but certainly a mortal world um that has um characteristic time scales and therefore um the free energy principle and everything that ensues from that including active inference um rests upon that primary assumption that you are describing a process a mortal process a a a a process that could have um a physics um although um the actual free energy principle itself is almost pre physics so you have to derive basy mechanics and classical mechanics and quantum mechanics from um the existence that you the the the mortality of this process cast in terms of you know time evolving systems so in that sense um you know there is no such thing as Immortal computation but I also I don't know if you said it out loud but I also liked your um more playful take on Mortal versus Immortal that if if software was Immortal uh and it had some autopoetic aspect to it then it wouldn't care about dying so there would be no wait there' be no way there' be no incentive to have characteristic States coming back to another big CA um uh so you know in that sense they cannot be immortal computation simply because that is removing the definitional uh stipulative definitional aspect of the characteristic states that we are trying to describe through the F principle and active inference and specific um VAR variance of that such as predictive coding but I just pick up on this crosscutting CC theme uh you also um um celebrated that with common cognitive models so again with we've got this this s notion commonality and uh um and you asked the question you know how How would how would one um apply these Notions and if you like sort of naturalize the common cognitive model under um a less anthropomorphic or functionalist active inference account and you gave the answer implicit in your question which is how would you simulate these things and I think Daniel asked the same you same question indeed the the whole point of this Symposium is about applications so I I just wanted to put a question to the two of you I was um impressed with the very um clear distinction between agent-based modeling and the common cognitive equivalent the co where the agents the members of The Ensemble were actually for example active inference agents that seems to me a very big move and a very difficult thing to actually do um most of the literature I know in this area um has at most taken baby steps into having ensembles of two agents with authentic agency in the sense that they can plan so I'm now limiting um the notion of an agent Beyond a thermostat or a virus to those kinds of systems that um have in mind or part of their Gena model the consequences of their their own actions so they can be future pointing and they can plan and select from alternative policies uh and of course the numerics and you know the the Practical difficulties of actually simulating doing agent-based modeling for example um with authentic agents in the mix in the ecosystem um I think is you know um is a really important Challenge and a really important distinction between classical agent-based models where you put lots of theats or rents attractors together and see what happens and of course they have a synchronization manifold and you can certainly box lyrical about um generalized synchrony and and and Joint free energy minimization of a generic sort but I think that misses the point I think Ma was making um um that you know part of the the multiple Loops of circular causality in sort of your know cultural Niche construction for example um you know really does rest upon um the capacity of any individual within an ensemble um to um act given they have selected a particular course of action particular path into the future which they have beliefs about the consequences of and you know to me that that that that is an authentic kind of agent so you mean perhaps well both of you d mean your your your ethology background um are you aware of anybody's actually sort of Bitten the bullet that was introduced to US during the workshop and actually started creating active inference agents that actually plan and learn from each other and carve out their own warington Landscapes you know finding those sort of joint joint Minima I don't know specific uh citation but certainly you've given a research Direction and it reminded me of in the shared protentions work where thinking about what what does a shared Ensemble what would it be for a planning Ensemble or for a collective it could be a shared protention as in there's the same plan disseminated across the ant nestmates it could be that they have different maybe nonoverlapping but complimentary parts that would be shared it could be having a shared reference but they could have different plans or it could even be a shared consequence without even a shared reference so there's so many ways in which especially when we consider ensembles that that are heterogenous they have different kinds of umelt different kinds of ways of engaging with the niche and and possibly some Comon niche as well as some uniquely accessible components there's so many flavors of Ensemble that I I can can only hopelessly or helpfully try to think about what tools and infrastructure and coordination would even give us a grasp to be able to sweep or design or visualize across that otherwise we might be locked in way too early to what it means for The Ensemble to have a collective protention and then that's not going to apply to every ant species and if it can't even apply to all the ants out there then it's not going to work for aliens Alex can I can I just respond to that and then ask ask for your your view on that one so I think it's a really important point and um one of the big takeaways for me on our paper on the variational synthesis use you're applying the renormalization group to the the coupling of evolutionary to um sort of phenotypic uh time scales was that it was not uh feasible or appropriate to talk about consp specifics in isolation that it was you had to have the all the varieties of predators and praise and things that we um that not even animate not even agents in the mix to actually um derive the um the you the non-equilibrium steady States at particular at particular scales which you know speaks directly it is not um it is not U useful to um as you know if you like take the idealized gas mentality of 20th century physics and now apply it to agent-based modeling to assume that all the agents are exchangeable and are identical in some sense and then are converging on some common narrative um or some common um uh is not the right way to you know to simulate these things you can't do Predator prey relationships you know for example you can't have um even simple economic games uh but furthermore you you you when people talk about depletion of resources you actually have to model the resource you know whether it's oxygen food or water as things in in that ecosystem so I think that's a really important observation which just speak to the challenge of using numerical studies to test out some of the hypotheses that uh that we' you know being being exposed to um you know you know if one just pursues agent-based modeling without introducing that heterogeneity I think one is making exactly the same mistake um that people make when they apply uh equilibrium physics to non-e equilibrium systems um you know you know a true gas is not an ideal gas and it only has the kinds of properties and breaks detailed balance and expresses Nona in virtue of the fact that there is no exchangeability you can't assume that every atom or every agent in our context um is exchangeable with with every other agent and to connect again to Commons and and that model of well within the firm we assume cooperativity and hierarchy in the market we have a flatness and a competitiveness and then there's this Commons that has sort of elements of each so then I think about doing ecosystem modeling it'd be like well we have within the ant colony we assume cooperativity possibly hierarchy among ant colonies you have red and tooth and Claw and competition and then it's like but can there be a unifying framework that would have the two genotypes and two phenotypes and two species and two adjacent niches and keep on adding in these levels of Distinction and seeing all those different variables in their joint Evolution across multiple time scales rather than the concatenation of a Cooperative hierarchical game NE with in a lateral competitive game it's like capturing what those models get the the width and the height of but bringing it into a framework where it's being seen as just one joint evolving process so C is the letter of the day Alex can you speak to Cesar fell what what that kind of heterogeneity and structured diversity would look like from a point of view of tissue for example or an organ okay wait what was the c word you said I just didn't hear it it's any any word starting with a letter C you can use but you have to oh I have to use okay this is the game um Community I'm gonna pick that one just because it starts with c um so okay Carl threw in organs and organel but I'm going to push that aside for just a minute to address I think one of the questions I heard emerge from yours and Carl's discussion which was uh is there anything like what you are arguing for which is avoiding the mistakes and by the way Predator prey model came immediately to your speaking Carl um about these classical models that assumed interchangeable agents and is there anything that looks at these heterogenous systems or heterogenous agent interaction and a couple of thoughts I don't think there's anything like directly what we would want and I think that ties or centers around are even the arguments you and I and many others in the community are making about uh why it's difficult to Showcase why active inference is so important even for places like machine intelligence um but I did write some thoughts quickly uh so I do think uh the first thing that came to mind weirdly enough was an application in the world of machine intelligence where they had some environment where they had a community of even though I say it with slight distaste large language models or Chad gpts or gpts that had some extra modality equipped to them and they had this like little Community now it was a virtual very simplified Niche if you will um but the idea is that they could observe this community um and so these different gpts would and I can try to dig up the paper and send it to Daniel later it's one of my students that actually pointed it out to me they would different roles they had different personalities if you will um and I know that they investigated that to some degree now of course that doesn't have any active inference in there and I'm not sure what degree of maybe perhaps light reinforcement learning they introduced but there is something like that and there is slight baby steps if you will uh in investigating some notion of embodi multi- embodied agent systems I think the AI Community is still sort of of barely starting to embrace embodiment let alone enactivism and let alone embeddedness which is the idea of a community to go back to the seword of multiple agents interacting with each other um and investigating how do social norms emerge or cultural norms and cultural constraints and relationships at this like more

Global level so I think there is a little bit of those Sparks of that thought and obviously I might not be aware of further follow-ups another place that uh I at least see the potential of no longer having these interchangeable particle like agents is in multi-agent robotics um I know that they have done things like soccer competitions and they have these little and again each of those robots they are the embod of what you and I want which is a version of a mortal computer so it's a little like assembly of mortal computers and there's a Cooperative nature so and then a competitive nature so you get a little bit of a very constrained real physical real world Niche um obviously it's in the constraints of it's a soccer game or some very simple task so I know in multi-agent robotics there is some development along this path and it'd be wonderful to have like a a a hardcore because I'm more of a simulation roboticist a hardcore roboticist to talk to us about the state of multi-agent Robotics I think that's the place where you could begin to test uh hypotheses and kind of or theories that are constructed even from a active inference point of view uh into these real world robotic multi- assemblies you know they have like we could analyze the competition Cooperative Dynamics I think that's our best bet in terms of what maybe exists at the closest to literal agents that aren't just very simplified even though they make simplifications in robotics too uh in order to have a little brain in there that you know does something meaningful usually learning is turned off which is my own criticism of Robotics um I think the last comment I wanted to make and again getting back to the Notions of what we need so again you and I have a fondness I think for some of cybernetics and cybernetics has this notion of complex emergent systems and we you know you have little uh building blocks at one level that follow their own physics or their own physical laws and then they maybe assemble or emerge into these higher level building blocks that are subject to their own laws then you have bottom up in top down causation which is a notion that you know comes from cybernetics and we always talk about that but then I think about like Santa you you might be familiar with the Santa Fe Institute and all their wonderful work like uh what was it not just Herbert Simon but uh uh John Holland and all the wonderful work on complex adaptive systems but they always did at the end of the day when they simulated it they did the particle thing this interchangeable particle simulation and then tried to analyze like those convergence properties or those emergent rhythms or patterns that kind of emerge in those systems with no to my knowledge and you know and I did study complex adaptive systems a while ago uh not with that heterogenity and let alone any of the complex agents that we could begin to design today even with deep learning regardless of its limitations and so I think that that maybe even turns into an answer to your question highlights an open opportunity Maybe revisiting and recasting Notions of cybernetics and complex adaptive systems in this idea of well can we actually build more complex agents uh for example that actually have neural substrate structures or something that we can simulate and then learn to acquire their own characteristics in their Niche I think the reason that maybe a lot of people avoid it or maybe why I don't perceive uh development in that direction is the problem also of Niche and you and I also had an an a not to try to highlight this paper but in the appendix we talked about the body Niche problem uh and that is this issue that uh you need a niche if you don't have a niche and you're designing these agents independently like deep learning typically does to apply to any Niche uh you're missing really half the problem which is understanding the physical laws and how do we simulate them such that people can then play with those models maybe you don't have access to a xenobot like Michael Levan does or I don't have access to organizes though I wish I did um but you know maybe we can simulate these and so I think that we need the community may this is a call to the community to develop uh viable niches virtual morphologies virtual uh areas that we can then connect these agents and then simulate many of them and try to analyze and understand do the things we predict even with these interchangeable particle models apply to these heterogenous communities that might emerge and again I guess that just to wrap up and I promise I'm done you brought up organel and organs and I think that highlights something that I wish I had more easy access to rather than uh being put into this position where I well may I need to build it which is you know these virtual organ systems or virtual organ organ LS and trying to understand how these dynamical systems interact and then I can kind of take that as that's my body or that's where my Mark and I can kind of start to put some instantiation of blankets and then okay I want to instill maybe a neural circuit maybe I put in a predictive coding model into this kind of virtual morphology so I can analyze and understand does do these interactions of the time scills actually come through when we actually build you know physical instantiations of these models and so I think that's going to be one of the biggest roadblocks to even my dreams of mortal computation which is this notion of phology something that we can actually work work with that it gets as close as we can to reality without reality and in Niche um because once you have those problems solved then you can start building multi and if you have an efficient way to simulate them and I'm sure lots of people that are watching might be worried uh because it's very hard to simulate you know real physical systems let alone allow machine intelligence researchers to play with them so I blabbed a little bit there but does that sort of give you some something to bite on based on your questions something to chew on something to consider um in the last little bit I'll I'll ask some questions so I'm going to read one from the live chat and then anyone else we'll just do some quick fun random questions sort of like sampling from the uncertainties expressed uncertainties of our live chat colleagues okay Pablo FM wrote is playing a simulation acquiring knowledge SL experiences through simulation of scenarios and how can we use active inference to do games how do you fellows make your active inference work playful and fun if when let you go Carl you go I'm not sure if there are two there there are two questions there um a practical answer to um why you might want to instantiate um a simulation environment with active inference agents or indeed just a single agent in the in the spirit of computational phenotyping but I think probably more interesting in the um in the context of um communities of Agents um is in the spirit of the sort of cognitive um agent modeling um is the ability to um parameterize the prior of the agents to best explain or replicate Legacy data from a particular ecosystem be it Financial epidemiological meteorological um purely ethological um and having optimized the parameters you then roll out into the future and in rolling out into the future you now have the opportunity to have a grounded in principle base optimal Plus in activism and embedded um um estimate of what is likely to happen in the future again in the spirit of weather forecasting but of course this becomes much more um if like useful um when you have control of the system so Financial Market forecasting or epidemiological forecasting or uh climate change forecasting in you're assuming that there's a substantive um um anthropomorphic um um anthropological um contribution to um climate and you all the other factors that uh you know um pertain to um resilience and uh resistance to things you know health and the like um as soon as you can intervene then once you've got a digital twin of this ecosystem that has been optimized in relation to um the past data in the sense that this now has the highest model evidence or marginal likelihood in relation to that Legacy data you've then got an opportunity to do scenario modeling with interventions so you can now ask questions what would happen if we um cut this Supply CH train chain what would happen if we impose this lockdown in this viral pandemic what would happen and so on and so forth furthermore because you've got um um uncertainty in the mix because this is effectively going to be um you're using variation procedures that we use in active inference to actually simulate um the behavior that best matches the um the Legacy data you've also got for free uncertainty quantifications and not only can you say if I inter been on this system and did this um then we would uh I can predict that in six months time or in six years time that and I can give you basing credible intervals o over that moreover you can now apply the principles of active inference to roll out under different interventions which now become policies and evaluate the expected free energy of each policy and select the one under the constraints the the characteristic outcomes U that you um commit to at this point in time and you can now use this scheme to make recommendations on the basis of the um the relative probability of this sequence of interventions versus that sequence of interventions so I think there are great practical importances of being able to put these principles in silico and basically do Evolution very very quickly so that tomorrow you can read the you can read the consequences probabilistically uh of evolution in this setting and that setting where the setting is actually something that we can intervene on or even if we can't just to know the consequences of our Collective Behavior Uh today so I think that would be um that's why you might want to make you know to invest in in these applications um with the fun and uh having uh and play I think that's a different issue I think that's basically um um you know one way of looking at the um the way that authentic agents behave under under the free energy principle which is basically to explore and gather the right kind of evidence that uh provides literally evidence for their generative models of of their world um which basically entails novelty seeking but now complement this with this um Bas red C we've been talking about also trying to find the minimal explanation for the data and the evidence that has been secured by exploratory playful Behavior arriving at particular insights which uh those are har moments that say oh I see it's just one of those or this one thing explains all of this playfully acqu acquired data and perhaps that kind of mechanism you know may be a really important aspect of the communal um um you know um cognitive agent based or yeah cognitive agent based I can't quite remember the acronym it was very nicely laid out in one of the I have to go back and look at the live stream a followup question from the live chat Arno wrote I agree with the approach to calibrate to pass data and forecast but it's already quite challenging for current models that only have a few parameters to be calibrated how then does one calibrate prior beliefs of millions of Agents I'm talking about epidemiology and economic models sorry I'm going to give a technical answer before Alex gives a gives deted answer uh you do it it's only difficult because 99% of agent-based modeling uh uses the sample distribution um you do not do that you use variational you actually um simulate and roll the um the um probability distributions themselves using variational calculus and that means that what was yesterday um really difficult agent-based modeling and certainly epidemiological modeling now becomes really trivial uh orders of magnitude more uh more efficient what you have to take care of is exactly the thing that Daniel was was highlighting before which is a heterogenity you need to put that into the mix and this manifests in a number of ways uh so when you apply um variational procedures to effectively um roll out population density uh probability density distributions over various states of being am I infected am I staying at home um have I died um you have to make sure that the heterogeneity is part of the generative model so this is sometimes called stratification so for example in epidemiological models you'd have eight bands for different age groups and if you notice the way that the epidemiological data was generated was usually age segregated so that people could stratify their models um another um way in which this heterogeneity becomes absolutely crucial um when you come to um agent-based modeling in the context of epidemiological or infectious disease modeling is called over dispersion now over dispersion is the fact that uh you get heavy tailed distributions that inherit from the heterogeneity so get this for free once you put heterogenity into the model but it does mean um for every if you like subgroup of um the population you do induce more parameters but remember the number of parameters doesn't really matter it's the amount of time that you take to effectively invert your model and if you try and do that using sampling based approaches which is currently the the norm you could take literally weeks to uh do something um whereas if you use variational procedures then you could do that within within minutes so it's a really important question a very practical one yeah to restate direct variational inference on beliefs about ensembles themselves being variational belief inference on agent level semantic beliefs rather than sample from the sample what the agents are sampling yeah you're you're you're invoking the The Twist yes how on Earth can you model a variational density over things that have variational densities you can do it under the f p room because the only observable things about an agent are on its Markoff blanket and of course they are not belief structures so there is a lawful relationship between you can um if you like simulate cognitive agents using traditional agent-based approaches but what you have to do is to amortize or to or to um um basically work out the um the mapping between the history of what something sees um and what it actually does but in principle you can do it by marginalizing out the um the beliefs of the agent in principle and and that's that's why I as that's why I'm so intrigued by this distinction because that will be a challenge Alex and then Carl just in our last minutes here what would you say was a favorite memory or moment in theory and practice for active inference this year and then what are you excited about for next year um I'll go I'll go first so a favorite memory of active inference let me think quickly well I I mean I'm going to be a little B EST just to give a highlight and a shout out and everyone will get to share in that memory if you stick around after the round table is actually some of the results from one of my PhD students um via and he's goingon to give a talk soon and uh the idea is that uh one of the things I was proud of is he's been working in the area of Robotics which I think is a really good playground uh is difficult and uh hard is it to democratize for everyone to work with robotics I think it really shows you how far we need to go and I think that's a good place to make strong advances an active inference um and one of my memories is that I remember him kind of showing me uh because he's working with partially observable Markoff decision processes and how difficult and unaddressed or lightly addressed it is in the field of working with raw sensory data to try to construct these continuous models and I remember him showing me some of his models rolled out trajectories in the future and I I'm always kind of in a fondest in my heart for generative models and I remember him playing out sort of these robotic memories of this robot trying to you know fantasize where it would want to go to kind of like move this block or to knock a ball towards a so that was one of those moments where we were really proud because the idea is we actually we actually made a breakthrough uh in actually showing that we can construct a very robust framework that applied across all these robotic control problems and so I was immensely proud of that and so I think my students drive a lot of some of my fond memories um and uh and then we got uh Chris Buckley shout out to him uh to you know uh kind of give us his insights into our work too and it was a really really nice work and so and I think the best way to say it is stick around if you want to know how that all works I was really proud of that so one of my favorite memories of the Year awesome Carl with the last words a fun memory and uh a direction or an excitement for next year right I'm I'm not going to tell you about my most fun moments that's that's very personal but my my my my excitement for next year is to see whether any of your hairs go gray after you do the master ceremony of these of not only these uh Symposium all the meetings because for those people who don't know Daniel has to run this almost single-handedly nonstop for a marathon sometimes 10 hours at a time I don't know how he does it at is unless you're using hair dye you're you're you're standing up physically remarkably well sir so I just see how long that lasts thank you Carl I look forward to the marathon changing continuing and flourishing in the coming year we appreciate you Daniel thank you guys it's been awesome okay thanks a lot see you soon nice talking to you guys bye everyone hello welcome back now we have the much anticipated student presentation that your adviser Alex recently told us will be awesome by Viet dwin on Dynamic prior preference learning for scalable robust deep active inference so thank you Viet take it away hello uh hello everyone um my name is vet I'm from uh the neuro adaptive Computing Laboratory um Department in computer science or CH technology um and I am uh Alex advis uh and today I'll be talking about Dynamic PR preference learning to scale B plus active INF so yeah first of all I want to talk about the motivation for this so nowadays you know from fun Shure work in robotics um excels in implementing robots behavior and planning for example um creating um we can you know creating autonomous robots that can actively explore in the environment um additionally um acquiring we can also uh do a lot of Robotics applications for example acquiring knowledge and learn skills continuously um and um this also envisioned um cognitive and developmental robotics um so there just a goal um for um robots robotics research um that's we can try to build a robots that can continuously develop through interactions with the environment and their learning processes um and it is also also better um to have interactions with their uh physical and social world in the manner of human learning and cognitive development so there's a lot of Robotics applications um around um and for example one application can be to build ASA robots and um building um social collaborative robots and stuff like that um so underlying um all these robot work um uh we it's like active inference research where we um often build a generative um um it's so the developments of Robotics actually depends a lot on um activ framework because um there's a great need for research um that integrates the dtive world model into robotics control system um as also deep learning is increasingly used in constructing complex action planning algorithms for solving control tasks um so um we can see that there's a need for scaling um active INF system spe especially and deep active INF system um to integrate into the real world applications so um resar in active inflence has led to many improvements um in terms of robustness uh of um model based agents that applies to both M processes um and partially absorbable MVPs or P MVPs so an example of an MVP environment should be Mountain car um uh for those who um knows or studies reinforcement learning already and um this m car environments um in mdp we only know POS velocity right um and that is the actual state of the environment um but when we get to PVP for example like robotic tasks um we as agents see only the image frame and we do not know the underlying states such as position and orientation of the block or um join states of the robots for example right um so basically active INF is basically a form of inference and learning that SS to balance first of all the um goal orienting objectives or highi preference um so this signal exploit um the instrumental signal um so that the agents the goal um and try to pursue its goal um and the other thing it needs to balance the epistemic terms or epistemic signal where the foraging uh it's uh cares about foraging and exploration driving signals um so researcher often actually encounter great difficulty in providing the agent with effective prior preference as well as Computing useful instrumental signal from the agents the priors um especially in the real war applications where um the environment is very complex the robotic control task and um the trajectory or partway um to be desirable um and task specific goals um often requires uh a very very long um trajectory of preference um so it is not that ABS practical to provide uh a prior preference distributions especially the um stream or roll out of goal images so that is the reason why we need to find a way to also learn to generate the prior preference signal in active inference so on the right here is just our um preliminary results um but we'll get to that in a moment so um yes so we actually uh given with that uh all of this uh right we need to um uh we need to predicts a u uh drawout trajectory of a goal um stat dynamically right um it can be a goal state or goal observations but basically the goal is um if you look on the figure on top here um the top row is the real trajectory um where you see the mountain car right there's a car that it is to go backward and to build some kind of a momentum and then go full to the right to get to the um to the flag if it is on only going to the right and it can it does not have enough momentum to to go to the top of the hill so um so in here uh is the um actual observations you can see that the trajectory is um a successful trajectory and the goal here is to um for each uh observation at each time step we predicts a um immediate predict predicted PR preference um image or goal at that single time step so that we have um the instrumental signal or idea of uh what we are trying to learn and then um from that we learn the actions planning uh on uh from the PRI preference signal so in this talk we we specifically analyze this issue um the issue of um uh prior preference learning um in the current literature um and discuss potential result solutions for scaling the instrumental signal inherence to active inference by introducing the um contrasted recurrent State prior preference or crsb um this is the model um learning framework that um we also have um done in the past so uh this methodology frames active inference agents in term of progressively constructing and adapt adapting um a prior preference um at each time step so this also facilitate the Dynamics emission of a useful or dense very dense um instrumental signal at each Town step so we also um study how our prior preference um adaptation scheme uh when operating jointly with the Laten space driven epistemic for aging terms uh offers active inflation um a very nice flexibility in challenging problem cont context um such as um spars to reward um environments um or and also composed of uh continuous observation and action space for example um for example robotic task um they only we often like do not know the sparse um do not know the dance reward um functions in the actual real environment and we we only know if the robot has risked the goal or not right um and also all these environments has you know varying goal States so um we can also see that providing the uh uh or manually craft the uh distributions of goal is very not very um practical so uh yeah so here here's the outline of this talk on the introduction I've had just covered um so we so first of I'll just assume that uh everyone is NE active inflence and I'm just GNA briefly cover active inference backgrounds um and deep active inference um and also so cover a little bit about um prior prior preference learning and problems um uh of all those prior preference learning framework and then I'm just going go through the actual methodology which is the dynamic Prim preference learning um and then uh first of all is the world model um next is the self-revision mechanism and then next is the um learning pre preference models and then finally um putting all of them together um to make agent um and then um I'll show the results and then discussion okay so first of all um I want to um cover a little bit about active INF um so just a brief overview about how active inference work and what is the underlying theories behind it um so um it is a ma mathematical framework that describes how intelligent systems are human intera with the environment um so it describe it also described the environment as the generative process um and the intelligence systems um or the agent or um us as generative model um so active INF built on top of the free energy principles um this means that we as agents or intelligent system continuously observe the environment and forming our and correcting our belief or our own representation of the world um so the generative process um defines a real world environment here right um this is treated as the data Generation Um process where the information P terms are sensed by the agent so you can see that's the um hidden State underly hidden States denoted as here um it's generate a sense of um some kind of um observations um denoted o and this is also um sensed by um our in intelligence system so on the right here is the generative model uh is uh the actual agent that we are trying to build on the intelligent system right um the intelligent system also has an internal State um which is trying to um estimate the process um States um so this is called um the prior or the internal relief prior to the observations of the environment at a particular time stamp um so this is denoted as um the random variable s or state and P of s here um is the distribution over the random variable s um so this make the agent generative model um the generative model represents the agent approximation of of this generative process um in this world um by continuously seeing the observations or making the observation o um and modified or updating uh its belief so we actually call the belief um as after uh being updated after uh making the observation or the posterior because it is the posterior after um making the observation right so in order to learn the model and estimate the posterior we first need to in introduce these terms um first of all is the prior over State um which is denoted as PS which is also the distribution over state right um P ofo here is the model evidence um it is the distribution over the current observations um denoting how well is our model at predicting the real data um and in order to compute this easily this is often written out the sum of probability uh of observation given every single state and then next one is p of O given s this is the likelihoods um and the probability uh this is the probability of U an observation given a States so this is a function it gives the probability given an input States and then the the last one is the posterior function functions uh p as given o which is um the posterior function that Maps the observation to the states um so uh normally in traditional inference um the per function would be uh the the transpose of the likelihood Matrix for example um and deep deep learning literature um it is often calls a encoder and the likelihood here is a decoder so how do we estimate the post now um we have to find a belief that maximize the modal evidence right because um at the current time points uh there there is only one single observation o here and it is real right so P of o p of that observation um it's going to be one right so it should be the probability of one um and then um we use the stat to estimate um the observations and then it has to be the uh the probability of one two right so uh that that arrives us at the conclusion that we have to um maximize the model evidence P ofo so maximizing p ofo is also minimizing equal to minimizing minus l p of O right because um it is is uh minimizing the surprise um so in the beginning we say that we continuously update our belief after making an observations of the environment um we call the belief after making an observation a posterior um this is Q of s right um so to compute posterior we want to put in a um we just want to um call this you know any kind of um distributions um over the states um then for example we can sample it we can randomize it right um and then we try to gradually trick this distribution so that it's used the highest model evidence or it's used the lowest surprise so how do we choose U the actual uh posterior um so that it bus the lowest so in minus l p of O here is the actual um uh minimizing surprise that objective that we are wanting to to minimize right so uh we can uh analyze it like uh we can uh do it with a base theorem this not base theorem but base formula or theories yeah um and then uh to evaluate the surprise minus log P of O we need to sum out um the hidden variable s from The Joint distribution PS um but well this summations there's a problem right because that we do not know like all the hidden States all the hidden variables in the environment um it is hidden from us right so it is normally very very hard to uh to sum up all these um State values so uh if we cannot sum it up then we can try to approximate it so first we introduce a posterior um in into this uh sum right we multiply both the denominator and the numerator by the posterior um so this is the belief that we are trying to tweak right so um in here we're trying to um took this posterior so that um we can um minimize this term so this gives us a we sum where s uh the ratio um P oos and um over the the distributions um Q of s is Red by the Q of s this is like we of su so according to the inequality of convexity or genins inequality says that's the convex function of a weed means is always smaller um than the weed means of the same convex function um so you can see the convex function here is minus lock of something right so minus L of something is just a hyper uh is just a parab something like here um and this is smaller than a what means of the complex functions also we put this Su outside right um so this is also apply to the context of probability Theory um and then we can arrive at this uh yeah this equations so formally it's called an upper bounds uh this one right uh a quantity that approach from the above right so this is also means that this is a free energy um when we find a posterior that minimize this free energy term we will have a chance of approaching the lowest surprise points um which potent increasing the uh modor evidence so after we have this terms right here we can just uh put the minus inside and then it's become this form this is the conventional form that we call variation of re energy um and after decomposing this we will have these two form uh which is the K Divergence of between the posterior and the prior minus the weighted sum of the Lo of um the um P of or given s which is the posterior functions or now this is likelihood so basically um the K Divergence term is called the complexity and the minus l p of our given s is called accuracy term where the complexity terms comput how um surprise are State um two State distributions posterior and the prior are far away from each other or not right and the accuracy terms uh defines how accurate our model is predicting um the actual observations right so this is also relatable to the variational auto encoder where um we comput the loss of the variation Auto encoder um to be the K Divergence of the posterior and the g distribution with a mean of zero and a standard deviation of one and then we also have the mean Square arrows of the actual predictions of the images right um and then vfp and MF is a little bit um different done from variational out on coder VF is variational free energy and MF is marginal free energy as in the um acence tutorial um the paper um so marginal fre says that um we comput the K Divergence between the firster and the prior and the prior is computed from the transition model um not the um not the normal distributions of uh mean of zero and the standard equation of one so uh here is the uh General model of trying to build a deal inference model right so here's first of all we're trying to encode right using the posterior functions So In traditional active inference this is the P given o is the transpose of the likelihood Matrix P of given s this is normally learns with the cical distributions and using dally conjugate prior um and then uh in the Deep learning rerat um this is often known as the encoder uh that's uh Often parameterized by the neuron Network um and then o here is often be an image um and state here is just um some hidden uh vectors that we are trying to compute and then next one is the um lik function right so normally we text in the states um of um the agents and then we're trying to estimate the uh predictions the observation predictions right um so in traditional active inference this is known as the likelihood Matrix or likelihoods um per se um this is also learned with the categor categorical distribution and using the rich conjugate prior um and deep learning lature um this is known often known as the decoder um that is often parameterized by a NE Network so uh as We Know the variation of free energies formula the complexity terms K Divergence here right and the um uh accuracy term is the minus L po right even s um so here we compute the K Divergence between the um encode States and the actual prior States here and then um next one we comput the um actual L probability the predicted observations and the actual observations right here um and that makes the um whole objectives where we comput the two uh objectives right here that's uh also be the um variation of free energy formula so we just talk about how inference um inactive instrument work but there's more we want to build an INF system that can interact with the world right so when it comes to interaction between the agent and and the environment we also need to have actions and um also roll out in town step right so basically we have another elements here which is at and remember I also mentioned the transition model right so after we take an action in the world um we um the the world um all the environments evaluates one step right and it uh it change it own State and then emits a new observations um and we as agents or intelligence system also try to predict um given the action in the current states what is the next States and what and this States is the prior over um the states of the next step time step where we have not made the observation of T plus1 yet so after we have the st+ one as prior um we do the same process as I have mentioned before so uh we make the observations and try to change um our belief uh after making that observations and then learns from the uh new observations to take our prior into the posterior so here we learn it um using the varation of fery or marginal fery formula so as you can see from the previous slides uh we do needs somehow U to somehow plan the actions to interact with the world so if we're trying to minimize or surprise um with the current observations um so in the action planning we aim to take actions that minimize the similar free free energy term in the future um this is often called as the expected free energy by the way um so in here uh we introduce some new terms right Theta um is the model parameters um the reason why this is here is because we want to also compute the um uh model the surprise over the modal parameters or the uncertainty of the modal parameters um so you can see that the expected free energy in the future it's kind of similar to the version of free energy formula it's only needs um another model parameters parameters and uh the given the pi or the actual PL policy or Cent actions so after a lot of derivations right I'm I not getting into like the real derivation here but um here's the end results that's um something something minus um minus the weighted means of the uh lock P of OT right so I'll just focus on this um term is because this is the um goal of the talk today so basically um this term lock of P T is the expected fre energy derivations um of the instrum uh instrumental term so basically uh this also means that um we are trying to maximize the now we're trying to take the action that maximize um the future observations that matches our um prior paraph distributions or what we want right so normally um to learn to take actions based um uh to learn to take actions um to align with what we want to see in the future um we of needs to craft and provides the priate preference manually in advance um and then we text the action based um we take the actions um that's maximize the likelihoods um given that distributions um but this is often not really practical in the realistic World um and because we can like craft manual distributions like every single time we encounter a new environments or encounter a new task right so what we can also do is we can also provide a goal States or goal observations directly to the agents which is more practical um but in the end the prior preference distribution is very sparse um and this provides little to no learning signal to the agent as you can see on the figure on the right and the left black right so black is a pror preference distributions and right is the single goal distrib go St distributions or goal State image so uh what can you do to dynamically uh produce um the uh uh prior preference right we can try to learn a prior preference model right that um produce a goal image at each time step so here is on top is the prior preference model that predictions right um and uh in the bottom here is the model prediction of the future um given the sequence of planned actions so we are trying to say that we um we want to take the actions um that have uh the roll out in the Futures uh predictions matching the prior preference uh Ro out in the future so how do we do that we first have to learn the model first right so in learning uh we can try and say that each time we achieve a goal then then we Mark the whole trajectory as the prior preference trajectory or the um positive um trajectory or expert trajectory um and then for each observation marks as goal we use a deep learning like another deep learning model to estimate startat observations um for example right and then um in inference time um we actually train when we train the policy planner uh to propose the actions right we make the policy planner proposing the actions first and then predicts the future observation based off of that um actions U that is planned by the policy planner and then um we actually learn the policy planner based on the signal that is emitted from how a line me the predicted future observation would be um to the predicted goal um the predict the predicted goal here is the actual prary problems model outputs right um then we train the policy planner with the um mean Square Arrow or um some kind of objectives um that's uh predicts uh uh some kind of objectives between the predicted goal image and the predicted future observation so there's this problem uh where uh uh it exists between um all these learning schemes because it is very computational expensive um so it is generally uh expensive because we have to work directly with the images so for example in learning we have to use a decoder of the um of the PRI prence model um to actually predict the go image at each time step right in inference time is also computationally expensive is because um it has to learn from the images itself and comput it down from uh the um the back propagations will have to flow through the decoder um back to the action planner so here it comes to the actual methodologies uh we actually use Dynamic preference preference um in the state space instead of the observation space right so first of all I want to talk a little bit about the world model so uh this uh model architecture is um basically what I have just um drawn in the background sections but this looks with robots images so in here we have the observations Ot Ot and here's also the observation the same ations OT minus one OT um and we text in the encoder encoder is the encoder posterior text in the encoder and we have the transation prior um and then we have the decoder right encoder encodes the imin have the states and then decode the um the image and the state back to the image and then computes the accuracy loss right accuracy loss uh is minus L prop of the P given S as we have discussed in the variation of free energy sections and then the complexity is the k k Divergence term between the what the area and the prior right so how do we compute the priors we comput from um the previous steps uh States and then we use the transation model um to estimate the next States prior right so that's um that's how we can get the um prior so the current work um and popular Works uh regarding the world model or generative models um often integrates um the model called um RSS Lang which stands for um recurrent States space model um in the translation uh model so it's to zoom out a little bit um we are trying to uh integrate some kind of temporal information Through Time right um so that's the state of the environment or the states that we're trying to predict be more accurate um and it carries more informations with time um so for example um we cannot um encod the uh enough information from just a single image because for example um there can be like uh position and velocity right the velocity is acquired um through multiple images um and cannot be acquired from a single image if does makes sense so what we often do is we try to integrate a recurrent neur networks that text in the previous States um or the previous recurrent States H my T minus one um and then we also text in the um encoded observations um and then with in transition um we produce the recurrent n network we produce the current recurrent State um and also predicts the outputs to to put into the uh transition model um and also it's the recurrent stat will also uh be served as inputs to the posterior Network um so that's basically uh the idea here and then uh the other way uh around we could just uh do uh the model as normal so uh in active INF agent takes in the actions um that's it Bel uh that would lead to its preferred outcome uh using instrumental signal um to construct this prior preference passord has um for example provide a go statement observations directly to the agents for example mazaka in 2021 with contrastive active inference um we can also manually craft a PR prence distributions um we can also dynamically produces the preferred observation that each time set um but as I mentioned already um all these methods is one way or or another very expensive um or it's very not practical or in the reward applications or um it is very computationally expensive for dynamically producing the actual observations um for for the goals so we can also use Behavior cloning but it its trajectory diverges from the training trajectory due to um small mistakes made by agent as well as environmental stochasticity um and also it is because of ident IND independent and identically distributed assumptions of positive exper data distribution so we can um we we try we will kind of avoid it because it's it's not going to um be uh the same right the two distributions of the actual um actual trajectories in the environment should not be the same as um every single positive trajectory of the expert data so to tackle this we try to leverage a small quantity of seat imitation data um to learn an uh a neur network that's dynamically produced the preference of State not observation but State at each time step so this effectively um produces an easily generated dense instrumental goal signal um and we also design the agent such as that it moves according to the trajectory that is is shaped toward its own estimation of the future States or future preferred States so it's trying to Notch the its own trajectory to watch the imitation or positive data distribution so as you can see on the figure on the left um so you can see that in Behavior cloning it's trying uh the actual trajectory is diverges away from um the actual imitation data where the prior preference over the states trying to um get uh trying to be shipped toward the imitation data itself uh using the model signal um and ra is our agent um is called robust active entrance um and it learns from both epistemic signal and instrumental signals but in here we'll just focus on the instrumental signals and how to construct so uh yeah so basically we use a decoder um or the likelihood functions of the generative model to only visualize on the preferred observations um not that in training or test time we do not use the motor learning uh we do not use the actual observation uh goal observations by itself we only use the hidden States so the goal here is to um get the states the goal States at each time step be predicted by the prior preference model right um You can see that's uh on the top here is the real trajectory and the bottom here is the actual um dream that the dream of the robots that it wants to be right and then we are predicting that dream right so here is the actual model of the uh um contast recurrent State pror preference learning um U so basically it has um it has also similar architecture to the rssm to the world model um it also has have an encoder um here it also have transition um but it does not have the decoder um so how do we actually uh learn this uh um PRI preference model so first of all uh we uh learn the world generative World model as normal right um but when we say whenever um it's uh whenever the trajectory is uh positive or a successful trajectory um we try to contract um the state of the uh crispy I just call it crispy um we try to construct the state of crispy to the um world model States and then um whenever the um trajectory is negative we try to push away the states um from each other so that the crispy model um can only predict the um positive States so how do we determin um if it is a positive trajectory or the negative trajectory without having a very sparse signal so we will have a thing called braw or the prior preference signal so how do we compute this RW signal so imagine um this is the real trajectory um and it is a successful trajectory for example right so we're trying to compute um the prior preference rate by uh having the um rate of the successful States is at one and it starts from one and then it decays backward so decays um very fast and um if so in this like really far away stat from here um they do not have um enough signal because there's no goal um like there's no direct go from that states and then there's real trajectory that is failed right um You can see that the mountain car cannot reach um the flags on the right here so whenever it's built um we can just PL PL the very very last observation as minus one and then we start to De Decay backward negatively um and then uh we have the prior preference rates uh signal as um very negative so the reason why it is it has to be last is because um we want to um we want to blame the L observations um hardest so um in order to uh in order for the agents to just push away from the failing States very very very strong um in the very last stage so in the real trajector that have like multiple successes um we try to put all the one in all the successful States and then just Decay backward um as similar to how we Decay backward in a single success case so with that s we now uh know how to compute um the roow and then this will serve as the dynamic signal to the contrasted loss and it creates a dynamic behavior of the contracted loss um that's trying to either push away or um contract um the two states of the first speed State and the world model state so for example in here um there's if there's positive row um the contracted log would be um cont Contracting all the state together and maximizing the similarity between the preit states and the world model States and then we also learn to U minimize the K diversions between the um complex uh uh the posterior and the prior uh of over the states of the crispy model um this is because we want to estimate the transations right so we transition to the um next goal State and then this goal state has to match the actual goal um that is the current state of the EnV of the rssm model or the world model um so in contrast right if row is negative then we're trying to push away um all these states away right so we're trying to maximize this um distance or minimizing the similarity between the um world model States and the crispy State and in here notes that there's no complexity objective is because we don't want to um predict the next goal that is um the failing State that's basically the the idea so that arrives that's the uh general formula for the objectives of the contrasted State PR preference model um basically uh it is just masking out the sign of the row and then multiply with the um the K Divergence or the complexity uh between the posterior and the prior of the crispy model um and then we uh minus the row and seam of U the actual predicted State and the world model state so same here can be very much um uh any kind of similarity functions for example cosine similarity um and then in the in our work we actually use the actual cosine similarity so in train time we have we have just talked about training time right so we have successfully assuming that we have successfully um learned the prisy model that predicts the future goal um and in inference time we use this uh crispy transition model to estimate future rollout trajectory of the goal right so we can see that um on the top here is the uh observation um actual observations and in the second row is the actual prior preference roll outs or predictions of the future goal uh given the um given the current observation so given the uh PL actions we're also training the PL uh the plant actions right so the actor trying to predict an actions given um the observations the predicted of observation in the future um so we're trying to say uh this is uh we're trying to compare uh the actual PR prior preference roll outs with the actual predictions given the actions um and then we are trying to uh learn the action planner based on the signals or how diverge um the future goal is to our future predictions of the world given the actions so uh there can be so many um there can be so many reinforcement learning agents and active INF agents that can utilize this technique um of Prior preference learning so active INF agents um I put my works here um it's ra and then there's also Chris bakley's work and baren M work about deep active influence in varal um policy gradients methods um and also Auto uh work is is the scaling of the uh act active inflence in PDP and not car environments and so on right so re enforcement learning agents there's all these sorts of um after prcts algorithms like dreamer series um soft a cretic dreaming um ddpg tqc um and TD and so on so there's a lot of agents that can utilize the framework because there's no change um in the original architecture there's just an add-on um to how um to how an agent would apply the priate preference learning signal um into the system so here the actual flowchart of U of the agents basically um we start the agents and for each episodes we evaluate um the environment um using the world model and the policy planner um actually basically we uh uh evaluates in the environment we let the robots play we let the robots do the task and then we collect the uh data and add them to the replay buffer for example right and then we get get the main training agent um logic um and then first of all we sample the training data from the buffer so in here we trying to have um we also have to use um the self- imited ation learning buffer um so uh yeah so self-imitation learning is very good technique um normally to balance the number of um successful and failing trajectories and we also learned um so first of all we learned the world model encoder decoder transation on rssm and then we uh train the epistemics model but in this case we don't need to care about it um and then next is the um constra recurrence State PR preference model um from the data and then um next we imagine the future trajectory using the world model and the policy planner right we try to estimate the future actions um to be um what is the optimized future actions for example we we estimates the future action based on how uh current state is and then we the world model will be continuing to BL out in the Futures and then we dream of the Futures and next is the part that we actually um infer on the uh crispy model we use a crispy model to imagine the future preference trajectory um given the goal um given the current observations and we predict the future goals roll outs goals right so after that we learn the policy planner right using the instrumental signal the instrumental signals here is computed from the um difference um between the future goal and the um predicted future dream of the world right and then with that the policy planner can learn from uh from the uh instrumental signal so here's our um preliminary results on um active INF agent so basically um we can see um the two top rows is the mountain and car environments and the two bottom row is robotics meta World um push button uh environment so let's look at the first row for example the observation here is a mountain par that have reached the goal and for each um observation here we uh we predict a preference roll out or preference um or a goal State at that single time that so in here we can see that um if the car has not have the momentum yet um then we have to build the momentum by predicting the car going to the left for example right and then after it has built some kind of momentum we trying uh the actual prior preference model predicting the car would go to the right go forward right when it goes here and the prence model will predict the car to be be um going closer to the goal something like this so the same applies to the um robots environments where the actual observation is the third uh row and the uh prior preference roll out is on the fourth row so we can see that uh it is also trying to estimate the dream or the dream of the goal of the robots is to actually push the button so um there's different kind of experiment setup that we are trying to learn um basically we have uh four trials on each environment and then we train the experts um using as sof critic or um po um proximal policy optimization um to collect um a small amount of seat data normally is about uh uh 20 episodes something um 20 to 30 episodes um so we test mostly on three Mains environments or suits um first of all it's m car there's only one task um metal wall is 13 task so here's um all R robots here is metal W and then the robot suit we have two task and then um the cam regarding camera setups uh we use um three views for met world and four views for um the robo suit so the green check mark here denotes uh that the robots has successfully solved task um here's the Robos suit door open door and then um here's Robos suit lift where it can successfully lifts the block and here's more uh solving agent tasks so basically agent with Crispy exhibits and ability to perform successfully early on um and uh so agent with without crispy uh struggle to actually succeed on soling the environment um and we also test on other environment as well for example genology and Robotics or and risk task um you can you can see in the middle image on over here so actually go to the discussion sections so actually um I want to make a note that um with respect to the crispy subm model um our model utilize contrastive objectives to adapt its parameters so basically it solely learns on um estimating the world Moto in coded States while pushing itself away from undesired World model States or trajectory so this does not require or the learning of the of the decoder so we we only make sure of the learned uh decoder in the generative World model um visualize the uh to visualize the preferred observations um from the estimated crispy States so experimentally we remark that this is an um that's uh we only use um this decoder um to as an auxiliary tools um to prove um that's the uh states of the contusive recurrence prim preference model uh is useful so theoretically Behavior clothing will be unable to achieve a great uh success in multiple multi goal environments due to its Reliance on a fix and finite size uh imitation sample pool um the expert trajectory in this fixed pool might further differ from the actual trajectories um that the agent needs to take to solve the problem at hand um for example there's a distributional gap in in observations and therefore and um it requires different sets of actions to be taken for example right um so on the other hand right crispy learns to produce dynamically the goal States and not the goal observation right um so we can empirically confirm in simulation that our agent um with Dynamic prior preference learning does not suffer from the goal Mis match problem u in the training phase so this is very applicable applicable to the multi-goal environments so um we can see that um this PR preference learning uh techniques has a lot of practical applications in different robotic tasks um with for example with potential dangers in the operations um and functioning for example when human safety must be considered right for example a trying to drive a autonomous driving car system where it has to have or it has to predict um the future U goal right the future goal is not to hit the human for example right um and we can also make a lot of advancement in imitation learning in research community and it can be um applied by the wide range of reinforcement learning and active inflence agents um and then the last one is to scale the the active entrance agents on the real world robotics agents so Future Works words actually considered um importing or integrating the dynamic PR preference learning in latent State space model um for example variational Dynamics discret variable Auto encoders or tension waiting models um and also uh there's another models uh physical neuro robotic system um which entails an embodied and active and survival oriented formulations of active perception and World model learning so in this uh talk um uh we have discussed a lot of ideas about uh being able to estimate the particular goal or preferred States um at any particular time step is very useful um for cognitive control agents so as our results demonstrates the agent can then learn to adapt the take actions that leads to um a estimated goal um and then um in contrast to the approach is taken in the impementation learning and behavior cloning literature um which train the agent's policy based solely based on the fixed collected expert data sets in our framework the expert signal is estimated dynamically um through an Adaptive prior preference model right closing this Gap the domain gap between the actual trajectories and the collected trading imitation trajectory or preferred trajectory um and then we also discussed the gener generative model um and encoder decod transations and how the varation of free energy work and then how to learn the model and then we finally we discussed the contusive recurrent State prior preference model um with the self revision mechanism and then discuss the model learning and then inference um objective um and how to uh integrate St into the agent planner learning system and then we show the results for the robotic task and then I mention how it can be applied to fields so that concludes my presentation um and thank you very much for um listening thank you that was amazing really great job with the background math done only like a PhD student can but it was great to see how you brought it all together and connected your research to it all at the end so thank you that's all for now see you later thank you see okay hi welcome back welcome back hi Denise hi thank you for for joining and looking forward to your presentation so take it away oh thank you so much and thank you for having me today um today I am going to talk about the potential impact of centralized AI through active inference and spatial web Technologies and what we're really going to be discussing is um the potential aspects of this next era of computing uh it'll be incorporating a lot of the research that you're hearing about here over the next three days um and decentral in my talk decentralized AI is kind of taking a main focal point here and there's a couple of aspects um in technologies that are evolving right now that uh I think it's really important to be aware of um and they factor into uh what I'm going to be talking about today so one of the main uh things that is occurring right now is our internet is evolving um we have a new protocol layer that's layering right on top of the other protocols that are part of the same internet that we're talking across today um the difference is is what these protocols hstp hyperspace transaction protocol and hyperspace modeling language is the programming language um what these are going to enable is for us to have programmable spaces so instead of just programming pages in domains with content we are going to be able to program digital twin spaces of anything and this is going to enable context to be baked into everything in every space so all of the attributes about various things in various spaces and nested ecosystems and all of their inter relationships to each other so one of the things this is going to enable is uh active inference agents to be distributed across our internet and I see this as the next wave of innovation and through the through active inference these agents will be aware of each other and the network uh they'll be able to sense the world in real time and learn from each other and this is going to provide unprecedented levels of interoperability um so what we're talking about here is first principles uh automated systems and distributed collective intelligence so first let's look at the challenges in traditional centralized AI structures um or architectures uh centralized processing constraint you have high energy demands and scalability issues limited adaptability to realtime changes and you know we're all familiar with the limitations of uh these deep learning models they require massive amounts of labeled data and processing power they have limited ability to generalize across different tasks and environments AI models uh lack adaptability to real world complexity they're only good for one thing at a time uh like one uh optimized for one type of an output and they lack interpretability and transparency in decision-making they have high computational resources leading to high costs and inefficiency and they struggle with generalization across different scales tasks now we need decentralized Ai and decentralization really matters because it increases resilience in scalability for autonomous systems it reduces Reliance on Central servers and data hubs and it enables autonomy across distributed networks so this bridge is the gap in AI we move Beyond centralized AI for scalable autonomous systems that can operate independently and locally now we're all familiar with this guy here Dr Carl friston and uh he he gave a great talk uh first thing this morning on uh renormalizing General generative models um so what I'm going to be talking about are various components in in some emerging Technologies along with active inference that potentially realize this research paper designing ecosystems of intelligence from first principles now you know the this uh paper was published in December of 22 and um it really laid out a new framework for autonomous intelligence that mimics self-organized systems of nested intelligence found in nature and in in this paper it described a cyber physical ecosystem of natural and synthetic sens making in which humans are integral participants what we called shared intelligence and this in the paper it says this V this vision is premised on active inference a formulation of adaptive behavior that can be read as a physics of intelligence and which inherits the physics of self-organization and in this context we understand intelligence as the capacity to accumulate evidence for a generative model of one's sensed World also known as self-evidencing and at the end of the abstract for this paper uh they also note we also consider the kinds of communication protocols that must be developed to enable such an ecosystem of intelligences and motivate the development of a shared hyperspatial modeling language and transaction protocol as a first and key steps towards such an ecology so what I'm proposing in this and what is proposed through all of this this uh technology together uh with all of the various parties that are involved in building these compon components is that the future of AI is shared distributed and multiscale an entirely new kind of autonomous intelligence able to overcome the limitations of machine learning Ai and as we know uh from understanding active inference this AI is knowable explainable and capable of human governance and it operates in a naturally efficient way there's no big data requirement and it's based on the same mechanics as biological intelligence it learns in the same way as humans so what I want to do is start with the spatial web protocol uh first because this is kind of the um the framework for these agents to be able to have this distributed intelligence structure so what is the spatial web protocol it's just the evolution of our internet into spatial domains um so people refer to it as web 3.0 it's just the next next evolution of the internet but this protocol has become a global standard a new global standard for the last four years uh this this standard has been in development with the i e uh which is the largest core standards body uh in the world and back in 2020 they deemed the spatial web protocol a public imperative which is their highest designation and this summer it was voted and approved um as the new global standard and it's in the final administrative uh cycle with the i before it becomes uh public so there's a couple of things that are happening with this through hsml the programming language for this new protocol all emerging technology whether it's iot or ar VR distributed Ledger technology um you know all of these Technologies they're all going to be able to uh converge across the internet because they will have a common language for interoperability and in that you also have a language that AI can understand and uh communicate together and communicate with these various aspects of Technology as well so this is going produce the next era of computing and deoe called it back in 2020 so it's the next evolution of the internet and it's going to be empowered by active inference AI now just to kind of understand what this means for the internet um let's take a look at what a website domain is today with the worldwide web a domain basically contains a website and these websites they serve up content so whether it's you know images or video or audio or you know information data um that is what websites are capable of of doing now a spatial web domain is quite different because a spatial web domain could be anything in any space right and it can be programmable a spatial web domain is an entity with a persistent identity through time that has rights and credentials so you're talking about going from a library of pages in our internet to the internet of everything and hsml becomes a common language for everything so it provides unprecedented levels of interoperability between all current emerging and Legacy Technologies uh it's a programming language that Bridges communication between people places and things laying the foundation for a realtime World model for autonomous systems and what you have are programmable spaces hsml is a cipher for cont context anything inside of any space is uniquely identifiable and programmable within a digital twin of Earth producing a model for data normalization and from this we get adaptive intelligence automation security through Geo encoded governance multi Network interoperability and it enables all Smart Technologies to function together within a UniFi system and then this produces a Knowledge Graph and digital twins of anything everything potentially our planet so as as this knowledge graph grows within the programming of all of these spaces across the network you get this Universal domain graph that enables a digital twin of our planet and all nested systems and entities within it we get nested ecosystems uh there there will be intelligent agents both human and synthetic across this network uh they will be sensing and perceiving continuously evolving environments making sense of the changes updating their internal model and what they know to be true at any given moment and acting on the new information that they receive so why do we KN need a spatial internet progr protol well the increasingly graph-like nature of global data the opportunity for Automation and uh autonomic activities using context aware cognitive AI the need for composable systems and applications including the governance of such systems the int the intrinsic need for secure transactions the rise of machine learning and neural network computation and Edge computing the need for explainable AI and robotic governance and the rise of iot and sensor mesh and some of the guiding principles of the spatial web standards are spatiality representing information as locations and relations in hyperspace ownership users own their data in digital property security secure data collection transmission and Storage privacy individual control over digital identity and personal Data Trust reliable permission driven validation of users assets and spaces interoperability seamless navigation and asset transfer across spaces and responsibility creating technology ethically for the benefit of humanity and what what has become of these Global standards is a so soot technical standard so the spatial web is a soci technical system of systems uh takes into the account the um so the development of the spatial web must address both technological as well as sociological considerations socio technical standards integrate and balance the technical social physical and legal use of technology and the spatial web standard is is a soci technical standard and what this results in is multiple scale cognitive Computing so the spatial web plays the role of a collective nervous system for smart infrastructure it hosts cognitive Computing uh inference perception learning and action selection by combining and abstracting distributed information the spatial web enables the formation of complex ideas and facilitates decision-making cognitive Computing occurs at multiple scales in the spatial web individual nodes host cognitive Computing functions and networks of nodes collectively perform emergent cognitive Computing so hsml hyperspace modeling language it's a human and machine read readable modeling language and semantic data ontology schema there are some key elements to it there are entities activities agents contracts channels credentials domains and it includes hypers space and time so hsml enables the expression and automated execution of Legal Financial and physical activities in the spatial web and all of these entities are identifiable and knowable by these intelligent agents through this uh language throughout the Network now let's look for a second at the agent aspect agents and activities so agents and that can be human agents or synthetic agents um but these agents are entities that sense respond maintain a model of their environment and take actions to achieve goals and the activities that occur they're single actions or partially ordered sets of actions that are performed by the agents and they unfold over time so agents interact with domains and other agents to perform activities in the spatial web so what the spatial web facilitates is a distributed ecosystem of intelligence if you look at current state-of-the-art AIS they're siloed applications they're built for optimizing specific outcomes and they're they're really unable to communicate their knowledge frictionless and collaborate with other AIS but this facilitates a a an ecosystem of intelligence that is a self-evolving system it's learning moment to moment and upgrading its World model and in that way it mimics biology while also enabling uh general intelligence so this is shared intelligence at the edge of everything so how active inference works in the spatial web it enables real-time decision making you have agents that are predicting and adapting to their environment from their frame of reference in the network uh they're continuously learning from their environment enabling realtime adaptation and unlike traditional AI which relies on pre-programmed rules and static models active inference continuously updates its models based on new data allowing it to adapt in real time so these active inference agents would seamlessly communicate in real time with each other sharing their knowledge and perspectives and learning from each other these active inference agents excel in handling complex uncertain environments by continuously minimizing uncertainty in fact they can quantify their level of uncertainty and this makes them suitable for applications where conditions are constantly changing so decentralized AI takes the processing to the edge to the edge devices you have a distribution of processing and power uh superpower gpus are not required the processing can take place on local devices throughout the network with infrastructure that already exists like your laptop or your mobile device or devices that haven't even been um invented yet um but we're already seeing a lot of those kinds of devices coming into play uh with you know um Apple Vision Pro and and you know the metal glasses and and different aspects of these Technologies so the processing takes place on these local devices throughout the network you have faster response times and reduced Network congestion and it uses real live data realtime data from iot sensors machines and everchanging context that's embedded to the uh in the network so this is the sensory information that's coming in for these active inference agents so that they can uh you know do the perform this action perception feedback loop and continually update their model of the world of their reference and it eliminates the need to tether to a giant database active inference agents throughout the Network learn and adapt from their own frame of reference within the network enhance data privacy by minimizing centralized storage and it minimizes complexity it uses the right data in the moment for the task at hand and with hstp with hyperspace transaction protocol you there's natural guard rails uh included in this framework so it creates a secure ethical and user Centric environment for AI applications fostering trust and reliability in digital interactions uh you have data privacy and sovereignty transparency and explainability security and authentication bias mtig mitigation ethical AI guidelines interoperability standardization user empowerment and control accountability and governance safety and reliability Dynamic policy enforcement and what this results in is in intelligence at scale uh through this network of distributed intelligence it enables the cognitive architecture of the collective intelligence of multiples of agents that continuously communicate coordinate and collaborate with each other you have individual and specialized intelligences together on a common Network speaking a Common Language hsml efficient and Powerful cross communication to perform tasks regulate systems and address problems in real time and it scales up and grows in tandem with humans through hsml we can take human laws and guidelines make them programmable for these agents to understand and abide by in real time so we're talking about unprecedented levels of cooperation between human and machine intelligence and what you get then is an Ever evolving collective intelligence uh decentralization is an imper imperative for intelligence at scale these agents represent scores of people from all over the world sharing their individual and specialized knowledge all coming together on a common Network and it's based on each agent's own world model you have unique perspectives from whatever frame of reference uh within the network nested ecosystems and levels of self-organization and Collective shared intelligence so the idea is that anybody can impart their knowledge and their wisdom into these Agents from wherever they are in the world and this actually acts to preserve all of the uh cultural differences all of the um the belief differences the governing Styles um you know people whether they're in uh Kenya or Singapore or anywhere else in the world they should be able to create these agents to solve their local problems and then share that with the world because then these agents can share their intelligence with each other so the spatial web protocol enables a decentralized secure and interoperable web with the potential to profoundly transform Global Industries and everyday human life now let's dive into um kind of the the potential that exists for this idea of rgms these renormalizing generative models and the spatial web and we heard a great talk this morning uh by Dr friston he uh you know he talked a lot about the you know the science behind these uh these rgms so I'm going to kind of uh maybe maybe break down some of the aspects of how these work as leading up to then what they can do within this uh potential potentially within this architecture so you know with rgms one of the interesting things is that it's a universal architecture um they're a UniFi unified scale-free framework for AI That's capable of reasoning learning and decision-making and they can generalize across multiple tasks handling small and large scale data efficiently they combine perception learning and planning into a single system and they can be used to learn scalable architecture over space and time similar to human thinking they require significantly less training training data than traditional models while achieving Superior results so what are these rgms they're a new class of AI models based on active inference they use renormalization to simplify complex data into hierarchical structures and they operate on the free energy principle to minimize uncertainty and adapt in real time so the significance is that they represent a Leap Forward from traditional deep learning methods they reduce data requirements and uh improve efficiency and it's one single framework for for all applications so the science behind rgms renormalization and multiscale learning renormalization is a technique from physics that's used to simplify complex systems to maintain consistency at different scales and rgms break down data into multiscale representations hierarchical structures rgms are actively learning from small details to large patterns simult mously and they analyze like with an example would be like an rgm can analyze an image by recognizing edges then objects then scenes and understanding the the concepts at every at every scale so key Innovations and advantages of rgms why they're Superior to traditional models they learn efficient efficiently with less data it's a unified system that handles perception and planning Sim multaneously and they understand conceptual relationships not just patterns and they can adapt to new environments with minimal training uh aspects of them are scale-free modeling renormalization discrete representation hierarchical structures fast structure learning unified perception and planning and explicit handling of uncertainty so if you look at how the brain proces data processes data um you know we we perform core screening all the time when we're learning right we break down sensory data into hierarchical patterns um that help us focus on the most relevant information um kind of this chunking of Concepts to to grasp different aspects of um whatever it is that we're we're learning or focusing on in the moment and rgms can mimic this approach by processing data hierarchically focusing on important patterns so it's a first principles approach uh using a way to simplify models without losing important information um they can model space and time Dimensions like we do and can learn the causal structure of information and it helps handle large scale problems efficiently using less data and less energy now when you can consider uh scale-free modeling you know um scale-free modeling these rgms they operate across different scales so from very fine details to highlevel con uh complex systems so unlike deep learning which struggles when moving between small and large scale problems this scale-free architecture allows rgms to seamlessly work on a variety of problems including small data problems and large scale strategic planning planning so when you when you're thinking of the concept of scale-free you know this is a great example here of a green apple you know you can you can see the green apple as a concept of just you know pixels in an image or a volumetric object or on a branch on a tree or in an orchard and you know the the way that these uh rgms can model these conceptual understandings of things is it can understand all different scales of what a green apple is so when you think of like um the supply chain or or factory operations or or anything like that these agents can understand the what's happening across the manufacturing line on a very uh local level but also what's happening across the entire distribution line uh throughout the entire system so they can predict both short-term movements and long-term outcomes so you know the fine details as well as uh large larger uh outcomes with the highlevel complex systems all at the same time and they don't lose accuracy when switching between local and global data scales so hierarchical modeling in rgms hierarchical data processing rgms process at multiple levels of abstraction from fine grain details to highlevel Concepts uh they use recursive learning they continually Loop over data learning at each level continually refining their understanding of Concepts as they process new information and you get hierarchical representations that they build models of the World by zooming in and out of different layers of detail and this is similar to how human cognition processes both micro and macro scales so you know think about when you're when you're thinking about the concept of driving a car if you're talking to somebody about driving a car or you know you're you're focused on the aspect of driving you're not thinking about the car as all of its various parts that make up that CL complex system like the brakes and the steering wheel and the seats and the every aspect of the engine you just call it a car right because that's all you need to know in the moment for whatever whatever you're uh doing with that concept and so this is kind of a similar idea um and this layered approach allows rgms to understand complex multiscale problems without requiring vast data sets so conceptual modeling for decisionmaking um rgms then focus on conceptual modeling versus just mere pattern recog recognition um deep learning can recognize patterns but struggles with conceptual understanding it can recognize correlations without understanding relationships and it can recognize patterns but it doesn't understand the underlying meaning and rgms on the other hand focus on conceptual modeling they capture multiscale hierarchical structured Concepts that represent both fine grained and Abstract features it builds a hierarchical understanding of the world capturing cause and effect relationship ships so rgms understand the why behind patterns not just the what and the key difference rgms are not just a better generative AI but a transformative AI framework that builds world rgms can reason about cause and effect relationships making decisions based on conceptual understanding that are more aligned with real world outcomes so how do rgms Le aage active inference by incorporating active inference Active Learning and active selection while incorporating realtime data to adapt its predictions and actions so what you get is realtime learning uh they adjust their beliefs based on new sensory input making decisions on the Fly optimizing both learning and decision-making processes and they also are able able to uh have unified perception and planning so deep learning typically separates perception like recognizing an object from planning like deciding an action rgms can combine both in a unified model they handle both within the single model and learn Concepts that guide both recognition and future decisions so for example you can have an autonomous robot that can recognize a door handle and also plan how to open it using the same model so how what does all this have to do with the spatial web architecture so the potential that I see with rgms is that um the nature of how they work fits very well with the nature of this architecture within the spatial web and just to kind of bring you back to what I was talking about earlier you have hyperspace transaction protocol this manages data flow between interconnected systems and it facilitates interoperability between different AI systems and the hyperspace modeling language provides a common language for these systems enabling uh computable context and decisionmaking and it provides a shared framework for context aware decisionmaking so this protocol aims to create a standardized way for different Technologies and AI systems to interact and share information across programmable spaces within our global internet and this unified model this becomes a unified model for diverse applications handling complex multiscale systems through unified perception and planning it's a single framework that can handle multimodal tasks so you have versatility they can perform across multiple applications whether it's object recognition natural langage anguage processing strategic planning uni using a single Universal model and it reduces the complexity uh it integrates the data analysis the perception and the strategy execution which is the planning within one model so you have cost savings it reduces the need for multiple specialized AI models it streamlines system management and saves resources so this how the spatial web enh es the rgm capabilities the combination of this scale-free active inference with the spatial web protocol could enable seamless integration of phys of the physical and digital these protocols allow rgms to operate across distributed networks exchanging insights and information between physical and digital environments enabling more sophisticated iot uh applications uh context aware Computing through the spatial web rgms would operate in context aware environments these agents uh adjust their behavior based on the physical and digital surroundings and interoperability multiple rgms these active inference agents and autonomous systems can operate in different environments whether it's your Home Smart City businesses and they could communicate collaborate and coordinate actions through through the hyperspace transaction protocol they uh energy efficient distributed intelligence instead of relying on centralized energy intensive data centers intelligence could be distributed across networks of smaller more efficient devices and this this creates a nested ecos system uh similar to a hallon architecture and if you're not familiar with a h what a hallon is it's something that can be both a whole in and of itself and also part of a larger system so the spatial web and rgms would form a nested ecosystem where each node in the network whether it's a domain device agent or subsystem functions as a hall on it's autonomous yet interconnected contributing to a larger system's goals as part of the larger interconnected Network like a smart City infrastructure um your smart home could contribute the contribute data to your neighborhood grid which informs the city infrastructure decisions all while functioning independently um if you if you look at this picture here of you know uh part of a supply chain image you know each item packed inside of each of those boxes is a hole in and of itself but it's subject to and it's part of the larger system the Box the box is subject to the the trucks that are moving it and distributing it or you know whatever's happening within the factory with those boxes and then that factory itself is its own hole that is subject to a larger ecosystem of the entire supply chain as a whole so when you're looking at what this next era of computing really enables it's just nested ecosystems that can be empowered by this natural intelligence and active inference Ai and the spatial web protocol can work together to create sophisticated adaptive systems at various scales so rgms for autonomy and self-organization with recursivity the scale-free models naturally form a nested hierarchy each level of abstraction in the model is complete in and of itself and yet part of a larger whole uh autonomy and cooperation active inference agents exhibit both autonomy in their decision making and the ability to cooperate with other agents and self-organization the renormalization technique in introduced in the paper allows the system to dynamically organize information at different levels conceptualization similar to the self-organizing nature of uh honic systems or these in these nested ecosystems so this last part um I want to explore the transformative potential of decentralized AI for the future of systems so the impact of distributed collective intelligence the impact on this next era of computing would be enhanced problem solving and Innovation increased efficiency and scalability democratization of knowledge and Technology resilience and robustness enhanced decisionmaking personalized and context aware Serv Services ethical and responsible AI collaborative platforms and ecosystems enhanced learning and education empowered communities accelerated scientific research and environmental environmental and sustainability Solutions and the implications for future systems and Society the convergence of rgms in the spatial web could transform Industries enabling decentralized context Ware systems that operate with minimal human intervention industries that could be impacted smart cities you would have autonomous systems that improve efficiency at all levels Global Supply chains rgms could optimize processes from local shipping to Global networks personal to critical systems homostasis so whether it's personal devices or critical infrastructure a continuum of intelligent adaptive systems can maintain homeostasis at each level in our Health Care Systems personalized treatment by integrating real-time data from wearables and medical records education we can have Aid driven adaptive learning systems that evolve with student needs and Environmental Management we can have ecological systems that could be modeled and managed more effectively from Individual habitats to Global Systems balancing local interventions with global goals and the future of AI and human civilization the development of renormalizing generative models within the scope of active inference and the spatial web promises to have farreach far-reaching implications more adaptive AI these systems are highly adaptable capable of operating in new environments with minimal training efficient Edge computing rgms enhance the power of AI on devices like smartphones and iot sensors reducing the need for large data centers improved human AI interaction with explicit uncertainty models these systems interact naturally with humans seeking clarifications as needed safer AI systems they recognize and handle novel or ambiguous ambiguous situations reliably preventing catastrophic fail in unfamiliar scenarios so they can fail gracefully and accelerated scientific discovery rgms expedite research in fields like genomics and climate science by rapidly identifying patterns and formulating plausible hypotheses thereby guiding further experimentation so the key takeaway here is that uh all of the all of this research and all of these Technologies together can provide us with a future where AI systems can interact seamlessly with the physical world autonomously optimizing operations and improving human lives so um yeah there's my website I have a podcast and I'll stop sharing now thank you

Denise awesome I don't know what time how how is how are we on we have uh 15 minutes and I'll read some of the questions in the live chat okay sure I'll start with Pablo who wrote how will all of this affect the gamification of processes in organizations well I mean the the interesting thing is is through hsml you know all of these emerging Technologies are are going to converge together and become interoperable in ways that we've not experienced before you know we've had this um kind of vision of of this augmented reality existence but the framework hasn't been there and this is going to provide that framework so I think when you're talking about gamification in every aspect it'll probably become kind of a natural part of of our existence and how we move through our day how we interact with with our work with uh brands with you know anything um you know I think that there's going to be a lot of of really interesting ways to kind of perpetuate that cool there's a lot of great questions in the chat so okay I'll read one from Tintin so they wrote it's impressive the obvious question I have is of alignment of this system in terms of that alien message from elazer gudowski for example and then they clarified what that exactly is so the key idea there is that simulated agents evolve and and came to think faster than their creators so at a moment they took control to ensure their own survivability it's the question of how to ensure that way more intelligent and faster systems keep aligned with our values right and so one of the things that uh is really interesting to me about this is through hsml we can actually collaborate with these systems in the sense that we can take human laws and guidelines and make them programmable so these agents can understand and abide in real time and you know um versus uh who you know Dr Carl friston is is their Chief scientist um they were involved in a a program called Flying forward 2020 and and they were doing a lot of uh testing around this and because it was it involved a drone project that I I don't know there were multiple countries throughout Europe it was with the European Union that were involved and you know testing whether the they could program in laws around airspace laws and you know things like that they did all kinds of scenarios of like you know Medical Supply delivery uh perimeter security all kinds of things and what they found was through hsml you could actually uh program these guidelines and these these agents abide by them so I think we are going to experience this unprecedented level of of cooperation and collaboration so we can grow in tandem with the you know with these uh agents and their intelligence and I think that we can also you know take some um solace in the fact of how you know active inference works and you know how our human knowledge grows you know I think we're going to see a similar pattern with this knowledge among these agents and the only difference is and and is just me speaking from you know my own you know my own uh perception and and perspective of this but you know they're going they're going to be dealing with real-time data and with real uh a real grounding of this programmed these programmed spaces and those are the things they're going to be operating off of to make their decisions right whereas if you look at a human you know we have so many other VAR going on and every when we learn through active inference it's a very subjective uh growth that's why you can have you know multiple children growing up in the same household and they all experience things differently right but when you have these agents and they're getting this they're getting their learning through actual tangible programmed information I think there's a stability there as well if that makes sense and and and and again that's just my own uh you know interpretation and feeling around this and I could be I could be wrong but you know just kind of throwing that out there that's how we move forward yeah okay next question from David Highland who wrote thank you for the informative presentation Denise can you share any thoughts or information about the ownership model for agents is it a decentralized market how we decide which parts are public goods yeah so with the protocol with a with hyperspace transaction protocol um because so it's interesting it really shifts to this uh kind of self-sovereign identity uh aspect of the internet because right now think about how web domains work you know if you want to interact with a web domain all the transactions are taking place under the umbrella of whatever centralized organization owns domain and you have two choices you can either say yes you can have all my data and you can decide what to do with it all or I don't want you to but then I'm opting out of being able to engage with your domain well in the spatial web because now transactions can be at every touch point of every entity within the network then permissions can be programmed in a very nuanced way right so it really puts that back in the hands of the of the user of of the entity within it you can you can program you can program guard rails around certain aspects of data while allowing permissions and Freer like Freer accessibility for other aspects of data so whether you're talking about a business you know they can guardrail off their data they can guardrail off certain aspects of it for certain Departments of employees some for their customer base some for you know um Partnerships whatever right um and we're going to be able to do that kind of thing too and the thing with this is this takes into account time the passing of time so you can set expirations on permissions as well so I think we're going to have a lot more self- soverign control in this in this area yeah that's a part that really excites and Peaks me as well like with the web domain or the email domain it'd be like saying well on one hand we're not going to look in your email let's just go with that for now but we do or don't accept letters from this area so it's kind of all or nothing and it's based upon like you pointed out the the custodianship of the all or none permission based around what boxes you can send where whereas having a more nuanced way to talk directly about belief sharing means that instead of just like plugging the USB into the port there could be even a active inference negotiation around what elements to share how to share when to expire all these other features which can can help like incrementally build rapport and and add to the relationship instead of doing all or none which can be just very prone to false positive or false negative in cyber physical interactions yeah you're so right Daniel yeah what do you think about learning trajectories and paths I know it's a interest of all of ours I guess like what what can we do to be learning and thinking about all these things well um so you know I I have I actually have put together a series of courses um to kind of give people a a real um kind of a prerequisite knowledge understanding of what this transition looks like and the different components and what it means from all aspects of like you know basic understanding of of active inference and uh you know the free energy principle and the spatial web Technologies to you know decentralized AI what does that mean convergence of all this Tech across the network what does that look like safe AI had do active inference benefit that and then rgms let's let's look a little closer at at that and what I try to do is I try to take the complex ideas but make them more understandable so that people can really wrap their heads around the these Concepts and then they can go dive into the more technical aspects of it and I know you with the Institute you have a lot of technical knowledge there so I think that you know between resources like you know some of my resources your resources and then I'm sure there are resources out there that maybe I'm not even aware of and you know I think there's going to be more and more resources available around this modeling these agents you know uh building out you know these these spaces uh within the network um I know that versus is going to be launching a platform that'll be the first interface for the public around this technology and I think it's coming to the public at the end of first quarter next year and with that there will be tools where people will be able to build these agents as applications within the network and and you know I think at that point it's going to get really exciting um so yeah I I just think over the next six months we're GNA see a lot coming about with this and if anybody is interested in the courses I've put together you know you can you can visit my uh my website and and sign up for them uh they're not available yet but they'll be available very shortly uh within the next week or two so yeah cool okay I'll I'll for last question ask the same thing that I asked uh Alex and Carl earlier what is a fun or a memorable moment from learning and applying active inference this year and then if you haven't already mentioned it what's something that you're excited for next year um this year honestly I I think that uh I have been on this Learning Journey in creating this course material around it you know and at the same time I've been building community and kind of having them be my test guinea pigs around it and stuff so I you know there's been this whole learning aspect for me around building this educational material so it it has been you know a constant uh feedback loop and next year I'm just very excited about the potential of um building these agents and what that means and and how we can tackle some of the the problems and the the issues and the challenges that we have um through this new exciting technology and this this uh framework so yeah well thank you for all of your great learnings and digested and and sharings and and all the ways that that what you talked about today I think connected a lot of the dots from hearing Carl's more technical view on the rgm on through what what does it mean to have multi-agent decentralized as opposed to just building up one agent so thank you again Denise thank you Daniel thanks for having me thanks everybody till next bye e okay the next talk is going to be by Bradley Alisia and colleagues purposefully non-standard cybernetics an alternative to purpose-driven control this is going to be a 1hour pre-recorded talk so thank you enjoy hello my name is Dr Bradley Elisa and on behalf of Jesse parent Morgan huff and Amanda Nelson I'm we're from the orthogon research and education laboratory and the cybernetics interest group and I'm going to talk to you today about per purposefully non-standard cybernetics an alternative to purpose-driven control so we start with a question what is the definition of behavior and this is varied throughout history according to the behaviorist Behavior was identified as a reflex elicited by stimulus associations the consequence of reinforcement and the current motivational state if we go forward to the cognitivists which is more or less the modern view they would say that behavior is a set of internal mental states that are responsible for observable behaviors with gold directed end States now it's of note that both behaviorist and cognitivist Views involve feedback this leads to learning a notion of what constitutes a state and predictive capacity so we start with this paper Rosen blue th all in 1943 Behavior purpose andology published in the philosophy of science and it's a formative paper in the field of cybernetics so in that paper they have this typology of Behavioral classifications so we're going to return to this typology throughout the talk and we're going to think about how we can reorganize this typology but it basically defines Behavior as a set of nested things the red text Non classes or they're basically all other things other than the uh black thing that we'll be talking about at each rung of this typology so behavior is active it's purposeful it involves feedback which implies some sort of kology or end goal it's also predictive which involves extrapolation and involves order of prediction so you might have first order second order or third order prediction so to understand what teeology is and where it comes from we start at the Middle Ages where thology was the end State or the end goal of some behavioral process was thought of as Divine causation we then move forward to teeology As Natural causation and then it branches off into two directions and the first was the vitalism of bergsen and evolution by natural selection made famous by Darwin leading from Darwin's work we had erar who talked about tonomy which is purpose versus history and we'll talk a little bit about that later in the talk kind of discontinous to this are two different streams so the first at the top is where behavior is viewed as habit that's by William James and Thorndike that leads us to the an molecular anti-y of Francis Crick and feedback is complexity which is summarized by control theory and dynamical systems at the bottom is our target paper that we just has talked about these are the views of Rosen Blu and Weiner and this is feedback relative to a goal this led us to cognitive science and reinforcement learning or at least the modern versions of this where people talk about gold directed behavior and variations on that in molecular biology such as molecular vitalism and the tame and U methodologies of leban janca and Ginsburg so we'll talk about a couple papers here and frame some of this in context so if we think about Behavior it's it's essentially a form of functional in the first paper here Rody and Germain they argue that functionalism is a pre-scientific relic so they actually argue that the whole idea of functionalism is something that comes from this era of divine causality biology is not designed so why should we expect to function and this is something that of course maybe is assumed about function that it comes from some sort of design or comes from some sort of divine intervention on the other hand toonomic systems are defined by spatial and temporal information and constant iterations Over States so we do have these toomic systems but include modern components only select behaviors can be defined as selection of a goal or derived from signals from the goal so that means that we can't necessarily paint Behavior with a broad brush of theology only some behaviors may be theological and others may be not not all behaviors and this includes biological and machine behaviors are goal selecting and while goal directed behaviors seem to be purposeful and and it may be true that some of these behaviors that are known as gold directed are purposeful are there Developmental and evolutionary antecedants so are those Place those uh ancestral behaviors or those basil behaviors or behaviors that occur earlier in development are those also purposeful and the answer may be no so Rosen Luth at all Define behavior as any change in an agent with respect to its surroundings this is much broader than the Neuroscience psychological definition and it sets up a contrast between all behaviors and a subset of purposeful behaviors so if a behavior is put under some sort of selection or if a behavior is selection of something in the environment this might lead us to a subset of voluntary behaviors which then some of those may be purposeful and so purpose is actually purposeful behaviors is actually a subset of all behaviors to demonstrate this we see that clocks both mechanic and biological are not purposeful as there is no final aspirational state mechanical clocks can keep time and you can set an alarm on them but that alarm is imposed by the outside and it's not a property of that behavior and of itself furthermore biological clock runs via physiological processes like a circadian rhythm and it can be entrained by external white but that again is not a final aspirational State more generally teeology requires a signal from a goal so it's some sort of signal from a goal that's specific to that behavior and purposes progress to that goal so again with our uh example of circadian rhythms it's entrained by light but that light itself is not a goal it's just something that's out in the environment and the system entrains itself to it we can also see this with the brain in that brains produced behavioral outputs but the exact mechanism is hard to pinpoint and this is the rust and lell article that we saw earlier and this asks the question signal from what to goal so there's a signal coming from something to a goal but it's not clear what that is this is something one of our co-authors Jesse parent works on this is a frontier map for teeology and related complexity Theory Concepts so the data from this is drawn from the complex world book by by CAU this is published in 2024 and it kind of goes over three different concepts that are interrelated so we have teeology organized complexity and emergence and so you can see that our paper Rosen blue 1943 is at the top and this is sort of the first paper that talks about teeology in sort of a scientific and in a manner of complexity and so this is the first paper that talks about teeology is a property of complexity and of complex systems so a lot of papers were influenced by Rosen Bluth this uh theology was inherited by Weiner's book cybernetics and control in 1948 two other papers adaptive systems in uh by John Holland in 1962 and organizations in complexity also in 1962 the good regulator of course in 1970 inherited this concept but it also inherited another concept from organizations and complexity which is organized complexity the good regulator then passes down this concept of thology to the book order and chaos in 1993 and there's a direct link between Rosen Bluth 1943 and the tame concept of Mike 11 in 20122 now we going back to the concept of organized complexity this comes from organizations and complex the good regulator in 1970 and the autopolis paper in 1974 both share this intellectual lineage the autop polies paper lends that organized complexity concept to this paper UND deiv in 1982 to this book order and chaos in 1993 and then to the tame Concept in 2022 finally we have emergence which starts with the paper by Anderson war is different in 1972 inherited by the autop polies paper in 1974 in turn inherited by the dissipativity paper 1982 in turn inherited by Order and Chaos 1993 so you can see that we have not just theology as a concept in which we've kind of talked about already but also a number of other related Concepts this is leevan uh this is the tame paper and then janca and Ginsburg this is the U paper and so Lan proposes the tame or technological approach to mind everyone Levan's principle here is it's cognition all the way down we can see cognition in terms of Minds we can see cognition in terms of molecular phenomena we can see cognition everywhere in between and so behaviors all those behaviors are cognitive to some extent and as a result we get these diverse Minds that have these different goal directed competencies that allow those systems to achieve embodied agency so in Levan's approach you know chology is a very important concept not just for cognitive type behaviors but behaviors that go all the way down to the molecular scale and maybe even down to the quantum scale second theology is a means to explain why agents develop towards goals and so you know we can look at uh agents such as artificial agents simple organisms like bacteria and we can see that like tileology is a way to explain learn how to move towards goals the next paper is this U paper by janca at all and in that paper they propose a concept called unlimited associative learning and this concept is a minimal marker for Consciousness so in this paper they also Champion teeology is an intrinsic reinforcement system you have this internally regulated model and service of attaining external goal andology plays a role in attaining goals so again in both cases there's a strong tendency to talk about goals a strong tendency towards too but we might ask the question is behavior at these different scales and any sort of behavior that's sort of motivated towards some sort of function telogical and purposeful or is it simply a product of the history of the system so you know we could talk about heliology and purpose and we've kind of set this up that you know this is something that has kind of come out of the natural study of behavior but sometimes you know we don't really fully you know sometimes we ascribe Theology and purpose to things that maybe are just simply a product of the history of a system first example I want to give is the basav jabotinsky reaction or the bz reaction and at the lower left you can see a image of the bz reaction and this is a deterministic internal model that we can model with differential equations and we can solve those differential equations and get interesting solutions that produces a cave chaotic behavioral output or the Dynamics of the system which are sort of the unfolding history of the system so if we take an example of some some sort of chemical soup we can achieve bellof jabotinsky reactions or bz reactions by setting up this sort of set of differential equations these ordinary differential equations setting up an initial condition and we get these spiral waves and score wings that result and so if we have slight variations in the initial condition we end up with very different systems very different types of patterns in these um in these systems and we can descri this with ODS and one might say okay there's a purpose to this but the purpose isn't necessarily clear when you run this simulation again and again with different types of initial conditions there's a chaotic Behavior at a variational dynamic dynamical output so this is an example of the tip trajectories of Bez waves this is from a paper I brought an angle in 1993 and it shows how there's a lot of variation in the pattern but there's a sort of um you know constant aspect of the pattern in terms of the tip trajectories so one might say okay the purpose is to produce these tip D trajectories instead of the actual pattern and the interactions which may be true but then of course you know that's something that I want you to think about as we go through the rest of talk we can also plug the bellof Jaa tinsky reaction into was cybernetic model with higher level feedbacks and so I talk about this in this dimensions of morphogenesis lecture on the D.A Worm YouTube channel and what I'm getting at in that talk is that there are different dimensions of morphogenesis there's onedimensional morphogenesis two-dimensional morphogenesis par dimensional morphogenesis and so you see this kind of output is and so I'm asking the question here is this purposeful or is this a product of sort of a chemical a dynamical chemical system with a history and so we can also look at this as a cybernetic model where we have feedbacks and we have other sorts of complexity and you know we can fit it also into that behavioral uh typology of Rosen lth okay another uh another example is this thing called an Al estasis machine so an Al estasis machine is is basically the output of an internal model and so an internal model of course as we talked about is something that has some sort of capacity for memory and learning and maybe some sort of output that looks Behavioral or cognitive in some way so what we can do is we can take this internal model we can have an oscillator which is a stochastic process or a deterministic process depending on our oscillator and we can have a memory with capacity so we have this oscillator that produces an output and we have sensory information that can drive the oscillator forward this output creates a trajectory which is an interaction with between noisy perturbations and a historical trajectory that's shown in this red line and the historical trajectory is stored in the internal moduel as a memory with a capacity so as this red line unfolds there's a memory of it in the internal model and it responds to external perturbations which drive this uh Red Line in different directions and the oscillator tries to recover to the original state and so you get these dynamics that may look purposeful but really have this stochastic and adaptive component we get more into this aspect of dynamical control Control Systems I look at intrinsic motivation in dynamical control systems so this is this tiamin paper in PRX life and so in this paper they talk about how you might model maybe something like an alist stasis machine as a set of actions generated without a specific reward signal so in this paper they talk about that as intrinsic motivation they model this with except States and as oft actions so an action maps to a number of states Ates and this results in a trajectory forced bya observation stochasticity this is entropy so you produce the trajectory there's observation that you have and that's the entropy of the system and so you maximize actions that increase susceptibility rather than entropy these are noisy observations that wead do in perfect observational States and you can compare arbitrary length sequences of such sequences in the future this gives your internal model the ability to sort of integrate information and to find sort of solutions to problems the idea is that you have the system that can work towards a goal but it isn't driven by any sort of teeology it's just driven by this memory component and the stochastic component and that's all you need a number of other papers here that I'm going to talk about I'm going to put this in the framework of anticipatory systems from our paper on tonomy and Benny and friston so to get to this we need to re kind of reformulate our behavioral typology so remember we talked about behaviors having these different comp components we have these non classes so in this case we're going from Behavior to active Behavior to purposeful Behavior but that's contrasted with historical or maravian behavior then we have feedback prediction and Order of prediction and so er Mar refined the Rosen blue F all theological category to two parts purposeful agentic and historical maravian so if we go back to the Benny and friston paper they talk about the free energy principle is managing different forms of of goal directedness and how you need negative feedback to attain homeostasis there's purpose in in cases where the goal is known which is a marvian system and surprisal is minimized which resultes uncertainty then we can expect purpose purpose isn't really just some sort of magical thing or it's not some sort of thing that requires a strict goal it's just certain conditions under which our behavioral trajectory exists we can also look at anticipatory systems it's the Louie paper and this anticipatory systems come from a generative model with depth so this is where we have temporal aspects of the system that lead us to teleology so it's more that the teeology is the history of the system and this depth comes from an internal model so basically if we go back to the internal model we have some process that generates Behavior we have this memory the memory is basically something that stores everything that the system is experienced and that's how we get toyology to kind of understand this a little bit more we want to distinguish between a marvian system and a non-markovian system so a Marian system is where you have in arrival times so there's a series of things coming in in time this is exponentially distributed in this diagram we have sensory information coming in we have some sort of encoding this encoding as a binary Channel with noise where we have these lines that represent some sort of sensory input or perturbation of the system and inal arrival time is the area between these lines so the inal arrival time is obvious viously not equidistant in this example but it's in between the red and blue lines or the red and red lines and so we can take those in our arrival times and if they're exponentially distributed we have a Marian system if they're not then we have a non-markovian system and so we've kind of developed this uh distinction between marvian and non-markovian sensory information and how that relates to an information measure in our gibsonian information work so gibsonian information is a markovian but can also be non-markovian information processing system and the citation is here so going back to that behavioral typology we have active purposeful feedback prediction and Order of prediction feedback is everything that is circular non-transitive this is the present past and future your behavior is controlled by a margin of error Corrections of agent location the time te relative to a goal position or state and so if we think about this in terms of a feed forward process feed forward process is where the past is your input present is your process and the future is your output in feedback however it's a little bit different your input comes in the past and then you have this present State here and then you have feedback which is the past and that Loops back to your input which means that your input is now the future and so the future is actually present with past feedback so you see the loop here you have input to the present you have this feedback which is the past and it goes to the Future so now you have this future State before the present State and if that keeps going out you get this all output with alternative Futures so your feedback loop can be positive or negative and in this case it doesn't matter we're interested in is sort of the order of past present and future so we end up with this input we go to the present we go to the past we go to the future we go back to the present so we're sort of interdigitating different types of causality here instead of just having this past present and future aspect to things we have this sort of convoluted aspect of past present and future if we go back to our historical map we recall that non- chological things can be characterized as feedback as complexity and so there are two papers here that talk about pit control that kind of get us at that idea so PID control is proportional integral derivative control and if we think back to our example of chemotaxis and phototaxis we find that PID control can be applied to chemotaxis and other types of biological control so this is The Balter and Bucky paper they talk about this in terms of variational free energy minimization given a linear generative model they talk about pit control resembling a reflex Arctic mechanism so this is The Reflex Arc we find in sensory motor control and the free energy principle provides a more direct account of control this is where we don't have any hidden states in the process of propioception so they're focusing on propioception AB but they also talk about chemotaxis and so they kind of give us this sort of uh variational free energy interpretation of this um in go all which is a more engineering take on this the pi and PD components or the proportional integral and the proportional derivative components and directed Network synchronization are all sort of uh components of this control mechanism in particular the pi and PD components of PID Define present past and future components in an internal model control system we can then also start to think of feedback is complexity in terms nonlinearity and complexity as well so this paper by Westerberg at all stimulus history not expectation drives sensory prediction errors in a melean cortex this is from the bio archive so the author has framed this in terms of predictive coding so predictive coding explains cortical responses in two ways a feed forward model where an internal model of expected offense occurs or in terms of an internal model of expected events or a feed forward approach and a calculation of prediction ER for unexpected stimuli or feedback approach the latter are actually Global oddballs which defy expectations in a local context so in Mouse and macak which are their two model organisms they see two types of signal they see a normal response where the internal model is matched by no stimuli and a Global odbo Response which is Emer which emerges as a higher order response to this normal response we also can look at this nonlinearity in terms of Behavioral output in general and then burstiness or nonnormality of so this paper burstiness of human physical activities and their characterization and quantifying postural sway Dynamics using burstiness and interent time distributions kind of gets at both this aspect of maravian systems and a behavioral system that exhibits sort of nonlinear complexity both papers focus on the burstiness of postural sway in postural sway we have this paradigm where someone stands on a plate and that plate is a forest plate that measures or the center mass of the individual and then we can measure this and we get this nice dynamical trajectory that has different features that we can analyze so it's a behavioral output that exhibits a lot of nonlinear characteristics and dynamical characteristics teuchi andano focuses on burstiness and postural sway in terms of adult housework and children's play behaviors they studied physical activities of children twage ages 2 and five and they found that burstiness is not an acquired feature it's something that's innate burstiness is present in healthy healthy individuals not in people with movement disorders or impairments we also looked at inter event time distributions and Associated fluctuations so we talked about interner event times as being key to Marian and non-markovian systems they found that they're bursty so they're the stochastic signals that have a noise component and sometimes this is a function of someone trying to maintain balance and sometimes it's just noise that's generated by the nervous system that's added to the stochastic signal in both cases this is not go directed it's generated by physiological system it's a very complex behavioral signal and it's not explainable by either a pan or gausian model which means that it has this very complex set of features so if we do some experimental cybernetics on this we can find that if we have the right internal model we can produce very interesting output signals or behavioral signals some of these T kind of look like they could be chological but they're not this is just simply running some stochastic one example is just a pure stochastic signal the other one is having a stochastic signal with feedback that then takes that signal and integrates it at a later date so what we have here are two Fe forward processes one is a spway stochastic signal in Blue on the left and then the other is a some signal that a sto a stochastic delay of that signal the signal on the right so on the right we have have the blue signal you can kind of see it under the red signal so the red signal is the one that's bursty and it's just basically where at random intervals we sum previous output and we add it to the present output so we end up with these bursts and so the delay which is this uh sum signal that we uh that represent our bursts is the buildup of the input signal it produces these bursts or what we might call avalanches and so we can see that we can produce these nonlinear signals from a simple internal model so this is the classic coopel paper from 1982 it actually has a couple of interesting things to say of cybernetic oriented things so we have this feed forward which is where you have a process that goes from A to B to C to D and we also have these systems that have strong back culing so we have a process that goes from A to B to C to D then we have these back couplings from D to C C to b b to and so what's more likely to be the case of like you know actual sort of complex behavior is not the feed forward signal but the strong back coupling and this is where we have ubiquitous feedback strong back coupling can be intractable to analysis but produces the most interesting aspect of network interactions and strong back coupling produces things like Collective Behavior emergent properties stochastic outputs and nonlinear outputs are Avalanche which we saw in our experimental cybernetics a few slides back we can think about feedback and we talked about this relationship between past present and future a few slides back and we also talked about maybe some of these types of behaviors that are non-random but maybe structured in a certain way and we can think about feedback just sort of revisit the idea of feedback that Rosen blue fed all gave us which is that feedback was purposeful so we can ask the question is feedback purposeful or simply non-random and so we can think of past present and future relationships as a trip around the feedback Bove namely that we can take our input we can take the present and have a feedback loop that gives us past information into a new present when we get a new time point and that ends up giving us a future that can then be recycled so this has implications for prediction we can go past present future to prediction or we can have inputs that produce these sort of differently ordered past present future Futures and use that as the output so feedback mechanisms reorder processes systems with a lot of feedback aren't teeology per se maybe they produce outputs that look to but they're not to logical and so going back to the different orders of prediction which were kind of like the final output of our behavioral um typ typology what are the different orders of prediction well we have our first second third and fourth or higher order feedback so we can have the these feedbacks that have PR past present future but we can Nest them together so that we can have a past of a past of a past contributing to the present or to the Future and so that's really interesting because what it's suggesting is these higher level feedbacks maybe are also not te logical but there's some sort of memory or maybe there's some sort of um you know inter prent interval and the way it's structured so there's some structural aspect of feedback that might be really interesting that isn't chological per se but it given some of these other examples might have some structural significance so if we build a a cybernatic model that is really kind of complex and has a lot of interlocked feedbacks and things like that we can get some really interesting results so our different types of higher order feedback you know we can kind think of these in terms of the derivatives of position so the derivatives of position revisiting that from undergraduate physics you have velocity acceleration jerk snap crackle pop and these are just where you start with velocity and you get a a derivative of velocity which is acceleration you get a derivative of acceleration which is jerk and so on and these can be thought of in this way of having these higher order feedbacks or these reordering is up past present future to give you these really complex Dynamics in cases of derivatives of position and feedback feedback if it goes farther into the past the effects persist farther into the future so if we go back a couple of feedback loops nested on top of one another and we introduced that to the new present or the new future those effects persist further into the future so we get these really longterm effects in a dynamical system instead of just being local how does this translate into the neuroc correlat of behavior well we go back to our idea about Behavior sort of being naturally te logical and we have to think about like kind of how we kind of assume that behavior does magic things so if we get a behavior like this with higher order feedbacks it tends to resemble the sort of magical aspect it could be something like emergence it could be something thing like burstiness or something like a set of avalanches that occur and so these are things that you know you know give us really interesting Dynamics but are no more sort of magical orological than any other kind of behavior and so more fully connect systems are perspective or might be sentient or conscious that's sort of the idea of does magic things so you know we don't want to us misattribute sentients or conscious to something that isn't and of course this is something that we've have a problem with with modern AI systems that people think they're STI Ander conscious when in fact they just have this higher order feedback and this sort of um you know behavior that isn't necessarily what people think it is magic things or you know what we talk about like complexity results from embodied function a good example of this is kinematic chains which we'll talk about in a little bit and now we introduce this concept of convolutional Cy cybernetic architectures or ccas and so we go back to our behavioral typology and we ask the question can we better understand the role of purpose or meaning and feedback or temporal order so here we have active Behavior purposeful Behavior theological Behavior and the non-classes of those are nonpurposeful or random behavior and non-feedback or non- theological Behavior perspectively but we're going to reorder this behavioral classification where now we have non- prediction leading to the order of complexity rather than prediction leading to the order of prediction so we start off with behavior that's active instead of passive then we have nonpurposeful or random Behavior instead of purposeful and non nonactive or passive and purposeful now non- classes then we have feedback which is non-teleological versus feedback which is teic theological feedback is now the non-class non- theological feedback is now the past to the order of complexity and then instead of prediction being extrapolation we'll have non- prediction which is also extrapolation and then we have our order of complexity first second and third order so the more feedback we have the more complexity we have so this is an example of a convolutional cybernetic architecture we basically have this toy example where we have an input and an output the input is 00 Z the output is 11 one and we have these boxes which process the present to the Future and then our feedbacks are give us like some past State and it feeds it back to the future or the present so we have a ubiquitous feedbacks here we have some feedforward elements we have elements of feedback that do certain things to the sequence and the question we ask here is what is the maximal number of steps needed to transform the sequence from 0 00 0 to 111 so we start with 0 z z we put that in we kind of move to our feedback loop our feedback loop does certain things to the sequence and then we end up with 111 and we actually want to know what the maximal number of steps is because we want to know you know how far we can go through these different uh feedback loops and get our answer this can also be restated in terms of temporal phases past present and future so we also have more explicitly this input of the present we have the future as an output and then we have these different uh Loops that give us the P give us past information and again we're interested in getting the most diverse information possible and so this is again this maximal Loop that we're looking for so we're trying to get the most for iety out of these uh convolutional cybernetic architectures instead of trying to find the most efficient path or the tightest path or the most go directed path we can further understand this or restate this in terms of semantic components so we can talk about this in terms of symbols reference and understanding symbol being the input reference being the feedback or part of the feedback loop and then understanding being the output and so we want to cumulate as many reference as possible for a symbol so that we get an output of understanding and again it's the variety that counts not optimizing the process or trying to find a goal directed aspect to this so if we think about how these ccas are assembled uh we can look at the different components of feedback and then we can start to assemble these more complex structures so let's look at first order feedback so CCA has input and in this case we're looking at past present future and maybe alternative Futures and things like that and they all produce a predictive output so first order feedback would be an input going to a future going to the present and naturally the future is just like the present plus the past so the present comes in as input it's passed to this feedback which is the past the present and past are added to the Future and then the output is some sort of predictive input so again we have this reordering a past present future in the first order feedback if we have a feedforward only case we have input which leads the present which leads to some complex output but of course as a feed forward only system we just only get the benefit of having the present we don't get to talk about the past and the future it's just like every input results in a an output and that output then is complex output within any context then if we put together a larger set of feedback loops we can see this sort of convolved mixed feedback where we have first order feedback second order feedback and a feed forward component so these are all different components producing two different complex outputs so we have an input that gives us the present we get a first order feedback that gives us some past and we add those together to a future state that produces complex output we can also have a feed forward part of that where the input goes to the present and then the present is a part of that state and then we might have a a a second order feedback where we have an input which is the present it passes A first order feed Loop uh feedback loop to the past and another first order or second order feedback loop to the past and those can both be combined into the present which is then informing a future state which involves a longer memory which has more information and that gives us another complex output but we also have this path from the second order feedback up to an alternative future which then has passed out as a complex output so you see what we have here is we have an input that goes to the present we Sub sub reference the past we sub reference the past again and that gives us an alternative future so happens for every cycle so you end up with these three different types of complex output and given this topology we can end up with these different types of characterizations of the system remember again with this convolution what we're introducing is this variety which is actually a concept from uh the ever good regulator theorem and a lot of the work of w ashb so we're doing with producing the maximum amount of variet iy so we you know we need variety to control systems so this is actually overall the appearance of purposes replaced with an onsemble of positive and negative feedback elements with diffuse effects so remember that we need to have not only a single Clos Loop feedback but we need to have multiple feedbacks and maybe some open loop feedbacks in there so we can have uh maximize a variety so the imperatives of such a network are to gain processes where to maximize the number of processes and this leads us to maximizing variety and of course variety was a concept that was introduced by W Ashby in the ever good regulator theorem and he talked about how variety is necessary control but it also creates the conditions for adaptability modularity modifiability so we have systems that are both not purpose ful but they also have this sort of imperative for control but also for other types of conditions in a in a system that allow it to adapt be modular and be modifiable and so we can also add in polarity we can have different types of effects resulting from feedback as we talked about if we reorder past present and future that's all well and good but this also has relevant to different types of processes that result in systems so in in complex systems we have these different effects like Avalanches runaway feedbacks and dampen feedback these are nonlinearities and we saw an example of avalanches earlier in our experimental cybernetics and that can result from a number of feedbacks of different polarity and this can produce unpredictable and stability we can have things like runaway feedbacks which basically just the product of positive feedback and that can lead to mass instability and we can also have something like dampened feedback which is a negative overly negative feedback which sometimes makes things more controllable and so we can have these effects that result from these processes that have a polarity that lead to these different complex systems phenomena that then lead to things that are maybe look they look te logical or purposeful that they're actually just the products of complexity and so if we kind of go back and look at like how we can model things like connectionist learning and reinforcement learning with cybernetics we can actually take the equations of connectionism and reinforcement learning and put them into this sort of cybernetic system so this is where you know we want to take modern learning systems or artificial Learning Systems in understand them using cybernetic approaches and maybe apply them into these kind of ccas and so this this is congruent with hopefield and Sutton's alternative Viewpoint which is that the weights of a network are reinforced by the activity and selection of previous inputs so this is where we have we're looking at Q learning here and we're looking at connectionist learning and we're looking at sort of the way equations or and we're thinking about how you know we can get learning from these systems of equations and you know this is a dynamical system you know as as much as a learning system so we want to understand and of how this is how this works and so we start at a random set of weights and lack structur due to selection we started a random set of weights and lack of structure due to selection we proceed to a structured set of weights and optimal policies unwittingly we kind of move from a marvian system to a non-markovian deterministic system and so there's a tar question here which is how do we go from something that does not resemble purpose to something that is purposeful the answer is is that purpose is a post talk determination so we can model this as habits with weights so we have this input we have these outputs and we have this uh CC C that's that has polarity and we can have habits that are the serve as the inputs and the weights are polarity and we have non-teleological behavior in a directed graph so this directed graph kind of moves towards the end points the outputs and so the lesson here is that purpose is basically whatever we determine these outputs to be so when we observe a behavior a behavior is generated by these CCA graphs we interpret the output the way we' like to interpret it and as I said before we can have these different outputs that are that look purposeful or they look peely logical but that's just really a post talk determination then we move to wier's Alternative Viewpoint so Wier is more of a cybernetics guy he also has an alternative Viewpoint and his Viewpoint focuses on communication and information which have both a control layer and a semantic layer so again we have to go back to the past present and future formulation so instead of thinking about this in terms of polarity we want to think about this in terms of not just the control but the semantics as well and so we have this past present and future we have an input which is the past we have a future which is the present with past feedback we have the present so the present goes to the Past goes to the future but the future also moves to a present where there's a present with past and future components and then an output so if we run this in time if we run this as a dynamical system we end up with this the present being sort of you know it's not really additive but it's basically you're concatenating things onto the present with information from the past and you're informing a future trajectory and you're you can actually generate alternate futures of a system from this and so this there's a control layer a semantic layer and this gives us information about the system both in terms of contingencies and synergies so connectivity and ordering provides both contingencies which is that nodes depend on other nodes and synergies for nodes working together so in this model we can't knock out any one node and have the system work we can't knock out the the feedback and have this sort of Rich self- reference we can only have you know a feed forward model and we've seen that that's much less interesting we also can't knock out the future and we can't knock out the present we have to kind of continue with all of our nodes but also these nodes working together produce the synergestic um trajectory that gives us rich information about time it gives us this present it gives us a a future but it also gives us these intermediate states where we get past feedback and we get integration of those this stands in contrast to the imperative of purpose andology is cybernetic control and behavior purposeful or stochastic activity tempered by recursively recombined temporal ordering and so this is again one of the ccas and CCA is also a series of steps that add recursive functionality so sometimes if we think about this in terms of our internal model memory this is memory with sub referential power so it allows us to have a memory again where we can reference the past of a past of a past and it informs the future of the present and adding present into future and things like that this creates a hierarchical system and high degrees of intentional order so again we can add these things together where we can run these ccas and we can produce this memory with sub referential power and the output can look very organized and hierarchical but the thing that produced that output is not necessarily ordered and it's not necessarily uh there's no overarching goal or any other sort of information going into it so once again thinking about habit and purposeful Behavior we have this paper creatures of habit the Neuroscience of habit and purposeful Behavior this was published in biological Psychiatry so this is where they talk about animal behaviors being composed of Habitual and gold directed categories habitual is non- theological but gold directed is theological they provide this sort of breakdown where you have teic versus non-te olical behaviors and deterministic versus stochastic Behavior if we think about te logical and deterministic behaviors we get gold directed behaviors those are things reaching for a Target or moving towards a goal but of course we know that some of these types of goal-seeking behaviors are chemo like chemotaxis or phototaxis are stochastic and so if we have theological and stochasticity we might have something like chemotaxis or phototaxis we have something that's non- chological and deterministic we have habitual work so we have something that's habitual it's not moving towards a goal but it's deterministic something like uh scratching or like a reflex we have non-ic on stochastic behavior however we know that that looks something like a random walk so we can actually characterize these different behaviors in this way the problem with chemotaxis and I put a question mark on the table so chemotaxis for example which is a movement along chemical gradient we already established that that was not toic the go directedness in that case is due to the environment not the internal model and so we can look to gibsonian information or we can look towards sort of thinking about why it's not te olical and so but it doesn't also doesn't resemble a random walk so that provides us with a problem which is where does this fit into this typology so a clue to kind of where chemotaxis and phototaxis fits into that typology comes from this paper brain like neurodynamics for Behavioral control develop the reinforcement learning and this is from the bioarchive from this year cot at all test two scenarios supervised or extent and reinforcement or developmental learning so we have extant and developmental learning only reinforcement learning produces gold directed behavior and neural activity similar to biology it provides robust short-term behavioral adaptations to perturbations and it produces an emergent emergence of policies that are not purposeful in an of themselves but lead to things that look the part so this is reminiscent of our ccas it's also reminiscent of our or discussion of chemotaxis and phototaxis building a CCA from a small Motif is a form of purposelessness or purposelessness reinforcement learning build up you build up to function and sometimes useless structures and that's okay another example of nonpurposeful behavior that kind of fits into our CCA model comes from fetal substrate cycles and biochemistry and so this is the paper stochastic amplif amplification and signaling enzymatic feudal Cycles their noise induced by stability with oscillations this is from pnas so this is where we describe these feudal Cycles as these feedback loops where you have an input of energy there's a enzymatic reaction and you get nothing that comes out except entropy or heat so you basically don't have an out put your output is entropy it's heat and of course this is a problem because we expect there to be an output we expect this reaction to actually do something and in fact it's creating heat but it seems like that's a lot of work for just heat so we have a biochemical pathway which is inclusive feedback it does thermodynamic work for no reason heat is produced from an energetic input but we don't get any biochemical products and this seems like it's a waste of energy that has no purpose but it's actually important for regulation and other processes it produces complex higher order Dynamics or amplification effects throughout a greater biochemical system so if we go from convolution to order we can see that the evolution of cybernetic control is a causally recursive chain of contingencies and so here we look at the contingencies in a CCA we go from 1 to two to three and then that's a feed forward system and then we can go from 1 to two to three prime to four which it feeds back to two and gives us this output four prime or we can go from one to two to 3 to four or 1 to two to three prime to four to five to six to 7 which is an output so we have these different outputs from these different numbered boxes and we have these uh resulting processes so let's take three prime as an example three prime is dependent on two and one and once feedback is operational four and five so four and five actually do feedback to three prime which of course is a dependency so if you knock that dependency out you lose your entire functionality of that part of the network seven on the other hand which is another output at the top is dependent upon six 54 3 Prime 2 and one there's no feedback from delay so you can see the different types of order and how there's a lot of recursive chains that lead to contingencies so again if we think about our feudal cycle our feudal cycle is like something at the bottom our her or feedback is like something at the top of this diagram and our alternate outputs alternate outputs are at the right and they are of course maximizing for variety and so so if we think about the evolution of cybernetic control as a causal recursive chain of contingencies we can think about the kinematic chain which is where we have these dependencies so D is affected by the activities of ab and C we get this chain it's this is this robotic arm that extends out across two joints and the activity in D with kinematic is constrained by a b and c and so we can describe it in that way if we go back to these kinds of contingent gencies we know that seven is constrained by its intermediates and thus it is with the kinematic chain in in the kinematic chain we can do this in a linear fashion we can add on conditionals as convoluted cyber as the convoluted cybernetic Network evolves this restricts the system to a deterministic path behaviorally this leads to interesting consequences for a developing sensory motor Loop and a kinematic chain is basically sensory motor physics with contingency so here are examples of a kinematic chain in robotics and we can think about uh embodied robotics here where we have these control systems for these kinematic chains maybe the CCA where we have different types of morphologies and we can design them appropriately we can design and we can design them appropriately we can design behaviors that aren't purposeful aren't te olical but still give us behaviors that we can that look intelligent we can also think about purposeful behavior and fore cognition so our approach to purposeful Behavior can be applied to Fore cognition so we have these different types of cognition some of them being embodied some of them being embedded some of them being extended which means outside the brain and some of them being inactive if we put this in the framework of Rosen glue that all's behavioral classification does for 4E Force us to rethink that behavioral classification as we did in a number of instances in this talk now there's one Insight of 4E that's interesting to us which is that the morphology and environment of a system is more important than its internal model so we've been talking about internal models and how these ccas model the internal state of behavior and how that shows that in a lot of cases teeology in per purp are sort of elery but if morphology and environment are more important than the internal model then maybe we need to incorporate ccas uh or or recast our ccas in terms of morphology and environment as well as an internal state in fact our CCA do that to some extent but it's uh sort of a a way to put this behavioral classification on its head and so in the case of 4E purpose is not ascribed to a mind feedback is diffuse and complex prediction its highest orders are not theological they must be open-ended we talk about this in a paper from 2023 and Royal Society interface Focus so what is behavior and what are its properties so yeah we we're kind of revisiting what roseen Bluth talked about we're looking at a a number of examples from neuroscience and from psychology and from other areas animal behavior Robotics and we kind of break down kind of what we're looking at here so Behavior can be intentionality it can be go directed and supervised and so these are things like control systems p word cognition and of course some of those are go directed some of those are regulated complexity then we have Rich higher order complex Dynamics and those are things like Dy reactions which we talked about and embryogenesis which we didn't really talk about but also qualifies in that category so these are outputs of dynamical systems we also have connectivity and convolution which are things like connectionism avalanches and open-ended learning and then finally we have behaviors that are hardw but recursive so with our ccas we're modeling some hardw Behavior behaviors in a recursive fashion we also talked about this in terms of non-markovian systems and habitual behaviors we want to go beyond the standard of teologico and purposeful Behavior so a standard way of thinking about behavioral regulation is that inputs or deterministic and outputs or Collective output should be Clos into so if we expand the basic closed feedback model we get an interconnected network of retrocausal ordering so this is something that we're seeing in the CCA where we talk about the order of causality PA present future we have this basic Lo the feedback model but if we expand that out we get much richer retr clausal ordering which leads to of course to variety but it also leads to these interesting paradoxes and interesting effects of high feedback behavior is often open-ended and this can be characterized by embodi Dynamics unpredictable which can be characterized by chaotic Dynamics non normal which can be characterized by non-markovian behaviors and systems Avalanches of different types power law Dynamics and so forth and variational which means that it's characterized by multiple goals and redundancies and so this requires a reassessment of Rosen blue theal typology and has wider consequences for the practice of active inference thank you for your attention uh one of our co-authors Jesse parent uh produced a lengthy blog post on the Rosen blue th all paper talks about its enduring insights and its con uh its context in cybernetics and then we have two papers we talked about in the talk from our group alist stasis machines and a primer on gibsoni information so I thank you for your all right welcome back here with Alexi tolinski a friend and colleague and collab wrer on this paper temporal depth in a coherent self and in depersonalization theoretical model so go for it well uh I'd like to say hello to everyone and thank you for being with us after such a long and fruitful day at uh 6 pm. on on the east coast and uh this is I appreciate your interest and thank you Daniel thank you active inference Institute for this fantastic Symposium um so uh uh we'll be working talking today about this paper on dissociative States and it's a collective effort uh this papers in peer review right now uh early uh preprint was posted uh on the site and it was uh uh work by Michael Levan Chris Fields Lan D Costa Rachel Murphy Daniel who's here uh David pinkos and myself and um this is the link to the paper if you'd like to take a look at the early pprint uh perhaps some of you would like maybe to get a feel for what we're talking about and this is not the patients account this is a poem uh from uh Kurt V and his song Pretty pumpkin I woke up this morning didn't recognize the man in the mirror then I laughed and I said oh silly me that's just me then I proceeded to brush some stranger teeth but they were my teeth and I was weightless just quivering like some leaf in the window of a restroom um and this uh poem was cited by Paul moleno he's a specialist on did in England on the Psychiatry and Psychotherapy podcast you can also find patients account on YouTube I couldn't post the video here for copyright reasons um so dissociations are a spectrum and they range from mild to severe transient uh to Chronic uh they can be benign and pathological and certainly they're not a bright line category uh the causes vary for them as well uh we call it iology in the clinical World they can be caused by psychological trauma neurological conditions substances seizures anesthesia and uh different sleep phases there's a subset of conditions here we call dissociative disorders typically in the history of these patients we see prolonged elevation of stress accompanied by repeated traumatic experiences in circumstances where where they feel There Is No Escape typically childhood abuse and neglect and we call that complex trauma which has a different presentation than acute trauma and we'll touch upon why this specific pathway of uh iology uh is is important for dissociative disorders um well I'd like to say why uh this matters what's at stake what's the motivation for this paper and we have many authors and they have variable motivations but I'm a clinician I see patients all day and um uh for me the main reason for that is these conditions are hard to diagnose many patients go undiagnosed for a long time they're very hard to treat uh and we're sometimes content with partial results and the risk of self harm or suicide is higher for these patients and there are many many reasons for that uh we think that perhaps one of the reasons is that there is no Unified theory of dissociations we have kind of um neurobiological ideas the corticol limic inhibition hypothesis we have uh psychodynamic ideas we have uh attempts to moral dissociations from first principles by anona and other colleagues but uh these different attempts are not tied together and certainly not tied together with the patients uh subjective experiences so we are trying to make steps in that direction and and we don't claim that we've built a unified theory of dissociations but what we're doing here is we're focusing on one aspect of dissociations one focal point that is a collapse of the temporal depth in any kind of dissociation regardless of the cause by temporal depth we mean how far into the future the person can plan what we're hoping for is that this uh uh focused approach has the potential to uh allow us to to to um perhaps develop uh more diagnostic uh uh procedures and and to possibly um help increase efficacy in in therapy with full respect to how difficult complex and messy this work is um uh additional point of convergence is we explore multiple theoretical perspectives in the paper and they all Converge on the same um point and here is the main thesis um the collapse of the temporal depth is necessary and sufficient for the onset of a dissociative episode so we're going a little bit further than colleagues who already mentioned a relationship between temporal depth and associations Dean and and an unic and others and Carl friston were saying it's causal that there is no dissociation without the temporal depth collapse and the collapse of the temporal depth will reliably lead to dissociative episode though we are um uh saying that temporal depth collapse can be conceptualized as a common pathway for the onset of dissociative symptoms regardless of iology and finally we suggest that the focus on restoring the temporal depth can be an effective strategy in therapeutic interventions one of the theoretical perspectives we explored in the paper is Micha 11's te framework technological approach to mind everywhere which I think is very useful uh and unfortunately I haven't seen many uh attempts to uh use it in clinical applications this is the title and the link to the paper uh there's so much in Mikel Len's work that is useful uh his uh paper on cancer which is based on his experiments is also one of The Inspirations for our paper here but here's another inspiration if you dive into Mike 11's work you'll probably see uh what he claims based on experiments that all intelligences are collective intelligences however at our subjective level at the level of the conscious mind we don't feel that way uh we perceive unity all our percepts Sensations thoughts feelings come together in a cohesive and coherent ho a single conscious experience in the here and now and uh we will talk about this seeming uh Paradox uh if you wish in in our paper um some of the core tenants of the team framework not all of them but some key points uh commitment to graduality there is no bright lines uh Michael suggests that intelligence is built gradually and uh by accumulation of complexity material Independence uh our model is not based on neurons or central nervous system we will talk certainly about a subset you know I'm a clinician I work with human beings so we'll talk about brain-based models but generally the model applies to synthetic agents and cells that are not neurons commitment to empirical uh research as opposed to just exchange of opinions on the subject um Dynamics the C are not fixed they're malleable and plastic and C can be nested and over lapping cooperating and competing both laterally and across levels uh this is an illustration of one of the key Concepts in Leven framework um uh called a light con and what you see here depicted is various organisms in the middle of the slide you see a tick and a dog a human being possible AI compound intelligence such as an ant colony and you see the sort of nested and embedded structure on the picture on the right and L con has two uh Dimensions there's a spatial Dimension and and a on the horizontal plane and the vertical Dimension is temporal and tick for example is not particularly concerned uh with things happening a mile away uh it cannot recall far from the past and plan far into the future while human beings are capable of planning things until uh that happen after we die a living will and uh with the use of language and culture and books we can uh use things that happened before we were born so we're capable of huge light cones and I'm highlighting this word capable of it doesn't mean that in our wakeful hours and alert hours we are walking around with the significant light codes they contast contract and expand depending on various circumstances but we're certainly capable of them well ticks or not and the vertical dimension of a lone is exactly what we mentioned before that is the temporal depth there is of course a perspective on the same thing in FB and I like this very sucing phrase by Carl friston who said because consequences follow causes the generative model acquires a temporal depth and more formally uh some agents are only capable of modeling their immediate environment and live in the Here and Now While others are capable of planning actions into the future and the generative models capable of planning are referred to as temporally deep generative models so temporal depth under fap is the length of the temporal Horizon that is considered during planning uh we're moving on to describing the if you wish healthy self um and we do mention that it is a materially independent U structure where you know in principle we can talk about the S of synthetic agents and you know cell collectives uh uh s depend on an organism's ability to infer and here we will uh provide uh a viewpoint on the two clinical terms familiar to clinicians such as derealization and depersonalization so the organism's beliefs about the world are the inferences about the external environment and the Continuum of issues in this process we will call derealization while the organism's beliefs about its own internal milu its body and mind we will call the self and the Continuum of issues of this process we will call depersonalization the selves are hierarchical composite nested and with boundaries uh so we consider self to be a system consisting of hierarchically structured beliefs which effectively makes it a dynamic hierarchical generative model and taking at its entirety the self is informationally separated from the external environment by a mark of blanket or a near Mark of blanket by boundary would like to emphasize I apologize if this is Trivial to the audience of the active inference Institute that the boundary is not Material like skin it is informational uh through which the self and not self interact so to emphasize the self as we have defined it is a model it is not an organism the organism constructs and implements it so the self is quite literally the construction of the mind this is an illustration of that from early work by Michael Leven I think this was his doctoral dissertation materials and what you see here is um dark embryos that started from oite and then you have blastoderm you after cell division you have a collection of cells and at the top left if nothing is done then this blastoderm will develop into a single ducky you see this green shrimpy looking thing that's a single duck however if you introduce a scratch in this blastoderm and the wound heals the same exact cell collected May develop into twins or triplets you see the red yellow and green duckies now think about that for a moment what happened here um this Collective of cells computed what is me and what is not me and the hardware did not change but in one case it made a decision that this is all me and the rest of it is not me and another one it says there's three Mees so you can see that this is not Hardware um s are also as I mentioned hierarchical composite nested and they have boundaries so now we're moving on to components of the self uh that are each component is separated by their own boundary and Mark of blanket and because we're using the word hierarchical uh we will uh Define that there is a core self a core component if you'd like a visual metaphor that Mark SS uses we can think about an onion and there's different levels of the layers of the onion and there's a core uh in the middle we will talk about that as apply to humans and mammals and rodents so now we are in organic uh in organisms uh this term is from effective Neuroscience from Yak bip and Lucy Bin's book um and also you know I believe SS and teral used it uh so core self is non-reflexive non-reflexive and dominated by raw feelings uh and part of purely effective core form of Consciousness so the effective homeostatic mechanisms allow an animal to problem solve in novel environments and I'll give you an illustration from Mark s um to maybe just um bring it home uh imagine that I'm in the room full of carbon monoxide and I've never been in before uh so I don't really know what to do but I feel badly I feel badly in a very specific way there's air hunger Suffocation distress uh and let's say I'm moving around the room but if I move to the left and I feel a little bit better and that's information and if I move to the right and I feel worse that's also information perhaps there's a door on the left and there's a draft underneath the door and I feel better so this Compass this internal mechanism the homeostatic mechanism allowed me to find out a possible solution and um we have a whole collection of these in know to mention you know the actual hunger uh for food and thirst and social homat such as rage and fear and panic separation distress Etc so the collection of that is what pip and s's conceptual as the core self uh uh therefore the set of predictions in the core self that all the life sustaining affects will be at or near their settling points which means in emotional terms contentment all my needs are met I am a okay thank you very much right and that is the minimum of the variational free energy if there are deviations from it such as hunger or thirst or uh air hunger then that would be a prediction error accompanied by a negative feeling like in the example that I used above uh in mammals we're not putting the equal sign but we're describing the anatomical structures that are uh participating in in uh serving this function we're talking about the upper brain stem regions including the reticular activating system and the periductal gray they mediate the functioning of the core self and you can see a lot more details in Mark's book Hidden Spring a journey to the source of Consciousness so um oror self is thought to be present in animals without the cortex such as decorticated cats that are actually more effective and not less and the cases of human children born without a cortex hydron calic children they have emotions when you know uh you put a Brother on other girl's you know uh chest she smiles and she cries so the core self does not require a functional neocortex the higher levels of the brain structures in humans including the The Vortex mediate the higher levels of both Consciousness and the self and sometimes it's referred to as extended Consciousness which allows us to reflect on what's going on such as to say I feel anxious right now uh there is an asymmetry in this uh um model where the higher levels the peripheral levels of the self and and Consciousness cannot exist without the core um well the reverse is not true the core can't exist without the higher Levits one of the other illustrations uh of that which PM cites this is based on Fisher's work in 2016 is that a tiny lesion a 2 cubic millimeter lesion in a parabal nucleus will reliably and IR irreversibly reduce a coma the patient will never become conscious ever again while you will never have anything like that happen in the cortex you know this small lesion in the cortex will not result in catastrophic outcomes like that this is one of the reasons why neurosurgeons don't usually like to operate around the brain stem this is too consequential too risky so everything we said about Consciousness the core and extended is applied to self as well that peripheral self components cannot function without the operational core self and we will illustrate that with specific examples in humans later on in addition to this onof where the thing is completely operational or not we can talk about the influence and the core self has the ability to change the regime of function in other more peripheral self components by inducing phase transitions in them and the Brain mechanism for that is generalized arousal um when the higher levels of consciousness are present and functional then we have representations such as fruit and vegetables and you know in the core self we have raw effects such as I don't know separation distress or hunger but with the higher levels of consciousness presence we are capable of uh uh thinking things like I want an apple or the aect the drive is in Freudian language and in s's language is attached to the object and similarly we can have these constructions about the meta object such as I uh such as I look pale or I am a pessimist and I here is a mental object and therefore I is a metacognitive construct an abstraction and it only exists at the higher levels of the self such as autobiographical self it is not used or even needed in the core self to reiterate core self as a system has the capacity to detect uh the effective prediction errors and uh attempt to minimize the variational free energy through action without any eye we're moving on to peripheral self components and we'll describe only a few uh and the terminology here is different from what you may read in wonderful book by Neil Seth being you so the first one is autobiographic iCal self and that's a system dynamically representing our life's history if you for example remember your 10th birthday and the cake is on the table and Mom is there and your friends are there and the sun is shining and everyone's smiling and it's Tuesday that's an episode that's an episodic memory and a collection of them organized through time is our sort of Life narrative um there is a bordly self which is a system dynamically building inferences about our body and various representations and interceptive processing there's a social self which is a system representing inferences about how we're seen by others and how we present ourselves and act in the environment each peripheral self component such as bodily self for example has its own Mark of blanket and each component is embedded into the whole self and also contains subcomponents creating a nested architecture that you saw on the T light cone picture before about the unity that in humans at the higher levels of the self such as at the autobiographical self level uh uh uh these these levels create an impression of a unified experience and this is just the experience of the autobiographical self not the entire system the autobiographical self claims to itself and others to be the entire self well it is not so Anil Seth refers to these things as delusion we have a delusion of being a unified I don't know Alex right but you know what we're saying is that it is that system that is diluted um uh the autobiographical self subsystem I'm ambivalent about using the term diluted I think it it's not the optimal one because we use it in the clinical world and I prefer it uh called inference or belief that we're unified stated differently we can say that a stable belief that I'm whole in the uh I'm complete in the order biographical self um its generative model contributes to us feeling as a Mantic self therefore the subjectively perceived Unity the coherence of the self is an inference related to coherence another thing we can talk about is continuity and nearly as a taty stemming from the definition of Mark of blanket ourself remain the same persistent time while all the processes or Communications across its Mark of blanket remain functional and we can um offer you another viewpoint on the same exact phenomenon from if you more neurobiological perspective the psychological perspective if you recall maybe you're at the train station or you know anywhere in the city and you see someone and you think you recognize them but you're not sure you know this sensation of recognition of familiar uh so the familiarity novelty calculations is what we're talking about here and depersonalization is a breakdown in that something that is familiar myself suddenly abruptly becomes very novel and not recognized but actually these calculations neurobiologically are not uh bright line dichotomus categories they are probabilistic and they are Spectrum but we course grain them in the mind often into yes or no familiar or novel so that subjective impression of yes I know this person or not is just the co grading of the continuous variable representational capacity is important uh the Deep generative model implies that the agent is capable of planning into the future which in turn requires an ability to generate store and retrieve counterfactual data generate store and retrieve is memory so therefore when the representational capacity is completely or partially impaired for any reason the agent completely or partially loses an ability to infer leading to depersonalization and derealization if we talk about health and pathology briefly the two key terms here are voluntary and involuntary you know again all of US during our day we can either sit down to plan things you know for you know 10 year temporal Horizon which expands our temporal uh depth or when we uh meditate or uh rock climb you know then we can be more in the moment in the here and now but uh trauma doesn't ask permission I mean perpetrators are you know there's nothing voluntary about it and so that's an involuntary collapse of the temporal depth uh which usually is leading to pathological conditions this is a perspective from the nonlinear dynamical systems you know if again I am in uh you know adult healthy alert functioning and I may have small deviations from ime say I'm daydreaming or something like that and the collection of these points from which I reliably return to ime can be described as the attractor Basin and the whole thing in terms of attractor landscape is the set point attractor regime I and ime would be the variational free energy minimum and that is a fairly stable system perhaps in the autobiographical s right now in order to for that system to change some energy uh needs to be applied and with external interference of sufficient power this attractor landscape can change when that power is applied temporarily there's a phase transition the attractor landscape evolves into something else perhaps during that transition you have a chotic regime and if we are in the case of trauma then the functioning can be chaotic for a while but then it may settle down so to speak into one of the possible postraumatic presentations there may be multiple postmatic presentations and one of the possible options is depicted here this is not again Universal but you know it's one of the possible options what you see here is a multiple coexisting uh point attractors represented by local Minima of the variation free energy and they can correspond to Alters in theid uh and in this example you also see that they're surrounded by a global minimum of variation of free energy now also if you look at the uh area the region between the local Minima then in dynamical systems that's called a repeller and uh repellers are uh notoriously chaotic so uh that suggests that you may not have very good predictive ility of how you will transition from alra X to alra y or alra Z to say a few more words about this when we switch between the altars then we are in regime of itinerancy this is the term Carl Frist likes to use a lot and you know justifiably so also when we have these transitions from X Ultra X to Y that the time is disrupted you know in the idea you don't have this flowing continuity of time there's a series of disruptions effectively then your uh coherent large light cone is broken into uh pieces here now imagine that this patient with with this condition reached out for help and they want to get better but this uh attractor landscape already acquired some stability and so again we need some interference to destabilize the regime the difference here would be that in therapy in the process of healing the this uh destabilization of the pathological attract landscape is voluntary controlled and guided ideally by a licensed and competent mental health professional and uh that's that's the uh the difference from the onset of uh acute trauma so Psychotherapy can be one such intervention and it's the treatment of choice we have very limited advocacy with medications and now it's very uh uh uh kind of um fash able to talk about augmentation with psychedelic treatment or perhaps neuros stimulation it's important to say that just destabilizing that landscape doesn't guarantee anything at all so when people present this idea that you just go to like I don't know and have some iasa or anything ketamine infusion and you're better is a myth because you know the important part is that the patient settles back into health and just the destabilization of this landscape doesn't lead to that you still need uh therapy you still need you know to process the trauma not just to destabilize the pathological regime um and that therapeutic process when you destabilize the the pathological attractive landscape is another phase transition temporarily temporarily chaotic with the idea that eventually it leads you to a healthier functioning either a trative landscape corresponding to cohesive uh system or at least a little more coherent system with less Alters partial Improvement let's talk a little bit more about the phase transition so before things happen in in in health we have a nested and embedded uh self and that can be described in in FB as a hierarchy of coupled interconnected attractors Carl friston you know uh wrote that slower attractor can possibly control the phase transition of the fast attractor and in our system the core self is is feeling BAS it's effective and it's one of the components uh of uh effects of feelings is generalized arousal and we know that the CH changes in generalized arousal level can lead to the phase transitions of the entire neocortex from periodic to chaotic State and back so this is one of the possible uh kind of you know uh explanations in our model how this works the phase transitions um you know a few more uh um thoughts about it trauma is not a um guided transition it's an effect of storm it's a flood of cortisol that is abrupt and severe and that uh abrupt increaseing generalized arousal leading to the destabilization of a healthy attractor landscape in the autobiographical self but then when there's an onset of a dissociative syndrome after trauma then uh the regime settles down to a lower energy uh perhaps effective numbness that patients talk about and in a a Psychotherapy not immediately but when the is ready which can take a year or more you know when there's active trauma work then there is again an increase in in energy level with with the idea eventually to settle down to a healthier regime of functioning a few clinical observations starting from dissociative disorders which are conceptualized traditionally as uh defensive response to prolonged and inescapable trauma clinicians talk about uh and patients report this this qual of emotional flatness or deadness or numbing sometimes they call it freeze so in our model it corresponds to issues with Communications between the core self and the peripheral self components for example that autobiographical self in this regime operates as if it is uninformed by the vital emotional informational flows that originate in the core self one of the possible ways it can be achieved not the only way but one of them is the frontal prefrontal cordex inhibiting the lyic structures that's the cortical liic inhibition hypothesis and in FB terms that means lowering of precision associated with effective prediction error messages um we're moving on to also severe dissociations but temporary such as general anesthesia propal and uh uh uh it's a very profound change in the body that influences the Gap Junctions between the cells and recovery from anesthesia is a very interesting process Michael Lan is fascinated with that it's actually a little bit of a miracle that we come back to be the same person we have been before and some people don't uh you may wake up a pirate uh and uh these conditions are called postoperative delirium dissociative Amnesia post-operative cognitive dysfunction this is also an illustration itself self is computed in plastic and meable it's not a hard um you know Hardware um and perhaps we get oriented with the use of our memory systems including the episodic memories another cases a seizure where not only autobiographical self is impaired but also the social and the bodily self and one of the reasons for that is the necessary resources for the functioning of the self components are not operational in seizure uh moderate dissociations uh in neurological cases but permanent ones are for example the perhaps one of the most famous neurological patients hm who had a bilateral hypoc compus damage uh and uh in Britain there's Clyde wearing and an Neil Seth describes Clyde wearing in his book being you these patients do not have episodic memory encoding and effectively they leave in about 10 minutes of the here and now uh there is no past and no future beyond that and if you read Clyde wearing's diaries that he talks about the series of Awakenings he said oh I just woke up and 10 minutes later oh I woke up and then I woke up and if you show this D I red to a clinician who is an expert in did they will which I've done you know I talked to Roxy wolf one of the fellows of the international Society for the study of f associations and she said that you know this patient looks perpetually depersonalized there is no continuity in so Coro patients they don't recall the past they can fabulate the past and in Alzheimer's in early phases there's issue with recent autobiographical memories but some access to the remote autobiographical memories all of these cases are discontinuity in autobiographical self um uh more frequent perhaps cases or substances ketamin ey does of THC or iasa lead to depersonalization and dualization and importantly if you read the studies then these patients actually report they tell you that their sense of time is impaired not only that the planning is impaired as well um if you have a alcohol blackout and the person tells you on Wednesday that they have no idea whatsoever what happened on Tuesday after 6 then uh this is a discontinuity this is a gap in episodic memory uh there's also something called traumatic Amnesia this very sad cases of high school shootings where you know the person literally doesn't remember you know 15 minutes of what was happening in cafeteria what happens in acute PTSD is that you know uh hippocampus is reaching cortisol receptors and there's a flood of cortisol and the effectively the episodic memory of trauma the cohesive context contextual time stamp memory is not laid down in a stable fashion you have shards of glass you have phobic memory semantic memories procedural memories but not the episodic one which is also discontinuity in autobiographical self and that is involuntary uh and we talked about mild benign everyday uh dissociations uh next time you wait up you know maybe tomorrow the day after you know try to monitor how what's going on uh and you may catch that pH of what met Syer calls you know a non-ic self where things are just flowing through your mind but your eye is not fully out borous here uh to summarize the level of Consciousness regime of Consciousness pathological States and all the underlying resources necessary for the successful operation of each self component collectively influence the complex dynamics of the system um a neurobiological perspective briefly the historically most famous one is uh cortical limic inhibition hypothesis by Sierra and barus who postulated that uh prefrontal cortex is uh inhibiting the lyic structures including the amydala and the interior singulate cortex ACC the subsequent studies don't fully support that hypothesis but partially support it I like the work by Ruth lenius and her colleagues and guess what you know her uh revision her refinement of the cortical limic inhibition adds Dynamics transience context and itinerancy all the things were like uh uh they talk about two regimes even in patients with dissociative syndromes over modulation and under modulation again itinerancy in not one regime but multiple regimes over modulation is indeed hypoactive amigdala and hyperactive prefrontal cortex which is described above but also these patients less frequently but but you can see it are in a reverse regime where amydala is active and and PFC is less active uh uh over modulation dominates in dissociative uh conditions they refined the anatomical Network and talked about different sub regions of amydala the central medial amydala based lateral amydala dorsal leral periductal gray ventral leral periductal gray Thalamus and other Reg and this network of regions collectively Works to create specific postraumatic regimes they also described uh the regime where patient has access to painful memories usually uh is accompanied by under modulation very active state with amigdala active and uh when they don't have access to painful memories they're more in a dissociative State we can also briefly talk about what are the anatomical substrates and the neuros psychological substrates of the temporal depth collapse and again acute trauma hpoc comple damage and dementia are affect the episodic memory system uh if there's an insulin damage then you can expect the disruption in the bodily memories and perhaps in ketamine use you may have a dysfunction of prefrontal Cortex and frontoparietal network which renders the work in memory uh of the patient dysfunctional you can imagine other circumstances when you have PFC in front of paral network impaired that usually leads to a psychotic state where you know it's very hard to think you know we need work in memory to be able to think um about dissociative disorders again the point made at the very beginning that prolonged inescapable exposure to highly stressful environments can be seen as lasting stress beyond the agent's ability to manage which results in the breakdown of um tame light going into the uh components fragmentation decoherent in discontinuity when the landscape uh attractor and repell Landscape corresponding to a pathological condition takes root and in contrast to that if you look at the single uh episode of camine intoxication then you have transient working memory disruption leading to transient dissociation again to summarize lasting or temporary severe or mild dissociative experiences are accompanied by the collapse of the self's temporal depth and we are ready for a discussion which I very much look forward to thank you Lexi great presentation thank okay where should we begin what would be interesting or or given where the work has gotten over the last few weeks where what do you feel like is relevant well perhaps any colleagues have comments I know hus is with us and he's a esteemed neuros psychiatrist and he sees a lot of patients and perhaps neurological patients and he can comment and some of the things we've mentioned I I had a uh clinical meeting with with uh colleagues who do a lot of dissociative works and I mean I'm seeking feedback and critique and so whatever people have whatever thoughts people have yeah if there's anyone in live chat they can write a question a few pieces that stuck out differently in how you presented it uh in comparison from working on the paper was the narrative self's role in maintenance of The Narrative of unity and about what is the experiencer that or is that all it can do why does it do that why does the autobiographical self have that kind of a voice or a cognitive function yeah is audio working okay can can we hear each other I I think why is a tricky question you know it might also be tell logical question I would say that it's probably functional you know uh for us to have coherent selves uh and um I don't know why uh but um we are observing it so we have descriptive level we have observation that I feel like the same elexi and it's a little bit of a miracle that as we know that the cells in the organisms they are they die and they're born but I still feel like the the same person even so uh we change developmentally I'm not the same person as I have been at age5 or age 15 but I still maintain this story if you wish this believe that I am the same and um I also protect the internal milu from external threats you know of the same organism and if we look at other species say an oak tree that you know uh grows from an acorn it is still the same Oak it's the same organism uh so um I I perhaps in biology as well there's a need uh for the system inside the mark of blanket to maintain that saying whatever is inside the mark of blanket is the same me but it is a belief you know and uh um I don't know if I'm answering your question maybe I'm just free associating to it so is that a clinically valid method um well it it reminds me of mortal Computing and there's a lot of focus on how behavioral activities in a niche relate to the material stability and I think you really Drew carefully on Mike Levan all's work to talk about the analogous Dynamics on a landscape of a narrative self and that's just kind of that was like a trampoline but then you also complicated that with multiple selves like the social and the narrative and so then there are these multiple disassociative or like it's kind of they each have their own unity and yet they can be discordant and unified yes and so then is there any general framework for different possibilities open-endedly of different regimes these are great questions or even maybe challenges to develop it further but I agree with you if we look in the bodily self right I don't perceive my hand to be a separate entity I believe I I feel very to be part of the whole organism myself my body right and so in the bodily self there's also a level of unity I'm just saying that this is again a computation within that specific subsystem that my body is one whole self and we can see how that gets uh perhaps distorted in somatoparaphrenia or even rubber handed illusion where uh this hand is perceived as not you know so uh and so we see that the system is plastic and malleable and and prone to change but and when when no uh interference is happening it continues to be cohesive as one body and one mind you know this autobiographical I I I've noticed you Ed the term narrative uh self I think that's Anil Seth's term and I prefer autobiographical for the reason that uh rats they have episodic memories and they don't have our verbal abilities um uh and so they they also have episodic memories which are what we when the contextual memories right and I think that uh there's the same eye in all autobiographical memories and that's unity uh yeah yeah thanks yeah a few comments it's it's it it it probably is more accurate to say it's a narrative cognitive capacity that encompasses autobiographical or aloe biograph graphical type narratives um and I'm reminded of a NPR radio discussion from a few days ago that I listened about Memoir and about what Memoir was and could it only be written when the person did or didn't change a lot but what or who doesn't change in every single one of these ways so that what would that mean like what so that that that was kind of a funny connection and then and then um do you have any thoughts on that or have active comment no I'm just enjoying your comment I don't know if I have a a thought on that yeah you you've see many clinical Memoirs and presentations of self- understandings and all of this so there's just so many and it gets brought out so sparsely like and indirectly and it makes a lot of sense to the person who might be saying but that that could you could be too or not confident enough about that interesting you brought it up thank you for that Daniel I think that you know when we theorize saying Dynamic theories then we are usually think about different sub agencies of the mind like Freud for example in his you know structural Mo model he had the eat egoo and super egoo where I want pistachio ice cream is the E and pistachio ice cream is fat and calories that's the super ego response and stash for ES is $4 and you have three and a half is the ego uh but there's a whole bunch of them we can name a sort of a sub agency of an internal judge you know which is sort of super eish or a neutral Observer or uh an accountant who's running Itali or and all of this so we all have sort of a jury there of different kind of regimes of operation if you wish but uh the patient is often surprised because when they for example self- criticize and you talk about that in therapy and say do you notice the voice of your judge it's kind of you know uh not very fair I mean there's no misdemeanor everything's a felony and judge is present all the time like no it's not it's me I'm doing it you know I screwed up you know so again you know the default regime is is me myself and I that I am you know I am bad and when we we discuss this things and we open it up and we show the different kind of uh we show a theater with different voices or a jewelry with different representations it's usually kind of an Insight it's a surprise but that allows a bit more control um for the patient in the clinical work so okay I'll read a question in the live chat Pablo FM wrote thank you very much I find it super interesting I wonder why humans seek for drug experiences having been so decisive in human history coffee could be a great example tea tobacco alcohol have also been very popular among civilizations it's a great question I don't know if I have a good answer it's a huge question and there's a good paper you can find it on Google schar a pbet by me zelner Mark SS and other colleagues I think Yak pip was alive it's called why depression feels bad and what do addicts really want and it's based on effective neuroscience and what it postulates is that uh of course there are individuals who have multiple drugs that they are you know using all the time so there's an upper and the downer and other things but there are usually some innate kind of predispositions where people who are wound up in tents they sometimes crave alcohol because it's uh binding to the gabble receptors and it's sort of you know sedating us a little bit it makes you a little more less kind of wired up people who are very painfully lonely and uh uh detached and they crave human uh comfort and you know to feel together and loved and accepted they may if they discover opioids that would be a you know lock and key match that would be a dangerous combination um and uh perhaps when somebody's oriented to Performance and they're very bored all the time and they want to something you know to do better and more they may gravitate toward uh stimulants and so there's these things uh and there's some common factors among different agents uh such as you know I believe that multiple substances they have the dopaminergic effect and you know a lot of motivations are subjective in individuals I've had patients say that I'm smoking po to slow down time I've had patients tell me that you know PO is my oldest friend I've been doing it longer than any person that I've been with uh some people seek this state of numbness and they want to kind of not feel as acutely they want to sort of feel desensitized so these states of dissociation can be desired um and that's briefly I guess what I what I wanted to say about it so well thank you I'm reminded of a comment from ma in the live chat earlier this morning and she was kind of pointing to how one mode of how we've seen active inference applied in these presentations is looking like athology at the sequences of behavior but not really looking at the self-maintenance as a behavior like especially talking about digital agents or or the it's just about what do they perceive what do they do and it's taken as a granted or a separate question how they exist yet also there's this self maintenance self reconstruction character that doesn't even need to have anything to do with the athology its Dynamics can just be like morphologically playing out not necessarily doing some cognitive task out in the world in any particularly effective way yeah it's a very interesting thought and for some reason I'll allow myself the Liberty to reassociate again it reminded me of yal Harari who when interviewed he said you guys go to the gym You observe your diet and you take care of your body and he said I take care of my mind I meditate an hour a day and uh he goes to silent Retreat a month and a year and the interviewer challenged him I think he was from The Daily Show he said how do you manage that and harar said I got no kids you know so but in addition to that he's also in therapy so there's a uh that idea of really maintaining the mind and not just the body with behaviors we're all oriented to what do I do do do what do I think think think which are sort of verbal virtual behaviors but how do you feel as my great colleague rajie uh Warrior says we're human beings not human doings we have a very tough time being we're all about doing more that's true and so uh yeah and also Harari talks about informational hygiene where a lot of stuff on the social media is noise and trash and a lot of it is consumed and we have pollution but you know so he doesn't do you know he reads books and he thinks in Millennia but I don't know I mean I went sideways from your comment but it just that's what where my thought went so yeah it it seemed like a very relevance when people are discussing the the the unity or the articulation of mind and body which is kind of what this unified promise on the cover of the textbook suggests is what what would it be to have a unified model of brain mind behavior and so on then the the analogy or the connection with the physical and the cognitive is just it has to be taken as a starting point not like a hot take yeah yeah but what you just said reminded me when I read Michael lans's work and got more familiar with it I actually had regret I'm like why were we not taught that that was a bit of a revolution in my mind it was a big pivot because if you go through graduate program in psychology it's all about the unity you know I am the same you know cohesive and coherent and the fact that we have intelligence at the level of individual cells and then the cell Collective and then tissue and organ and all of this thing that I think is one monatic Alexi is a pyramid with Communications bottom up and top down and laterally it's is a revolution I think that you know I I wonder what models of psychology can be built with that and they don't exist yet it's all about one big block what happens to it I am depressed you know there's no collectives this is one big block right yeah it's a really important in I think that's the kind of frictions and and experiences we'll start to learn more about and see produce in different ways like just seeing how you connected the topics to specific quotes translated the ontology re reworded descriptions using kind of like a translation sometimes really just as simple as a lookup of words just authors using one word and being able to see an otim onology term equivalent or roughly related it's uh and and what what does that look like or I guess as your last comments what do you think after the paper after this now like what are the directions towards which people would say okay that's applying active inference in a clinical setting I my goal eventually I think is to get to Therapeutics and for example you know um actually I probably will not use that example it's it's a good material for the second paper that we'll follow up with uh but I I do think that you know the question is what do we do in therapy differently how do we understand how do we formulate cases differently with this information I I haven't heard it clearly said out loud in the did literature there's a lot of perspectives there's a lot of endless complexity and there's a lot of nuance but if we can highlight a central point and say there is a temporal depth collapse I think it allows us to do something with it in clinical work and in diagnostic work and that's my hope it may not materialize but this is what I'm hoping for so well even from a researchers ethological perspective it's kind of like that's the transferable and the interesting research question how could you be observing not on the inside and make some kind of inference or assessment that would matter clinically with no map territory pretense not that it's 100% reliability just taking a clear look at Diagnostics available rethinking it from dynamical systems perspective maybe different uh Biometrics maybe pupil it's like it's not it's just kind of that's the open question and then where it provides clinical utility that's where it matters yeah thank you perfectly said yeah I agree okay thank you Alexi see you thank you Daniel see you byebye for e all right that has concluded the live presentations section of the Symposium I'm now going to play the pre-recorded videos so one second to rearrange get it in position now we'll continue with a pre-recording videos hopefully all of them concatenated into the next few hours okay this talk is by Leonardo MOA Rodriguez Groove as the musical glue for bonding ladies and gentlemen hello welcome to today's talk on Groove as its musical glue for social bonding my name is Leonardo mul Rodriguez I'm a PhD student at Sim and this is a neuroscientific and crosscultural perspective Ive now I'm a bit self-conscious by the fact that the only audio you'll be hearing today is my voice no music but do books sing No we're doing a deep dive into the scientific literature not musical theater maybe musical theater in the future of uh my active inference well my my name is Leonardo Müller Rodriguez I'm a PhD student at the Cambridge Institute for music therapy Research Center and my focus is on sleep music four dimensions and if you're curious how does this tie in with danceability where relates to musical entrainment U more specifically slow beats I'm also a composer music producer producer and musician so how does groove Foster social bonding there's a little outline of today's presentation Groove and its relationship to Spotify stability the Neuroscience of it the social bonding the crosscultural aspect flow States in music what musical features Define Groove an active inference application of GrooVe and a later update to spot uh suggestion so what is GR in essence it's the Apex of an inverted u-shape relationship between rhythmic complexity and grooviness so and pleasure if you wish uh grooviness is at a moderate level of um rythmic complexity which is um rythmic complexity is if you use the musical term syncopation is the amount to which the Rhythm deviates from the meter or the beat or alternatively you can think of the tempo the beat is a Tempo and it relates to how certain you have your predictions of that Rhythm so Groove is a pleasurable sens sensation of wanting to move the body to music driven by rhythmic patterns that balance predictability and surprise here's another study looking at the same thing and we'll see into the neuroscience in a bit that proves this now spotify's danceability is a measure they use and they Define it as so we don't have the the actual computation that they do it's the suitability of a track for dancing based on a combination unknown combination unless you work at Spotify of tempo Rhythm stability beat strength or pulse Clarity and overall regularity and this data set um someone put this on on Reddit and they've looked at a million tracks and was like okay you beat me in terms of data sets um and so they've categorized all the musical genres several musical genres in terms of uh as a function of dance ability and based on the discussion and it makes sense um on Reddit it's more about rhythmic Simplicity seems to analyze because it's more more danceable music is techno deep house and then you have salsa music which is lower and people in the comments who are salsa daners like what what is going on and especially metal heavy rock or hard you know metal fans that were thinking like wait is comedy more dancable than my favorite runer so they're a bit offended with that um I least sleep music makes sense that it's kind of low um yeah so it lacks this relationship between the complexity and predictability fun fact if you listen to all the million tracks it will take you seven years and N9 months if you listen to it on 24/7 basis here's my data set I compared sleep soundtrack sleep playlists um that are either very popular on YouTube or that I've created my own self or playlist that I have millions of subscribers on Spotify to high energy music such as Jim music that has this beast mode playlist has a mill 11 million uh subscribers or this regon uh music with regon is a Latin popular dance music very popular 11 million and this has about five million and you see the dance ability is in general higher for this higher energy music compared to sleep music and here compar veillance and high energy music is also more positive apparently uh sleep music is more negative anyway let's dive into the Neuroscience of it and here the main thing I want you to focus on is the fact that beat perception is present in the brain and uh frequencies increase their strength at the onset of the pulse here there are different tempos that were used or different speeds uh between uh pulses and as you see there's always an increase just after the uh pulse regardless of the tempo it happens all the time and it happens at the level of beta and Alpha brain waves and gamma brain waves these higher brain uh frequencies higher frequencies and um it activates these uh brain regions here and we'll go a bit more in detail that seem it's quite wide range and it turns out that we can call these the musical Rhythm brain Network because it makes use of the dopaminergic and rhythmic sorry the dopaminergic reward system in the brain um for beat perception and positive effect of groovy music and this involves a supplementary motor area premotor area nearby the pman as well as for pleasure and the urge to move to the music is a nucleus acaban and this connection to the medial orbital frontal cortex because we're talking about an auditory stimulus um auditor regions must be involved and uh the cereum also for fine motor movements turns out that musical Rhythm predictions and surprises are measurable in the bra in the brain on in two forms so the first one is about short-term habituation to a rhythm via rep ition of a certain Rhythm and then the deviation from that the surprising change to that rhythm is seen here in this paper where this's this measure called the mismatch negativity which measures prediction errors in relation to Rhythm deviations changes or Omission you don't have that note that used to be in that Rhythm and the other form um of surprises and predictions is by the recruitment of long-term memory or uh we can talk about it in terms of implicit music theory syntactic knowledge where here regardless of you're being told in advance that there will be a non-traditional chord sequence um that will be played you will have this sort of mismatch negativity which is more specific because it recruits long-term memory here is called the electric response in the auditory nerve era and yes that's an interesting aspect um by traditionally I mean traditional music western music theory um for harmonies here again we're just looking a bit deeper into this whereby beta waves we saw earlier and gamma waves The High Frequency uh waves us for early in terms of beat perception are responsible for top down so the beliefs the predictions better and the updating the surprises um through the gamma high frequency uh um activity in the brain so another aspect of of music is that it's usually done in groups such as musical performance um appreciation music therapy or learning about music and it turns out that music in groups synchronizes brains and this is called interpersonal brain synchrony and in addition it recruits the same musical Rhythm Network that we just saw with a you know a bit more broadening of it it has the amygdalin insula which is usually involved in visceral U activity and more generally the prefrontal cortex the Moto areas uh you know could involve actually moving if you're performing and the basil ganglia in general which incises all the dopaminergic reward system which nuclear acents uh and dopaminergic system Etc put in cor we saw that earlier now music the real function of music beyond the auditory cheesecake that Stephen Pinker says it's not just pleasure not just report systems it has to do with social bonding and building trust music is a participatory um activity and dancing actually is a way to participate without any musical virtuosity or even familiarity to the music this is why I call it embodied Democratic music participation open tool and fosters joy and social bonding and music strengthens this interact uh interactive rituals you can talk about it in terms of ritual and it's a social tool for enhancing uh groups so as as you can see here you start to predict you the beat you see repetitive motions this induces Groove induces dancing to it and this enhances social bonding within this sort of you can frame it within an agent environment system moreover uh seeing others dance and move encourages dancing so this reinforces the social bonding aspect and this was examplified here where eyes open uh with low Tempo High Tempo High roof actually increases the collective movement um in a group of people listening to music and this is what they were measuring here with this cross wavelet and uh dance promotes also here interpersonal movement interaction promotes self other integration it enhances social cohesion through synchronized movements and more specifically moderate uh syncopation and low as well for also there an implication for parkon disease low and moderate syncopation which is groovy music um does so as well he encourages self other inclusion with social bonding and and we're talking about music The Groove and is it applicable to a crosscultural uh definition of Music well first there has been this is with Japanese music actually there has been an optimal Groove Tempo range identified um which seems to be a u inverted u-shape relationship non linear relationship with Tempo is this seems to be from 100 107 to 126 BPM beats per minute uh Tempo and The evolutionary interpretation for this is the fact that it's near 120 BPM which is the optimal tempo for walking and dancing so this is where Groove might come into sensory Moto synchronization is most optimal in that around 120 BPM um and there are also universals cross-cultural universals of music which has been identified using statistics um and the main aspects that are crosscultural is Beats dancing making music in group percussion repetitive melodic sequences these are musical features that are Universal just going a bit deeper into this they used a music for all around these spots it's quite geographically varied data set although only 304 recordings you know there's millions of music but I think it's nice General generalization we can make based on the Black Box the black box here are musical features that are at least 50% uh shared in globally and the most uh which are the most you know the whiteness is is shared um the most shared beats subdivisions by binary tary subdivisions or scales that have up to seven notes so this is interesting and this was control phenetically meaning that this ensure that the these musical patterns and associations are not solely due to Shared cultural or evolutionary histories another aspect that is crosscultural musical bodily Sensations and emotions what I want you to focus on here is this correlation they asked um Asian participants 5 500 people and Western participants 500 people to associate emotional labels to musical features and there's a very strong correlation .94 uh between these so there's a sort of crosscultural aspect here obviously it's not all cultures maybe African Latin and uh Middle Eastern cultures do uh feel differently but here's just a Latin um influence that's indeed instrumental sinking in syncopated music which is uh groovy music which here was Samba um encourages the urge to dance so Groove at least is present in Latin American music now musical flow is not restricted to groove what is flow flow is a state of absorption total absorption in an activity in other words the feeling of losing yourself in the moment uh flow state is usually characterized as as a state of emotional resilience against trauma and here emotional expressivity is one of those not Groove uh uh musical features that induces musical flow and here the two uh main things I want you to focus on is the blue um histograms which are states of flow here were performers piano performance and this is a musical feature and so there was higher flow during swelling Dynamics so crescendos de crescendos which is volume um fluctuations as well as um repetition of um Melodies and not only just any Melodies but stepwise Melodies that are not too far apart in terms of distance on the keyboard um or in terms of notes so the closer they are blue is again flates the closer they are the The Melodies the more flow inducing they provoke um another musical element that provokes flow that is not Groove is the narrative Arc storytelling emotional Journeys in music and this interesting paper by colleague Annie hesit identified stages emotional stages of the hero's journey which is a a narrative Arc found in myths films popular films yeah and stories narratives in general they've identified emotional stages of the heroes journey in patients and clients specifically in this case was eating disorders but applying to just more generally undergoing music therapy so most they were look they were doing guided imagery with music which is a type of music therapy and depending on what they're saying the clients these themes were identifiable within this uh the hero's journey narrative Arc so that's very interesting um and this is very important for music therapists because this uh informs further the scanning aspect that they usually do in gim which is that they try to identify using the music where is the emotional stage that the client and the therapist should start their um work emotional work and so this is actually a emotional journey is are actually a concept that is applied by DJs they usually talk about energy arcs for their DJ sets um and it's also very prevalent in film soundtracks um you know film soundtracks are using so music is being used to guide the emotional journey of audience now flow as neural entrainment I think it's neural entrainment is evident for beat perception we've seen it several times here's more evidence like accurate beat perception increases neural entrainment here is just saying that beat sensitivity uh there's a sort of tempo range where this sensory motor synchronization decreases uh with very slow Tempe um but here again beat perception this the beat is being um being identified in a neural activity and even if you imagine certain rhythms these frequencies appear also in the brain activity so neural train with beat perception is clear but how about for emotional expressivity and um emotional Journeys well this study um reconstructed the music that was listened to by the uh participants from their neural patterns it was a combination of EG and fmri technique that made use of EG fmri and deep neural networks to reconstruct the music based on the neural activity and you can see that not only very impressively resembles quite a lot in terms of spectrogram information but obviously there's not just Rhythm activity or yeah Rhythm information that's being displayed there's more this frequency activity so this takes us to this study that demonstrated that spectral flux is actually the most um accurate measure for neural entrainment to music and you can see this with this red line is the highest um and with a neural entrainment and in addition they've uh demonstrated that it is a metric that encompasses amplitude Envelope as well as beats they they have mutual information here and spectral flux or spectral novelty is called is a measure of Novel energy increases in the frequency spectrum um so that's an interesting aspect now nevertheless as you can see other strong neural synchronization musical features um in addition to spectral flux are Tempo range so uh roughly as you can see here with the orange uh line 60 to 120 BPM uh High familiarity with music and pulse Clarity we we to perceive the Beats so let's keep it simple what are the groove's musical features two things moderate rhythmic complexity within an optimal range of around the walking tempo of humans um this these two Encompass most of these characteristics syncopation Tempo micro timing event density this has to do with and even beat sence has to do with rhythmic complex complexity soap of attack is a variant of beat saliency a low frequency band is the base enhances dancing uh volume and then this is the core progression and that now keep in mind that sensory motor synchronization is not limited to this uh Groove Tempo range the sensory motor uh synchronization Tempo range is shown to be between 60 to to 180 BPM below and above this the sensory motor synchronization which is usually finger tapping or dancing to the Beats decreases um which is probably linked to biomechanical reasons um however probably training increases your capacity to do so uh to synchronize however the beat perception still can occur Beyond those uh temp this Tempo range uh I recommend watching this YouTube video by Adan NE which uh goes into this uh the extremes of be perception let's apply active inference into two Groove but first before we do that we need to understand that the brain is constantly generating predictions about how the world Works in order to minimize the uncertainty about how the world Works two the brain is a layer thing and it balances top down down and bottom up influences where higher layers send beliefs about the world down to lower sensory layers which these sensory layers as we can see here um actually send back up sensory stimul to adjust the uh beliefs through the mechanism of minimization of prediction errors or minimization of free energy now talk about Groove roof balances confident prediction uh confident predictions with surprise about the Rhythm uh just reminding you it is an optimal playing with belief updating about musical rhythms now listening to groove is accessible to groups of people and therefore can become a shared stimulus for motor and emotional synchrony and therefore under the inference uh framework um generates generalized synchrony uh which means that people have shared World models the listeners have shared World and this is depicted here um uh with this quote it says neither a model of my behavior or your behavior but a model of our Behavior quote by friston um and generalized synchrony actually requires flexibility in terms of your higher uh beliefs um and because you're trying to model uh flexibly your Collective identity and Collective behavior and because of this and the social bonding aspect I place musical groove on the spectrum of the Rebus hypoth hypothesis which is a relaxed belief under psychedelics meaning that it's not as uh the the flattening of the free energy landscape is not as Extreme as psychedelics are supposed to do but musical Groove can achieve this through the the social bonding so dancing to the same music not only increases trust and social bonding but it also allows for individual creativity compare an expert dancer and a novel dancer um a novice dancer and an expert won't be having m mimicry but they will be sharing emotional synchrony and this uh provides the mechanism for trust and bonding while considering uh creativity of individuals and you can make use of a Federated belief multiperson system um model Groove as the shared marginal likelihood that people nonverbally communicate um so because idiosyncratic dance moves have been shown to even increase more uh the social bonding and Trust I encourage you next time to take the risk and bust on some moves on the on the dance Pro all right so what are what I want you to consider right now is the current models of the context that encourages dancing yes they say musical properties they have temporal predictions that you can do induces pleasure induces arousal and this together with regularity chis the urg to move to the music and um this will generate motor planning or policy generation under the active influence uh framework and and this desire to move and in training um body movements create social connection I like to add a bit more to this in terms of contextual factors BAS on the literature review we're presenting today this is a hypothetical contextual Factor um framework for danceability um basically Groove is the mechanism for uh dancing but there are several contextual layers that people individuals might need to cover um before starting to dance for instance I have two example individuals a white individual here this white Arrow who requires um seeing other people seeing their friends to before starting to move whereas this red individual just requires to feel a change in emotions to start wanting to move and these layers uh have a nonlinear relationship with the groove meaning that uh your entry level contextual entry level to start dancing um might be seeing a friend starting to dance and you might not need familiarity with the music to start dancing to start connecting to the groove uh whereas DJs sometimes make us a familiar music in their DJ sets to encourage people to start dancing they say getting those first three people to dance is always the most difficult part and uh once you have them the crowd joins and you have a crowd dancing so this was modeled with a ruction of entropy to do relatively simply uh whereby depending on the distance to the groove there's a higher um entropy level that has a nonlinear relationship uh with grooving and K has is based on personality and the relationship between each layer there's another way of conceptualizing this as a the gravity of GrooVe the layers uh are done like this so implications for spotify's danceability we saw that is more likely has to do with rhythmic complexity comedy is more dancable than uh Blues apparently or or heavy metal head banging is not as you know dancing as as laughing veence as well the most uh joyful music seems to be ASA and the less joyful music is sleep maybe Spotify is inspired by uh the rapper Nas who has a lyric saying sleep is a cousin of death so let's go beyond this uh and uh uh Beyond Rhythm and Tempo and incorporate what we've been seeing today into Dan ability so if you want to generate a formula that goes beyond theability unknown equation which is most lik rithmic Simplicity we convert uh we Sorry first uh want to point you to the study which asked musicians to perform groovy music uh with simple and complex M Melodies and they show that there's about 30% occurrence of fast notes specifically 16th notes um when people perform groovy music and this was also confirmed in terms of listeners saying yeah that's groovy um and so here I've just uh used the optimal Groove range and transformed it into seconds so you can have an idea and ah too quick for Spotify they can't no I'm just kidding uh you get to see the the the equation um so what I've done is I made use of uh spectral flux which is this spectral novelty makes use of for transform and then logarithmic compression to enhance uh weakened spectral components and then you compute the difference between time frames in terms of positive energy increases and so it measures positive increases in Energy Event density is another measure that's been used in these studies and it's a number of events onsets per window so here's a Groove uh formula I suggest which is open to to your feedback um which is 30% of energy onsets uh that is in the 16th of a note um time window um divided by a measure that's dependent on the tempo and it measures four times the beat because most music is on 4x4 meter music so I want to give you this quote dance is an energetic mode of of participation for a large population regardless of age familiarity with music or instrumental singing virtuosity to summarize we found that Grove is a musical mechanism for social bonding through movement and therefore through movement is danceability right and we can make use of the active inference to model group as a musical mechanism for generalized synchrony even uh as Federated beliefs to um model individual creativity further Works um would encourage the usage of uh um other social bonding metrics in music such as group singing uh chanting is seen around rituals around the world and this has implications for group music therapy life performances and team building music recommend systems Etc so please do comp contact me I'm open to collaborate for this uh Groove um project and discuss any of your ideas uh we could discuss about musical uh uh you know exchange music if you want and I just want to show here my YouTube channel which explores thanks to the YouTube channel I've actually learned how active inference works you know they say teaching is the best way of memorizing and in here for instance I made explored Chris Fields um work in active inference with the holographic boundary um for social interactions which is which could also be a a further uh direction of of trying to to conceptualize musical Groove in that manner but shared marginal likelihood is probably very effective as well uh thank you so much please do ask asked me for the references um I will send them they have been sent or yeah I should be on the chat by the way asking answering some questions thank you very much for your time see you very soon byebye thanks all right I'm going to play next a short video from Anna Pereira and active inference Institute research fellow hey an A Prayer here I was disappointed uh that there was an resolvable conflict uh during the time of the Symposium but wanted to share uh or contribute um a small update nonetheless uh for those of you um who haven't had a chance to hear from me before um we've been looking at accelerating and disseminating active inference for the Humanities uh we're actually just starting to call it aih uh and we continue to make progress um and quietly inate in the background we're not really at a point to do um a large more solid update uh but we wanted to quickly say hello and and say that we are incubating uh we continue to be available uh for collaborations or conversations um others find useful uh and also to provide an update that uh we've been really grateful for the opportunity to incubate within the larger aii structure um has provided several mutualistic affordances um has been a really great um construct within the the larger a uh space so we look forward to being in touch uh at Future conferences and symposiums and available for conversations um in the meantime uh wishing you a great Symposium and uh be well all right the next talk is by aswin Paul bridging biology and AI active inference for biologically plausible decision-making models this is a 15-minute video hello everyone my name is ashin Paul I recently finished my PhD from monach University in Australia and is currently working as a machine learning researcher in versus AI so today I'm here to talk about using active inference to model biologically plausible decision making uh and thank you active inference Institute for organizing this Symposium let us start uh by mentioning a recent Cutting Edge experiment that was developed by C labs in Melbourne so what they managed to do is they developed a chip a silicon chip and they cultured real biological neurons like mice cortical culture and human cortical culture and they managed to couple this chip with a computer game like the game of pong that you see on the screen right now so the experimental result is that given meaningful feedback these neural cultures actually demonstrate relative improvement over time in playing this game as you can see from this graph right like the MCC and secc neurons actually demonstrate that they can actually improve in game uh play when compared to Baseline groups like the silico um agent or the control groups so given such advancements in experiments from a theoretical perspective active inference is a great candidate to model such experiments and biologically plausible decision making so we have emerging literature and a lot of evidence um suggesting this so when we actually try and do modeling there is a clear need of scalability in the active inference literature where we are used to um modeling very small environments with very low number of states uh and uh very low number of actions Etc so in order to actually um use active inference in such scenarios we have to scale up the existing active inference algorithms and in the first part of the talk I will be talking about some algorithm and techniques that we can use to scale up modeling decision- making and active inference and in the second half I will talk about using these algorithms uh to actually uh model the particular experiment that we saw earlier so when we talk about um scaling up active inference or scaling up decision-making in active inference the first technique I want to talk about is evaluating um decision making backwards in time and not forward in time as we usually see in the active inference literacy so when we Define the expected free energy um and um compute which action should be taken given a particular observation uh rather than evaluating expected free energy or Computing It Forward in time as we see in techniques like sophisticated inference we can actually use dynamic programming and model decision making backwards in time and this result is actually published in paper uh in this particular paper if you are interested in this particular algorithm and a more General version of this algorithm was recently published as the inductive inference algorithm um by um our colleagues in versus Ai and um in that paper we note that um inductive inference algorithm scale up decision making drastically and can be applied um to um bigger generative models um another technique that we can employ in this regard is to actually learn prior preferences and and in order to do that we actually take um inspiration from the optimal control literature uh and this particular paper and this particular paper introduces an algorithm called said learning that um improves um scalability and um efficiently computes um the optimal action um when compared to um Cutting Edge algorithms like the Q learning algorithm and a particular Matrix in this Al called the said Matrix um is actually very similar to uh the prior preferences in active inference and we can actually learn this set metrix orders of magnitude more efficient than um classic reinforcement learning algorithms and what we do in in um in this paper is that we show that we can actually learn prior preferences using a similar learning rule uh and once we learn this PR preferences then decision making is really easy because we don't really have to plan a lot forward in time and we also demonstrate that learning is robust in this particular algorithm and um talking about computational complexity both these methods one is DPF method that computes um expected feny backwards in time and the other one that um learns by preferences are actually orders of magnitude more efficient than the classical active inference we see in the literature or the recent um sophisticated inference and both these methods actually learn um um or scales linearly with time against the exponential computational complexity of decision making that we see in the active influence literature um so we tested these two algorithms uh in terms of its scalability and adaptability in um State spaces um as high as 900 uh and what we saw is that when we actually um introduced stochasticity into the grid these algorithms are more adaptable uh than um Cutting Edge reinforcement learning algorithms like Dina Q uh and we demonstrate that these algorithms are scalable which gives us confidence to go ahead and model um experiments like this and when it comes to modeling neuronal Dynamics we already have existing active inference algorithms uh that model neuronal dynamics that is from um CBS ricken developed by Professor is and colleagues and in one of their Works they actually talk about um a similar experiment where um neural cultures are actually performing a perception task uh which is called B blind Source separation and in a more recent work they also talk about decision- making in neural cultures uh and um develop a decision- making model uh that can be equated to neuronal Dynamics so going into details um we have this algorithm called factual learning algorithm uh and uh they develop a version of the variation free energy uh that is term to term equivalent to a neural network cost function uh and this is valid for a particular class of neural network and more details can be found in this particular paper and there are two terms of interest uh in this algorithm one is a Time dependent risk parameter um and a Time dependent State action mapping that this particular agent um uses for decision making and because of um how it is developed it is biologically plausible and can be shown that every term is oneon-one related with the cost function that a particular class of neuronal networks actually minimize so in one of our recent Works uh we um generalized this algorithm to have memory and we tested it in a similar um control task like the card P game that we see on the screen and we can actually draw comparison with the card Pole game and the pong game because both of them are reactive task where are where where the agent is supposed to actually react to the environment uh and take decisions quickly uh to balance this pole or to deflect the ball back um so we observe that um compared to planning based algorithms um like DQ and DPFE the counterfactual learning algorithms perform really well in this particular um task and gives us confidence to go ahead and model uh the biological um neurons or the or the kind of experiments that we saw earlier right and in addition to being able to model this uh we also have um the advantage of explainability in these algorithms unlike deep learning algorithms that are black boxes so if we actually putb the environment say in a particular episode we can actually see that the risk term is shooting up in that ex episode and um this um particular term captures um changes in behavior of the counterfactual learning agent and we can actually try to explain how decisions are being taken at every time Point uh unlike blackbox approaches so given these um several algorithms that we have can we use them to actually model um the manifestation of n natural intelligence uh in experiments like disin right so we for that we actually develop uh an experimental based generative model that is based on this particular experiment so for example um this is the pong game environment that they use in the experiment um um precisely so for example um we have the x coordinate of the ball that being communicated to the dish brain chip uh through 38 distinct frequencies in the experiment um and uh the y coordinate of the paddle and the ball being communicated through a different voltage levels in the experiment um so for example like we have three observation modalities uh in this experiment and we note that these are the number of states and the ways of communicating that to the agent and we um uh enumerate the state space of our generative model um using this experimental details uh and match our generative model to be exactly like um the experimental generative process and we also have like the active inference parameters um in our generative model like uh the transition dynamics of a pom DP uh and counterfactual learning based State action mapping the risk term and the prior preference distribution Etc that we discussed in the talk earlier and given that we have this experimental based generative model we can model decision- making um of inico active inference agent and hope to compare them with the experimental result to draw insights and so on right so first of all we uh perform simulations using the see that counterfactual learning agent um with memory Horizon um as low as two is actually uh performing at par with the HCC group in terms of performance so here CFL um T stands for um CFL memory Horizon T So as expected with more memory Horizon it performs um better uh and uh we can actually look at which memory Horizon is performing similar to the experimental group and so on and in one of our recent works we knowe that the active inference agents are incredibly data efficient compared to deep learning methods and it is only the active inference based model that is counterfactual learning uh learning method that demonstrate data efficiency at par with the biological neurons when compared to deep reinforcement learning algorithms like a2c and poo if you are familiar with the Deep reinforcement learning literature um so we also compare the planning based methods that we mentioned earlier like the dynamic programming method so in this particular task or the pong game um planning is not really useful so with increasing Horizon of planning um we don't really see um Improvement um and um planning Horizon of performs as good as HCC neurons and planning Beyond a point is not useful in this particular task uh because of the nature of the task and we also not um the explainability of our models for example if we look at the average risk um in the counterfactual learning method we see that the risk is actually only reducing um significantly uh with a memory Horizon of four and which and this can also be correlated with the performance Improvement that we saw um in the results graph where with memory Horizon 4 there was a significant jump in performance and these kind of explainable insights um that can be drawn from our uh models is a clear advantage over black Mo methods like um deep reinforcement learning uh and we we can also look at um the entropy of the CL parameters and we see that the entropy is decreasing over time that is the evidence in learning and parameters reveal um differences in behavior of different agents and that is a clear Advantage here and to summarize um we actually developed an experimental um informed generative model and simulated decision making in the particular um uh environment uh we tested different active inference based algorithms in this setting and uh we also noted the similarity of memory augmented counterfactual learning method um to the biological neurons and we are interested in investigating similarities um of the memory based CFL method more to the biological neurons and also insights like this can Inspire more experiments where um aspects like memory can be studied well uh there are several um limitations to this work where we haven't conducted um a really um involved statistical anal is in comparing these agents we are right now qualitatively comparing them um with um average parameters but we would like to do more statistical analysis uh and we would also like to um Inspire um experiments from our theoretical predictions in our future work so I thank um all my collaborators in this work and if you're interested in this particular um work and the algorithms that I discussed uh you can actually U get access to all the papers here and uh feel free to contact me if you have suggestions or more questions about this work so thank you for your time uh and um I look forward to meeting you all when when I have a chance next thank you so much all right the next talk is by Bert Burgers towards autonomous Urban digital twins this is a 20 minute video for the next 20 minutes I will talk about transport policy planning and how active inference could be applied through digital twins urban planning is a very broad field encompassing everything that occurs in urban regions from the design and operation of infrastructure to social programs to scope this I will focus on the transport system giving a rough sketch of what active inference May entail for this domain modeling the transport system relies on operationalizing decision making many individual choices lead to busy roads therefore the main story line focuses on the difference between active and passive decision making the presentation is structured in three parts first is the context giv an outline of urban digital twins I will focus on livability as it is an objective of Transport policy next I will present some domain specific building blocks which are then combined in the final part finally I hypothesized that renormalizing generative models can be used as transport models concluding that active inflence based modeling might require A New Perspective on trans transport planning first I will give some context in order to frame how active inference transport planning and digital twins come together the field of Transport planning is moving towards the development of digital twins these twins enable quick calculation of effect giv an intervention hard than performing a comprehensive study per project which can take months a digital twin can be reused many times an example application would be the construction of a new neighborhood what are the effects on congestion air quality and inundation risk there are roughly five levels the first introduces a compelling visual interface often involving a 3D representation of the real urban region next is the addition of information on top of the visualization and the geographic data works the third step not only Maps data but incorporate different physics based models hence it is called predictive transport models are one example others are airflow or water simulations high performance computing speeds up the scope of parameters which can be tested the fourth level Builds on top of the predictive models by adding an interpretive layer the added value is scenario based planning where a range of outcomes is calculated for a hypothetical intervention instead of configuring parameters exactly you could propose a preferred outcome to the computer and that would only simulate what is necessary as it understands relations between models the last level introduces autonomy not only are physics based models configured intelligently as in step four if the T actively goes out to construct its Niche transport policy evaluation is going to somewhat of a change and that before the development of Transport models allowed the Paradigm of predict and provide we could forecast traffic flows and accordingly invest in infrastructure to prevent congestion more recently the field has acknowledged that models trained on past data are inherently flawed in predicting in certain Futures as such it is moving towards scenario based planning sometimes described as desired and provide along with this change comes a reflection on the philosophical context for Simplicity I rely on two Frameworks the first is utilitarianism it relies on the Assumption of Ral Choice enabling the quantification of benefits and costs in terms of util utility and disutility between every origin and destination there are several alternative Roots each has attributes such as the distance traveled and the cost combined with preferences we get generalized travel cost the rational Choice assumption enables visibility this allows us to take the ratio of estimated B parameters in utility functions these ratios can then be used to calculate for example the monetary value of travel time in practice we always use monetary value as the final measure so that we can compare the cost of implementation against the benefits experienced finally utilit ISM leads to aggregate increases in welfare at the ignorance of the distribution not only does this apply to wealth but accessibility too take for example the discrepancy in infrastructure spending between core cities and the periphery to address these concerns contemporary policy evaluation attempts to move away from utilitarianism instead of attempting to relate all preferences to monetary value we leave them separate not only do we reject the visibility of indicators but we also consider many more than before shown in this wheel spending from living environment accessibility health and safety finally the distribution of costs and benefits is explicitly considered and much work is now spent on reworking existing modeling Frameworks to make these distributions explicit there are three main objectives in planning infrastructure accessibility safety and livability capturing liveability is much more difficult than accessibility and safety livability is generally defined as the fit in an ecological relationship between resident and the living environment with regard to the spatial temporal scale of this relationship livability is local and sustainability global for practical reasons fit is operationalized through valuation the more an environment can satisfy one's preferences the greater the fit theoretically it has been long known that fit could also be operationalized as a continuous transactional process however little progress has been made and this is where active inference comes in these two approaches to livability are best defined by contrasting them to other on the one hand we have the static approach which is concerned with measurable comes on the other hand there is a dynamic approach in which livability is viewed as the hidden driving force behind that which is measurable by Framing liability is comprised of three components indicators percepts and needs desires I can show that the location of fit determines which is which either it is a weighted function between percepts and our preferences through valuation or liability is a weighted function between indicators and percepts which are biased towards our preferences finally these two approaches might be complementary and this might be useful for modeling purposes there are many different transport models when it comes to those used for urban planning these can be distinguished into micro and micro simulations the micro scale simulates individual vehicles and their interactions meanwhile the micro scale applies constrainted optimization over a network preferences constrain this optimization here I will focus on micro models which combine transport and L use patterns fundamentally we rely on seeing the transport system as a market Travelers generate demand while the Network provide Supply the four-step model forms the backbone of microscopic transport modeling and I will go over each step first is trip generation this step determines how many trips start and end in each Zone this about determining the row and column sums see also the top right figure for trip production we look at residential zones and chuse social demographic factors like household size car ownership and income levels for trip attraction we look at the number of jobs shops or school capacity trip distribution iteratively balances row and column sums to fill the origin destination Matrix balancing is done using a gravity model where the number of trips increases with the mass but decreases with the travel resistance with the full origin destination Matrix we can split these flows per travel mode used the utility of each mode can be weighed against our order in the choice set logit models can be estimated for different segments of the population uh that is the figure in the the equation in the bottom right finally we assign trips to Links network according to wardrop's First principle Travelers select roots to minimize the travel time leading to an equilibrium where no user can reduce the travel time by changing the entire workflow is iterative because assignment affects travel resistances due to congestion not shown here is the modeling of land juice which affects zonal data next I will Outline Three conceptual building blocks which will be combined in the synthesis as we just saw modeling transport networks requires a representation for example the travel time per link additionally we would like to approximate human process ception for example by using images or sounds the field of urban representation learning attempts to automatically capture the content and structure of cities luckily there is a strong inductive prior from geography similar things are closer together as such major design choices revolve around which measures of distance to include examples are ukian distance travel volumes or accessibility generally an application of urb representation learning relies on three choices first is the need to discretize the Uber region into spatial units each acts as a clause the figures on The Right Use hexagons secondly is the type of data data used where traditional transfer models only only use tabular data we can now also use street view images and text last but not least is the way in which parameters are structured and learned craft neural networks are widely used as they naturally represent the transport Network Additionally the loss function can be used to steer training accessibility is a major field of study in transport planning that connects well with active inference Concepts in urban an urban representation learning we aim to capture spatial similarity in edings accessibility provides a theoretic iCal foundation for which similarities to capture intuitively accessibility is akin to Landscapes of affordances which are shared by everybody living in an urban region location based accessibility is theise from constrainted Maximum enthropy and can be configured to equal utility based accessibility the calculation is straightforward for each Zone we Sim the product of destination attractiveness and the deterrence function based on generalized travel cost the deterrence function captures how travel cost relates to similarity as shown in the maps accessibility patterns differ significantly between bike and car networks Hy is a wellestablished element of Transport networks think of how local streets feed into high into arterials and highways each operating at different speeds and distances where local streets facilitate access main roads enable throughput to capture this hierar key in urban representations we start by defining the study area creating Columns of course hexagons and then mapping them to the final resolutions I have used a unet to capture both bottom up and top down information each block in this net is a a graph convolutional neuron Network a precomputed accessibility graph sparely connects centroids of hexagons where the edges are the accessibility value reconstruction loss ensures we capture input data while consistency loss ensures Columns of hexagons are integrated combining context and building blocks I will now go over the synthesis we started by discussing digital Twins and two approaches to livability static and dynamic the dynamic approach frames livability as Niche construction when it comes to policy the problem boils down to the objective function used good free energy acts as an objective function for policy evaluation unlike utility free cannot be sum the cost decision makers perception is itself is subjective meanwhile with utility we all live in the same fully observed world if Urban regions engage in Niche construction perhaps we can create digital twins that embody the n construction process autonomy then becomes about influencing an existing process rather than creating something from scratch there are three potential approaches agent based models on a hex crit simulating individual Travelers simulating each hexagon as a cell with the expectation of emergent Behavior or use a renormalizing generative model with a fixed hierarchy the third option is the most promising as it requires a single model than the combination of many smaller ones in line with this third approach I want to consider a fixed hierarchy and renormalizing generative models are perfectly suited to this purpose the idea is that these might already be transport models as they upap Origins to destinations using cost functions sticking with the notion of a market equilibrium there's a demand and a supply side to the operation of renormalizing generative models just like in a four-step transer model the demand side determines the landscape of afforded embeddings and these embeddings capture the spatial similarity the supply s side aims to calculate the distribution of flows through the network it is just that here the network is a hierarchical structure not the physical transport Network on the demand side the landscape of a forness is or the landscape of afforded embeddings is perceived through the combination of top down prior and bottom up data from the top down we impose our cost function on traversing the lateral graph attributes on edges are valued against each other for example travel time and the price of petrol we also impose our preferences in term of the content through which to see so the content of the embeddings for example the concentration of emissions air from the bottom up we receive information for this lateral accessibility graph like the travel time at this moment as well as the content of discrete spatial unit and this content is in the form of embeddings Ander from high dimensional data using neural networks so that they can so that this process can be automated on the supply side we have a landscape of affordances at each layer in the hierarchy and we need to pass messages them supply is about narratives of attention either about errors flowing up or predictions moving down to move laterally you have to go up and over in doing so carve the transport system at its joints for example when you need to do groceries you probably do not zoom out to the courses layer instead you will find CP which more quickly and walk a few blocks using active inference as a framework to understand decision- making in urban regions is still speculative the standard workflow of xante evaluation to study the benefits of a project might not be applicable anymore as the digital twin goes out to construct its Niche on its own the separation between technical calculations and political motivations in planning becomes blurred how do we capture both travel choice and policy objectives and one model accessibility is already a central topic yet it does now gained additional consideration due to the reframing into Landscapes of performances these affordances may not only be in terms of T tabular data but can be high dimensional like images accessibility heavily draws upon constrainted maximum entropy basan mechanics treats this the same as free energy minimization it therefore seems worthwhile to reframe accessibility in this way too active inference is useful not only for autonomous Urban digital twins but also for level four twins to refresh level four twins use physics based models with the addition of an interpretive layer and this additional layer contains an understanding of the sensitivities between models which allows an uran planner to quickly Converge on a set of designs rather than simulate thousands of Alternatives therefore a practical starting point would be to take level three digital Twins and upgrade them with an inference module to explore the parameter space efficiently the apparent complimentarity of static and dynamic approaches of liability suggests that we could use representations learned through valuation so a supervised um way of learning Urban representations to then use them in learning the structure of an active inference model in s it is fair to say that autonomous Urban Digital Trends still need a lot of work regardless considering free energy as an alternative to utility is a valuable exercise in understanding decision- making in urban regions lastly I would like to thank the team of Symposium organizers for the time and effort all right the next video is 23 minutes long iness hippolito dysfunctional markof blankets from stuck states to adaptation hi everyone I'll be presenting today uh dysfunctional Mark of blankets from stock states to adaptation this is a paper that has been um in progress for quite some time and it is um a collaboration with ma alasa Alex Kea and Daniel fredman so I want to start by uh highlighting that there are stuck systems or stuck St all around us and that we find that example in neurocognitive um levels such as in OCD patterns in PTSD trauma Cycles or in addiction circuits then on the social or ecological scale uh we find it for example in poverty traps in eamers and Market uh monopolies and climate tipping points then in AI we also find them for example in a large language models Loops who isn't been stuck in a I apologize kind of cycle then uh in reinforcement learning local Minima in recommendation filter Bubbles and in neural networks training fails so what um these kinds of systems uh share is that uh they are some sort of critical threshold uh they have um or present um self reinforcing loops and uh they have this persistence to uh change so they tend to remain in the same state hence why they are stuck so then the core problem is going to be a sort of a paradox so um the view is that systems adaptive or complex systems um coupled with the environment they should adapt but sometimes they don't so then uh from this perspective a few questions um become Salient why do flexible systems become so rigid and do not have that flexibility to adapt to the environment how do these adaptive boundaries uh between the system and the environment break down and what maintains that dysfunctional State what is there that must be so strong that um the system cannot uh change to a different uh or look for or act for uh to attain a different more preferable state so then of course that this presents um these are presented as Challengers because there is an everchanging world and the system that needs to adapt to that everchanging World um and uh it seems like um existing approaches um such as dynamical systems theory and network uh science um still have some limitations to quantify and address this kind of phenomenon that we call a dysfunction so then from our perspective uh we need um a framework that unifies that uh unifies this explanation so we need three things first we need to identify and characterize this functional States then we need to provide a mechanistic explanation for the Stu States and finally we need to offer a principled approach to designing these kinds of interventions and that's what we set up to do uh in this paper and that I'll be presenting you so um our framework Bridges um theoretical Frameworks uh from complex systems theory and the free energy principle the main or core um Concepts uh they are going to be useful are Notions such as emergence self-organization and adaptation from complex systems theory and coupling self-maintenance and free energy from the free energy principle by uh utilizing these two Frameworks uh we developed uh what a construct that we uh defined as dysfunctional Mark of blankets and with that uh we hope that we will offer a formal theory of a systems dysfunction and we hope to develop quantifiable measures for this phenomenon of stuckness and finally practical recovery strategy so that we can act upon uh the system that is stuck in a therapeutic way and we uh uh hope to kick the system out of that stck state into a healthy State okay so as systems we all know that um they are comprised by components which interact and from that interaction we observe some sort of emergence um for example in the case of an immune response we have different immune cells each one with a specific role and they interact in the form of sell signaling Rec recognition of events and that that interaction then gives rise to emergence that we observe in the form of coordinated defense or system memory then uh from the free energy principle which we uh are very acquainted with uh in this conference uh we learn that systems that exist act to minimize their free energy and um further to that um the understanding or the Insight is that system are coupled with the environment and they act upon the environment to precisely uh attain or look for more preferable States for their self-maintenance and we can explain um that behavior through um active inference then another very useful uh construct is Mark of blankets of course the framework is a dysfunctional Mark of blankets and the mark of blanket uh as we all know is a statistical boundary between what is inside of the system and what is outside but further to that uh an important uh Insight is the conditional Independence between internal and external states which give um the system sort of a very important boundary um to have a distinction between the self and the no self and that can be seen also through the the notion of it's an operationally closed system A system that self organizes and maintains itself for that maintenance it's also quite important that it is thermodynamically open it is an open system it's not a closed system and that's what allows them to remain alive by adjusting to the environment and here you see for example we have a a neurons in the form of internal states that could be part of a network so a key Innovation concept that we bring in is the notion of selectability and selectability is a systems capacity to explore navigate and select among possible States these can also be seen as the system having multistability many points um that are afforded to the system in the environment such that uh the system can be healthy in that navigation so you can see that from example that comes represented here in the form of state A B C or D different points that the system can explore within the environment and select the most preferable ones and um the notion of selectability allows us to um think about State space accessibility or affordances the adaptation capacity and the boundary flexibility to adapt rather than being rigid and stuck in a particular state so then uh within the free energy principle we can think through that Paradox it becomes if systems minimize free energy if they Accord to the principle why would they get stuck does the principle not work or is it the system that is uh not in a stuck state so then uh what we realize is that um what is not what is really within the Paradox is that we cannot explain this persistent maladaptive States and there is not really a formal treatment of the dysfunction so then there is a missing link between

Theory and intervention and our solution is the dysfunctional Mark blankets there are two types of boundary dysfunction that we can observe uh in these systems that are not adjusted attuned adapt to the environment so they can come in the form of stuck States or blanket breakdown and stock States I mentioned this in the first slide and uh for example in the case of depression what we observe is rigid kinds of thought patterns reduced emotional flexibility impaired Social uh information processing and shortcuts bypassing healthy emotional regulation and in a more General way we can Define these stck states as the sil reinforcing kind of Cycles being trapped in deep energy Wells and being resistant to environmental perturbation then on the other hand we have blanket breakdown and an example of this is the immune system when the immune system attacks its own tissue and what happens here is a very interesting phenomenon of excessive permeability uh on the boundary of the blanket right loss of system Integrity so the self no self distinction the system doesn't know what's self and what is not self that's the loss of of system Integrity because of the boundary dissolution okay so then we arrive at uh the definition of the dysfunctional Mark of blankets and this this is a statistical boundary that um contrastingly has lost its adaptive capacity for Effective mediation between internal and external States and the key characteristics of um the dysfunctional Mark of blankets are that the system is persistent in High free energy despite attempts and what we observe is the breakdown in information processing and the loss of effective environmental coupling I will unpack this further but let me just situate us um for a moment so uh we will go from the framework to practical tools to cross scale applications that's where we would like to go uh so to do that we need to First Define quantifiable measures then um identify a dysfunction index and finally the recovery metrics then as to practical tools we need adaptive pations in order to kick um the system out of the valley and Recovery strategies and these on more therapeutic kind of site and hopefully we'll be able to apply these cross across scales so now looking at the framework itself the first thing that we need to do is to um find that function index and what that is going to help us do is to measure and identify and delineate the this function it will measure the information flowing through the wrong channels like a diagnostic kind of tool that um spots the system bypass and the higher the value the more dysfunction for example like a cell with a compromised membrane then uh we will need also a recovery metric and this will be a pro a progress kind of indication indicator for us so it shows a dysfunction reduction and the thought here is that um we ask ourselves are we fixing the bypass problem so then uh we will need uh to uh in order to advance without we need the landscape uh reshaping kind of like bringing out a new GE graphy of points uh that are afforded to the system making sure that we don't get stuck again and hopefully we attain the systems adaptability where we find the question can it be explore can it explore and respond flexibly to uh the environment so why are these two measures relevant well they are they will uh serve the role of being Universal tools across scales we will be able to apply them uh to uh all scales from cells to ecosystems and they will help us diagnose and track recovery and guide our next stage which is the intervention design okay so now moving to practical tools adaptive perturbations and Recovery strategies uh we have adaptive pations here playing a crucial role and um an adap itive perturbation is going to be a controlled intervention that temporarily disrupts a system state to facilitate adaptive change and what we mean by this or our goal is uh not simply to push a system out of being stuck but it is to help it to become more resilient precisely to make sure that it doesn't come back to the stuck State as you can see an illustration here on the graph below so uh then we Define the Adaptive perturbation design as in the first term as traditional intervention approach where we would like to nudge the system towards uh these better states such that it adapts based on the system's response then with the middle term we can reshape the energy landscape have a different sort of more um flexible uh geography that makes it harder to get stuck again like smoothing out the deep valleys and then finally uh we can promote exploration and that means help the system uh discover new affordances new possibilities hence enhance adaptability um there are kind of like two examples of adaptive perturbations that um I just want to mention and uh one is by the parameter changes where for example you can think of that as a temperature modification in cellular environments and the other one is the external force or input an example of that is transcranial uh magnetic uh stimulation so why is this adaptive perturbation relevant well it combines three intervention strategies self adjusting all parameter parameters will adapt over time um and it uh includes a balance between immediate Improvement longterm perturbation sorry longterm prevention and future adaptability the Practical impact of um an Adaptive peration is that it will help us guide for Designing interventions and work across different scales and prevent the kind of uh so to speak Band-Aid Solutions right so then why do parations matter in the end well for the stuck States they're going to be very relevant to help us overcome energy barriers um break self-reinforcing cycles and enable the system to have more States and the capacity to navigate um and explore uh the the environment and select the states that are going to be much more preferable to the continuation and maintenance and self-maintenance and self organization of the system itself in the case of the blanket breakdown which is this is the case of for example the immune system that has lost uh the capacity to differentiate between what is and what is no self and attacks um its own tissue then utilizing an Adaptive perturbation uh will be uh useful to restore boundary function to stabilize information low and to reset System Dynamics and therefore uh repair uh the capacity of the system ultimately to recognize what's self and what is no self what is internal States what is external States okay so uh I will recap uh the journey now uh from the dysfunction to the recovery um so first we need to measure the dysfunction then we need to track the rec y then we will need to design interventions and we do that through the Adaptive perturbation to implement the recovery finally we need to monitor that kind of recovery such that we now have um been able to uh get the system to be adapted to the environment uh rather than uh just um in a very short term so key relationships that come up within this journey include going from higher dysfunction to Stronger perturbation that is required going to better recovery to reduce intervention strength and going from more selectability to Greater adaptation capacity future directions are where we need to go is to extend uh these mathematical formalism to non-erotic systems to multiple time scale Dynamics and coupled systems interactions in order to do that we also need to uh develop measurement tools such as real-time dysfunction metrics early warning indicators uh that can predict a Tipping Point or a point of no return and then we can apply a preventive measure and cross scale validity tests some of the implemental implementation challenges that um we can see um include of course measurement intervention validation and scaling up and in order to address them we will need uh better tools more precise tools standard tests and infrastructure the potential impact of uh this framework um is like I said across skilles uh in healthcare for example we can look into uh Therapeutics for uh any sort of disease kind of cycle uh it includes mental health and aging patterns in climate action uh we can apply um this uh framework to tipping points to as a preventive measure and to Restoration in AI we can apply to uh develop more robust AI we can also apply it to uh make more smoother ethical and governance uh measures for safety in AI as as well as um can help us design bio inspired AI such that it's a system that is uh more adapted uh and hence more effective finally in Social systems we can also apply this framework for uh policy designed for economical stability and social cohesion and um coming to the end uh call for Action um so uh research priorities include developing standard standardized standardized metrics and build shared data sets create open-source tools and establish cross-disciplinary teams so please join us in uh theory development tool creation application testing and knowledge sharing conclusion and takeaways so we went in this journey from uh these functional systems that we observe that are stucked in a local minimum and we want to develop a perturbation in order to help the system become more resilient and adapt to the environment out of the stuck State and hence a full recovery in the long term so then key takeaways uh are that we hope that we um have developed the foundations for a formal theory of system dysfunction quantifiable measures for the stuckness phenomenon and a practical recovery strategy that can be applied that can be further in the future applied to scales thank you so much for your time please do not hesitate to contact myself or any of the authors uh if you'd like to uh engage or learn more about uh this work all right keeping with the co-authorship theme next talk is a 25 minute recorded talk by Alex kefir resilience and active 2024 this is Alex kefir bu this is Alex kefir I'm here to present some ideas on resilience and active inference um so this is largely just a unpacking in a careful way of some Concepts from this paper from 2022 that I was involved in um but also will explore some some novel directions um that this uh this thinking has taken since then so the concept of resilience really if you look at the emology it's just it means like the capacity to recoil essentially or to or to to bounce back or just to survive through hardship and change um so obviously this concept is sort of a fundamental one if you're looking at uh life life forms um dynamical systems more generally um in the context of like um ecosystems and sustainability it's an important concept um so that's why we're interested in this um it's actually also sort of a fundamental Concept in Material Science so so since this paper has come out I've been approached a couple of times by by journals of Material Science for um for follow-ups and such um so as I said the the fundamental concept has to do with recoiling or bouncing back um but there's sort of there are some peripheral sort of different facets of this of this meaning and we explored all of these in this paper um so roughly speaking you can think of resilience in sort of three ways um so first there's inertia um so inertia I mean I guess literally just means really uh sort of not changing um so persistence uh but you can also think of it as resistance in the sense that right unless you're unless you're exists in a vacuum uh then you're a system that's embedded in some broader system and inertia is basically continuing on your way despite changes uh that might that might occur or interactions with that broader system um so if you think of the red arrow as representing the path of a of a subsystem or an agent through its uh State space then and The Black Arrow is some kind of potential disturbance um then the top diagram here illustrates inertia um the central meaning again of resilience arguably is elasticity or the ability to bounce back so in that case you undergo some perturbation or change um and you end up recovering your initial doesn't have to be that you recover your exact initial position but the point is you make your way back to a reasonable state in terms of the kind of system that you are um then there's also this sort of third thing called we call plasticity or adaptation um so in this case case you begin in one configuration uh the red arrow at the beginning here and after the perturb uh the perturbation or disturbance you end up in a slightly different configuration so that should be read not as just that you change your state like you update your um your beliefs on the time scale of infers or something but you actually you actually wind up somewhat adjusting your model in response to this change um so this is more like something like Al estasis um whereas El C is returning to the same state um so I'd say there are some close relationships just to to get into some philosophical conceptual analysis for a second here um the first thing uh inertia probably doesn't really exist in its pure form right in nature in a way or it it if it does it it reduces to sort of elasticity so you know I mean presumably if you uh if you perturb a system um then you have to cause some change in the system however slight um otherwise there hasn't been a physical interaction right and inertia is going to look in practice just like the capacity to sort of instantly recover um from that Tiny from that tiny change um there there is in the limit case there is sort of a different meaning here because again there might just not be a change but I think um I think in the context of realistic systems there will always be some some interaction um but AR ARA these two these two concepts are very close um plasticity seems like um on the face of it like a different thing because um well first of all uh the ability to be to be plastic in this sense sort of explains how you could come to have this property of the elasticity um so if you look at the example of neuroplasticity for example um it's the ability to adapt to change and to basically update your your synaptic weights to arrive at a new model that allows you to be uh in part allows you to be resilient in these first two senses um but there's another sense in which if you just assume that there's some property of the system that doesn't change through this um through this process which I think is a reasonable assumption in the case of uh you know biological systems um then there's going to be some kernel of the system that survives through change and then you're back to your sort of original um okay so if we look at this concept the lens of the free energy principle um I mean first of all it turns out to be a very fundamental at you know as is often the case when you try to unpack um sort of uh fundamental properties of of systems in nature in terms of the FP and active inference uh you find that there's a fairly direct mapping from the properties that the FP attributes to systems or um the types of systems that are well described by the FP and this uh and this concept of resilience um so I mean Carl firston has said in discussions of this that essentially uh systems well described by the fdp are resilient um uh full stop so uh but we can look at at the three uh facets of the meaning of resilience here so um certainly systems described by the Fe are they have this property of inertia in the sense that um by definition right they're things that have managed to persist so they limit their exposure to uh to unlikely States unlikely observations um but they're also sort of by definition elastic systems um because because they have this attracting set so um having an attracting set of States means that um you know of course there might be changes or perturbations that are sort of catastrophic that destroy the system but as long as the system is not destroyed then by definition it's going to make it make its way back to this attracting set of States um so resilience in the core sense is really a um almost a defining feature of systems that are described by the FP uh and then finally plasticity you can think of as something like Basi and model selection um on some time scale and so that that might seem like something extra but it's only extra if the multiscale aspect of the FP is extra and I would argue that it really isn't because at its heart the FP is a is a scale free description of things and this multiscale and scale free are sort of uh sort of um connected Notions um so that's plasticity um okay but so most of the interest I'd say of this lies in the the mapping of resilience onto this this formalism um of active inference so uh a key concept uh in this analysis is the notion of degeneracy and this is a uh somewhat technical term um here we're following we mainly follow on the definition of and and in fact the analysis of degeneracy in terms of active inference in uh SJ at all 2020 this is a great paper on this topic um so here we have just the expression for variational free energy um in the top row um is a expression in terms of basically a Helm Holtz free energy so it's an energy term minus an entropy term uh uh so if you just look at that for a second um the we've identified uh again following this paper from 2020 we've identified degeneracy with the entropy term um so we'll get into why that is in a second um but basically uh if the concept of degeneracy is useful redundancy um so this you can think of this as like complexity that that earns its keep or that's worth having um so complexity in this this bottom breakdown of variational free energy uh you uh you analyze it in terms of complexity which is a the the difference the K Divergence between the prior belief uh overstates the the generative density Over States States and the approximate posterior so this is kind of like how how difficult is it or how much change has to uh does the system have to undergo in order to represent the states that are um inferred from some observations um what's what's what's the departure between that posterior distribution and the prior distribution Over States uh that's complexity um minus accuracy um the the the expected uh log probability of observations given States um so what's the connection between complexity and this idea of useful redundancy uh sorry of just redundancy um there's there's a assumption here that complexity and redundancy are closely related um I think the basic idea is that so redundancy uh in this context is quite literally just duplication of States um so the I'd say the classic example is like an error correcting code so if you're trying to transmit some uh bit of information over for a channel uh you might you might send 10 copies of a bit in case in case some of those bits get flipped or corrupted uh and you want to make sure that the that the message gets transmitted um so that's redundancy it increases the complexity because um uh well one way of looking at it is that it uh it it increases the description length uh right this this K Divergence is like how how how much how much more do you have to say in order to communicate that an event drawn from this Q distribution has occurred um given that you assume this P distribution um so uh so you can think of it as well what these these extra bits are are the Redundant bits um so there there's a sort of conceptual connection between complexity and redundancy um and then degeneracy on this first definition is just the complexity that serves some purpose and we'll get into that in a moment um it's just also important to note that there's this definition of um degeneracy as partial redundancy in the in the literature uh and this is partial overlap and function so um the idea is that um redundancy exists in a system uh sorry degeneracy resists uh exists to the extent that multiple states in the system serve some function in common but then those States also have um otherwise uh have distinct or disjoint sets of of functions um and in practice I think these two almost always line up so there might be sort of uh limit edge cases in which um in which they're different but in but in practice um right if you have multiple States they'll always have some different properties and therefore um perform some different functions okay so so this is one of the core things that we pointed out in this paper um of ours so um this this actually sorry this breakdown is from the sjad all 2020 paper uh but we sort of have a different gloss on this C Center term here um so this is the complexity term in the variational free energy you can break this down in terms of um further break this down in terms of this uh expectation under Q of the negative modog probability of States under the generative model um so this is called a cost term in the 20120 paper um we describe this in epistemic terms as as as an certainty term so this is U this is basically uh you can you can read this as the Shannon uh minimum description length of the of the states given the generative density um but it's sort of the expected surprise um how surprising are are these states that I that I infer uh using the variational posterior under my generative model um so that's an uncertainty term uh and then we subtract this degeneracy or the entropy over um over the variation of posterior Over States um so I mean first of all sorry I realized that I didn't quite make the the link here between um degeneracy and entropy so why is this the case uh why do we identify these uh it's it's because um this means if if the if if this H distribution Over States is higher in entropy it means that there's a um degenerate mapping in the likely Ood from states to observations right so meaning for multiple States uh you can get the same observation um and so then when you invert that likelihood uh mapping in order to recover this posterior um this means that for one observation you might have several um several different states and the the spread of the of the distribution Over States of course is just the entropy so that's that's the link to to genery there uh so so this complexity term is uncertainty minus degeneracy um one reason that's interesting is that when this complexity is zero um it's not as if the system is completely certain of what's going on um the uncertainty term actually remains it's just balanced by this degeneracy in other words all of the uncertainty in this ideal case um is is useful um and so I think this really this really captures what what degeneracy in this context amounts to um it's uncertainty that useful and is it's useful precisely because there's some amount of uncertainty that's inherent in the source of the information you're trying to model right so um here p is the the in this context is the generative model that the system has of um of it's the sort of internal generative model of of the external States um and so there'll be some right there'll be some entropy some Source entropy associated with that system that that you're trying to model and and um so long as you match your variational posterior matches that degree of uncertainty um then you're in sort of an optimal um an optimal model so in other words for maximum model goodness try to ensure that um P equals Q um so uh this ties nicely into the idea of plasticity resilience is plasticity so your goal here is to adjust the posterior to match the generative model essentially um so you can do that um you know in in like perceptual inference uh you can try to infer a state uh a q distribution this is again an approximation to the posterior the true posterior over over those States under the generative model um but equally you can adjust the generative model in in a in a multiscale setting you can learn learn or adjust update the parameters of the generative density in order to match what's in the external World um so right this this complexity term um is affected by changes to p as well as changes in Q and so I think the fundamental idea that I had here was um in an ideal uh sort of optimal system under the free energy principle you should expect um first of all P will be adjusted so as to model the environment for sort of um basic you know um free energy principle SLG good regulator theorem types of reasons uh and then Q also be adjusted to match to match P um so this involves um this cybernetic principle called The Law of requisite variety which is sort of a um um it's closely related to the good regul good regulator theorem um but basically it just says that you know if you're trying to regulate or control some states then you need uh at least as much uh variety or uh in probabilistic terms that that translates to variance um or or entropy um you need as much Variety in the controller as you have in the in the environment and so it's a fairly intuitive thing because you basically need sufficiently many degrees of freedom in the controller uh to be able to map to to model what's in the environment um so you can see this as this degeneracy um property of resilient systems or uh any system under the FP as as an instance of this law of of requisite variety it's also clearly by the way a an obvious case of constrained maximum entropy so there's work um showing that that many many lines of thought really showing that the active inference uh sorry the free energy principle is um is equivalent to a principle of constrained maximum entropy uh so here the constraint is this uh you can you can think of this as constraint by accuracy um but you you basically maximize this entropy term subject to that constraint uh subject to the generative model okay so um looking just a bit more closely at this notion of degeneracy um so I've said that it's Central um one pointed question you might ask here is well are all do all resilient systems uh have this degeneracy property uh and so I want to say the answer is yes um so the type of degeneracy that we've already talked about is this degeneracy and the likelihood mapping uh so this is multiple States mapping to the same observation um and again that's uh that can be cast as you'd expect to see that given um that you're performing maximum entropy inference um so that that is um crucial but equally important I'd say is degeneracy and the Dynamics uh so this characterizes just systems that again that have an attracting set of States so if these STS are states at a given time and then St +1 is the state or set of States at the next time step um so this in order for for a system to be resilient um it really has to be the case that this transition um uh that these transition probabilities or the Dynamics of the system are such that um that that mapping itself is degenerate um so you know this means that basically you can perturb the system in several different ways and you'll still arrive uh will still end up at the same at the same uh state or or set of attracting States um obviously if the if the perturbation is too large then that might not be the case the system will diverge and you'll destroy you'll destroy it but by definition the systems that survive through change are the ones that have property um so there is a very deep link between resilience and degeneracy if you consider Dynamics as well um just a couple more points here one is um something that I've recently been thinking about it's it's it's actually a theme that's been on my mind for some time but I've recently written about it in some other work um and this is what what does the principle of requisite variety and um what do these concepts of degeneracy and resilience look like in in the context of multiscale active inference um but specifically here I don't I don't just mean um the fact that excuse me the fact that model parameters as well as states can change but I mean U multiscale active inference in the sense that you've got potentially higher level agents that are composed of lower level agents um so in that kind of setting uh requisite variety ends up looking like the the property of higher level agents that they're composed of a variety of different types of lower level agents um so I mean I'd argue that if you combine this law of requisite variety with multiscale active inference in this sense then you really sort of automatically have something like a a diverse ecosystem in which there are many different types of Agents that participate um so here in this diagram we're illustrating first of all just the same degenerate State observation mapping that we saw in the uh in the previous slides um but here the nodes are individual agents that compose this higher level agent um also as pointed out in this paper um by by um Carl friston Daniel fredman at all um the in this kind of cont text different agents can be expected to and I think empirically in some simulations are shown to take on the roles of these different types of States so of course the breakdown here is in terms of active and sensory and uh active sensory and internal States so here you've got active sensory and internal agents uh and that's one instance I would argue of this um application of the of the law of requisite variety um to this higher level uh system um I also show here the the policy selection based on expected free energy for completeness um but that's sort of the main the main point is this this specialization and diversification of Agents um in this in this setting um in terms of the second definition of degeneracy you can also think of these agents as playing um somewhat distinct but partially overlapping roles um and there's some some lots of interesting work on um on that sort of multi-agent perspective from the point of view of activing um so that's really it I'm just going to conclude with some with sort of a philosophical question that I think is is kind of important for um for tying this this together and and sort of checking that it that it just makes sense um so I've said earlier that um any system well described by the FP is resilient so hold on a second uh not all systems are equally resilient though uh right teacups by are sort of the Paradigm of a fragile as opposed to resilient object so since the FP describes things in general doesn't it mean that all things are resilient um and doesn't that contradict this obvious fact so I think the answer here is that it's simply it's a matter of scale so yes all things are resilient all things that is things that are reidentified over time and that persist through change um all such things are resilient but it's a matter of scale so if you if you introduce a very slight perturbation to this teacup um very slight you know um relative to a certain a certain reference frame or a certain scale but large enough right relative to the scale on which that system is resilient then by definition it will survive right and so obviously again in a in a a real Universe in which there are interactions and you're you're always interacting in some way dealing with some sort of um random fluctuations um then yes for the system to persist through time to be inert and elastic um it has some right the material has some uh coherence that allows it to to persist through those changes so it is a resilient system it's just that if you introduce a change that's um on a scale that's That's too great for the system to handle then it'll be destroyed um but sort of the just the basic coherence as a spaciotemporal object of this teacup is itself a form of resilience uh and then when we when we get into more complex forms of life more complex generative models that's when we when we when we begin to see more um impressive forms of resilience and shading it even into into plasticity and the ability to to really survive through more significant changes so thanks for for listening slw watching um I'm happy to participate in this Symposium um thanks to all the other uh the co-authors of this um paper by Mark Mol 2022 that was the departure point for this work uh yeah thank you all right the next video is going to be 28 minutes long it's from ma alasen sustainability under active inference hey everyone my name is mar baras a PhD candidate at the University dubec and director of research strategy adversus today I'm going to be talking to you about budding work on sustainability under active inference which we had started with multiple authors that I listed down below uh starting from the core components and scaling up to a computational model we are pursuing this research in multiple collaborations ADV verses around sustainability ultimately this is meant to Charter a path to much larger projects so feel free to reach out if you want to collaborate so let's begin with the most fundamental aspect of sustainability namely resilience resilience refers to an agent or systems ability to persist and adapt in the face of challenges or perturbations under active inference resilience is a multifaceted concept involving different strategies that allow a system to maintain mainin or recover functional stability resilience can be broken down into three core components inertia elasticity and elasticity so let's start by defining inertia it describes an agent's capacity to resist change when subjected to disturbances it really is the stability and robustness of a system in active inference inertia is represented by high Precision beliefs which can be thought of as highly confident low entropy beliefs about the system's current state inertia is reflected in the Precision weighing of prediction errors higher Precision implies stronger resistance to updating beliefs maintaining a stable State u means that Precision is implemented as a scaling Factor so here higher Precision really implies increased resistance to change the problem is that inertia alone makes systems brittle errors they cannot adapt so then we move to the second component elasticity elasticity describes the system's ability to return to characteristic or retractor States after perturbation it corresponds to the con concept of homeostasis so the capacity to adaptively recover in active inference elasticity is expressed in terms of an agent's ability to minimize free energy and seek stable States it can be mapped onto the relaxation back to attractor states which is driven by minimizing free energy as the system follows a trajectory in doing so restores the system to a preferred State analogous to returning to an attracted plant so for example in nature biological systems really exhibit elasticity through homeostatic feedback mechanisms like temperature regulation or blood glucose balance these systems have the ability to change but they ultimately return to where uh they were without necessarily learning and this may yield the same errors and ultimately can crap Under Pressure much like a rubber band pulled too hard so let's move on to the third plasticity it refers to the expansion of an agent's repertoire of adaptive States it allows systems to adapt not only by returning to previous States but also by exploring new States enhancing robustness so in active inference plasticity is related to functional redundancy and structural degeneracy having multiple Pathways to achieve the same outcome structural degeneracy means that you have different components perform overlapping roles allowing the system to adapt to novel challenges but we will dig into this a little later so here plasticity is captured through posterior beliefs that can change flexibly based on new sensory input or structurally if you actually add a new state we can use the variational density such that the agent can represent a diverse range of potential States maintaining a dynamic model of its environment think for example of neural plasticity where different neural circuits can adapt to perform similar function after damage to understand how plasticity comes into play we really have to hone in on certain key concept underlying it so complexity refers to the uncertainty or diversity of System Dynamics in active inference complexity is associated with the range of possible States an agent consider complexity is linked to entropy of the system State distribution higher entropy reflects greater uncertainty and the number of potential States the agent must account for variational free energy is a way of balancing two important things how complex our beliefs are and how accurately they match what we observe and so we can U balance this in a formula that goes a equals the difference between the complexity of updating our beliefs and how accurately these beliefs predict what we observe so in this formula the first part um dkl measures the complexity or the cost of updating our beliefs to it line with reality second part written as EQ represents how well our current beliefs can predict the observations we make so we said we were going to turn to degeneracy and redundancy and

I'm obviously here leveraging work um by no and Carl friston for degeneracy what we're considering is that the presence of different components are capable of fulfilling similar roles with in a system in her paper they talked about for example left hand right hand and redundancy means that you have similar components performing the same function which enhances robustness against failure the redundancy here involves overlapping functions in state transition matrices B which indicates similar transitions across different states and degeneracy can be Quantified by the entropy of the posterior distribution which represents the number of alternative configurations that lead to similar outcomes and so managing complexity through diverse beliefs really means that the agents can adapt to changing conditions degeneracy means the agent can switch strategy when one pathway becomes unavailable which promotes adaptation and redundancy ensures the continuity of function when certain components fail which adds resilience to system so degeneracy here helps reduce the complexity cost by Distributing probabilities across multiple States thus maintaining lowf free energy and supporting robustness in diverse conditions but while resilience is really critical for maintaining stability it doesn't fully capture the dynamic interaction between systems and their environments resilience focuses on an agent's ability to withstand or recover from perturbations but it lacks insight into the broader System Dynamics without considering systemic interdependencies focusing solely on resilience can lead to shortterm adaptation without promoting long-term stability this is work that Mark Miller and inish poo have uh written a lot about so for sustainability we really need that um systems not only adapt but also ensure the long-term viability of their environment and resource base we can through active inference integrate resilience and sustainability and and show the need for systemwide coordination for harmonious interaction as Carl likes to say active inference really is just the story of resilient and sustainable systems so let's Define sustainability a little more sustainability is defined as the enduring capacity to meet needs without depleting crucial resources and here resources is taken a little more liberally we can expand Beyond material resources and Encompass social ecological and System Dynamics sustainable systems are those that persist over time and do so in a manner that promotes stability and health across all Dimensions environmental economic and social we achieve this under active inference by longterm adaptability and interconnectedness across multiple scales sustainability emphasizes the creation of conditions that allow for continued adaptation without exhausting resources effectively ensuring that all parts of the system work cohesively to support resource renewal and plasticity so sustainability can be framed as optimizing the expected free energy over deep temporal scales which is work that Casper hesp uh has pioneered balancing immediate need needs with long-term system viability so the optimization of expected free energy can be represented as um f of expected uh SMA equals the risk plus the ambiguity of X of f expected s so risk quantifies the Divergence between the predicted paths I.E the future states of the system given an action sequence and past observations and the preferred paths which are ideally sustainable and align with long-term goals this encourages the system to pursue states that meet its sustainability preferences over time ambiguity measures the expected uncertainty in mapping states to observations and it ensures that the system continuously learns about its own Dynamics and the environment this helps the system reduce uncertainty and improve its predictive Dynamics um over time the components of expected free energy balance the drive to fulfill preferences with the need to minimize uncertainty and so this balance supports plastic responses that enable the system to adapt to change but also actively guide itself towards more sustainable and self-organizing configurations over extended time scales ensuring resilience at both individual and Collective levels but sustainability isn't just about survival or at least it's not a simple story it must incorporate UD dionic well-being focused on growth fulfillment and development and honic well-being which captures immediate satisfaction so in active inference this means that the system should balance present experiences with long strm strategies that enhance overall capacity for adaptive growth so when we talk about a dionic well-being we represent growth and the system uh the system's ability to find diverse beneficial States and it aligns with long-term sustainability by ensuring that inures survive and thrive through extended adaptability and resource renewal it represents the growth and systems ability to achieve its long-term goals this this concept aligns with active inference because it emphasizes long-term sustainability uh by optimizing internal models that allow agents to find those beneficial States and So within active inference agents act to minimize expected free energy which involves the balance as we saw between risk minimization which means that the outcomes are aligned with the agents preferences and ambiguity minimization which ensures that the learning uh continuous vious ly happens so that's what enables the agent to grow and Thrive and so UD dionic wellbeing can be conceptualized as the successful maintenance of favorable States and the expansion of the system's potential allowing it to reach higher order goals over extended time Horizons and so this Behavior reflects structure learning where in agents adapted changes in the environment by restructuring their inter internal models which lead to growth and a diverse set of beneficial States now honic well-being involves immediate moment to moment satisfaction which is necessary to maintain ongoing motivation and effective response to environmental demands it focuses on immediate satisfaction and so the agent's capacity to respond adaptively to current needs uh it can be seen in active inference as the system's ability to reduce momentary free energy and maximize instantaneous rewards so we can see this at the lower level say of higher equal models so you minimize prediction erors in the present you seek sensory states that satisfy immediate needs or preferences like finding food to satisfy hunger and this corresponds to reducing free energy associated with specific observations the internal reward or momentary satisfaction that the agent achieves after fulfilling immediate needs corresponds to the agent reaching a preferred State as predicted by its internal mod this is crucial for maintaining motivation and ensuring effective responses to environmental demand here we also have to factor interdependencies within the system these inter dependencies can be represented through mutual uh information equations which uh ultimately um quantify the connections between different components the plasticity at different at individual levels translate to the sustainability of the entire system aligning short-term actions with uh long-term goals so sustainability is an emergent property that depends on the balance between all elements in a system this really involves presenting resource depletion maintaining well-being and promoting resilience through self-organization and learning so it's a continuous effort to increase resource redundancy and avoid overe exploitation thereby ensuring sustainable practices over time you can think of it as avoiding a local minimum the sustainability of a system depends on harmonious interactions among its different components which range from Individual agents to the broader Collective this means that the system must align both internal and external goals to ensure stability over extended period periods enhancing resilience and sustainability so for example uh the resilience components of inertia elasticity and plasticity will work synergistically to support adaptive actions that contribute to sustainable outcomes the emergent stability is promoted through decentralized interactions where each component of the system self organizes and responds to environmental feedback which reduces the complexity and increases adaptive capacity and allows the system to maintain a dynamic but stable equilibrium also sometimes referred to uresis so we can think of attractor states which emerge as agents minimize their V variational free energy over time and this process supports the alignment of internal models with the environmental demands ensuring that agents remain adaptive while reinforcing the renewal of resources and overall system well-being so these are all nice ideas and you know we think they hold up mathematically but we wanted to test them out and obviously this is the beginning of the journey so we built a small and very simple agent based models to illustrate sustainability under active inference at the level of one single agent within one environment this is extremely simple and the following studies will cover more so the model here simulates an agent managing food consumption in different environments to ensure long-term availability our objective was to show that uh the agent's goal is to maximize well-being while ensuring resource sustainability over time and this involves balancing immediate needs with uh future availability of resources so we started with a validation case um in this case the agent makes decisions uh based on current food availability without affecting future availability and so the agent observes food and decides whether to consume or not uh based on its current state and PRI preferences so we have the generative model uh we have hidden States representing food availability and saity and the preference uh C which encoded a strong preference for sa while maintaining food availability as as you can see these two preferences are technically contradicting and so in a static environment which was the first case the agents decisions are really straightforward they just maximize short-term well-being without needing to manage resource Dynamics so as you can see uh in the tiny little plots here uh the we can see the the agent's Behavior specifically uh whether the agent chooses to eat or not over a series of time steps and so uh the middle plot shows that the amount of food over time which is represented as food left and the last bottom plot shows that uh the uh you represent the agent's level of SA or how satisfied or full the agent feels over time higher values indicate higher levels of sa so uh the agent uh has a strong preference for high sa uh since food is continuously available the agent keeps eating at the beginning which increases its saity level and since the environment is static the food remains available indefinitely and so the agent successfully increases its a uh we then wanted to show whether or not what we did was just an artifact of the Simplicity of the model so we screwed with its matrices and so incorrect matrices caused the agent to misjudge the effect of eating which leads to an unstable sa level so we we showed how errors in perception and action models can significantly impact the ability of an agent to achieve stable and desirable States even in an extreme simple uh setup so then we tried something a little more complicated which is a dynamic environment where the strategies have to change so uh the environment changes based on the agent's actions so resources deplete when consumed but replenish over time when not used and so the agent's challenge was that it had to learn to balance consumption with replenishments uh which effectively had to demonstrate some degree of sustainable Behavior so the model includes the transition Matrix that Maps the state of food available based on the agent's actions and expected free energy here minimization uh guides the agent to consider future State and ensure availability um so this was the case as you can see it hasn't changed much the only thing that has changed is the Dynamics of the environment and the ability for the agent to ultimately learn those Dynamics so the agine had to adopt adaptive strategies that balanced immediate rewards with long-term stability uh and this was showing us how active inference can foster sustainable behaviors in fluctuating environments um so when we look at this plot um we could see that the agent uh has learning turned on or not and we wanted to see whether an agent that didn't learn did just as well as an agent who learn just to see again whether it was an artifact of Simplicity of our model and also the agent can die if the resources are depleted and it has high sa it is not satiated over time it dies which you know ethically might not have been the best thing to do but whatever so agents that have learning quickly improve their survival Times by refining their internal models and strategies and so the rapid Improvement really shows that learning enables them to find efficient and sustainable ways to interact with their environment without learning there's no significant increase uh in survival time uh as they aren't able to really adapt their behavior based on past experiences so they end up uh in suboptimal decision Loops leading to a lower average survival so we can really show that learning is crucial for agents ability to adapt which in turn allows for a longer more sustainable uh presence in the environment and ultimately uh you know the also the environment's ability to actually be sustainable in itself the agent overcomes initial limitations and eventually converges to strategies that maximize their survival which really shows the power of dynamic adaptation in un certain environments so we saw that the agent's behavior in both static and dynamic environments really shows how well-being can be optimized while maintaining resource availability and in the dynamic environment the agent learns to moderate consumption it is able to understand the tradeoff between immediate satisfaction and future resource sustainability the agent has the ability to create adaptive strategies that adapt uh that align sorry with the principles of resilience and sustainability and so here our model illustrates in a very very minimal way how minimizing free energy can lead to behaviors that promote long-term viability rather than short-term gains and so these simple models are just laying the groundwork for modeling more complex systems such as ecosystems or Urban environments where sustainability requires coordinated interactions between multiple agents and resources so to recap we started by understanding resilience with an active inference we broke it down to inertia elasticity and plasticity we then explored sustainability building on resilience and we incorporated well-being as a critical component we demonstrated how agents can balance immediate needs and future viability achieving sustainability again all through active inflence and so it Maps systems that can adapt sustain and Thrive by balancing short-term adaptation with long-term sustainability but we still need to show how active inference can effectively be scaled to real world applications involving multiple interacting agents and resources uh involving the ability for the environment to fully deplete and therefore how to actually manage such possible um such possible patterns and how agents might learn over different areas um we have to transition from theoretical models to more practical implications that involve challenges in data acquisition model complexity computational scalability um we also need to factor Mutual interactions and the effects of each subsystem on each other and the larger system such that they promote more resilience and plasticity so all of this is extremely seminal we have to move further and actually demonstrate in much larger scale and more complex simulations this these ideas but it has critical potential applications achieving all this we can use active inference to model and predict adaptive strategies for mitigating climate impacts at different scales or improved decision-making processes in organizations uh enhancing resilience and sustainability in economic and contexts so active inference can be extended to other complex adaptive system and we can support initiatives aimed at fostering sustainability in health care Agriculture and urban planning which is work that again Mark Miller and Ben White teni poo uh are really pushing forward so thank you for listening and I hope enjoyed all right the next recorded talk is 50 minutes long By shager Ramen Humanity story a search for moral Clarity hi everybody so I'll be going over what I'm calling Humanity's story a search for moral Clarity and I'll get into kind of what that cheeky title means um going over kind of myth morality narrative uh self- understanding and um the meaning of life and no that one was too far but I guess not not too much far than one I'm claiming um so yeah before I get started I just want to give a big thank you to the active inference Institute um you know for overall the work they do um it's been super rewarding um and engaging to to be a part of uh I put my information here so feel free to reach out if anything picks your interest or you know bothers you or you have to correct anything feel free to um give me shout so we'll be discussing um you know a few things as I mentioned so one of the big roles um that I feel active inference can um you know one of the ways that I can at least intuitively scale into human culture first of all is through our use of symbols of what makes us unique as a species um can be understood you semiotically or through symbols um and the road there uh as I make the case in myth of objectivity is through morality um largely through bodifying norms and expectations which again ties very nicely into active inference I believe and the last way is to narrative and specifically I'll be talking about uh kind of at a meta level The Narrative of modern narrative and you know why are we why are we doing this you know it's you know epistemic reasons interesting as it is but also I think that kind of applying a r rigorous framework to humanity specifically Bly as we get into narrative um we'll see it more broadly tackle um some other areas of humanity um this is necessary I feel for cultural cohesion um something generally lacking these days um you know with partisan um and in polarizing um the nature of our culture at the moment in in our politics uh and so it'll achieve hopefully or at least understand the outlines for cultural cohesion uh and things like social and personal uh integration and I'll do this by as I mentioned by expanding upon my myth of objectivity hypothesis right and so my own road to to um becoming interested in the free energy principle and active inference was being disillusioned with the economics Paradigm um finding it to be as as I'm sure other folks were um insufficient to describing human behavior even through the kind of quote unquote narrow lens of the marketplace conceiving people as a neoclassical framework does as youtility maximizing um individual agents who are well-versed in what their own preferences and interest are um you know it's kind of quite unrealistic to say the least uh not only that but behavioral economics didn't really um do it for me either um it also held up this idea this neoclassical idea of rationality um as the kind of goal standard and you can see that with some of the recommendations you know the way we should um you know default constantly default 401ks um you know maybe that makes sense but it does it comes up with that advice it justifies that advice using this kind of principle of aity meaning an agent um preferences should not change over their lifetime which of course is not not realistic and so of speaking of um know behavioral paradigms that are lacking um you know at first I thought well the neoclassical model on its own isn't interesting um maybe we can marry that with you know some views of moral philosophy because I'm sure moral philosophers have you know a much better view of of humanity that's what I thought uh so here are a few uh few different moral philosophies that maybe some of you folks are aware of maybe others not um but kind of briefly um deontology utilitarianism are you know traditional from the enlightment ER going back to Kant um and Stuart Mill um paradigms of moral philosophy um interpretivism is a bit more recent but um I think still been around for for quite uh some time um and the last one is uh just a couple examples of some Eastern philosophy and so when we think about I guess the question I'll POs to you know those listening is what type of thermodynamic systems in terms of um their adaptability um you know how do we characterize these various different systems you know are they single equilibria are they steady States um are they maybe chaotic maybe they don't reach any equilibrium um you know how do we think about these um so firstly deontology and utilitarianism um and you know if not familiar with them um don't worry I'll kind of explain it but um the onology and utilitarianism um I view as single Lake album even though they are uh different um they seem to have one um kind of one unifying uh mistake which is you know the environment or kind of context doesn't change or in other words there is no uncertainty in these systems that they assume you know by their prescriptions of morality and moral philosophy so deontology assumes there are certain set of rules that you should never violate that um never change over time um so you maybe thou shall not kill thou shall not steal etc etc um there's no conditions under which those should be violated um and utilitarianism this is all of your actions are based their morality at least is um evaluated based upon the impact that they'll have on the the aggregate welfare of others right so um this is kind of featured for instance in uh the good place for those um watched it but the problem with both of these is that um you know in a word it's uncertainty right there's no epistemic foraging there's no model for how Agents come to learn okay what are the rules um the maybe the deontic rules that I should follow what are the ways that my actions impact others like what is that process of um socialization that kind of awareness those you know the empathy um you know all the things that you know you know should I read a book should I read fiction um you know how should I speak to people um you know all these are essentially epistemic tasks um of learning tasks that are not featur Ed into you know these twin systems um and and I'll get into perhaps what this reveals about Western Society um um of spoiler but this is the kind of analytical bias of Western Society I think revealing itself in the onology and and utilitarianism specifically so interpretivism um this comes out of postmodernism I believe or at least it was um rebr into um the dialogue following modernism into postmodernism te Etc so it says essentially kind of rejecting these uh analytical reduction single equilibrium as I presented it here um you know frames of moral philosophy and says that well it's based on the agents interpretation of their behavior you know what are the context for their actions also how do they interpret their actions etc etc well of course it's an important element of morality um the kind of classification that at least I would label is chaotic right so if if we're just saying it's up to someone's interpretation well how should they interpret it um should everyone interpret it differently is everyone's interpretation equally accurate how should we you know yada y y the point is that it's not super helpful for society um as a approach to morality all right so final two confusion and dosm um uh I see them as being adap of it being steady state um focus on the kind of rituals and behaviors not content specifically not focused on particular deontics um and not kind of hyperfocused on the impacts of behaviors to others although that can be a component as can you know interpretation but um it's really about the emphasis of these two paradigms um kind of an inherently acknowledging um that the world is changing that Society is changing and yet the goal of um morality at both personal and social level um is to find coherence so it's kind of still privileging um some kind of equilibria um some kind of Harmony but not kind of how you get there and I'll get into how it it does um with active inference so you know before active inference what I would say is that of Daoism and um Confucianism they come from a tradition that um Carl Yung would label as a the mytholog mythological construct of the cosmic human or Cosmic man um aka the symbolic self which is the you know term I'll use for kind of going forward kind of ties into um you know my theme of symbolic agents Etc so the point is that uh you know Carl after serving um doing a crosscultural analysis of various different mythology IES and and traditions found that this particular what he called AR pipe reemerged uh and essentially the cosmic human was the perspective that we all kind of part we all kind of connected to one another um not only to other people but to the cosmos and to the world um in general and our actions do have consequences because of that inherent interconnections um moreover we do have a certain responsibility to you know the world as a whole to other people um because of our place as a kind of a node in this hierarchy this some signaling where I'm going with this um but it allows as you can see um an area both for the kind of characteristic uh state so the ways to minimize uncertainty kind of using both that you know epistemic and exploitative um you know Paradigm of active inference um but kind of does so specifically um by conceiving this kind of broad view of the self right and so here of course is how it connects to active inference so active inference unlike the neoc classical framework and unlike other um single equilibrium um approaches to social science and to science in general it allows um nodes to communicate unicate um both with spatially and temporally and so kind of over time an individual um passes information to its or to kind of quote unquote its its future self and um in some ways it's its past self in terms of goals um but the self is not a stationary construct it is something that is constantly seeking um who what other nodes are kind of included in my hierarchy um what other are including kind of myself um and the symbolic self is an example of our ability to exate all these different nested nodes so our the result of us being a symbolic agent is that every time we meet someone say you know who says they're Canadian um well if if we Haven met this person before our model of what it means to be now include that person we now have a you know instance of the con the general concept of a Canadian of that kind of General archetype so that means that the model we have of Canadian whether or not we are Canadian um we have the ability to compute to constantly compute what the self is um and by virtue what the other is and so this the kind of Road the tra the trajectory that I outlined that kind of brought us to here and what it means and all that um I draw from our social bias so what are kind of some instinctual impulsive elements that first um started to see Human Society expand kind of outward laterally or even over time well at least laterally at least kind of um within a specific time uh I make the case that it is social bias something that we are inherently born with um you know although we are not um you know born racist per se um we are born with a specific bias and that's accent bias or shth it does seem to be something that emerges in utero and we have preference for um pronunciations for vocal accents uh over others those we are more familiar with and while that's all negative it does have clear biological benefits of course that's why it exists and that's why it you know persists today um and secondly what what do we do with a society that's based on people with the same accents um well that's inherently un instable one when that's not um super adaptive Al is extensive and allows for greater cooperation um it's unstable because allows for deviant free writing things like that okay so enter morality enter morals um and so here now we're connected in this um symbolic node Way by codifying those Expectations by minimizing surprise using morality right so this is my myth of objectivity um you're supposed to behave cooperatively you're supposed to follow these Norms by virtue of where you where you've grown up you know that is social bias that is shth um evidently you are not doing that um so how do I deal with that surprise well over time of course that model you have um the shared model you have of um you know that's based on identifying with those like you starts to wear thin starts to break down and so in comes morality as a way to stabilize um to offer kind of a hood that you place on um deviant and that you can then you know use to shame and guilt um into aligning with the shared model um or of course there's punishment noiz and other things right but that's what the the tool that morality probably had in the p and continues to have with us today all right okay that third leg um okay so if this is the self um what are we modeling off of right so we have the internal model Act of inference we have the Marco blanket that screen between us and the world so then we have the external States so what once we have kind of symbolized itself what is the world what exactly are we modeling off sometimes it's the world you can say okay we think about um farming think about you know someone who's born and raised as a farmer the land will feature very prominently in their model of the world think about a hunter the land wol figure um you know the world itself the physical world will um you know will mean something um however it won't mean everything so what are some other things that will constitute the world probably the other right so social bias of course Works into that there's those who sound like you there's those who's not or those not like you um so by virtue of having a bias means that you're biased away from other people and that's why we don't like to admit that we have biases um and that's why we were troubled when um people will tell us that we're biased so that's one example over the trouble um with the other and you can think about you know the Cold War think about any time that you know a country is um you know under attack the other becomes prominent becomes Salient um the in group binds together and you know for a while um you whether it's 911 or whether it's World War II um there's kind of great social cohesion trouble of Wars end um enemies become alent um and then what happens well this is you know how we have this is you know enter kind of Mythology not these things are entirely separable um there's a lot of overlap but um you know for our purposes if we think of you know mythology and we start to think of um narrative what we're talking about is uh in the language of Mel in fields we're explicating the reference rame we're exting the measure the kind of yard stick of what's good what's bad um where is you know according to mik 11 the frame of reference that allows agents whether you know bacteria or complex organisms to judge whether they should approach something whether they should avoid it is it an internalized um coordinates coordinate plane or frame of reference um what we're doing eventually once we get to myth once we get to um explicit morality once we get to narrative is we're exting that frame of reference we're saying okay um you know these group of people say are evil um you know we we're all descended from this um ancestor you know from you know Hercules from beol from this of half man half god um we explicating these reference frames to gain cultural solidarity to gain cultural cohesion right and so um I mentioned this uh this is kind more going into my um myth of objectivity hypothesis but I'll just going to briefly show it because you know I would love to get some feedback from the folks listening to this um so this is how I have um I hope to formalize um you know how we um you know became symbolically literate um if you will so that kind of passed from shith to morals um to you know full symbolic agent um and so I just have kind of an agent here um assuming that another Agent B has a fellow X and therefore shares the model of the world has some error know the error reaches some threshold um and then suddenly you know if then else um you end up um with a new model eventually leads you to um modeling the prediction error so here I have um a somewhat familiar equation but kind of modified but instead of um uh taking the where the posterior is kind of conditional on um traditionally conditional on the model um here in place of that I have you know an objectified error um so the point is that you have these categories of where you know typically the prediction error will just be whatever kind of noise that you experience that are not in line with the prediction um and either you'll select a different prediction model um different model at that point um or you just assume it away as um as noise in this case there is an explicit model of so when someone decides to steal food when someone decides to um not help with uh childbearing when someone decides um I can't think of any um moral deviations um but um the point is that there is a specific set of deviations from the typical predictions versus just being um surprised generically so it's a modeled version of error um that then allows uh an agent a to view a given agent um as possessing the ability to select between adherence to a higher level or superordinate model um now an objective model um of the group or a narrow one so they can comply or they can deviate that's a C and D um subscrib and the point is that um or you know I guess one one takeaway is that okay this isn't just metacognition um that of upper level superordinate G um and this is transc meaning that it's not um purely objective um it's framed as objective from um the point of view of the agent but that only means that transcending their own subjectivity and the subjectivity of the agent by conceiving that they have the ability to select between a higher level or a lower level model and so in this kind of simple way at least at this level of explanation we can see how active inference point of view we could have gotten into something like objective morality and more importantly something like Free Will and of course the various different significance and so just um again a couple you know I view this as um the kind of hypothesis hiding in plain sight why because morality is something that we're so impulsive about um something we have so many emotional um ties to um you know for instance um we're constantly looking for vict um we're constantly claiming to be the victim we're constantly looking for um people who victimize um and yet we can't really agree on what the specific contents of morality should be even though as I'm when I mentioned deontology I'm sure there was folks listening who had ideas of what could be deontic meaning forever forever there um so went over the some of this um you know there's you know social biasing uh morality there's impulse to do this um and it's some kind of research that I've done um in the economics going back to that neoc classical Paradigm although I was kind of dumping on it earlier um what these single equilibrium analyses allow for uh is the ability to say okay where does this model break down where does this model fall so you might say it falls down everywhere well that's not quite true um people are uh Salish they do have um stationary preferences at least for um a long enough time for that model to be useful um however it does fall down in certain interesting ways one interesting failure of the single equilibrium economics Paradigm is the need for social trust um and so social trust means that we have certain expectations we have expectations that company selling something sells what they promise to sell um it means that you can you know trust the market system trust the institutions um in the market that you inhabit and engage in economic activity um so you think that this at all levels you you essentially need it you essentially need to take medicine and understand that hope the people are um you know so ethically inclined um you know because of course if um they do something bad they might not got caught for it yada yada on point is that there needs to be some sort of um agreement on social trust so it's an example of how kind of morality seeps its way into something even as um even as intended to surgically remove all kind of humanity from the Paradigm all all morality all essence of morality from econ um the study of Economics social trust and morality seeps back into it um the same thing with um Linguistics um won't get into this one too much now but the de of transcendental signify so it's a way of structuring otherwise of elusive um set of symbols linguistic systems um we always have something there's always as I mentioned there's always God there's always some sort of myth that's tying everything together um even in secular society we have not um you know we don't have kind of uppercase God G God the way you made might have in the past you know there's a lot of different religions there's a lot of Institutions and probably agnosticism and secularism is kind of increasingly dominant um so what do we have in this case we have democracy we have um science we have kind of other um essentially kind of mantras if you will um that require this kind of shared belief um element which kind of has very moral um characteristics but is is kind of tied back into um a kind of shared belief system um then Heathers our uh you know norms and expectations too it allows for us um they kind of have you know social and political dialogue um yeah right um I yeah I spent a bit of time on the hypothesis um quickly um I don't know if I'll have time for uh this entire run through um but once we have established that um we achieved a symbolic capacity um of what is uh the next step do we just kind of walk around shouting symbols um no we have kind of narratives that uh attached to our symbols it's the way we kind of piece them together it's how we connect one symbol to another at a very um kind of crude way of describing this um it was a recent paper on narrative Theory and active inference um which I encourage um you know folks to go out there and and um and seek um it covers episodic memory it covers narrative identity as so some interesting topics throughout on narrative um the one thing that I wanted to add is um it makes the claim that narrative is for the future it's for prediction just as many things in um active inference are however I think that you could equally say that in order to establish um you know a productive prediction um we write narratives about our past so we can separate you know the person we were in high school um the person we were um you know in our 20s our 30s so we can we just tell stories about that about our last job in order to draw a boundary between us um and that person so it's a way of um kind of inheriting in in um inhabiting various selves um but not identifying not overidentifying with the past one so I just want to add that because I don't think it was covered in the uh in that paper um and so here I have a couple different of the core elements of narrative and why I think that it it SPS so well in the The Narrative Paradigm so I think the story describes as a set of you know the underlying events um so not explicitly what's in the narrative so if we read a novel um and you know the the novel is a short story about um you know Harry Potter it's uh exactly what's happening in that version of Harry Potter maybe he's um you know already at school doesn't kind of go into his backstory necessarily um um but the but the the story is all the things all the underlying events that happen that are related that are necessarily depicted in that narrative um and so this is essentially the the generative process right so it's the outside world that's um you know that's being modeled in um The Narrative and in the kind of causal chain that is the plot um so the point is that um uh I kind kind of quickly but the point is that when discuss or when of analyzing um analytically the core elements of narrative um you know a lot of the themes that emerge are kind of similar to um at least some of the key elements of um active inference skip this one for now um yeah um one thing I think it's important to notice is that there is implicit versus kind of explicit um narratives um and so there's this there's the idea of of feature placing um so this is essentially how um vocal vocalizations you know emerged prior to symbolic CU so you know other animals have um verbal Expressions they're non symbolic um a feature placement is um kind of one single verbalization um that might be some way that we can think about how um language began um and so essentially the kind of going back okay Concepts need uh context if we just s fire um what does this mean does it mean that I'm fired from a job does it mean that something is burning um or does it mean that you know I'm just um you know someone said something goes really cool and I'm just saying fire uh in that kind of slang context um so explicating the explicit so um here uh I just wanted to raise um the instance where creating narratives um although everything can be narrated um not everything um is narrated um so there could be unconscious events that happen in your past or childhood or even during the day that you hadn't really thought about so um various different therapeutic techniques um address that so internal failing systems is one where you address um know different emotional states or different psychological States and try to ify those and try to kind of give them um uh you know abstract structure in your mind by engaging with them you know through journaling or through uh dialogue with a therapist um so that's internal family systems there's also expressive writing similarly and so if we think about um one of the impacts that um narrative can have that that that's one and sometimes it's good to um you know not sometimes it's good or sometimes it's just um a fact that we don't have narratives like we don't have narratives on all of our feelings every day I don't think we would be able to kind of narrate everything that's that's happening to us um if we think about meditation and mantras the ideas there is that you know for a time at least we shouldn't narrate everything or at least we shouldn't accept the first kind of order narration um which we think about mindful minutes meditation and kind of focusing on their breath as opposed to fall following our thoughts um you know if for anyone just practice meditation you know you'll know that um sometimes we should just you know be quiet and not listen um to the narratives that that show up in our brain have a couple different obviously examples um right so here what I'm talking about is and as I mentioned on the previous slide or I had it written down I don't know if I um spoke to it um but what we're talking about here is that the problem of in promise of cultural narratives as is most known as Master narrative or grand narratives um that traditionally would would have been the purview of Mythology and religion more modern times um is kind of constructed um in other ways and so um as I mentioned from our nomadic multi-band past um to even agrarian societies um and then kind of coming into the universal age these um started from these fully integrated narratives where a particular nomadic um HRA tribe would have a single kind of Master cultural narrative um or at least that's what we understand as our kind of best guess is that this was the case or wasn't um you know which religion am I part of I'm part of my tribe's religion um you know why did uh we catch food today and and and not yesterday there was you know an answer from the shaman for that you know this this it was fully integrated um the second kind of epoch was the leading up to kind of the the axial age when the universalizing religion you know the setting the stage for Christianity um you know Hinduism um you know Daoism that that I mentioned um so for a very long time um this is kind of following from you know 10,000 years ago when we um started farming um perhaps can tell recently we had these dominant religions um that essentially kind of took over the the cultural sphere um those are the the universal narratives religious Nar I'm sure folks are aware of and most recently in um weird Societies or Western societies it's kind of Western educated industrialized rich and Democratic um that's the an acronym for for weird there um and my case here is that these are you know we're larg in a culturally fragmented environment so earlier when I mentioned single equilibrium analysis neoc classical economics but specifically um deontology um and utilitarianism those are all examples of from I think a thermodynamic systems point of view from active inference point of view this would be kind of a um a very um single equilibrium um you know kind of ways of looking at the world um and why are they well um you know according to Joseph enri and I think According to some similar analysis from um Ian Mel Christ um this is because Western people are distinctly individualistic um they're a bit more left brain um centered than um at least the axial age portion of um Society so those um in traditionally of settled societies um that are more justult and more R brain oriented um would think of things a bit more holistically rather than as kind of reductive so the good uh of course there's um know many benefits for left brain thinking it's um more individual it's more analytical uh and it'll be a bit more geared towards science um versus mythology versus um stories and narrative um but what is it missing well it can make a lot of um mistakes and not random mistakes um a specific type of uh mistake which is um leads to kind of reductionism um and which is how we kind of ended up with which might be according to these views how we kind of ended up with um some of the paradigms that we did and so the the fragmentation we can understand along kind of three different these three different dimensions right and so one is aesthetic uh the next is kind of institutional and the final the last is uh kind of objective and so um this is essentially how I see our Master narrative currently broken down um and of course it could be additional ways but here's of three ways as a starting point to understand the kind of cultural narrative um and how specifically it's it's fragmented so aesthetic we have the kind of modern era the postmodern era and um now what some people are calling the metamodern era it's maybe a bit too soon to um turn on it however I do think it's super interesting especially in the light of the free Eng principle um which I'll hopefully have time at the end to kind of um touch on um the second one is our institutional um narrative so uh e economic framework um you know how we should distribute wealth um you know what does that say about us and and kind of what's the most quote unquote Vision Way um but what are the narratives around um our economics what's our narratives around our politics um and finally there there's these idea of uh of you know objectivity still kind of creeping in so um what are the kind of quotequote true narratives out there um and so so you know reasonable question is okay this is you know we all have kind of our own stories um perhaps there's still vague senses of a master narrative and I think you know there always um are uh there always kind of might be um at least some emergent um Master narratives by virtue of us kind of all occupying the same at least you know digital space the same kind of time um well Converge on some things um however without a um kind of coherent or a more of like a religious story that we um perspective that you know we had in the past where um you the narrative is not just um okay we're this you know Third Rock From the Sun we're revolving around um we revolving around space and you know we're you know victim to two laws and um this is what happens to be happening with Society um what we lose with the kind of grand Cosmic narratives of the past um the type of psychological um coherence um so where do we fit into this world you know the idea going back to kind of cosmic ban you know where is our place in the universe um what is the kind of personal meaning for me that's where a um a tighter narrative kind of more intentional Master narrative um comes into play so our current fragmentation um leads to personal anxiety can lead to social anxiety um and it can lead to kind of the polarization uh and partisanship and the results are fragmentation institutional misalignment whether the company you're working for um shares your values um whether they're working on something that is you find deeply meaningful that is deeply meaningful for your identity um and your values um that's um what you mean thereby misalignment okay so in terms of Aesthetics um as I mentioned um the per own is is sometimes called metamodernism um and so that kind of gets into the idea of genres meaning that we have kind of plenty of good um you know television so we have plenty of good Nar is available um for us to consume um that does include um kind of classic genres um but the problem is that they're all genres so we're all kind of consuming um content we're all consuming it through different media um even in times past um not even too far past uh we would watch TV at the same time you watch the same movies because because they're in the movie theater that's where you could that's the only place you could watch them um that b so recently um now we're watching movies at different times moving from movies from um 20 30 years back um from 50 years back and the essentially the fragmentation um has gone up an explanation as I mentioned there there's an analytical um orientation here um I think it's it's worth noting that along with the fragmentation that has occurred in weird societies um along kind of side that that kind of loneliness that happens um with the rise of uh the analytical world is some great some great literature um and so one of the reasons I argue that this came about was specifically because um once theology once religion stopped serving the role that it did um that left in particular artists in particular those kind of aesthetically inclined to seek out um art to create art in some cases to fill those gaps for um offering a narrative for answering the questions the big questions that were typically answered by those institutions yeah um and so kind of going back to my quest what we're kind of after here is we're looking for greater um cultural cohesion we're aiming for better personal and social integration so I argue that these pieces are connected um and I'd argue that if the free if the freeable told us one thing um it that agents minimize uncertainty however it tells us another thing is that one way to verify kind of any Claim about um a self-organizing system it's to not look at one Paradigm assignments to look at is to synthesize various different views so what we can glean from that is essentially a at a modern approach um which is to not necessarily ensure that everyone in culture everyone in a particular society is consuming the same content or media but it's to highlight certain aspects of it that each person can find meaning and the right type of meaning in their own content and even if we achieve that through different areas we'll all be experiencing something similar with each of our traditions of and once we've identified the certain components or make the case for certain components um that answer maybe some important questions from acers point of view and allow greater visibility into that uncertainty minimizing process at a cultural scale um so here I think poetry is a great example because of it um you know it reaches everything from the senses from its sounds to its visuals um to its more deeple deeper symbolic meaning and I think great great example of that is TS Elliot um the great many references he makes in his work um also um the kind of poetic conventions he aderes to makes it very uh in some cases accessible um works that he offers and so whether it's TS Elliot or Kendrick Lamar um in each case what these artists constructing is a explicated symbolic self an explicated um a number of different nodes in a hierarchy um that are connected with all these various references in a particular song um that are that can be made clear you know for instance in a um this website it's genius so you can see where the different um lines in this case Kendra Kar has and what the references were and so in a sense what you experience when we read poetry or we read a work of fiction that has um deep references deep um cultural references like this it's constructing an extensive self um bringing coherence to it and to everyone who listens to it um and by kind of calling it out you're able to share in it almost as if you know it's something you know it's a greater work of literature in the past um as folks would and so that I think is is of a few examples of um ways we can start to apply a rigorous framework to the humani that tells what kind of literature does for us why it does what it does for us um and parse through some of the components that make it meaningful to us and kind of allow us to um use literature um in ways that I think it was originally intended to in past times has so leave it there and um you know once again feel free to um reach out to me with my contact information and um and please you know follow my um you know substack and and let me know if um anyone has any feedback and I just want to thank Daniel Freedman again um and you know all the folks at active inference Institute and Carl priston for um the this content and um the discussions here so uh thanks again all right the next recorded talk the penultimate recorded talk is by Bradley Alysia earlier Bradley gave a talk on purposefully non-standard cybernetics and this talk is going to be an overview of Open Source and open science so that's really cool and thanks for bringing this important complimentary topic to the Symposium for all all who are working on this okay it's 51 minutes long hello my name is Dr Bradley Alisa and I'm a member of the active inference Institute scientific Advisory Board today I'm going to give you an overview of Open Source and open science I'm my home institution is the orthogonal research and education lab I'm also affiliated with the open worm foundation and the University of Illinois champagne or champagne I'm also with particular interest in the orthogonal research and education lab is our open source interest group and I'll talk about that throughout this this talk this starts with a question what can we do with open source and open science and I pair these together because sometimes we're building software sometimes we're doing scientific discovery which kind of relies on the open source software and so for doing both we can actually build uh systems that can sort of help us do better software engineering and science so they're kind of grouped together so in terms of open Source you know it it's maybe a trism that open source is about software and putting it out into the world but that's not enough and you really want to what you want to do instead of just putting software into the world is building a community around technical standards and code and this is true if you want to build a community of users or an open source community that helps build the software because you need to have some interaction with the software to make it sustainable you also want to have good technical standards and and with academic softwares we see that's been lacking this also allows us to open up the organization's processes and products so having these communities and having these technical standards are not only good from Just sh this stamping of sharing software but it also opens up our processes and products and allows us to serve our communities better our academic communities open science is where we share data share results share research artifacts of the public so we do a lot of sort of transparency related things we give people our results and our data and it's software and papers and presentations and we're very open about it we can give them access to these things so when I say open science I also kind of mean Open Access and it's they're kind of in the same bucket open science and general allows us to open up our organization's research capabilities so that people can benefit from our discoveries and we can benefit from their feedback so a lot of organizations have proposed different sort of road maps or statements on open science and what that involves so not every organization's view of this is going to be the same but I'm using the young European research University's Network statement on open science at the link below and to them you know they have a very strong emphasis on collab collaboration and citizen science so this is where you have collaboration around the world around the scientific community and also involving say citizen science where people just want to contribute their own um analyses sometimes it's data sometimes it's observations things like that there's also visibility and idea sharing so know being able to get to the outputs and read them and understand them is important open education is important idea sharing because you need to have you know uh focus on collaboration alternative models of publishing you need to be able to publish the work to get it out there into people's hands incentives assessments and research Integrity so those are all kind of part of this open science tree now again every organization who Champions open science will have a different view of this we don't really have a formalized view in the orthogonal lab but it follows this kind of outline line close quite closely and so if you want to know more about the where the open or where the orthogonal research lab is coming from you can read this paper and I put this out in 2020 and he needs to be updated but it's a good overview and this is building and distributed virtual laboratory adjacent Academia um and and so there are a number of things in this paper it's just basically it's very similar to the model that the active inference lab uses were much more focused on sort of educational opportunities and opportunities for people to sort of pick up the load and build uh research artifacts and you know having students come in and do short-term projects but it's it's it's a very similar process so one of the processes or one of our goals is to integrate open access in our open source practices for advancing distributed research projects so we want to be able to help people make these things open help them build research projects the other thing is that we want to encourage practice and open working through the creation of digital research artifacts it's like papers or presentations and when I say open working I mean I borrow this term from uh the Mozilla foundation and their approach to this whole Enterprise is basically working open so it's working in these collaborative distributed groups and how to make the most of that that kind of experience you know things like note taking and meetings and managing sort of the human capital so that's also part of it open source and open Sciences managing human capital managing these work groups so one of the big thing other big things about open source and open science is this idea of practice and so you know it's it's very practice driven as opposed to sort of theoretical or philosophically driven and so you know practice means that you need to establish practices procedures routines workflows and

Technical Solutions for working open so this means you need to adopt open practices it could be Community review of code releasing code under a license pre-printing manuscripts and generating open data sets so these are all things that involve you know planning but also practice and being able to produce something at the end related to that are what we call open workflows so when we want to release an open data set we can't just pop data out onto the internet we have to put open data sets into a production line model where we you know clean the data we analyze the data we produce metadata and then we publish that formally in a repository with metadata attached we also want to have Community standards for open participation so we can't just expect people to pick things up and run with them without giving them some standards for working so you know you know usually in academic work you're trained a certain number of years it becomes kind of intuitive to you what to do but a lot of people especially students undergraduates they come in and they don't know what to do they don't have like those standards ingrained and so you have to have Community standards but also because sometimes the incentives aren't there for people to follow long lengthy procedures and so they don't in any case this helps you standardize this e these efforts then we have open Technical Solutions so open source and open science are built or enabled by Technical Solutions you couldn't really have open source in open science without having open Technical Solutions like GitHub live streaming tools and graphic tools and of course those things don't make something open source or open science by themselves but they enable those practices so this is a nice graph where I kind of talk about vision of sustainable open science and so I have a couple of categories here to the left and the right which are sort of suboptimal areas and then where we want to go is we want to go to this Zone this elusive Zone in the red box so the elusive Zone the red box are the set of tools that would allow us to do sustainable open science and Open Access and the stuff to the left and the right are kind of like things that we do now that are suboptimal so we start with the suboptimal and we talk about black Open Access which is kind of an fancy word for piracy or informal sharing and so each of these types of Open Access are colored black just means it's kind of related to under the table or you know piracy or something so one of the examples are is silab where you know it's I my position on piracy is it's Justified given the incentive structure in science but you know you have systems where you can just get access to things and then of course sometimes we share PDFs by sending emails to our colleagues and sending them the paper and it's all black Open Access because it's very informal and it's not really sustainable because you're relying on people to have you know a you know someone has to get access to these and then share it with someone else um then you go over to the other side you have two other areas of Open Access you have gold Open Access and diamond platinum Open Access so gold up and access is where typically an author will pay an APC fee which is a publication fee it's usually a fairly decent sum of money to a publisher to make a paper open access to all lot of tend journals will restrict access to paying customers if you are a member of an academic Library you can get access but if you're not then you don't have access and that's one of the things about like places like the active inference Institute or the orthogonal lab is not everyone has that academic Library access and so it's really hard to get things get access to papers and things that you need so this becomes a very important set of points here gold Open Access then of course is is beneficial to the Publishers but not to the authors Di and platinum Open Access is more beneficial to the authors there's no APC fee but someone has to take that fee and you know they have to absorb that fee so that's not optimal either so this elusive zone is a way to sort of build on sort of the lowcost but still like sufficient resources we need to provide access to our work we need to open it up give everyone access to it and allow them to interact with the work in a way we couldn't even do with any of these other types of Open Access so the first thing we have is green Open Access which is pre-printed this is where it's like not the gold standard but it's also you know you can post a paper on a pre-print server there's a pre-print server that gets funded through like a foundation and you know that's that makes it open de um there you know formal sort of you know there's a DOI which is a formal link it's a permanent link so your work is in a place that has you know some level of reputation with with a mark of approval on it you can also take that preprint and go through what's called open peer review where you open up your paper to peer review not to like three referees you've never met but to the entire community and the thing about pre thing about peer review and I won't get into this uh any more than what I'll talk about here but if we do peer review with like three people on an editor you know you pick the people who are maybe the most available or maybe the most visible people in the field or maybe people with the most time and they have certain opinions about what you should do what you should add on to a paper to make it publishable and sometimes it's not in the best interest of the science furthermore given that they still don't really have a very good handle on detecting things like mistakes or even fraud so you can have a paper that goes through a three reviewers and an editor and still be retracted for mistakes or fraud so that's not good either open peer review allows us to have people many eyes on the paper many comments you can incorporate the comments by generating a new version of the preprint and you have this continual feedback of your work and you can eventually Maybe publish it somewhere but you can also update the pre-print continually and have that process going um so that's you know Green Open Access and open peer review now a lot of papers also have data and we can include open data management in this elusive Zone because we also want to have a strategy for managing and sharing data and this goes along with the peer riew and the access of the preprint that that's the link to it and that the data can then be reviewed and worked with independently of the preprint but also you know it it's transparent and people can actually work with it far beyond the LIF span of a single paper then we have overlay journals so we can have preprints we can have but that's sit on a pre-print server we can have multiple versions but sometimes we want to build our own journals because we want to narrow the focus of the preprints funnel them down into things of interest for certain Fields so say for example we had an active inference overlayer it would take preprints in the active inference area only of those preprints and present them as Journal content so what journals do besides inferring status on people is they give you this curation of the literature of of research so that's what overlay journals can do but without the overhead of a journal so you can have like your preprints you put them on this with the skin over over the top that gives us this emperator of uh you know what a journal would give you so that's the elusive zone of Open Access um and I just wanted to point that out because I think it's you know a good way to think about sort of the production of Science and the production of academic work I posted a blog post for Open Access week 2024 and it's kind of a critique of open accesses that exists and this is why I kind of went through this and basically I argue in this post that open access is essentially failed at least up to this point that we've don't have the right incentive structures we don't have the right intrinsic and extrinsic incentives to do open science correctly Open Access and so we need to change those and so you can read that post on my blog synthetic daisies so in that blog post I talk about the intrinsic and extrinsic motivators for doing Open Access sustain so this you think of this more generally as a reinforcement learning of working open or reinforcement learning of open so when I talk about intrinsic motivators I talk about the internal drive to do something so you know the idea is that you would do things that are meaningful to yourself that's an intrinsic motivator or what motivates me in terms of a scientific question what motivates me in terms of how I can contribute to the field and then how can I benefit from or contribute to a team these are all intrinsic drives these are things that sort of you would generate and you interact with the group based on those things by contrast extrinsic motivation is the external encouragement to do something so this could be money paying people this could be cost and benefits so if there's a cost to opening my data set I do it if there's a benefit to opening my data set I will do it and that leads to an inequality within the scientific community and then of course Prestige which is of course if something is prestigious I'll do it but if something isn't prestigious I may not do it so it interacts with intrinsic motivation but it also is this external force that that sort of is encouraging people to like a you know an a reinforcement mechanism extrinsic motivations can also be where things tie into existing activities so one example is where a lot of funding agencies and journals will mandate that people share their data if you want to publish a paper you have to share your data so this is like tying data sharing and doing this to publishing the problem is there's no intrinsic motivation to do this it's not meaningful to everyone and so people will do it to differing degrees of completeness or of usefulness so if you're forcing people to do something versus it's something that people want to do and think is a good idea it becomes uneven and you don't have the same quality everywhere sometimes there's you know of course there other types of mandates and other T and there's been General uh an overemphasis I think on mandates and sort of top down motivation to do things in open science and Open Access um trying to get people to adhere to certain standards or trying to enforce more uh uh trying to enforce moras that may or may not fit the situation so this is a problem in open science Open Access and even open source where where people you know you have to kind of figure out the psychology of the people doing and enabling a lot of these open practices so if we look at this graph we see that their intrinsic motivations and extrinsic motivations that interact so at the middle of this in the gray box you have intrinsic motivations things like I want to do science I have Theory building goals I need to create tools these things feed into extrinsic motivations such as funding Community Building uh adhering to the standards of the field and adhering to rigor standards of rigor so this is all kind of these these intrinsic and extrinsic motivations interact and then contribute to how you carry out your open science Open Access and eventually open science so now I'm going to shift to sort of how we approach this in open source and so this paper from neuron it was published in 2017 it's called the commitment to open science and open source it's called the commitment to open source in neuroscience and uh P gleon hrew Davis R Angus silver and Georgio ESC were the authors they're are involved in an organization called incf which is the international neuro informatics Corp and they've all kind of done work in this area and they're contributing to a project I'll talk about in a little bit um that's open source sort of solves a lot of problems in Neuro Imaging uh and how to do that so one thing that they point out in this paper is that code sharing is useful for a number of things in science but they make a claim that releasing code can be used to replicate results so one reason why we do code sharing is so we can replicate our results we can be confident that those results can be replicated because we have the code and everything is you know perfect um which is not always the case but this is the claim this is the motivator that they're talking about and so sometimes you have to be strategic about how you release your code sometimes you know you just release your code as a matter of you know requirement but sometimes you can actually benefit from releasing your code and in this CA in the paper they talk about releasing core scripts and libraries before publication it will offer early feedback from the community so you can release your code just as a matter of sort of requirement or you can benefit from it by releasing your scripts and libraries early so people can debug them and help you if there are any problems with them or whatever and so you end up with this nice um optimal situation another paper that I'd like to highlight here is this paper from nature technology a technology feature in the journal Nature about six tips going public with your lab software so you know it's not enough to just write the best quality programs and to sort of put it out there you need to really kind of think about a lot of other things going on under the hood so one of those things is that you need to sort of have carve out some time for maintenance so if you put software out into the world the reason you're one of the reasons you're doing so is to sort of serve your community you know sometimes you want to be the standard in the community other times you want to give this gift this to other researchers so that they do better research sometimes you want The Prestige of having like the top software in your field but whatever the reason you want to be able to maintain the software so that it has a longivity and so different sources that they talk about in this paper estimate anywhere from five to 15 hours a week to 20 30 20 to 30% of the time of the project to be spent on maintenance maintaining the software now as you might guess you know there isn't a lot of motivation for that unless you know your software is really successful and so it's it can be a problem if you you know take the leap into putting your software out there but don't devote the time to maintenance it ends up hurting you later another problem is that you need to simplify runtime so a lot of times you'll release software and people will try to install it and fail because they don't have the right dependencies they don't know how to run the software and especially with a lot of academic software there we kind of build a clu of plugins and other types of software on top of software so that's not ideal and that's a problem that can't be solved by necessarily by in incentives and motivations some of it is going to depend on how we implement the software and I'll talk about that in a little bit you also maybe want to spend time in interface design automated testing a documentation to help users who you know when you're in your own research group you're sort of an bubble where you understand what you're doing you've used the software you understand the specific problem for which your software is written well when you release it into the public you might get people who are not as experienced or simply don't understand you know what some of the variables are or whatever and so there are a lot of ways that you know making this easy to use is a is an imperative and then a need for transparency in community building so if you release software you need to have users to make it worth your while and if you have users you want to build a community around that so that you can do things like have a more you know have a user base but also have a community where you know you can solve problems collectively and address open scientific problems and really for everyone to benefit but also transparency because we need to understand every 's methods and we don't want to make it you know obtuse for people when they use the software to understand the methods that went into generating the result that they're getting so you need to think about all these things when you open up your software when you go open source with academic software especially this brings us to another point which is the need for research software Engineers so in Academia we usually homebrew software we end up putting it out there and you know we maybe find time to maintain it we find time to build a community but the software is not up to the standards of say professionally designed software and so one of the things we don't have is our research software engineering skills in Academia a lot of academics know the technicals matter but they don't know how to build software so software needs to be maintained and needs to undergo quality control and so those things are you know separate skills so the problem of course is that in Academia we tend to have graduate students or open source contributors build packages or whole software platforms and their contributions are well appreciated but their involvement is itinerant so that means that they might build something and then leave the lab in two years where they may you know contribute to an open source project and you may never see them again and so they built something with no documentation and obious viously no maintenance and it just kind of sits there and Withers so I've seen this a lot in my career where people will build software for one purpose one use or like for a single paper and then it goes away because it's not maintained I mean there a lot of code I've written when I was a graduate student say where that was the case and I wish that we could recover that software and maintained it but you know who has the time so really what you need to do is you need to include research software engineer roles in your organization in your academic organization and sometimes open- Source projects will have this expertise but sometimes they don't because there's no requirement for it it's also a different skill set than say like the academic skill set that people building the software might have so this is something where you can either cross Trin people and they're programs where you can cross Trin research software Engineers but more often than not you just have to help people acquire those skills and so there all sorts of roles for RSC in different you know academic and open source organizations things like intentional software design continuous integration release Cycles bug fixes and maintenance strategies and those are all things that are separate skills from the academic world separate skills even from like you know if you even if you're computer science researcher you're not necessarily a software engineer so you have to have that set of skills or at least you know have a way for people to acquire them okay now we talk about open data so open data is also a part of open science and as as well as a part of Open Access so Wikipedia defines open data as openly accessible exploitable editable and shared by anyone for any purpose so it's an data set that you can access you can exploit or change you know you do analyses on or whatever you can edit it and you share it and so anyone should be able to do this now when they say any purpose they mean that it's licensed under an open license so an open license is a permissive type of liability protection and copyright for people who put these things out there that they work on and they want to benefit from in some way but they also want others to benefit from as well so it's it's a legal regime to maintain you know to manage liability but also to manage the social relationship between the person making the data set producing the data set and the person using it open data allows their data to be easily shared and reused but requires governance issues of ownership and adherence to community standards so this you know part of this is you know open is in sort of producing data sets for public consumption the other part is working on what they call an open data stack so an open data stack is a management strategy that is needed for sharing reusing And archiving data so we typically start at the bottom with some sort of interactive interface for uh working with data we need a metadata format we need a Cloud Server to host our files on to host our data and we also need standards so we can have these higher level standards for data production for open uh science production in general things need to be findable accessible interoperable and reusable so we need to have those four principles embedded in our workflow and then we need to have a strategy for archiving data and for that we might use something with dryad which is a a data repository so we have these techn this technology layer a standards layer qu security access layer and an archival layer so put together that's called a data stack and you want to build a data stack that's you know with tools that you know how to use and standards that you know how to adhere to but also things that last you don't want to pick something that will last a year and you have to migrate to something else so you want a sustainable data stack with sustainable tools with things that are easy for your team to use and so this also holds true for our Open Access publishing strategy we don't want to pick a strategy that isn't sustainable because we'll stop doing it now in the sea elans community and this involves my work with the open roomm foundation you're fortunate in that you have many open databases so the neod seans is often used in biomedical research and there's some basic research on it as well and it's a tightnit community where there are many many open data sets meaning people produce data they share them uh with the community and they have tools that are curating a lot of these data sets and papers and and things like that so we have all sorts of data sets uh available for all sorts of modes of investigating seelan biology and behavior so we have a number of open databases covering Anatomy genetics gene expression regulatory elements Behavior conomics so this includes like the worm wearing site which includes data for anatot conomics so we know in seans we know all of the neurons in the conecto we know a lot of their connections either through Gap Junctions or through synapses and we have worm wiring diagrams for both the male and the hermaphrodite so we have very rich data on worm brains we also have worm Atlas which is an atlas of structural Anatomy so on top of the connectome data you can fit structural Anatomy data and that's at worm atlas.org the seans database uh run by MDC in Berlin features a lot of data on proteomics and RNA and so this is another database that we have available senen is a database of gene expression and celegans and the nerve system so we can you know build a model of the connectome we can have the worm wiring data set senen data the worm Atlas the mdcc Elegance database and we can actually you know produce a study from that so we have all those layers of data that we can include in and say a model or an analysis we also have this uh resource wormbase which is where we can explore the world of celegans Def find mutants so cgans has a number of defined mutants for specific genes they specific mutations that we can backcross and so if we have a strain of say some mutant defin mutant we have this worm that has a certain mutation that might be expressed in some Behavioral or anatomical deficit and so we can compare that to the wild type and say things about mutations of function and so this this is a really useful resource both for experimental work and for looking at the celan genome and finding you know like alignments and other types of data like in one of papers we recently did we found uh different types of we aligned different uh open reading frames within genene so it's really a good resource and then of course uh nature methods this paper uh by people in the open worm Foundation um and this is called an open ource platform for analyzing and sharing worm

Behavior data and this is um on something called the tpsy tracker which is a movement tracker so they can actually take microscopy images of celegans they can track the worm's movement and they can characterize different movements they stereotypical movements worms make and they can actually track the behavior of the worms and classified so that's you know that that's the kind of data that's available in seans and so if you think about the active inference Community you know what kinds of data sets and what kinds of resources could be built around things involving active inference so you might for example think about you know and and some of these resources are just more General resources that you could bring to bear so you might say for example you know go to a neuroimaging resource and then go to a behavioral resource or build a behavioral resource there are all sorts of ways that you can build these um open data repositories to help the community Aid the community in solving problems and answering questions so this is a talk uh by eof freed this is called open science is an antidote to poor practice cynicism so one of the things he advocates for on this video is to use open science as a way to combat poor practice like we said open science is about reinforcing practice and when you reinforce practice of open science you reinforce good scientific practice in general because you're focusing on the process you're focusing on you know cross checks and and you know other things that involve teamwork and so there there's a lot of this sort of advocacy for helping science through open science but also combating cynicism where when we see open when we see poor science we often think well this is just the state of science that it's actually failing or it's bad and so you can fight disinformation as well with open science this is a nice set of views so in the orthogonal lab we have done a lot of work modeling open source communities and their sustainability so I talked earlier about Open Access sustainability now I'm going to talk about open source sustainability and so we've done a lot of work with modeling Frameworks to understand this problem so why is sustainability important well in the case of Open Source many projects fail for a host of reasons sometimes they're related to people's projects you know not being of good scope where they weren't asking very good questions sometimes the team falls apart the team is dysfunctional sometimes the project runs out of money so there are a lot of reasons why projects fail and sometimes open source projects especially have problems with maintaining containers problems with people contributing so they fail for those reasons as well so you can have things like we're scoping of a project we're trying to do too many things we're trying to do too few things we might have bad management or a lack of participation or we might have no resources in terms of time or money so these are all things that we want to be able to figure out how to manage at a reasonable level and to do that we have to understand sustainability so in these you know in these models that we've built we use different methods including active inference reinforcement learning and large language models to model open source communities and understand what makes a sustainable so the first one is this preprint on that kind of includes a lot of the work that we've done uh especially in terms of reinforcement learning models and active influence models this is our research search gate and then there's this newer work on com combining agent based models and large language models this is sarasta Wall's Google sumar code project from this past year so she actually presented in front of the Google sumar code um uh Symposium and and she presented her project there and so this work so this work on large language models involved using large language models to generate issues and then generate agents that could solve those issues and see which kinds of you know almost create fake projects and then create issues to solve and then see how the agents were able to solve them and so you know we used the combination of agent based techniques and large language model techniques to sort of work this out a very exciting project another approach is to take the multi-agent reinforcement learning route and so you see this grid down here on this figure and all these dots and they represent agents that are doing different things sometimes they have different statuses open source Community sometimes they're of a different level of Competency and you can actually run these agent based models you can Implement them in Misa as a python package and run these agent based models and then use reinforcement learning algorithms to sort of optimize the strategies that they're using so this is you know again where are using these uh sets you're creating these projects you're using these sets of issues and you're measuring their ability or their competence to solve those problems and so uh shomon RV Raja Galan and hu shle were all involved in this work over the past several years we've been doing these kind of reinforcement learning models for understanding open source sustainability then we've uh done this work on Collective cognition for AI ethics this is where I applied active inference and we modeled agent behavior in terms of active inference using a package called interactively and this is the GitHub repo here interactively this applies a partially observable partially observe Markov decision process or a pmdp model uh to the problem we also used a moraly arm banded approach to model explore and exploit Behavior so explore and exploit is sort of this tradeoff that agents make between exploring a project exploring issues to be solved and exploiting specific issues solving them so it's like making you know coordinating agent behaviors and then figuring out what strategy they're going to use and modeling all of that so it's a very interesting project uh it needs more work on it so if people in this group in in the active inference Institute are interested in following up on it would welcome and so there are a number of issues about maintenance that are very important to sustainability and this is a special in the academic sphere because academic organizations and open source typically don't have the resources that private organizations might so maintenance and sustainability issues are multitude in open source so I'll talk about a couple of them here so the first one is Log forj and this happened a couple years ago where there were a number of Open Source projects that got hit with the security threat and it was because their security fell at a sync with the current state-ofthe-art so in open source we don't maintain things as well as say like commercial products so you have these holes that can be exploited by malicious actors so love 4J was so serious that they had to um consult with the US government to try to help fix this problem and so this was a big problem um with just kind of maintaining the security of software but also it exposed a big hole in the open source model and so this is still something that probably requires meta governance you know some organization between like the federal government and open- Source organizations to solve the second is called the Mak or taker problem which is basically where some organizations make open- Source software and other organizations fork or copy the software for their own purposes thus being a taker and the problem here is where that relationship is not reciprocal that you know the makers will develop systems that encourage forks and then the open source contributors or the takers are expected to contribute back to the makers some sort of you know maintenance or something like that but they failed to do so and so this is the problem where you need to coordinate this reciprocal relationship and so an example of this is the word press for controversy so WordPress is the maker in this case and like I think it's WP engine is the taker in this case WP engine was forked from WordPress and it resulted in a fight between the two organizations one being the you know the maker and one being the fork or the Aker and they've been fighting over commercial issues and fair use issues that have served to break the Norms of Open Source so there's this primary issue of maintaining sort of a reciprocal relationship between organizations that rely on basically rely on each other to survive but you can have all sorts of you know fights and disagreements about different ways that the software can be used commercially and other types of use that you know are although are stated in the license sometimes evolve away from the license and their practical concerns that you know isn't going to be solved by just some sort of reciprocal relationship or even intrinsic and extrinsic motivations and then finally the final example is XZ backdoor autotools this was an exploit that was made in in an open source repository where it was it resulted from lack of maintainer vigilance so this maintainer in this particular case was burnt out and they didn't feel like really kind of closely inspecting every poll request which is where a contributor will submit a change or contribution to an open source organization and so malicious malicious actor was able to get exploits into the software and it became a huge security problem it's kind of like l forj and the problem here is it's you know the failure here is at the level of maintainer and it wasn't the maintainer fault necessarily it's that sometimes people in open source organizations as well as Open Access organizations open sites have become burnt out especially when you have to do all these things all these practice oriented things to realize the open vision and you know also you have disincentives that lead to problem so if you're not paying your maintainers enough or they not Mo you know if they don't have that intrinsic motivation high enough they can't do the prop the job that they need to do to make things work or the extrinsic motivations aren't well good enough you don't really pay or maintain her very much and therefore disincentivized to sniff out exploits and so these are our problems with maintaining open source in an especially in an academic environment this is a second uh YouTube video I wanted to point out this is behind the scene behind the screen there's a word play here which is open source software is both a tool and topic for research this is from William Schuler at Santa Fe Institute and he talks about sort of the history of software and how you know you can research open source software but also use it as a tool it has a lot of interesting issues in here so I recommend watching this video now I'm going to talk about how to sort of implement open software or like a purpose or an organization so we talked about you know one of these points about Distributing open academic software is being being able to run it properly or run it in a way that doesn't rely on too many dependencies and so this is where we can use something like Linux which or you know some flavor of Linux to package data analysis tools so that people can use them in an environment where they don't have to worry about dependencies they don't have to worry about having the proper software requirements so there are two examples of where we can use a Linux dist to package data analysis tools and present them to people in the community sometimes with fairly low technical standards so the first is narof Fedor and so this was created as free and open source software this is where we have a version of the fedora this Dr of Linux and we in Fedora you package a bunch of neuroimaging analysis tools and you put them in and you just give people the package they install it on their machine and then they can run those tools without worrying about dependencies or getting thing you're run on Windows or Mac or whatever the other example is n Debian this is run by incf this is the group led by por gleon and and Angus Angus um run by run by or Leon and and colleagues and this is where they have a Debbie in drro of Linux and they have these neuroimaging tools in Debbie and so you have Fedora and Debbie and you can choose between the two dros and you can run these neuro iming tools and these environments and that's that makes it easier I think for neuroscientists to do this kind of work to work with these pack which is uh a similar thing could be done for active inference and especially in Fedora you have these different flavors of Fedora like for artists or for musicians where you can package tools for those professions and they can just you know use it out of the box so focusing on the neurop Fedora uh project this is a special interest group whose goal is to provide Neuroscience researchers and enthusiasts with a strong easy to use ready made free and open source software platform for their work and anchor syha has let the taken the lead under aladora Morgan Huff is part of the orthogonal lab has also worked on this as well then we have neurodes so neurodes is not an operating system but it's a virtual machine it's a virtualization or a containerized environment where you have a basically virtual machine on your existing computer and you run everything inside of that and it really paves over a lot of the operating system differences and system requirement differences that you might find trying to install open- Source software on your home computer so like you know this is where you've basically created a controlled environment for the software so it really helps people run it when they have very little technical expertise and just get it to run that's all you need to do so neurodes uh is a containerized data analysis environment specialized for neuroimaging research and this is something that you could easily do with the active inference it's where you package the software code with the operating system libraries and dependencies required to run the code in a single lightweight executable so this is platform agnostic it's just you open up the container you run it and you can do everything inside there so thank you for your attention um I especially want to thank the members of the open source interest group group where Bradley alisah Jesse parent Morgan Huff Sarah Bala Brian mccorkel hu shle rvy Roger gopalan and Shu pomon um we have other people who cycle through the group from time to time and you're welcome to join us uh please join the worth Al research lab and then be on the lookout for our open source activities we usually do the interest group in the summer if you're interested get in touch other wise happy infering all right that was a great presentation this will be the last of the recorded presentations for the Symposium and last video on this part stream we have JF claer an experiment in artificial agency an experiment in artificial agency I'm Jean FR CER I'm a research fellow at the Active infin Institute and I do research in in cognitive robotics one of the principal question I ask myself is what does it take at a minimum for an autonomous robot to learn to survive in a world it knows initially almost nothing about other questions which follow is are can a robot act for its own reasons does it have agency can its agency be programmed and would such a robot be an active inference agent but one way to find out is to actually build an artificial agent uh and by artificial agent I mean a sense making adaptive and self-directed robot so this project is a continuation of a personal project that I started quite a few years ago and three years ago here's where I was I had a Lego robot uh with uh quite a a few sensors um like ultrasonic uh sensors infrared sensor um a color sensor luminum sensor um Motors of different types all this uh running uh from a Raspberry Pi with a board that that provides an interface between the Raspberry Pi and the various Lego motors sensors and the robot also had a 360 degree Beacon that allowed one robot to see another through uh the other robots 360 Dee Beacon Seeker so let's look at them in action I Nam them Carl and and D for I think quite obvious reasons the Carl had gone through quite a lot of training runs so it was better at finding the food which is the piece of paper on the floor um with a a beacon infer Beacon providing the scent the attractive scent we can see that Andy is uh having bit of a hard time and they see each other so that if one robot starts acting panicky which they do if they Collide too often the other will notice the pattern of movement and if one robot makes a beline then the other robot will perceive that and say aha that's where the food is and try to move in the same direction but we see them uh moving about and trying to get the food and uh eating the food and Paul is clearly uh more apt at this task than ND and what they've learned through various trials are the policies that um are most successful at finding the food so that's where I was um three years ago the way I implemented uh the cognitive capabilities quote unquote of these robots uh was based on a uh architectural model uh described by Marvin Minsky in his book The Society of Mind this architecture posits that interesting behaviors emerge from many simple actors that interact in simple ways so we're looking at cognition as a collective intelligence the Imp mentation was a hierarchy of these simple processes I call them cognition actors and they formed an abstraction hierarchy with at the very bottom um cognition actors that simply wrap sensors in uh effectors Motors and then at the very bottom we have cognition actors that care about very uh simple things like the location food the then at a higher level information obtained from the food location cognition actor feeds into the food approach cognition actor and this one is in tries to figure out am I getting closer to the food and so forth and so on other cognition actors uh are concerned with avoiding collisions and avoiding Dark Places now each cognition actor as I said is a separate process and each such process uh does very simple things it makes predictions about what it will observe next it then updates its beliefs based on uh predictions that were successful or prediction errors received and then selects a policy randomly in the beginning but then based on uh what was successful in the past and applies that policy Acts and then decides and then predicts what and what the next uh incoming Sensations will be and so forth and so on essentially implementing an UDA Loop observe Orient decide and act there's one big problem though that this Society of Mind these this hierarchy of cognition actors is predefined it's hardcoded for each robot it's the same the only thing that's learned is what policy is more effective over time based on experience so the robot acted for my reasons for the programmer's reasons not its own the robots had autonomy but no agency in 2022 I joined the active Institute and continued this project under the uh symbolic cognitive robotics um tag and I asked myself what if what if a robot started with only a rudimentary Society of Mind what if it didn't have that pre-built Society of Mind this this hierarchy of cognition actors it started only with the the the most basic ones the ones that wrap sensors and motors then the Society of Mind uh would grow from experiencing autonomous engagement so the robot would start and with all these very basic um prior cognition actors for sensing and affecting and then would create new cognition actors and these cognition actors uh would be very um very simple in the sense that they would take their uh Sensations from the the uh uh the sensors themselves and would act uh by uh engaging the motors and then uh observe how it changes its its its perceptions of say color or distance and whatnot Etc so cognition actors would be added to the Society of Mind BT them up to make sense of observations and actions now each new cognition actor would select an umelt composed of cognition actors one below and how do we know that a a a a particular level of cognition actors in This growing hierarchy is complete it's when all cognition actors one level below uh belong to an envel now it would not only grow but it would also shrink because whenever uh con actors were recognized to be hopelessly ineffective they would eventually be removed and new cognition actors would be would take their place and the hope would be that uh as this cognition hierarchy grows uh becomes more uh competent we would have a robot that starts acting uh intelligently and and in survivable ways so the Society of Mind would grow shrink and evolve as needed to survive now that's that's a key Point here um this is an implementation of um a form of mortal Computing obviously the the robot is physically the wheels uh the sensors themselves uh are not mortal they're they're like a a a a computer on Wheels you turn it off you turn it on it's it's still alive it's still it hasn't died by from having been turned off however the Society of Mind itself is Mortal it grows it evolves in such that the robot will be competent at finding the food it needs to keep the Society of Mind alive so what we have here is a um Immortal Society of life of of mind trying to um achieve a level sufficient level of Competency to maintain itself this is work in progress this is um system that I'm in the process of building and it's composed of these subsystems ageny where uh the smarts are for growing shaping uh and evolving a Society of Mind there's another subsystem that it runs on the robot itself which I call Body which provides a um an interface to uh activate uh Motors and to read sensors there's also a a a version of the body subsystem which will run in a simulated world and the subsystem world implements that simulated environment will also um animate an a real environment and then there's a subsystem that I call Observer which provides me or any uh other human being with a a window into what's going on with the within the Society of Mind of the robot the agency subsystem runs on a powerful computer the the body subsystem the physical one runs on the brick pie which is powerful enough to run the significantly complex programs but not two computationally intensive ones and um world and observer and and body simulated run on a um a separate computer the karma body subsystem as I said gives access to real and virtual sensors and affectors and here I have all the um the sensors I want to use and the motors um removed from a a the robot just like connected directly to the uh rickp and they are uh they can be uh activated they can be uh reached uh via an um arrest API which is essentially um they exposed a a a web interface uh to the agency subsystem the karma world subsystem uh here we see the simulated environment it's a grid World um the the gray tiles represent just the the surface and with a brown tiles representing obstacles so the the robot will bump into these obstacles and will perceive them with its uh simulated distance sensors the green patch is represents a Food Patch and the robot needs to find food in order to consume the food and replenish its its resources if it runs out then it dies the uh we can see that some ts are are uh darker others lighter and they this gradient this luminous gradient corresponds to um a a scent gradient so the closer the robot is to the food mood the lighter the tiles are there will also be um poison patches which will be red so detectable as a a red surface on on the simulated floor the robot must learn to avoid poison food because that will harm it and if it's uh armed uh if it's um excessively harmed then it will also die when a robot spends a enough time over the food the food is consumed and uh another Food Patch appears somewhere else so the the environment is dynamic and it changes with the uh with the its interactions with the robot that's an important uh element here now the um the karma agency subsystem is the most complex and in the the key subsystem and it implements a um generative cognitive architecture uh for the robot implementing agency so we're going to it's going to implement um cognition actors as we've mentioned already it will Implement something I call metacognition actors a very limited uh understanding of the the the concept of meta cognition one metacognition actor per level in the hierarchy and the cognition actors are responsible for adding removing uh cognition actors as uh needed as required then very very importantly we'll have well-being actors and these actors are represent the um homeostatic values that the Society of Mind must keep in and within a acceptable bounds in order to survive we have fullness integrity and engagement we'll talk in more details about what they are later and then very importantly uh there's an app perception engine actor uh last year at the Symposium I presented the um my implementation of an a perception engine and this is the um very important part of uh component of the Society of Mind it is responsible for making sense of observations in order to be able to make predictions we'll go into more details well now let's go and look at these well-being actors they Implement uh homeostatic uh States and I have three one is fullness which represents how much energy the uh the agent as a whole robot has available to to do things that could be Computing doing some comp uh expensive computation or um just moving about so there's a an energy budget that gets depleted and also gets replenished when the robot finds food and spends enough time in a Food Patch there's also the Integrity well-being actor and that represents uh the health of of the uh agent the if the agent the robot suffers too many collisions then Integrity goes down and integrity is is recovered over time think think the think about lives in in a video game that's pretty much the same concept and finally a little bit more abstract uh well-being actor is engagement and it represents how to what degree the robot is engaging its environment we want to avoid the dark room situation where the robot U stops doing anything uh stops moving stop engaging its environment it h it it has a it it feels quote unquote an imperative to engage its environment now the agent as a whole meaning the Society of Mind of the agent strives to maximize well-being for the entire Society of mind now low well-being levels will signal risk to the survival of the Society of Mind so that information will be available to each individual cognition actor and to the metacognition actors and will modify uh constraint uh constraint what they do key concept here is that well-being imparts normativity to beliefs beliefs acquired in the context of a um low fullness or low Integrity becomes a uh unpleasant undesirable belief now this this this normativity Associated to beliefs will focus attention and motivate action as we'll see later on so well-being actors aggregate metrics uh received from the various cognition actors and then aggregate these metrics and broadcast uh wellness events back to the cognition actors creating a give essentially informing them of the um ostatic State uh of the agent as a whole now these metrics include the work that's been done the food that's been consumed these these two will impact the fullness um values then uh poison consume Collision count will impact the Integrity overall Integrity of the agent and then uh Pol predictions made predictions received prediction errors sent policies executed we'll see that in more detail later all these are indicative of the uh Society of mine engaging its environment ment now metal cognition actors their role is pretty straightforward they're responsible for adding cognition actors that are missing and to remove underperforming cognition actors each cognition actor is assigned one level in the abstraction hierarchy and it looks at the conion actors under its supervisions supervision and and determines if all the con actors below them are part of one or the other's umelt so the notion of umelt here a cognition actor can only observe or act within its umelt and what is the umelt of a cognition actor it's other cognition actors of a lower level abstraction so at the the first level of ction actors the prior level uh where we have the um where they wrap sensors and Motors um they don't they they don't have a novelt well they do it's the actual external environment of the robot they don't have an internal enel anything above this level uh the cognition actors have as part of their umel other cognition actors and they constitute essentially the internal umelt of of within the Society of Mind so if a a one layer of the um Society of Mind is complete it covers all the conion actors below then it's time to create a cognition actor at the level above create a new layer of the Society of Mind and when you create a cognition actor you give it or the metacognition actor gives it its umelt mostly randomly with um the constraint that there's going to be overlap between umvs of various cognition actors and making sure that they have access uh to the motors directly or indirectly now when when well-being is low within the Society of Mind when the well-being actors are reporting low well-being levels this might accelerate the removal of cognition actors the metacognition actors being aware that we're not doing well now we'll say okay which cognition actors are underperforming our expectations and let's remove them and in order to for them to be replaced with new Cog actors which hopefully will do better and if uh well-being is low this will also slow down adding new cognition actors so there's an influence of well-being on the the actions of the mid factors now I want to say adding and removing um it's important to uh to to understand that adding is bottom up and removing is top down for example if I have a a meta if I have a cognition actor that has let's say two other contion actors in itselt I cannot remove a con actor in its unv because it depends on it but I can remove a cognition actor that is not in any others umelt so that removal will be from the top to bottom and addition will be from the bottom to top what does a cognition actor do well a cognition actor observes its umelt of other cognition actors it generates beliefs from analyzing its obs ations and then it generates policies in order to modify or validate its beliefs and it does this this observe believe and act always within itselt and one time frame at a time so each cognition actor goes through its more or less UDA Loop one time frame at a time now the the lower the lower a conition actor is in the uh Society of Mind the more concrete it is the less abstract it is um the shorter this time frame will be it will cycle faster it's important also to realize that there there's no there's um there's no synchronization between cognition actors each starts its time frame independently of other cognition actors and so that the they they may overlap but they don't coincide they don't overlap completely and the more abstract the cognition actor is the higher up it is in the hierarchy of cognition actors within the Society of Mind the longer its time frame will be which means that very abstract cognition actors will update their beliefs more slowly and will take actions uh not as quickly as a a more concrete low-level cognition actor so we can see here the the cognition actor the large one surrounded by its lowlevel cognition actors the what does it observe what does the con cognition actor observe in itsel well it observes it observes the beliefs of its of the C actors in its zelt so as a more abstract cognition actor I observe the beliefs of the more concrete cognition actors in my unel then based on these beliefs and as we'll see based on the normativity associated to the beliefs are they good beliefs are they bad beliefs the cognition actor will uh um decide on what its goals are in this current time frame the goals would be modifying a particular belief or validating a particular belief then it will generate policies to attain this goal and then emit this policy to its envel and say okay do what I want and we'll see in more details what do what I want means So within each time frame a cognition actor Will predict itselt after having made sense of it that's a key uh aspect that we'll we'll go into more details it will update its beliefs by analyzing past and present observations so a a a cognition actor has a memory of the current observations in this the current time frame and observations made in previous time frames and it will update its beliefs but from analyzing current observations and past observations then it will select a goal affecting a belief in it tends to impact and construct a policy and recruit its invel to realize it so if you look at the little U redesigned UDA Loop here first there's predict make predictions about beliefs of of other cognition actors in uh my umelt in order to predict I need to have a causal model of my observations that's what a perception will do we'll see in more details then the as a cognition actor I update my beliefs select goals act and back to predicting in the next time frame a perception is defined here as the process of assimilating sensory information into a coherent unified hole so let's look at uh predicting what are the observations of a cognition actor observations are predictions it makes about the beliefs of other cognition actors which are in its umelt plus that plus prediction errors so it makes predictions of itselt of contion actors in sunv about their beliefs and the umel cognition actors receive that prediction look at it does it corresponds to my current beliefs if yes not saying anything I'm not contesting the prediction if it doesn't match then I emit a prediction error and the cognition actor accumulates these predictions that are uncontested and the prediction errors and together they constitute the Cog actor's current observation ations so we'll see that not all observations receive the same attention that they're not subject they're not equally subject to prediction there's a prioritization there's a notion of attention we'll see that a bit later attention attention prioritizes which observations to update so the cognition actor starts a time frame with observ observations made in previous frame and if there and it wants to make to make those observations current to the current time frame as many of them possible but there's a limited amount of time in which to um update these observations so it the Cog actor will need to prioritize which observations from the previous time frame to update into the more current observations so which one will it try to update first well it will look at predictions that were made by higher level predic uh cognition actors about its beliefs and knowing which observations support the beliefs will say ah these is are the more these are the observations that um more uh abstract more important cognition actors are making about me my state my beliefs so I'm going to I'm going to try to make sure that I have up-to-date observations that are of interest to my parent my my parent conition actors so that's the first one I care about the first observations I care about the ones that are of interest to my parent function actors actor sorry or actors the second most important observations in in terms of uh updates are the observations that support the strongest desirable and undesirable beliefs remember that uh beliefs are the result of analyzing observations current and past if I have very strong desirable or very strong undesirable beliefs then I will want to verify that those OB observations are still current so there will be the next uh observations that I will try to uh update then finally the the observations that uh support weak or neutral beliefs and then whatever remaining observations they are so the cognition actor will take will prioritize the observations it it it that existed from the previous time frame and say okay I'm going to now make predictions um that these observations are still valid or not valid and if those there are obser prediction errors then then the the the observations are updated to the prediction error and if the observation is not contested then the observation is is is um the one that is current this implements a predictive processing uh architecture because as you can imagine a high level cognition actor will make predictions about beliefs uh of the the its children actors this will trigger predictions about the observation supporting dis beliefs one level down to the conition actors in the layer below and all the way down so predictions flow down and prediction errors flow up through the entire hierarchy of uh the conition actors in The Society of mind now in order to make predictions we need to have made sense of uh observations that's what Apperception does so we need to make sense of observations in order to predict them now in order to to explain the concept of app perception in which in last year's presentation I went into quite a lot of details but just I want to give a an overall understanding here let's take the example of observing uh two lights light a and light B and the light can either be on or off and at different steps in time the lights may change their state from on to off off to on or not so here we have a set of observations of light A and B over time and sometimes we may not be able to observe a light like uh light a in in the fourth state is a gray rectangle we don't know but we have a good enough set of observations and now the goal of the app Apperception engine is to find a causal theory that explains how the observations change time why are we seeing some light a turn on after having been off or Etc in in order to formulate a causal Theory kind of a little program that when run recreates what we've observed the Apperception engine may need to imagine another light it might need to imagine some kind of connection between the lights so the Apperception engine will in in formulating a causal Theory will do some abduction some imagining of unobserved latent objects so we have a number of observations that are remembered from previous time frames we may uh we we don't accumulate all observations forever we forget the older ones and then we feed these observations into the a perception engine out of that comes a little program a causal theory that when applied to the latest observation will produce the next incoming expected observations so that's a way of of of making predictions of incoming uh observations from current and past observations from the current observations given a causal Theory so you feed the latest observation you feed all the observations into the Apperception engine it gives you a causal Theory you feed the latest observation into the the causal Theory as input the output is incoming predicted observation so an app perception Apperception means finding a causal explains by all the rules of the theory and predicts observations so a cognition actor accumulates observations wants to make sense of them in order to predict incoming observations and for other reasons as well which we will see and it will what it does is okay it it gives the app perception engine actor its observations over time and says please explain to me what I'm observing give me a causal Theory so I can predict future observations given this causal theory if if one is found successfully then the con actor can predict in coming observations and what does it observe it observes the beliefs of cognition actors in itselt more more concrete contion actors and then if the prediction is correct if the causal theory is correct and accurate we don't get a prediction error if the causal theory is partially correct sometimes correct then we get a prediction error so getting prediction errors undermine confidence in the causal theory that the con actor has received from the aips and engine to explain what it sees in its umelt so it's quite possible that there's a moment in time when the observations when the causal theory is is quite effective and it's accurate but then something changes in in the state of the umel function actors such that the causal theory is is is missing is is very hit and miss and it may come a point where the contion actor says I don't like that causal Theory anymore it's it's producing too many prediction errors so it's going to ask the a perception engine actor give me a new causal Theory based on the latest observations that I've accumulated In My Memory okay so we looked at um how cognition actors predict income observations in order to uh accumulate those new observations to update those new observations based on whether or not these uh predic the predictions uh are contested by prediction errors or not so now once we have a new set of current uh observations within this time frame that we are operating in it's time to update beliefs beliefs are if you want um integrate are are the integration of observations into a a a more aggregated um analytical uh assertion so a cognition actor updates its belief from analyzing remembered observations and also uh remembered actions that it that that that it took so and we'll see what kind of of integration of observations and Analysis of observations lead to what type of beliefs but whatever type of belief it is a belief will be desirable or undesirable depending on how it correlates with the agent's well-being if a belief is being held mostly while the a the agent as a whole is feeling bad it has low resources or low integrity or or or limited engagement then these these beliefs are going to be tainted by the uh wellbeing um levels of the agent at the time those beliefs are held now that's what makes a belief desirable or undesirable what makes a belief strongly held is if it is supported by observations those observations from which analysis the belief is derived if it is supported by observations that are predicted by an accurate causal Theory we have an excellent causal Theory it's it's it rarely misses so um the the cognition actor is feeling confident about its observations and it makes the the whatever belief is is is produced from analysis of these successfully predicted observations better supported more strongly held now if a belief persists over many time frames thanks to or in spite of actions taken action taken to validate or invalidate it if the belief persists then it's even more strongly held it's it's it's there's more there's increased confidence that the belief is uh represents uh what's really going on in the umelt of the cognition actor now there are different kinds of beliefs beliefs are are are generated are produced from the analysis of uh current and past observations and okay so there's adduction one type of belief is is abduction and it's a direct consequence of um having requested a causal Theory and using a causal Theory remember in in a causal Theory May um imagine a a latent object or a latent relationship between objects well that's that's a form of belief I believe there's an object that I don't see but must exist in order for me for for a causal Theory to make sense of actual observations I made so an abduction belief would be like I assume there's a hidden thing next to the one I see another kind of belief is just counting I'm I'm I'm next to two things I have made two observations that represent uh say being next to something and so I have two of them so I'm I can have a belief that I'm next to two things account belief now a trend belief is a it's a belief that represents the the the um discovery of a trend in a c certain observations over time for example uh if um the contion actor observes distances to some obstacle and those distances are getting smaller um over time then that's a trend my distance to a thing is getting smaller and then there's another kind of belief is is some observation ending stopping my getting closer to a thing stopped so that's an ending kind of belief and then if the conion actor has uh taken some action well there's the belief that it tried to do something like I I tried to stop my distance to a thing from getting smaller so if the action was trying to end the belief that the belief in a trend as you can see now that beliefs are are are composed they're getting more and more abstract so I tried to stop my distance to a thing from getting smaller is a kind of another kind of belief it's an attempt kind of belief so we have all these different kinds of beliefs and when a Cogan actor synthesizes the belief from analyzing observations it assigns of belief a globally unique predicate name parent cognition actors the cognition actors in which umelt this cognition actor belongs um the parent conition actors observe predict the beliefs of cogn factors in in in their owns and reference them by name so if the there's a belief in umel actor that some Trend exists the um we're getting closer to an obstacle well the this belief will be exposed will be made available for for observation to the parent as a named uh predicate why and how that belief uh came about what the observations are by the umel K actor that led to that belief it's it's it's opaque there's a name and that's the name that's available for observation there's an a statement a predicate a name predicate that's available for observation by the parent cognition actor only the cognition actor that holds the belief it named knows which of its own observations support the belief and how so if I say to you I believe I I I believe getting closer that's something that can you can observe as a parent contion actor ah my child con actor believes in getting closer but the parent cognition actor doesn't know why I believe we're getting closer doesn't have access to the set of observations that led to me holding the belief that we are getting closer all these distance observations that led to me having holding the belief that getting closer all that parent knows is that I hold the belief called getting closer and it's able to observe it observe it you know ending or or starting Etc but that's that's that's a key um a key concept that beliefs are are essentially named um there are predicates their names if you wish that uh hide the C the reason why the belief is held to parent function actors that observe my beliefs so there's information hiding going on so a cognition actor observes the beliefs of its children cognition actors these these cognition these these beliefs that are observed are the consequences of themselves observations about from of of actors one level below so my belief is is uh another if I'm a cognition actor my belief is another cognition actor's observation so it's really beliefs it's it's beliefs about beliefs about beliefs so it's beliefs all the way down that's that's a maybe a little tricky concept if I'm at the top as a cognition actor I observe beliefs of connection actors under me these beliefs of connection actors under me are supported by their observations of beliefs of cognition cognition actors under them so beliefs about beliefs about beliefs time to open a little parenthesis here maybe take a break the implementation that I'm working on of agency is a symbolic implementation so it's its agency realized from symbolic reasoning all the various agency actors the the cognition actors the mid cognition actors the well-being actors the uh the perception engine actors all run prologue programs prologue is the programing language and those pro programs do symbolic infome a cognition actor as we've seen invents symbols invents gives names to beliefs it discovers from analyzing observation so it invents symbols and it Sy synthesizes these beliefs as symbolic statements getting closer to XYZ the app perception engine as we've seen very kind of briefly its job is to discover caal models and these caal models are symbolic programs they're little logic programs so agency is built on symbolic reasoning in my implementation closing parenthesis now let's talk about precision and belief strength the more accurate a causal Theory the more precise it are are are considered its predictions so we saw earlier now precise predictions if they're not contradicted by by PR errors do strengthen the beliefs the predicted observations support so you have a bunch of observations held by the observations in the moment remembered observations these observations are the result of predictions and or prediction errors and we analyze these these these observations coming from predictions to generate beliefs I'm getting closer to an obstacle from observations of distances now if my observations of distances were predicted by a a really top-notch causal Theory one that almost never misses then I have confidence in the beliefs that have derived from my predictions I I feel confident that I am indeed getting closer to an obstacle given all the the the the distance observations that I've accumulated because I've accumulated them by making predictions using a very accurate empirically accurate causal Theory so precise predictions strengthens the belief that our der from these from observations from predicted observations now the stronger beliefs are because I have may have strongly held beliefs and weakly held beliefs the strongly the more strongly held beliefs are the more attention they will receive When selecting when this the cognant actor needs to wants to select one and say I don't like this belief It's associated with poor well-being and I want to I want want to change that belief I want I want to impact it I want to I want to get rid of that belief I want to make that belief not true or if it's a very pleasant belief oh I want to make sure that it is I want to validate it I want to keep that belief uh I want to keep holding that belief because it's a good one but I care whether this the belief is strongly held or weakly held because strongly held belief will get more attention will be the ones that'll be selected for validation or invalidation above the the the weekly held beliefs so how do we impact beliefs that's the next step in the ud next two steps in the UDA Loop of our cognition actor within each time frame it predicted based on its understanding of past observations it updated its belief from uh from analyzing predict predictions that were realized or or errors so those predictions I.E those observations it updated its beliefs and now it says it needs to select a goal and or want may need to select the goal that say I want to get rid of this belief or I want to validate this belief and how am I going to accomplish this goal I'm going to select the policy and then and and then ask my unal to act to realize this policy so that's what we're looking at now so a cognition actor Acts via policies it formulates a cognition actor strives to invalidate unpleasant beliefs and to validate Pleasant beliefs but it will prioritize strongly held beliefs whether unpleasant or or Pleasant a cognition actor will impact the belief validate or invalidated it holds by uh promoting or inhibiting the observations that led to it now I have a belief that we are getting closer to an obstacle and somehow it's it's it's an unpleasant belief maybe uh It's associated with in the past with um a a decrease in in in um Integrity leads to bumping into things so I want to I want to put an end to getting closer to an obstacle I want to I don't want this belief that I'm getting closer an obstacle to go away now I hold this belief because I had observations of reducing distances over time to some obstacle so I want to change these observations basically I want to I want to stop getting close having um having uh distances that get smaller I want to stop this so my observations of distances happen to be beliefs of function actors in my umelt so what I'm going to do I'm going to say okay I'm going to select a number of of observations that support the belief I want to get rid of and I'm going to formulate a policy as a list of goals to my invel CA I'm going to say I want you to get rid of the fact that I have small distances to an obstacle that's that's that's your goal so I'm going to I'm going to set a number of goals me the cognition actors who believes we're getting closer to something and it's not does feel good I'm going to tell I'm going to formulate a policy such that my umelt con actors will try to um impact their own beliefs that we're we have a small distance to an obstacle for example so that's that's a little bit subtle here so I'm as a cognition actor I want to modify a belief that belief is based on observations these observations are beliefs of umelt cognition actors so I'm going to say to my umelt cognition actors I want you to to modify your own beliefs that I've observed I want you to change your beliefs beliefs that I'm observing that lead me to believe that I'm getting closer to an obstacle so I'm I'm setting up a bunch of goals for my umelt function actors so that if they realize if they if they modify the these beliefs then it will modify indirectly my own belief that I'm trying to get rid of okay so I'm basically telling my my umelt these are the goals I want you to realize so that I it I would indirectly realize my own goal okay so this Pol a policy that I put together are a set of goals for my umelt actors to realize now A nelt C actor will then given the goal it receives formulates its own policy because we're I'm asking a con actor in myelt to modify some some belief or other okay says the con actor then I will formulate my own policies to do that to to achieve the goal that you that my parent C actor sent to me so a cognition actor will receive goals from parent cognition actors and these goals received from IR up will have higher priority that whatever goals the cognition actor might want to formulate on its own because the umel actor has its beliefs has belief that it wants to validate has belief it wants to to to validate and left on its own to its own devices will do just that but if it receives goals from a parent ction actor these take priority and it it will put aside its own concerns to uh address the concerns of the parent cognition actor now a cognition actor can execute only one policy at a time which is why it would prioritize the ones that it receives from its parent and when all is said and done the cognition actor that initiated the policy will determine will will verify if this was successful or failure over time if the belief they wanted to impact was indeed impacted and a policy policy that's known to work will be reused that's how contion actors develop habits okay so where does the causal theory that the con actor uh holds guide the formulation of its policies because there was CA caal causal Theory serves two roles one is to predict incoming observations and the other role is to formulate an effective policy so let's say that cognition actor is a belief that wants to impact because uh the well-being me levels are are

are low Associated to that belief so he wants to modify that that belief get rid of it well it needs to formulate a candidate policy which of the observations that I made that I've analyzed into this belief which which observations that support this belief I want to modify so which policy do I want to formulate well what it will do is we'll say okay what do I need to do given what my what I'm observing right now so that I will observe something differently that will either remove the belief or or reduce it so it will use the causal Theory so what would happen if I did X if I I I I I if I asked my umelt to do XYZ so so the causal Theory will answer that question we say well plug that into the causal Theory turn the crank and this is what you would be uh be observing next if you were doing this if you if if you executed this action this is what you would observe next and then given what you would observe next you can calculate ah this these will be the anticipated beliefs out of these anticipated observations might the policy work with would the belief that I want to impact be impacted the way I want to impact it if so yeah let's use that policy otherwise let's look for another one so the cognition actor searches for an effective policy given its understanding of how things work in its umelt its causal Theory and and then applies the ca theory predicts what would happen if it took this action if it it it it if it requested this goal to be realized looks at the the the the resulting anticipated beliefs and decides was this a a a good effective policy and then and then it would Emmit that policy and ask it umelt please achieve these goals so that I achieve my goal so and and that percolates down because each COG actor being given a goal from the parent con actor says ha how should I best realize that goal what policy should I formulate for my own child my own umelt in order to realize the goal that was given to me by my parent so it's policies all the way down and it bottoms out when at the level where um effects are produced moving Motors turning moving forward moving backward whatever okay so that's a lot of um a lot of content describing this cognitive architecture this generative cognitive architecture what I hope will happen is through this growing shrinking evolving um um defining of this Society of Mind composed of of cognition actors um with uh in the context of of of of homeostatic measures provided by actors I'm I'm hoping that with policies going down beliefs percolating up predictions going down prediction is going up all this this this this changes and evolution of the Society of Mind all these all these activities from all of this would emerge agency and I I'm looking at that as as a form of operational closure with uh constraints flowing down constraints flowing up and eventually hopefully the um the Society of Mind will explore a space of possible organizations and and and and and and and space of of possible uh ways of defining cognition actors relating cognition actors together such that it leads to behaviors of the robot that preserves the life of the Society of Mind that Society of Mind never reaches a point where it runs out of fuel where it is damaged Beyond repair that's that's the overall goal I think of the Society of Mind as kind of autopoetic with all these um elements of the Society of Mind leading to other elements for example a a causal Theory will lead to predictions predictions will lead to prediction errors prediction errors will lead to observations uh observations will be fed into apperceptions which will lead to an updated causal Theory and and observations lead to to beliefs um beliefs lead to policy um a causal Theory will determine which policy is more likely to work policy leads to effect effects lead to New Sensations Sensations lead to new observations and so forth and so on so this I look at this as a a kind of a um a a set of of of processes that support each other and in in something of a of a closed set of complex Loops leading to a Society of Mind that survives over time now there's a number of pH philosophical stenes that are taken by this project and by philosophical stance I mean uh practic pragmatically Justified perspectives a way of seeing world first I look at active inference as a normative framework I don't think that's controversial at all I mean that's that's what it's defined to be and uh to persist an agent must behave as if it acts to minimize surprisal now clearly I'm not implementing the uh Society of mine using the mathematics of the free energy principle but I'm implementing it hopefully such that doing an active inference analysis of the robots observable behaviors in states will demonstrate that the uh Society of Mind the agent as a whole is acting such as to reduce surprisal so I'm implementing an agent and also its environment as a system of generative processes the active inference analysis will be done in terms of generative models that are inferred uh from observing the behaviors of the robot and uh observing the states that are visible to an outside Observer other uh philosophical stenes I'm just going to go through them very quickly each one is worth an entire presentation I I I take the stance that cognition arises from an embodied agent engaging its environment that sense making is the Discovery of unified causal theories that meaning is grounded in the agent's drive to survive no mortality no meaning that intelligent emerges from interactions between a collective of parts that's the Society of mind and that the parts constrain the whole and the whole constraints the part so constraint closure I'm implementing I'm in the process of implementing this this this this system that will hopefully demonstrate artificial agency and here's the current progress there are three phases in this project phase one I developed in cognitive architecture that's what I presented today uh implemented a one version of the of the app perception engine and that's what I presented last year I implemented uh the subsystem to uh realize a to connect to sensors and actuators that's the karma body and I implemented a simulated grid World in which a simulated um agent would would operate and and and interact that's to uh speed up uh implementation because it takes a long time to uh if I were to do this only in the real world the um the the edit run debug Loop would be extremely long and so I'm in the process right now of implementing uh the agency uh capabilities in phase two um which would be about six months to a year from now I will have a a redesigned robot body I will um uh be testing in a real life and I will be collecting a lot of data through a a The Observer subsystem and in the third and final phase the uh data gathered from runs both in real life and in virtual world will be subjected to active inference analysis to uh validate that my agent does survive because it minimizes surprisal that's it um if you are interested you can contact me at uh this email address JF do Cloutier active. Institute you can also uh reach me on uh the Sim Cog robotic channel of the institute's Discord and if you're interested in the source code it is on GitHub so I'd like to uh thank uh the active INF Institute for uh support and encouragement